PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)

(Estd. Under the Panjab University Act VII of 1947—enacted by the Govt. of India)

SYLLABI

FOR

B.Sc./B.Sc. (Hons)/B.Sc. (Hons. with research)
Major & Minor Courses

THIRD YEAR
(SEMESTER SYSTEM)
EXAMINATIONS, 2026-27

(SEMESTER : FIFTH AND SIXTH)

i.e

Fifth Semester: November/December, 2026
Sixth Semester: April/May, 2027

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AGRICULTURE

NEP 2020 COLLEGES SCIENCE TEMPLATE: STUDENTS OPTING FOR SINGLE MAJOR AND SINGLE MINOR
B.Sc., B.Sc. Honours and B.Sc. Honours with Research in Agriculture (Table 1*)

V	L-300	AGR_DSC-9 Maj-501	Cultivation Practices of Vegetables	3+1=4	24
		AGR_DSC-10 Maj-502	Agroclimatology	3+1=4	
		AGR-DSC-11 Maj-503	Animal Husbandry	3+1=4	
		AGR-DSC-12 Maj-504	Soil Physics & Conservation	3+1=4	
		AGR-DSC-13 Maj-505	Food Technology	3+1=4	
		AGR-INT/COM-506	Internship/ Community Work	4	
VI	L-300	AGR-DSC-14 Maj/Min-601	Fruit growing	3+1=4 (x2)	24
		AGR-DSC-15 Maj-602	Diseases of Crop Plants	3+1=4	
		AGR-DSC-16 Maj-603	Dairy and Poultry	3+1=4	
		AGR-DSC-17 Maj-604	Landscaping & Floriculture	3+1=4	
		AGR-DSC-18 Maj-605	Sericulture and Apiculture	3+1=4	
36x2 (Maj)+32(Min)+9(MDC)+8(AEC)+6(VAC)+9(SEC)+4(INT/COM)+4(PBI/HCP)					144
The student successfully securing 144 credits will be awarded B.A./B.Sc.degree					

Notes:

1. In the first semester student is offered three different subjects A, B & C.
2. At the end of second semester each student has the option to choose any one of the two streams namely Single Major or Multi-Disciplinary. The student choosing Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student who chooses multi-disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3-year degree) shall be eligible for 4 YEAR DEGREE B.Sc. HONOURS with research, on the condition that the Institution is a PU approved Research Centre
5. The student opting for Single Major must secure 50% credits of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for Minor in a particular subject, the student must attain atleast 12 credits in that subject(s).
6. The student exiting the programme after one year securing 48 credits will be awarded UG Certificate provided he/she secures 4 credits from summer internship/community work. Further, the student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secure 4 credits from summer internship. The student successfully securing 144 credits will be awarded BA degree.
7. Language paper chosen as MIL-1 and MIL-2 should be in the same subject.
8. Once Major and Minor papers are chosen in 3rd semester, the same will continue as Major and Minor subject in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write Dissertation of 08 credits in one of the two Major subjects.

AGRICULTURE

B.A./B.Sc.(General) Agriculture Major/Minor

Course Title: CULTIVATION PRACTICES OF VEGETABLES **Course Code: AGR_DSC-9 Maj/Min-501**
Semester: Fifth

Objective: To give student basic knowledge of principles and practices for raising vegetable crops.

Max. Marks: 100 **Course Credits** = 3+1=4

1.	Theory: 75	Written Exam: 67	Internal Assessment: 8
2.	Practical: 25	Written Exam: 22	Internal Assessment: 3

Periods per week: 1. Theory- Four of 45 minutes duration (Three hours)
2. Practical- One of two hours duration

Instructions for Paper Setters:

Question Paper will have four units. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY** question consisting of 10 short answer type questions of one and a half mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

Instruction Aids: Teachers can use lecture, lecture-cum-demonstration, project, assignments, problem solving, inductive and deductive methods and can use L.C.D., L.E.D., and multimedia projector in addition to black/white board.

UNIT – I **12 Hours**

Scope, Importance and Types of vegetable growing.

Climatic requirement, Soil requirements, Improved varieties, Agronomic practices (land preparation, seed rate & seed treatment, weed control, fertilizer application, irrigation) and harvesting of Summer squash (Chappan Kadoo) , Pumpkin (Halva Kadoo), Bottle gourd (Ghia Kadoo), Bitter gourd (Karela) Sponge gourd (Kali Tori).

UNIT – II **11 Hours**

Climatic requirement, Soil requirements, Improved varieties, Agronomic practices (land preparation, seed rate & seed treatment, weed control, fertilizer application, irrigation) and harvesting of Cucumber (Kheera), Long melon (Tar), Round gourd (Tinda), Tomato (Tamatar), Brinjal (Baingan), Chilli (Mirch)

UNIT – III **11 Hours**

Climatic requirement, Soil requirements, Improved varieties, Agronomic practices (land preparation, seed rate & seed treatment, weed control, fertilizer application, irrigation) and harvesting of, Sweet pepper (Shimla Mirch), Okra (Bhindi), Onion (Piaz), Garlic (Lashan), Pea (Mattar), Carrot,

UNIT – IV **11 Hours**

Climatic requirement, Soil requirements, Improved varieties, Agronomic practices (land preparation, seed rate & seed treatment, weed control, fertilizer application, irrigation) and harvesting of Cauliflower (Phulgobhi), Radish and Turnip, Palak, Coriander (Dhania), Kasuri Methi, Potato (Aloo).

GUIDELINES FOR PRACTICAL EXAMINATION

Time- 3 Hrs **Max Marks: 25**

Practical: 22 **Internal Assessment: 03**

1. Vegetable crop identification based on seed and morphological characteristics.
2. Preparation of the field for sowing/ transplanting of various vegetable crops.
3. Practice in different methods of fertilizer application, hoeing, other cultural operations
4. Sowing of important Vegetable crop.
5. Preparation of calendar of operation for various vegetable grown in Punjab.

Suggested Readings:

1. Dhaliwal M.S., *Handbook of vegetable crops*, Kalyani Publishers, Ludhiana, 2008.
2. Das, P.C., *Vegetable crops of India*, Kalyani Publishers, Ludhiana, 1993.
3. Chauhan, D.V., *Vegetable production in India*, S. Ram Prasad and Sons, Agra, 1993
4. *Package of Practices for Vegetable crops,* P.A.U. Publications Ludhiana, Corresponding year.

B.A./B.Sc.(General) Agriculture Major

Course Title: AGROCLIMATOLOGY **Course Code: AGR_DSC-10 Maj-502**

Semester: Fifth

Objective: To give student basic knowledge of weather parameters and their influence on agriculture.

Max. Marks: 100 **Course Credits** = 3+1=4

1.	Theory: 75	Written Exam: 67	Internal Assessment: 8
2.	Practical: 25	Written Exam: 22	Internal Assessment: 3

Periods per week: 1. Theory- Four of 45 minutes duration (Three hours)
2. Practical- One of two hours duration

Instructions for Paper Setters:

Question Paper will have four units. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY** question consisting of 10 short answer type questions of one and a half mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

Instruction Aids: Teachers can use lecture, lecture-cum-demonstration, project, assignments, problem solving, inductive and deductive methods and can use L.C.D., L.E.D., and multimedia projector in addition to black/white board.

UNIT – I **12 Hours**

1. Importance of weather & climate, Different branches concerning atmospheric phenomenon. Division of climatology.
2. Agro climatology:-definition, scope and agroclimtology as an interdisciplinary science.
3. The earth and its atmosphere:-structure & composition of earth. Motion of earth and their importance. Atmospheric characteristics, physical properties and composition of atmosphere.
4. Role of atmospheric air in agriculture. Vertical layers of atmosphere & their important characteristics, Vertical distribution of temperature and pressure.

UNIT – II **11 Hours**

1. Elements of Climate and weather (solar radiation, air temperature, pressure, moisture & wind). Latitudinal and Seasonal distribution of temperature and precipitation.
2. Factors of climate. Effect of temperature, pressure, wind, moisture, solar radiation on crops. Climate and crop growth.
3. Role of weather in sowing, growth, maturity and harvesting of crops. Climatic requirement of wheat, rice, barley, bajra, maize, cotton, gram sugarcane, sunflower, millets, tomato and potato.

UNIT – III **11 Hours**

1. Weather hazards (frost and high temperature, drought, floods, storms, tornadoes & water spouts, lighting, blizzard and earth-quakes).
2. Effect of weather in agriculture industry.
3. Modification of weather hazards. Field climate modifications. Role of shelter belts and mulches in microclimate modifications.

UNIT – IV **11 Hours**

1. Climatic classification based on temperature, rain fall, vegetation.
2. Climates of world with special reference to India. Agroclimatic regions of Punjab and India.
3. Role of weather and climate in sustainable agriculture- role of temperature, solar radiation, precipitation on sustainable agriculture. Combined effects of different climatic parameters on crop production.

GUIDELINES FOR PRACTICAL EXAMINATION

Time- 3 Hrs **Max Marks: 25**

Practical: 22 **Internal Assessment: 03**

1. Installation and use of various meteorological instruments for measurement of different weather parameters.
2. Study the principle and working of various meteorological instruments.
3. Study of wind direction and climatic water balance.
4. Measurement & significance of atmospheric humidity, rain fall and evaporation over an area. Interpretation of climatic data in relation to crop weather relationship.
5. Study of climatic requirement of important different crops & their distribution and yield in India.
6. Study the important crops yield sensitivity to weather fluctuation.
7. Recording, processing and presentation of climatic data on Climatographs, Charts and Maps. Study of different factors in soil formation.
8. Meteorological satellite technology in weather forecasting.
9. Climatic maps of India and Punjab.

Suggested Readings:

1. Mavi, H.S., *Introduction to Agrometerology*, Oxford and IBH Publishing Co Pvt Ltd, New Delhi, 1994.
2. Ahrens, C.D., *Meterology Today*, Thomson Teaching Inc. USA, 2003.
3. Critchfield, H. J., *General Climatology,* Prentice Hall of India Pvt Ltd., New Delhi. 2002.

B.A./B.Sc.(General) Agriculture Major/Minor

Course Title: FRUIT GROWING **Course Code: AGR-DSC-14 Maj/Min-601**
Semester: Sixth

Objective: To give student basic knowledge of growing fruits plants.

Max. Marks: 100			**Course Credits** = 3+1=4
1. Theory: 75		Written Exam: 67	Internal Assessment: 8
2. Practical: 25		Written Exam: 22	Internal Assessment: 3

Periods per week: 1. Theory- Four of 45 minutes duration (Three hours)
2. Practical- One of two hours duration

Instructions for Paper Setters:

Question Paper will have four units. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY** question consisting of 10 short answer type questions of one and a half mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

Instruction Aids: Teachers can use lecture, lecture-cum-demonstration, project, assignments, problem solving, inductive and deductive methods and can use L.C.D., L.E.D., and multimedia projector in addition to black/white board.

UNIT – I **12 Hours**

1. Fruits, their importance and scope (Horticulture and branches of horticulture, nutritive value of fruits, Area & production of fruits in India, Scope and problem of fruit cultivation),
2. Selection of site and soils for planting an orchard.
3. Preparation of Land and Layout of Orchard (Land Preparation, Method of planting fruit plants, high density planting)
4. Propagation Methods. (Sexual and sexual propagation methods, Micro propagations)

UNIT – II **11 Hours**

1. Nursery Production (Care of young nursery plants, weeding/hoeing and plant protection measures).
2. Orchard cultural practices (Soil management practices, Fruit plant management practices)
3. Flowering of fruit plants (Blooming periods, Mode of pollinations, agencies of pollination, pollen viability)

UNIT – III **11 Hours**

Cultivation practices of Tropical Fruits viz: Mango, papaya, Guava, Sapota (Origin, history and distribution, uses, climate requirement, site selection, varieties, planning/planting, orchard cultural practices, training/ pruning and important plant protection measures

UNIT – IV **11 Hours**

Cultivation practices of Sub-Tropical Fruits viz: Citrus, Grapes, Litchi, Pomegranete (Origin, history and distribution, uses, climate requirement, site selection, varieties, planning/planting, orchard cultural practices, training/ pruning and important plant protection measures

GUIDELINES FOR PRACTICAL EXAMINATION

Time- 3 Hrs **Max Marks: 25**

Practical: 22 **Internal Assessment: 03**

Practice in propagation, methods, layout of orchard, transplanting, manuring, training and pruning of fruit plants.

BOOKS

1. Singh, R., *Fruits,* National Book Trust of India, New Delhi, 1969.
2. Hayes, W.B., Fruit growing in India, Kitabistan, Allahabad, 1945.
3. Bal, J.S. Fruit Growing, Kalyani Publishers, Ludhiana, 2021.
4. Lal, G., Siddappa, S. And Tandon, G.L., *Preservation of fruits and vegetables,* Indian Council of Agricultural Research, New Delhi, 2009.

B.A./B.Sc.(General) Agriculture Major/Minor

Course Title: Diseases of Crop Plants **Course Code: (AGR-DSC-15 Maj/Min 602)**
Semester: Sixth

Objective: To acquaint with diseases of crop plants and their management.

Max. Marks: 100		**Course Credits** = 3+1=4
1. Theory: 75	Written Exam: 67	Internal Assessment: 8
2. Practical: 25	Written Exam: 22	Internal Assessment: 3

Periods per week: 1. Theory- Four of 45 minutes duration (Three hours)
2. Practical- One of two hours duration

Instructions for Paper Setters:

Question Paper will have four units. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY** question consisting of 10 short answer type questions of one and a half mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

Instruction Aids: Teachers can use lecture, lecture-cum-demonstration, project, assignments, problem solving, inductive and deductive methods and can use L.C.D., L.E.D., and multimedia projector in addition to black/white board.

UNIT – I **12 Hours**

1. Symptom, causal organism, disease Cycle, management & control of diseases of Wheat (Black or stem rust, Brown or leaf rust, Yellow or stripe rust, Loose smut, Flag smut, Karnal bunt, Foot rot, Powdery mildew, Leaf blight).
2. Symptom, causal organism, disease Cycle, management & control of diseases of Turmeric (Leaf Spot, Leaf Blotch. Rhizome Rot)
3. Symptom, causal organism, disease Cycle, management & control of diseases of Sunflower (Root rot or charcoal rot, Leaf blight, Rust, Head rot)

UNIT – II **11 Hours**

1. Symptom, causal organism, disease Cycle, management & control of diseases of Cotton (Wilt, Verticillium wilt, Anthracnose, Bacterial blight, Leaf Curl Disease).
2. Symptom, causal organism, disease Cycle, management & control of diseases of Rice (Blast, Brown Spot, Narrow brown leaf spot, Sheath rot, Sheath blight, False smut, Stackburn disease, Bacterial leaf blight, Bacterial leaf streak, Rice Tungro Disease, Rice Grassy stunt disease, Rice yellow dwarf disease).
3. Symptom, causal organism, disease Cycle, management & control of diseases of gram (Ascochyta blight, Rust, Wilt. Powdery mildew, Anthracnose).

UNIT – III **11 Hours**

1. Symptom, causal organism, disease Cycle, management & control of diseases of Sugarcane (Red rot, Smut, Sett rot or Pineapple disease, Wilt, Rust, Gummosis, Red stripe, Sugarcane Mosaic, Grassy shoot, Ratoon stunting).
2. Symptom, causal organism, disease Cycle, management & control of diseases of Pearlmillet (Downy mildew, Smut, Rust, Ergot or Sugary disease).
3. Symptom, causal organism, disease Cycle, management & control of diseases of Maize (Downy mildew, Leaf blight, Rust, Charcoal rot, Bacterial Stalk rot).

UNIT – IV **11 Hours**

1. Symptom, causal organism, disease Cycle, management & control of diseases of Groundnut(Tikka leaf spots, Rust, Collar rot or seedling blight or crown rot, Root rot, Groundnut bud necrosis disease).
2. Symptom, causal organism, disease Cycle, management & control of different diseases of winter vegetables (cabbage, potato, onion and peas)
3. Symptom, causal organism, disease Cycle, management & control of different diseases of summer vegetables (brinjal, Chilli, cucurbits)

GUIDELINES FOR PRACTICAL EXAMINATION

Time- 3 Hrs **Max Marks: 25**
Practical: 22 **Internal Assessment: 03**

Detailed study of symptoms and host parasite relationship of representative diseases of the above mentioned crops. Collection and dry preservation of diseased specimens of important crops.

Suggested Readings:

Gupta VK &.Sharma SK. 2000. *Diseases of Fruit Crops*. Kalyani Publ., New Delhi.
Pathak VN. 1980. *Diseases of Fruit Crops*. Oxford & IBH, New Delhi.
Singh RS. 2000. *Diseases of Fruit Crops*. Oxford & IBH, New Delhi.
Walker JC. 2004. *Diseases of Vegetable Crops*. TTPP, India.
Package of Practices for Vegetable Crops, P.A.U. Publications Ludhiana, Corresponding year.

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BIOCHEMISTRY

NEP-2020 COLLEGES SCIENCES TEMPLATE: STUDENTS OPTING FOR SINGLE MAJOR AND SINGLE MINOR (TABLE 1*) AND STUDENTS OPTING FOR MULTIDICIPLINARY / INTERDICIPLINARY (TABLE 2*)
B.Sc., B.Sc. HONOURS AND B.Sc. HONORS WITH RESEARCH IN BIOCHEMISTRY

Table 1*						Table2*					
Sem	**Level**	**Course Code**	**Course Name**	**Credits (Th+Pr+Tut)**	**Total Credits**	**Sem**	**Level**	**Course Code**	**Course Name**	**Credits (Th+Pr+Tut)**	**Total Credits**
V	L-300	(BCH-DSC-6 Maj/Min-501)	Molecular Biology and recombinant DNA technology	3+1=4	24	V	L-300	(BCH-DSC-6 Maj/Min-501)	Molecular Biology and recombinant DNA technology	3+1=4 (×3)	24
								(BCH-DSC-7 Maj-502)	Physiological Biochemistry	3+1=4 (x2)	
								BCH-INT/Com-506	Internship/ Community service	4	
VI	L-300	(BCH-DSC-8 Maj/Min-601)	Immunology	3+1=4	24	VI	L-300	(BCH-DSC-8 Maj/Min-601)	Immunology	3+1=4 (×3)	24
		(BCH-DSC-9 Maj/Min-602)	Clinical Biochemistry	3+1=4				(BCH-DSC-9 Maj/Min-602)	Clinical Biochemistry	3+1=4 (×3)	
36x2(Maj)+32(Min) +9(MDC) +8(AEC) +6(VAC) +9(SEC)+4(INT/COM) +PBI/HCP)					144						
Thestudentsuccessfullysecuring144creditswillbeawardedB.Sc.degree											

Notes:

1. In the first semester student is offered three different subjects A,B&C.
2. At the end of second semester each student has the option to choose anyone of the two streams namely Single Major or Multi-Disciplinary. The student choosing Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student who chooses multi-disciplinary stream will continue with the three subjects and complete his/ her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (upto3-year degree) shall be eligible for 4 YEAR DEGREE B.Sc.HONOURS with research, on the condition that the Institution is a PU approved Research Centre
5. ThestudentoptingforSingleMajormustsecure50%creditsofthetotalcreditsintheMajorsubjectandobtain24creditsfromtheMinorsubject(s).Toqualifyfor Minor in a particular subject, the student must attain atleast 12 credits in that subject(s).
6. The student exiting the programme after one year securing 48 credits will be awarded UG Certificate provided he/she secures 4 credits from summer internship/communitywork.Further,thestudentexitingtheprogrammeaftertwoyearssecuring96creditswillbeawardedUGDiplomaprovidedhe/shesecure 4 credits from summer internship. The student successfully securing 144 credits will be awarded BA degree.
7. Language paper chosen as MIL-1andMIL-2 should be in the same subject.
8. Once Major and Minor papers are chosen in 3rd semester, the same will continue as Major and Minor subject in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write Dissertation of 08credits in one of the two Major subjects.

BIOCHEMISTRY

Semester V

Major/Minor

INSTRUCTIONS FOR PAPER SETTER AND STUDENTS:

1. Examiner will set total of NINE questions comprising TWO questions from each unit and ONE compulsory question.
2. All questions carry equal marks.
3. The compulsory question will consists of 12 short questions of one mark each covering the whole syllabi.
4. Besides Question No. 1, all other questions may contains 2-3 sub-parts.
5. Questions should be uniformly spread over the entire syllabus.
6. The students are required to attempt FIVE questions in all, ONE question from each unit and the Compulsory question.

(BCH-DSC-6 Maj/Min-501)/: Molecular Biology and Recombinant DNA Technology (3+1)

Marks: 75 (60+15 Int.)

Objective: To make students aware of Molecular biology principles especially replications and proteins synthesis and genetic engineering.

Learning Outcomes:
Upon successful completion of this course, students will be able to

- Comprehend the basic molecular level interactions between DNA and proteins
- Learn basic types of experiments related molecular biology

SECTION-I

(Lectures -10)

Prove for DNA as genetic material. Watson – Crick Model of DNA. A&Z-DNA, structure of chromatin topology of DNA. Organization of different genomes. Extra nuclear genomes.Biochemical basis of mutations. DNA replication: semiconservative, semicontinuous, bidirectional & from a unique origin. Enzymes & proteins involved in DNA replication. Mechanism of DNA repair.

SECTION –II

(Lectures-10)

Transcription in prokaryotes and eukaryotes: promoter sequences, initiation, elongation, Rho dependent and rhoindependent termination, regulation, processing, alternative splicing, mRNA editing & Inhibitors of transcription.Regulation of eukaryotic gene expression. Transcription factors: Zn fingers and leucine zipper.Hormonal control of transcription.

SECTION –III

(Lectures-10)

Translation in prokaryotes and eukaryotes: Genetic code, Wobble hypothesis, structure of tRNA, amino acidactivation, initiation, elongation, termination and inhibitors of translation, posttranslational modification of proteins. DNA bindingdomains of protein. Concept of operon, regulation of lac operon, trpoperon, phage infection (lysogenic & lytic mode)

SECTION –IV

(Lectures-10)

Recombinant DNA Technology: Gene cloning different types of vectors, libraries & isolation of gene. Yeast gal gene and mat locus.Analysing genome, genome size, genome sequencing, microarrays, DNA finger printing, SNPs and applications indiseases and forensic.

Suggested Books:

1. Freifelder's Essentials of Molecular Biology Malacinski GM 4th edition Narosa New Delhi 2005
2. Molecular Biology of Gene Watson JD 7th edition 2007 CSH New York 2013.
3. Genes XII, Benzamin Lewin, 2018 Oxford press Oxford.
4. Molecular cell biology Lodish H et al 8th edition 2016 McMilian Publishers, London.
5. Genomes IV Brown TA. Garland Science New York 2019.

PRACTICALS:

(BCH-DSC-6 Maj/Min-501)/: Molecular Biology and Recombinant DNA Technology

Marks: 25 (20 + 5 Int.)

1. Separation of serum proteins using electrophoresis.
2. Isolation of chromosomal DNA from E. Coli cells.
3. Demonstration of Western Blotting technique.
4. Extraction of RNA from yeast.
5. Estimation of RNA by Orcinol method
6. Agarose electrophoresis of DNA
7. Extraction of DNA from Onion

Major/Minor

INSTRUCTIONS FOR PAPER SETTER AND STUDENTS:

1. Examiner will set total of NINE questions comprising TWO questions from each unit and ONE compulsory question.
2. All questions carry equal marks.
3. The compulsory question will consists of 12 short questions of one mark each covering the whole syllabi.
4. Besides Question No. 1, all other questions may contains 2-3 sub-parts.
5. Questions should be uniformly spread over the entire syllabus.
6. The students are required to attempt FIVE questions in all, ONE question from each unit and the Compulsory question.

(BCH-DSC-7 Maj-502): Physiological Biochemistry (3+1)

Marks: 75 (60+15 Int.)

Objective: To make students aware withbiochemistry principles involved in the animal physiological processes.

Learning Outcomes:

Upon successful completion of this course, students will be able to

- Comprehend the concepts of human physiology and various physiological processes.
- Learn basics of diseases related to physiological diseases

SECTION-I

(Lectures -10)

Brief Organization of the body, homeostasis, Structure–function relationship in cells, Blood & Body Fluids: Plasma proteins (albumin, globulins, fibrinogen), Hemoglobin structure–function, O_2 & CO_2 transport, buffer role, Acid–base balance: biochemical and physiological regulation, Blood coagulation pathways biochemistry and physiology.

SECTION –II

(Lectures-10)

MusclePhysiology: actin–myosin interaction,excitation–contraction couplingEnergy metabolism in skeletal, cardiac, and smooth muscle, Muscle fatigue & metabolic disordersCardiovascular: control of heart rhythm andcontractilityRole of ions (Na^+, K^+, Ca^{2+}, Mg^{2+}) in cardiac physiology. Biochemical basis of atherosclerosis and hypertension.

SECTION –III

(Lectures-10)

Hormone chemistry & classification (peptide, steroid, amino acid derivatives), Mechanism of hormone action: receptors & second messengers, Hypothalamic–pituitary axis (neuroendocrine integration), Thyroid, adrenal, pancreas, gonads biochemical regulation of metabolism. Integration of carbohydrate, lipid, and protein metabolism, Fed vs fasting vs starvation states, Role of liver, muscle, adipose tissue in energy homeostasisExercise physiology.

SECTION –IV

(Lectures-10)

Renal Physiology include Filtration, reabsorption, secretion: biochemical transport mechanisms Renalregulation of electrolytes, acid–base balance, biochemical regulation of BP**.** Digestive Physiology: Biochemistry of gastric, pancreatic, and intestinal secretions Digestion & absorption of nutrients, Gut hormones & microbiota metabolism interactions. Pathophysiological Integration: Diabetes mellitus: biochemical–physiological basisObesity & metabolic syndrome, Molecular mechanisms of aging & oxidative stress

Suggested Books:

1. Guyton, A.C., & Hall, J.E. Textbook of Medical Physiology. Elsevier, 2019.
2. Barrett, K.E., Barman, S.M., Yuan, J.X.J., & Brooks, H.L. Ganong's Review of Medical Physiology. McGraw Hill Education, 2023.
3. Murray, R.K., Bender, D.A., Botham, K.M., Kennelly, P.J., Rodwell, V.W., & Weil, P.A. Harper's Illustrated Biochemistry. McGraw Hill Education, 2022.
4. Nelson, D.L., & Cox, M.M. Lehninger Principles of Biochemistry. W.H. Freeman and Company, latest edition.
5. Widmaier, E.P., Raff, H., & Strang, K.T. Vander's Human Physiology: The Mechanisms of Body Function. McGraw Hill Education, latest edition.

PRACTICALS:

(BCH-DSC-7 Maj-502): Physiological Biochemistry

Marks: 25 (20 + 5 Int.)

1. Methods of Blood Collection, Processing and Storage of Blood
2. Total RBC and WBC Counting
3. Differential Leukocytes Count
4. Clotting Time of blood
5. To estimate the Hemoglobin Content in a Blood Sample
6. Determination of Ratio of Albumin and Globulin
7. Determination of pH of Urine using pH strip
8. Measurement of Blood Pressure and Pulse Rate
9. Studies of Histological slides of Connective Tissues, Liver and Spinal cord

BCH-INT/Com-506 Internship/ Community service

It would be compulsory for students to secure 4 credits from summer internship/community work**.**

Internship Statement

As part of the degree requirements in Biochemistry, the student will undertake an industry or community-based internship program. During this training, the student will be placed in a relevant industrial setting, research facility, healthcare organization, or community project where they will observe, participate, and learn practical skills beyond classroom knowledge. The internship aims to help the student understand how biochemical knowledge is applied to solve real-world problems, contribute to industrial development or community well-being, and develop professional work ethics.

Internship Outcomes:

The student is expected to:

- Engage in activities that connect biochemistry with industry processes, healthcare, environmental improvement, or public benefit.
- Acquire technical skills, teamwork experience, and problem-solving ability.
- Understand safety standards, ethical practices, and regulatory requirements.
- Contribute meaningfully to the host organization's goals and learn how science benefits society.
- At the end of the internship, the student will prepare and submit a comprehensive final report describing the training undertaken, skills learned, observations made, and potential applications for societal or industrial benefit. This report will be evaluated as part of the degree completion requirement.

Semester VI

Major/Minor

INSTRUCTIONS FOR PAPER SETTER AND STUDENTS:

1. Examiner will set total of NINE questions comprising TWO questions from each unit and ONE compulsory question.
2. All questions carry equal marks.
3. The compulsory question will consists of 12 short questions of one mark each covering the whole syllabi.
4. Besides Question No. 1, all other questions may contains 2-3 sub-parts.
5. Questions should be uniformly spread over the entire syllabus.
6. The students are required to attempt FIVE questions in all, ONE question from each unit and the Compulsory question.

(BCH-DSC-8 Maj/Min-601: Immunology (3+1)

Marks: 75 (60+15 Int.)

Objective: To make students aware with Immunological principles involved in the animal defence against the foreign invaders.

Learning outcome:

Upon successful completion of this course, students will be able to

- Comprehend the basic principles of immunology and its role
- Learn basic types of experiments related immune system

SECTION-I

(Lectures -10)

Basics of immune system. Cells & organs involved in immune response. Molecules associated antigen, hapten&immunoglobulins- their structure, properties & functions. Innate & adaptive responses. Humoral & cell mediated immune responses.

SECTION –II

(Lectures-10)

Immunological techniques-their principle, method & applications: precipitation, agglutination, Immunofluorescence, radioimmunoassay (RIA), enzyme-linked immunosorbent assay (ELISA) and immunoblotting.

SECTION –III

(Lectures-10)

Complement system: components, types and functions, pathways, regulation. Immune response to infectious agents such as bacterial, viral, parasitic & fungal.Mechanism and types of immunodeficiency diseases, allergy and hypersensitive reactions.

SECTION –IV

(Lectures-10)

Vaccines: immunisations, types of vaccines, merits / demerits generation of monoclonal antibodies, abzymes, flowcytometry, magnet /fluorescence activated cell sorting, T cell cloning, proliferation, apoptosis, different types of animal models.

Suggested Books:

1. Guyton, A.C., & Hall, J.E. Textbook of Medical Physiology. Elsevier, 2019.
2. Barrett, K.E., Barman, S.M., Yuan, J.X.J., & Brooks, H.L. Ganong's Review of Medical Physiology. McGraw Hill Education, 2023.
3. Murray, R.K., Bender, D.A., Botham, K.M., Kennelly, P.J., Rodwell, V.W., & Weil, P.A. Harper's Illustrated Biochemistry. McGraw Hill Education, 2022.
4. Nelson, D.L., & Cox, M.M. Lehninger Principles of Biochemistry. W.H. Freeman and Company, latest edition.
5. Widmaier, E.P., Raff, H., & Strang, K.T. Vander's Human Physiology: The Mechanisms of Body Function. McGraw Hill Education, latest edition.

PRACTICALS:

(BCH-DSC-8 Maj/Min-601: Immunology

Marks: 25 (20 + 5 Int.)

1. Use of different anticoagulants.
2. Estimation of haemoglobin in blood.
3. Purification of Immunoglobulins
4. Assays Based on Precipitation Reactions
5. Assays Based on Agglutination Reactions Blood Typing
6. Demonstration of Enzyme Linked Immunosorbent Assay (ELISA)

Major/Minor

INSTRUCTIONS FOR PAPER SETTER AND STUDENTS:

1. Examiner will set total of NINE questions comprising TWO questions from each unit and ONE compulsory question.
2. All questions carry equal marks.
3. The compulsory question will consists of 12 short questions of one mark each covering the whole syllabi.
4. Besides Question No. 1, all other questions may contains 2-3 sub-parts.
5. Questions should be uniformly spread over the entire syllabus.
6. The students are required to attempt FIVE questions in all, ONE question from each unit and the Compulsory question.

(BCH-DSC-9 Maj/Min-602): Clinical Biochemistry

Marks: 75 (60+15 Int.)

Objective: To make students aware with Immunological principles involved in the animal defence against the foreign invaders.

Learning outcome:

Upon successful completion of this course, students will be able to

- Comprehend the basic principles of clinical biochemistry and its role
- Learn basic types of experiments related clinical diagnostics

SECTION –I

Laboratory organization, selection and use of laboratory test, biological material sampling of specimens, interpretation of laboratory test and clinical assessment of diagnostic test and quality control and c biochemistry lab test. Clinically Important enzymes, use of isoenzymes in diagnosis.

SECTION-II

Disorders of metabolism:Carbohydrates- glycogen storage diseases, galactosemiaAmino acids- disorders of glycine, sulphur containing amino acids, aromatic amino acids, histidine,branched chain amino acids and proline, disorders of propionate and methylmalonate metabolism.Disorders in urea biosynthesis.

SECTION-III

Lipids: hyperlipoproteinemia, hyperlipidaemia, Tay-Sachs Disease (Gangliosidosis), Neimann Pick disease, Gaucher's disease, Krab's disease, Metachromatic leukodystrophy and Fabry's Disease,Wolman's Disease.Disorders of porphyrin and heme metabolism.

SECTION-IV

Renal: GFR and its clinical importance, clearance tests (urea and creatinine) Intestinal: Malabsorption of fats, carbohydrates and proteinsFunction tests: Hepatic: Tests based upon the metabolism of carbohydrates, lipids, protein Diagnostic enzymology-enzyme determination in serum/plasma, urine and cells.

Suggested Books:

1. Kasper, D. L., Braunwald, E., Hauser, S., & Longo, D. (2005). ***Harrison's Principles of Internal Medicine***(16th ed., Vols. 1–2). McGraw-Hill.
2. Walker, G., Rae, S., & Ashby, P. (2006). ***Clinical Biochemistry*** (7th ed.). Blackwell Publishing.
3. Stanbury, J. B., Wyngaarden, J. B., Fredrickson, D. S., Goldstein, J. L., & Brown, M. S. (1983). ***The Metabolic Basis of Inherited Disease*** (5th ed.). McGraw-Hill Book Company.

4. Marshall, W. J., & Bangert, S. K. (2008). ***Clinical Chemistry*** (6th ed., international ed.). Elsevier/Mosby.
5. Burtis, C. A., Ashwood, E. R., & Bruns, D. E. (2006). ***Tietz Textbook of Clinical Chemistry and Molecular Diagnostics***(4th ed.). Elsevier.
6. Bishop, M. L., Fody, E. P., & Schoeff, L. E. (2004). ***Clinical Chemistry: Principles, Procedures, Correlations*** (5th ed.). Lippincott Williams & Wilkins.
7. Lieberman, M., & Marks, A. D. (2009). ***Marks' Medical Biochemistry: A Clinical Approach*** (3rd ed.). Lippincott Williams & Wilkins.
8. Bhagavan, N. V. (2002). ***Medical Biochemistry*** (Latested.). Academic Press.

PRACTICALS:

(BCH-DSC-9 Maj/Min-602): Clinical Biochemistry

Marks: 25 (20 + 5 Int.)

1. Estimation of sugar in blood
2. Estimation of proteins in blood and urine
3. Estimation of uric acid in blood sample
4. Estimation of SGOT/SGPT
5. Estimation bilirubin
6. Estimation of creatinine and urea
7. Estimation cholesterol in blood.

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BIOINFORMATICS

COURSE OUTLINE & SYLLABUS FOR BIOINFORMATICS

Students opting for Single major and Single minor (Table 1) and STUDENTS OPTING FOR MULTI-DISCIPLINARY/ INTER-DISCIPLINARY (Table 2) B.Sc., B.Sc. Honours and B.Sc. Honours with Research (FYUP Programme) Semester V and Semester VI

Sem	Level	Paper Code	Paper Name	Credits (Th+Pr)
V	L-300	BINF-DSC-20-Min-505	Drug Designing and Delivery	3+1=4
VI	L-300	BINF-DSC-26-Min-605	Database Management System	3+1=4
		BINF-DSC-27-Min-606	Introduction to Proteomics-I	3+1=4

SEMESTER – V

BINF-DSC-20-Min-505: DRUG DESIGNING AND DELIVERY

External Theory Marks: 60
Internal Theory Marks: 15
Total Theory Marks: 75
Exam Duration: 3 Hours

Course Duration:

1 Semester = 15 weeks which will have 45 hours of Theory Lectures (3 hours/week) and 30 hours of practical classes (2 hours/week)
1 Credit = 15 hours of theory lectures or 30 hours of practical classes
Total Credits: 4 (3 Credits for Theory Lectures and 1 Credit for Practical)
Each credit is equal to 25 marks

Learning Outcomes:

To understand the principles and techniques of drug design for identifying and optimizing bioactive compounds.

General Instruction:

- The question paper will have seven questions, and each question have 12 marks. The first question would be compulsory having sub-parts covering the entire syllabus in the form of short and objective type questions.
- The remaining six questions will be set out of the Units I and II consisting of three questions from each unit.
- A candidate is required to attempt five questions in all by selecting two questions, from each unit and the first compulsory question.

UNIT – I

Drug handling by the body: Basic concept of Pharmacokinetics- Absorption, distribution, metabolism and elimination.

Introduction to Drug Designing and Discovery: An overview of drug development process, concept of therapeutic drug targets, hits, leads and drug. High throughput screening

Introduction to Computer Aided Drug Design: Basic concept, advantages, concept of drug-likeliness, Lipinski's rule of five

Approaches of Computer Aided Drug Design: **Structure Based Drug Design** – Binding site prediction, docking based virtual screening.

Ligand Based Drug Design- QSAR Basic Principle and applications, 2D QSAR and 3D QSAR. Concept of molecular descriptors (1D, 2D and 3D).

UNIT – II

New drug approval process and clinical trial design: FDA, overview of drug approval process, drug discovery and lead compound selection, preclinical testing, new drug and abbreviated new drug applications. Clinical investigations-Phase I, II, III and IV clinical trials.

Post approval activities: Safety monitoring and changes to an approved product.

Clinical trial planning and design: Selecting trial objectives, trial designs and controlling of bias. Regulations governing the conduct of clinical trials. Drug blinding- Techniques and considerations for blinding of drug products.

Conventional drug therapy: Potential problems associated with multidose therapy. Modified release therapy: Terminology and potential advantages.

Rate controlled delivery systems: Diffusion, dissolution, osmotic, mechanical, controlled release by stimulation.

Targeted delivery systems: Collodial drug carriers- nanoparticles and liposomes. Bioadhesives, prodrug and ligand mediated targeting of drug. Protein and peptide drug delivery.

Recommended Books:

1. Text book of drug designing and discovery (2002) by Povl Krogsgaard-Larsen, Tommy Liljefors (CRC Pres).
2. Burger's medicinal chemistry and drug discovery, 6 Volume Set, (2003) edited by Donald J. Abraham (John Wiley & Sons, Inc., New York).
3. Smith and Williams' Introduction to the Principles of Drug Design and Action, Fourth Edition: edited by H. John Smith, CRC Press.
4. Felton L. Remington: Essentials of Pharmaceutics. Pharmaceutical Press; 2013
1. Applied biopharmaceutics and pharmacokinetics (2004) by Susanna Wu-pong, Andrew B. C. Yu, Leon Shargel (Mcgraw-Hill Medical Publishing)

BINF-DSC-20-Min-505: DRUG DESIGNING AND DELIVERY (PRACTICAL)

External Practical Marks:	20
Internal Practical Marks:	05
Total Practical Marks:	25

1. Searching drug target using therapeutic drug target database
2. Exploring DrugBank
3. Evaluating drug-likeliness using pkCSM
4. Toxicity prediction using ProTox
5. Docking based virtual screening

BIOINFORMATICS

SEMESTER – VI

BINF-DSC-26-Min-605: DATABASE MANAGEMENT SYSTEMS

External Theory Marks: 60
Internal Theory Marks: 15
Total Theory Marks: 75
Exam Duration:3 Hours

Course Duration:

1 Semester = 15 weeks which will have 45 hours of Theory Lectures (3 hours/week) and 30 hours of practical classes (2 hours/week)
1 Credit = 15 hours of theory lectures or 30 hours of practical classes
Total Credits: 4 (3 Credits for Theory Lectures and 1 Credit for Practical)
Each credit is equal to 25 marks

Learning Outcomes:

Understand the concepts, design, and management of databases for efficient data storage, retrieval, and manipulation.

General Instruction:

- The question paper will have seven questions, and each question have 12 marks. The first question would be compulsory having sub-parts covering the entire syllabus in the form of short and objective type questions.
- The remaining six questions will be set out of the Units I and II consisting of three questions from each unit.
- A candidate is required to attempt five questions in all by selecting two questions, from each unit and the first compulsory question.

UNIT – I

Introduction: Basic concepts, Advantages of a DBMS over file-processing systems, Data abstraction
Database Languages and Data Models: Introduction to Hierarchical, Network, ER, and Object Relational Model, Data Independence, Components of a DBMS and overall structure of a DBMS, Multi-User DBMS Architecture, System Catalogs

Data Modeling: Basic Concepts, entity, attributes, relationships, constraints, keys, E-R and EER diagrams – Components of E-R Model, conventions, converting E-R diagram into tables, EER Model components, converting EER diagram into tables

UNIT – II

Relational Algebra & SQL Relational Query Languages

Relational Algebra Introduction to SQL: Characteristics and advantages, SQL Data Types and Literals, DDL, DML, SQL Operators,

Tables: Creating, Modifying, Deleting; Views: Creating, Dropping, Updating using Views, Indexes, Nulls SQL DML

Queries: SELECT Query and clauses, Set Operations, Predicates and Joins, Tuple Variables, Set comparison, Ordering of Tuples, Aggregate Functions, Nested Queries, Database Modification using SQL Insert, Update and Delete Queries, union, intersection, minus etc. in SQL queries, sub-queries, equivalence of queries. Database Architectures: Centralized and Client-Server Architectures, 2 Tier and 3Tier Architecture, Introduction to Distributed Database systems

Recommended Books:

1. Database System Concepts, Silberschatz A., Korth H., Sudarshan S., McGraw Hill.
2. Fundamentals of Database Systems, Elmasri R., Navathe S., Pearson Education.51
3. Database Systems, Connally T., Begg C., Pearson Education.
4. An Introduction to Database Systems, Date C., Pearson Education.
5. Database Management System, B.C. Desai
6. Database Concept, Korth.

BINF-DSC-26-Min-605: DATABASE MANAGEMENT SYSTEMS (PRACTICAL)

External Practical Marks:	20
Internal Practical Marks:	05
Total Practical Marks:	25

1. Design a Database and create required tables. For e.g. Bank, College Database
2. Apply the constraints like Primary Key, Foreign key, NOT NULL to the tables.
3. Write a SQL statement for implementing ALTER,UPDATE and DELETE
4. Write the queries to implement the joins
5. Write the query for implementing the following functions: MAX(), MIN(), AVG(), COUNT()
6. Write the query to implement the concept of Intergrity constrains / create the views
7. Using the following tables:
   a. PET-OWNER (ownerID, name, phone, email, street, city, state, zip)
   b. PET (petID, name, type, breed, DOB, ownerID.
8. Write SQL Statements to do the following:
   a. Write Create Table statements to create the tables shown above
   b. Write SQL statements to add at least 3 rows to the PET-OWNER table.
   c. Write a SQL statement to display the Name, Breed and type of all pets on file.

BINF-DSC-27-Min-606: INTRODUCTION TO PROTEOMICS-I

External Theory Marks: 60
Internal Theory Marks: 15
Total Theory Marks: 75
Exam Duration: 3 Hours

Course Duration:

1 Semester = 15 weeks which will have 45 hours of Theory Lectures (3 hours/week) and 30 hours of practical classes (2 hours/week)
1 Credit = 15 hours of theory lectures or 30 hours of practical classes
Total Credits: 4 (3 Credits for Theory Lectures and 1 Credit for Practical)
Each credit is equal to 25 marks

Learning Outcomes:

Understand the large-scale study of proteins, including their expression, structure, and functions.

General Instruction:

- The question paper will have seven questions, and each question have 12 marks. The first question would be compulsory having sub-parts covering the entire syllabus in the form of short and objective type questions.
- The remaining six questions will be set out of the Units I and II consisting of three questions from each unit.
- A candidate is required to attempt five questions in all by selecting two questions, from each unit and the first compulsory question.

UNIT – I

Protein Isolation: Selection of a protein source, Methods of Solubilization, Stabilization of proteins, General Strategy of protein purification.
Methods of protein separation: Ion Exchange chromatography, Affinity Chromatography, HPLC, Native PAGE, Isoelectric 27or acti
Determination of molecular weight of proteins: Gel Filteration chromatography, SDS PAGE
Introduction to Proteomics, Proteome Analysis: 2D PAGE- Sample preparation, Solubilization, Limitations, Applications

UNIT – II

Analysis of Proteome: Mass spectrometry-based methods for protein identification. De-novo sequencing using mass spectrometric data.
Protein-protein Interactions: Principle and methods of determination of protein-protein interactions- Yeast 2-hybrid system, Phage display
Prediction methods for protein-protein interactions: Phylogenetic profiles, Gene neighborhood, Gene fusion
Protein-protein Interaction Databases: BioGrid, BIND, STRING

Recommended Books:

1. Malcolm Campbell and Laurie J. Heyer. Discovering Genomics, Proteomics and Bioinformatics2nd edition – Cold Spring Harbor Laboratory Press 2006.
2. Pennington, S. R., & Dunn, M. J. (Eds.). (2000). Proteomics: From protein sequence to function. BIOS Scientific Publishers.
3. Voet and Voet, Biochemistry (2004) John Willey and Sons Inc.
4. Wilson, K. And Walker, J. (2010) Principles and Techniques of Biochemistry and Molecular Biology. Cambridge University Press, Cambridge.
5. Baxevanis, A. D., Bader, G. D., & Wishart, D. S. (2020). Bioinformatics: a practical guide to the analysis of genes and proteins. Fourth edition. Wiley

BINF-DSC-27-Min-606: INTRODUCTION TO PROTEOMICS-I (PRACTICAL)

External Practical Marks:	20
Internal Practical Marks:	05
Total Practical Marks:	25

1. Determination of Theoretical PI
2. Proteome 2D PAGE
3. Searching protein-protein interactions using STRING Database, BioGrid
4. Prediction of protein-protein interactions

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BIOTECHNOLOGY

NEP2020COLLEGESSCIENCETEMPLATE:STUDENTS OPTINGFORSINGLEMAJORANDSINGLEMINOR(Table1*) and STUDENTS OPTING FOR MULTI-DISCIPLINARY/ INTER-DISCIPLINARY (Table 2*)
B.Sc., B.Sc.HONOURS AND B.Sc.HONOURS WITH RESEARCH

Table1*						Table2*					
Sem	Level	Course Code	Course Name	Credits (Th+Pr+Tut)	Total Credits	Sem	Level	Course Code	Course Name	Credits (Th+Pr+Tut)	Total Credits
V	L-300	BIOT-DSC-9(MAJ/MIN)-501	Plant Biotechnology	3+1	24	V	L-300	BIOT-DSC-9(MAJ/MIN)-501	Plant Biotechnology	3+1(x3)	24
		BIOT-DSC-10(MAJ)- 502	Animal Cell Culture	3+1				BIOT-DSC-10(MAJ)-502		3+1(x2)	
		BIOT-DSC-11(MAJ)- 503	Enzymology	3+1							
		BIOT-DSC-12(MAJ)- 504	Basics of Bioinformatics	3+1							
		BIOT-DSC-13(MAJ)- 505	Advancesin Immunology	3+1							
		BIOT-INT/COM-506	Internship/Community Work	4				(BIOT-INT/COM-506)	Internship/ Community Work	4	
VI	L-300	BIOT-DSC-14 (MAJ/MIN)-601	Aiin Biological and Clinical data analysis	3+1(x2)	24	VI	L-300	BIOT-DSC-14 (MAJ/MIN)-601	Aiin Biologicaland Clinical ataanalysis	3+1(x3)	24
		BIOT-DSC-15(MAJ/MIN)-602	Animal Biotechnology	3+1				BIOT-DSC-15(MAJ/MIN)-602	Animal Biotechnology	3+1(x3)	
		BIOT-DSC-16(MAJ)- 603	Bioprocess Engineering and Technology	3+1							
		BIOT-DSC-17(MAJ)- 604	Intellectual Property Rights	3+1							
		BIOT-DSC-18(MAJ)- 605	Genomics and Proteomics	3+1							
72 (Maj)+ 32 (Min) +9 (MDC) +8 (AEC) +6 (VAC) +9 (SEC) +4 (INT/COM) + 4 (PBI/HCP)					144	36x2(Maj)+32(Min)+9(MDC)+8(AEC)+6(VAC)+9(SEC)+4(INT/COM)+4 (PBI/HCP)					144
The student successfully securing 144 credits will be awarded B.Sc.degree											

Notes:

1. In the first semester student is offered three different subjects A, B & C.
2. At the end of second semester each student has the option to choose any one of the two streams namely Single Major or Multi-Disciplinary. The student choosing Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student who chooses multi-disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6(up to 3-year degree) shall be eligible for 4 YEAR DEGREE B.Sc. HONOURS with research, on the condition that the Institution is a PU approved Research Centre
5. The student opting for Single Major must secure 50% credits of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for Minor in a particular subject, the student must attain atleast 12 credits in that subject(s).
6. The student exiting the programme after one year securing 48 credits will be awarded UG Certificate provided he/she secures 4credits from summer internship/community work. Further, the student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secure 4 credits from summer internship. The student successfully securing 144 credits will be awarded BA degree.
7. Language paper chosen as MIL-1andMIL-2 should be in the same subject.
8. Once Major and Minor papers are chosen in 3rdsemester, the same will continue as Major and Minor subject in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write Dissertation of 08 credits in one of the two Major subjects.

BIOTECHNOLOGY

B.Sc. (Hons.) Biotechnology (Semester System) Examinations
(Session 2026-2027)
Semester-V
BIOT-DSC-9 Maj/Min-501- Plant Biotechnology (Discipline specific core course)

Objective: The objective of this course is to familiarize the students with different aspects of plant molecular biotechnology and techniques for plant genetic manipulations and their applications.

Course Duration
45 hours of lectures for Theory + 30 hours of practical
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)

Credits : 4 credits (Each credit is equal to 25 marks

External Theory	Internal Theory	External practical	Internal practical
68	07	22	03

Theory Paper: The exam of this course will be of 3 hours duration.
Instructions for paper setters and candidates:

i. The syllabus of this paper has been divided into FOUR units.
ii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. The students are required to attempt one question from each unit. **Compulsory question will be of 12 marks. Rest all questions will be of 14 marks each**

Unit – I

Structure of plant gene. An introduction to chloroplast and mitochondrial DNA.
Plant Promoter (35S, Adh1, Actin 1); Plant Selection markers and reporter genes.
Plant Vectors: Binary and co-integrate vectors, viral vectors (CaMV, TMV, Gemini virus) and their benefits.

Unit –II

Transformation of plant cells: Modes of gene delivery in plants
Particle bombardment, electroporation, microinjection; Advantages and disadvantages
Agrobacterium mediated gene transfer, natural pathogen mode of infection: Role of vir gene, Ti / Ri plasmids functions.
Screening and selection of transformants; PCR and hybridization methods, Transgene selection and silencing, Generation and maintenance of transgenic plants.

Unit – III

Transgenic plants: Genetic modification of plants for herbicide resistance (Glyphosate, Glufosinate). Pest resistance (Bt), virus, Bacterial and fungal resistance.
Delayed fruit ripening (Polygalactorunase), improved protein composition.
Explain Bt cotton, golden rice as examples.

Unit – IV

Plant cell as factories for: Production of industrial enzymes (Phytase, Cellulase), biodegradable plastics, antibodies, edible vaccines;
Manipulation of metabolic pathways for: Production of fatty acids, industrial oils.
Secondary Metabolites: terpenoids, flavanoids.

Reference Books:

1. ***Plant Breeding 2050: Next-Gen Crops*** *(2025) P. M. Priyadarshan & Rodomiro Ortiz*
2. ***Plant Biotechnology and Genetics: Principles, Techniques, and Applications*** *(3rd ed., March 2025) C. Neal Stewart (ed.)*
3. ***Green Biotechnology for Herbal and Medicinal Plant Metabolites*** *(due mid-2025) Parul Johri, Pedro Lopez-Sanchez, Sachidanand Singh, Mala Trivedi*
4. ***Plant Biotechnology: The Medicinal and Phytopharmaceutical Aspects*** *(Sep 2025) Durgesh Nandini Chauhan, Nagendra Singh Chauhan, Deependra Singh, Manju Singh*
5. ***Plant Biotechnology and Food Security: Adoption, Farming Systems and Sustainable Development*** *(Nov 2025) Jyoti Prakash Sahoo, Kailash Chandra Samal*
6. ***Biotechnological Advances in Horticultural Crops*** *(Nov 2024) Ajay Kumar Thakur, Pankaj Kumar, Dinesh Kumar Srivastava.*
7. ***Plant Biotechnology: A Textbook*** *by S.M. Paul Khurana & Narendra Kumar*

Plant Biotechnology (Practical)

1. Preparation of stock solutions of MS basal medium and plant growth regulator stocks.
2. Micropropagation of suitable plant by leaf disc culture.
3. Isolation of plant genomic DNA by modified CTAB method.
4. DNA check run by Agarose Electrophoresis.
5. *Agrobacterium tumefaciens*-mediated plant transformation.

BIOT-DSC-10 Maj-502-Animal Cell Culture (Discipline specific core course)

Objective: The major emphasis of this course is to introduce the students to the field of Animal cell-culturing and its importance to mankind. The students will also learn the techniques involved in animal cell culture.

Course Duration

45 hours of lectures for Theory + 30 hours of practical
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)

Credits : 4 credits (Each credit is equal to 25 marks

External Theory	Internal Theory	External practical	Internal practical
68	07	22	03

Theory Paper: The exam of this course will be of 3 hours duration.

Instructions for paper setters and candidates:

i. The syllabus of this paper has been divided into FOUR units.
ii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. The students are required to attempt one question from each unit. **Compulsory question will be of 12 marks. Rest all questions will be of 14 marks each**

Unit – I

History of development of cell cultures, simulating natural conditions for animal cells, metabolic capabilities of animal cells in cultures.
Biology of cultured cells: Culture environment, cell adhesion, proliferation, differentiation, signaling. Equipments and materials for animal cell culture.

Unit II

Introduction to the balanced salt solutions and simple growth medium. Role of different constituents of culture medium, Role of carbon dioxide, Role of serum and supplements.
Serum & protein free media and their application.
Animal cell culture Techniques: Dispersion and disruption of tissues; primary cultures, anchorage and non- anchorage dependent cells; secondary culture.

Unit – III

Established/continuous cell lines, commonly used animal cell lines, their origin and characteristics. Growth kinetics of cells in culture,
Measurement of growth and viability of cells in culture.

Unit – IV

Cell fusion and production of monoclonal antibodies. Transformation and immortalization, Cryopreservation.
Principles of Bio-Safety & Bioethics.

Reference Books:

1. ***Animal Cell Culture: Principles and Practice*** *(2023) Shalini Mani, Manisha Singh, Anil Kumar*
2. ***Freshney's Culture of Animal Cells: A Manual of Basic Technique and Specialized Applications****(2021) R. Ian Freshney, Amanda Capes-Davis*
3. ***Animal Cell Culture*** *(1st Ed.)* ***(2025) Animal Cell Culture*** *(1st Ed.)*
4. ***Animal Cell Culture & Technology****, 2nd Ed (2024) Butler M*
5. ***Animal Cell Biotechnology: Methods and Protocols*** *(4th Edition) Springer protocol editors / contributors*

Animal Cell Culture Practical

1. Fumigation of animal cell culture laboratories.
2. To study Sterilization techniques used in animal cell culture.
3. Preparation of Balanced salt solution &growth media for cell culture.
4. Isolation of lymphocytes for culturing.
5. Measurement of cell count & viability using Hemocytometer.

BIOT-DSC-11- Maj-503-Enzymology (Discipline specific core course)

Objective : The main objective is to provide a comprehensive understanding of enzyme structure, function, and mechanisms of action and to develop the ability to analyze enzyme kinetics and the factors affecting enzyme activity. This will foster skills in experimental techniques used for enzyme purification and characterization and to explore the applications of enzymes in biotechnology, medicine, and industry.

Course Duration
45 hours of lectures for Theory + 30 hours of practical
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)

Credits : 4 credits (Each credit is equal to 25 marks

External Theory	Internal Theory	External practical	Internal practical
68	07	22	03

Theory Paper: The exam of this course will be of 3 hours duration.
Instructions for paper setters and candidates:
The syllabus of this paper has been divided into FOUR units.
Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. The students are required to attempt one question from each unit. **Compulsory question will be of 12 marks. Rest all questions will be of 14 marks each**

UNIT – I

Structure and functions of enzymes: Historical background: Structure and general properties of enzymes, concept of active centre, binding sites, stereo specificity and ES complex formation, activation energy, Evidences for enzyme-substrate complex; Lock and key, Induced fit and Transition state hypotheses, Coenzymes and Cofactors- Prosthetic group, coenzymes involved in different metabolic pathways.

UNIT – II

Factors Affecting the Enzyme Activity: Concentration, pH and temperature. Kinetics of a single substrate enzyme catalysed reaction, derivation of Michealis-Menten Equation, significance of Km value, Vmax, Turnover number, Kcat. Enzyme activity, international units, specific activity,

Concept of Enzyme Inhibition and its types

UNIT-III

Enzyme Regulation: Feedback inhibition, Organization of enzymes in the cell, localization, enzymes in membranes. Acid-base catalysis, covalent catalysis, Metal ion catalysis, multienzyme complexes and ribozymes, catalytic antibodies, Allosteric enzymes.

UNIT IV

Applications of Enzymes: Immobilized enzymes, industrial applications of immobilized enzymes, Thermophilic enzymes, amylases, lipases, Proteolytic enzymes in meat and leather industry, enzymes used in fermentation processes, cellulose degrading enzymes, Metal degrading enzymes.

Enzymes as thrombolytic agents, Anti-inflammatory agents, Isoenzymes

Reference Books:

1. Enzymes: Catalysis, Kinetics and Mechanisms (2nd Edition)(2025) Narayan S. Punekar
2. Laboratory Guide to Enzymology (2024) Geoffrey A. Holdgate, Antonia Turberville, Alice Lanne
3. Enzymes: A Practical Introduction to Structure, Mechanism, and Data Analysis (3rd Edition) (2023) Robert A. Copeland
4. ***Biocatalysis in Asymmetric Synthesis*** (Foundations and Frontiers in Enzymology series) (2024) Edited by Gonzalo De Gonzalo, Andrés R. Alcántara et al.
5. Advances in Enzymology (2024) Dr. Keshavamurthy M., Avinash K O, Kiran K S
6. Practical Enzymology (2004) By Hans Bisswanger (Wiley- VCH, Weinheim)
7. Introductory Practical Biochemistry (2005) by S.K. Sawhney, Randhir Singh (Alpha Science International).

Enzymology Practicals

1. Estimation of alpha- amylase enzyme activity.
2. To study the Effect of pH on activity of enzyme.
3. To study the Effect of temperature on activity of enzyme.
4. To study the effect of substrate concentration on enzyme activity.
5. To estimate K_m and V_{max} of an enzyme and plot Line-Weaver Burk plot.
6. To study the effect of metal ions on enzyme activity.

BIOT-DSC-12- Maj-504-Basics of Bioinformatics (Discipline specific core course)
Objective: Computational methods for Biomolecules structure and sequence analysis to equip students with specific tools and methodologies.

Course Duration

45 hours of lectures for Theory + 30 hours of practical

(1 credit = 15 hours of theory learning based on 3 hrs./week)

(1 credit = 30 hours of practical based on 2 hrs./week)

(1 Semester = 15 weeks)

Credits : 4 credits (Each credit is equal to 25 marks

External Theory	Internal Theory	External practical	Internal practical
68	07	22	03

Theory Paper: The exam of this course will be of 3 hours duration.

Instructions for paper setters and candidates:

The syllabus of this paper has been divided into FOUR units.

Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. The students are required to attempt one question from each unit. **Compulsory question will be of 12 marks. Rest all questions will be of 14 marks each**

UNIT – I

Introduction to Bioinformatics, Biological Databases and Sequence analysis
Introduction, overview and needs of bioinformatics technology.
Primary Databases: Primary Sequence database i.e. GenBank & EMBL.
Secondary Databases: SwissProt/TrEMBL, Pfam.
Molecular Structure Databases: Protein Data Bank (PDB), SCOP, CATH. Understanding the structure of each database and using it on the web.

UNIT – II

Sequence Alignment
Introduction to sequence alignment and its applications.
Pair wise sequence alignment: Concept of global and local alignment, Dot Plot, algorithm for pair wise sequence alignment (Needleman Wunsch), Introduction to BLAST, types of BLAST, algorithm of BLAST and interpretation of its result.
Substitution matrices: Introduction to substitution matrices: PAM and BLOSUM matrices, concept of log odd ratio.

UNIT – III

Multiple sequence alignment: Methods of multiple sequence alignment. Introduction to consensus sequences, motifs and profiles.

Phylogenetic Analysis: Introduction to phylogenetic analysis and its application, phylogenetic tree topologies, methods of phylogenetic tree construction and tools.

UNIT IV

Genome Annotation: Concept of genome annotation, methods of gene identification. Tools of gene identification: GenScan and Glimmer.
Protein Structure Prediction: Concepts and strategies of protein structure prediction, methods of secondary structure prediction.
Structure visualization tool – RasMol.

Reference Books:

1. Unraveling New Frontiers and Advances in Bioinformatics (2024) Amit Chaudhary, Sushanta K. Sethi, Akarsh Verma
2. ***Advances in Bioinformatics*** (2nd edition) ***(2024)*** Vijai Singh, Ajay Kumar
3. Bioinformatics and Beyond: AI Applications in Healthcare (2025) Moolchand Sharma, Deepak Kumar Sharma, Deevyankar Agarwal (eds.)
4. ***Bioinformatics Concepts Skills And Applications*** (2nd Ed) (2025) S. C. Rastogi
5. Encyclopedia of Bioinformatics and Computational Biology (2025) Shoba Ranganathan, Mario Cannataro, Mohammad Asif Khan (eds.)

Basics of Bioinformatics (Practicals)

1. Exploring NCBI and its resources.
2. Searching literatures over PubMed.
3. Performing DOTPLOT on web.
4. Retrieving amino acid and nucleotide sequence from sequence databases using Entrez.
5. Performing BLASTp/n and interpreting its results.
6. Performing multiple sequence alignment using Clustalw.
7. Using GenScan to identify exons in nucleotide sequence.
8. Download protein structure form PDB and visualize it using RasMol.

BIOT-DSC-13- Maj-505-Environmental Biotechnology (Discipline specific core course)

Objective:

The course focuses on an introduction to environment, major threats to environment by various polluting agents and the remedies for the same, incorporating design and monitoring of waste treatment processes. As well as learning environmental technology fundamentals, with special focus on biological treatment processes,

Environmental management. The course is use of biotechnology to design cleaner manufacturing processes and to solve pollution problems. It is ideal for under graduates just embarking on their career, or scientists and engineers who have been working for a few years and wish to develop their career in this direction.

Course Duration
45 hours of lectures for Theory + 30 hours of practical
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)

Credits : 4 credits (Each credit is equal to 25 marks

External Theory	Internal Theory	External practical	Internal practical
68	07	22	03

Theory Paper: The exam of this course will be of 3 hours duration.
Instructions for paper setters and candidates:

i. The syllabus of this paper has been divided into FOUR units.
ii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. The students are required to attempt one question from each unit. **Compulsory question will be of 12 marks. Rest all questions will be of 14 marks each**

UNIT-I

Basics of Environment and Environmental pollution, air, water, soil and noise. Air – Types, Sources & Effects,
Soil – soil pollutants (fertilizers, insecticides fungicides, pesticides). Noise pollution, its control and impact on human health.
Modern Fuels (gasohol, hydrogen, wind energy and solar energy) and their Environmental Impacts.

UNIT-II

Water pollution and its management: Measurement of water, pollution, sources of water pollution. General strategies for wastewaters treatment.
Microbiology of waste water treatment, aerobic processes, activated sludge, oxidation ponds, trickling filters Anaerobic processes: Anaerobic digesters, upward flow anaerobic sludge blanket reactors.

UNIT-III

Integrated Pest management
Biodegradation of environmental pollutants: pesticides, hydrocarbons, Azo dye.
Biofertilizers for clean environment– Nitrogen fixing microorganism, enrichment of the soil with assimilable nitrogen

UNIT IV

Introduction to solid waste and municipal solid waste management: Sources, types, composition. Land fills, Bioindicators for detection of pollution

References Books:

1. Environmental Sustainability and Biotechnology: Opportunities and Challenges (2022) Naveen Kumar Arora, Piyush Pandey, Jitendra Mishra
2. Biotechnology for Environmental Sustainability (Interdisciplinary Biotechnological Advances) (2025) Pradeep Verma (Ed.)
3. Integrated Biotechnological Solutions for the Treatment of Industrial Wastewater: For a Healthy and Sustainable Environment (2025) Dwivedi, Naveen; Shubha, Shubha; Shah, Maulin P. (Eds.)
4. Soil Improvement and Water Conservation Biotechnology (2025) Israel Valencia Quiroz
5 Biotechnology Innovations for a Sustainable Future: Integrating Clean Energy, Life on the Planet, Clean Water, and Climate Action (2025) Ranbir Chander Sobti (Ed.)

ENVIRONMENTAL BIOTECHNOLOGY Practical

1. Detection of coli forms for determination of the purity of potable water.
2. Determination of chlorine in water
3. Determination of total alkalinity of water
4. Determination of dissolved oxygen concentration of water sample.
5. Determination of biological oxygen demand (BOD) of a sewage sample.
6. Determination of chemical oxygen demand (COD) of sewage sample.
7. Isolation of phosphate solubilizing microorganisms from soil.

Reference Books:

1. Microbiology: A laboratory manual (2007) by Cappuccino J.G and Sherman N. (Publisher Pearson Benjamin Cummings).
2. Biotechnology Laboratory Manual by Swamy, P.M. Rastogi Publications (2008).
3. Practical Microbiology by Dr R.C. Dubey and D.K. Maheshwari, S.Chand Publications.

SEMESTER-VI

BIOT-DSC-14 Maj/Min-601-Genetic Engineering (Discipline specific core course)

Objectives:

Genetic engineering refers to the process of manipulating the characteristics and functions of the original genes of an organism. The objective of this process is to introduce new physiological and physical features or characteristics. The students will learn how the genes can be isolated and inserted from one organism to another and what its implications are.

Course Duration

45 hours of lectures for Theory + 30 hours of practical
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)

Credits : 4 credits (Each credit is equal to 25 marks

External Theory	Internal Theory	External practical	Internal practical
68	07	22	03

Theory Paper: The exam of this course will be of 3 hours duration.

Instructions for paper setters and candidates:

i. The syllabus of this paper has been divided into FOUR units.
ii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. The students are required to attempt one question from each unit. **Compulsory question will be of 12 marks. Rest all questions will be of 14 marks each**

UNIT-I

Introduction to genetic engineering. Why gene cloning and DNA analysis is important. How to clone a gene – What is clone, Overview of the procedure
Tools in Recombinant DNA Technology:
Restriction and modifying enzymes, Type I, Type II and Type III enzymes and their characteristic features; restriction sequences, isoschizomers, rare cutting enzymes, enzyme cutting similar sequence in different manner.
DNA Modifying enzymes: Characteristics and applications of Nucleases – Dnase and Rnase, DNA-Pol I, Klenow fragment, T4DNA polymerase, T7 DNA polymerase, T4 Polynucleotide kinase, Phosphatase, Reverse transcriptase, Taq polymerase and Ligase. Terminal deoxy nucleotidyl transferase.
Polymerase Chain Reaction: Types and applications

UNIT II

Basic biology of plasmids and Phage vectors Basic features of plasmids, plasmid classification, Bacteriophage λ, lytic & lysogeny, linear and circular forms of lambda vector, DNA cloning with single stranded DNA vectors.

DNA cloning vectors Cloning vectors for E. coli- Nomenclature, pBR 322, pBR 327, pUC 18, pGEM3Z. insertion and replacement vectors; Vectors based on M13. Methods of identification of recombinants: Insertional inactivation, blue/white selection. Cloning vectors for yeast- Yep, Yip, Yrp.
Advanced Vectors: cosmid, phagemid, Bacterial Artificial Chromosomes (BACs), shuttle vectors, yeast artificial chromosomes.

UNIT-III

Preparation of genomic and cDNA library: Partial digests, Choice of vectors, Construction and Evaluation of a genomic library. cDNA library: mRNA enrichment, cDNA synthesis,Random, arrayed and Ordered library.
Finding The Right Clone: Gene identification, Nucleic acid hybridization, Screening Procedure, Probe selection, 42 or ac screening, functional complementation. Southern blotting, Northern blotting.
Preparation of nucleic acid probes: DNA and RNA labeling techniques, nick translation, random priming, end 42or acti, radioactive and non- radioactive labels.

UNIT-IV

Site directed mutagenesis (cassette, primer extension, RT, real time, multiplex, inverse),
DNA sequencing (Maxam-Gilbert, Sanger, pyro).
Production of Protein from Cloned Genes: Special vectors for expression of foreign genes in *E. coli*, General problems with the production of recombinant protein in *E. coli*. Production of recombinant protein by eukaryotic cells.

Reference Books:

1. Genetic Engineering (2-volume set) (2023) P. M. Priyadarshan, Rodomiro Ortiz (eds.)
2. The Genetic Book of the Dead: A Darwinian Reverie (2024) Richard Dawkins
3. Genetic Engineering (Text Book) (2023) Dr. K. Gopi
4. **Artificial Intelligence (AI) in Cell and Genetic Engineering (2025) Sudip Mandal (Ed.)**
5. **The Risks and Ethics of Human Gene Editing: A Philosophical Guide to the Arguments —** Madeleine Hayenhjelm & Christer Nordlund — Springer VS, 2025.

Genetic Engineering Practical

1. Demonstration of PCR.
2. Demonstration of Southern blotting.
3. Preparation of competent cells.
4. Transformation of competent cells and blue/white selection.
5. Spectrophotometer analysis of DNA.

BIOT-DSC-15 Maj/Min-602 –Animal Biotechnology (Discipline specific core course)

Objective: The major emphasis of this course is to introduce the students to the advances in the field of Animal cell culturing, tissue engineering, transgenic technology and their importance to mankind.

Course Duration
45 hours of lectures for Theory + 30 hours of practical
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)

Credits : 4 credits (Each credit is equal to 25 marks

External Theory	Internal Theory	External practical	Internal practical
68	07	22	03

Theory Paper: The exam of this course will be of 3 hours duration.
Instructions for paper setters and candidates:
The syllabus of this paper has been divided into FOUR units.
Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. The students are required to attempt one question from each unit. **Compulsory question will be of 12 marks. Rest all questions will be of 14 marks each**

Unit-I

Organotypic and histotypic cultures: Organotypic culture: Gas and nutrient exchange, structure integrity, growth, differentiation, advantages and applications. Methods, advantages and applications of histotypic culture.
Three dimensional culture and tissue engineering: Concept of tissue engineering, components of tissue engineering, cells imaging in 3D construct.

Unit-II

In vitro fertilization (IVF) in Humans and Embryo Transfer in Livestock.
Cell culture based vaccines: Cells as virus host/cell culture based vaccines, cells as protein factory/cell expression system and cells as antigen presenter/personalized vaccine.
Scaling up of the animal cell culture: different methods of scale up at laboratory and industrial level.

Unit-III

Transgenic animals and their applications: Concept of transgenics, Methods of gene transfer, selection of clone containing DNA insert and application of transgenic animals (Food, environment, recombinant proteins, drugs etc.). Safety and ethical issues of transgenic animals.

Unit-IV

Production of various products of human use using animal cell culture:
Antibiotics production Human Growth factors Insulin and other Hormones

Reference Books:

1. ***Animal Biotechnology*** — R. Sasidhara (2024)
2. Animal Biotechnology Models in Discovery and Translation (2nd Edition) (2024) A. Verma
3. Biotechnology for Animal Health (2025) Chandrabhaga Desai
4. **Animal Biotechnology: Models in Discovery and Translation (2nd Edition) (2022)*by Ashish S. Verma & Anchal Singh***
5. **Biotechnology for Animal Health*by Chandrabhaga Desai (2025)***
6. **Advances and Applications in Biotechnology: Volume 1 – Prospects of Farm Animals and Future Developments*Edited by Arti Gupta & Ali Asghar Rastegari (2025)***

Animal Biotechnology (Practicals)

1. Growing the cell monolayers, in vitro.
2. Trypsinization of the monolayers and cell counting using hemocytometer.
3. To check the viability of the cells using Trypan Blue dye exclusion assay.
4. Checking the cytotoxicity of the compounds using MTT Assay.

BIOT-DSC-16 Maj-603- Bioprocess Engineering and Technology

Objective

The course aims to introduce the knowledge and skills to design, optimize and manage processes that use biological systems to produce valuable products. To acquaint students about the use of bioreactors, various fermentation systems, and downstream processing techniques.

Course Duration

45 hours of lectures for Theory + 30 hours of practical
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)

Credits : 4 credits (Each credit is equal to 25 marks

External Theory	Internal Theory	External practical	Internal practical
68	07	22	03

Theory Paper: The exam of this course will be of 3 hours duration.
Instructions for paper setters and candidates:
The syllabus of this paper has been divided into FOUR units.
Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. The students are required to attempt one question from each unit. **Compulsory question will be of 12 marks. Rest all questions will be of 14 marks each**

Unit – I

Fundamental principles of biochemical engineering.

Sterilization of air and media sterilization, design of batch sterilization process. Delfactor, sterilization cycle, continuous sterilization process

Unit–II

Microbial growth kinetics Simple kinetics of microbial growth (Batchand Continuous, Feed Back System),yield coefficient, doubling time, specific growth rate, internal and external feedback systems, effect oftemperatureand pHon theproductformation

Unit – III

Design of fermenter:

Types of fermenter (Batch, Fed-Batch, Continuous)

Components of Fermenter, Aseptic operation of the fermenters, control and measurement equipment of fermenters, pH and Doprobes, impeller and spargers,

Unit– IV

Downstreamprocessing

Removalofmicrobialcellsandothersolidmaterials,foamseparations,filtration,industrialfilters,centrifugations and industrial centrifuges, cell Disruption, aqueous two phase extraction system, super criticalextraction,whole broth process. Effluent treatment (Breweries, Antibiotic industries)

ReferenceBooks:

1. **Bioprocess Engineering Principles (3rd Edition)** Ross Carlson, Kate Morrissey, Pauline M. DoranElsevier, March 2025
2. **Integrated Bioprocess Engineering** Clemens Posten De Gruyter, 2024
3. New Bioprocessing Strategies: Development and Applications (2024)
4. **Bioprocess Engineering Principles** Rohit Manglik: EduGorilla Publication, July 2023
5. **Bioprocess Engineering and Technology: Three-Volume Set**, Biswanath Bhunia, Navneet Joshi, Muthusivaramapandian Muhuraja: Routledge, 2025
6. **New Bioprocessing Strategies: Development and Applications**: Springer, 2024

BioprocessEngineeringandTechnology-Practical

1. Demonstration of componentsand sterilization of fermenters and other accessories.
2. Determination of doubling time, yield coefficient for growth of microorganism.
3. Effect of temperature and pHandgrowth and product formation.
4. Determination of Specific growth rate and maximum specific growthrate
5. Cell disruptionbymethods.

ReferenceBooks:

1. Cappuccino J.G., Sherman N. (2007). Microbiology: Alaboratory (Pearson Benjamin Cummings).
2. PlummerD.T.(2004).An introduction to practical biochemistry (Tata McGraw Hill Publishers Co.Ltd., NewDelhi).

BIOT-DSC-17 Maj-604-IntellectualPropertyRightsand Biotechnology

Objective

The course aims to introduce the fundamentals and significance of Intellectual Property Rights (IPR).It develops understanding of patents, copyrights, trademarks, designs, and related legal frameworks.Students will learn Indian and international provisions including TRIPS, WIPO, and major treaties.The syllabus emphasizes patent drafting, filing, infringement, and compulsory licensing.It builds skills in literature search, claim drafting, and trademark searching.Learners will explore IPR in biotechnology with ethical and disclosure considerations. The course highlights the role of creative commons, plant variety protection, and geographical indications.

Course Duration

45 hours of lectures for Theory + 30 hours of practical
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)

Credits : 4 credits (Each credit is equal to 25 marks

External Theory	Internal Theory	External practical	Internal practical
68	07	22	03

Theory Paper: The exam of this course will be of 3 hours duration.
Instructions for paper setters and candidates:

i. The syllabus of this paper has been divided into FOUR units.
ii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. The students are required to attempt one question from each unit. **Compulsory question will be of 12 marks. Rest all questions will be of 14 marks each**

UNIT – I

Introduction to Intellectual Property Rights and their importance. TRIPS and patentable subject matter.
Patents: Introduction to Indian patent law and amendments, conditions for patentability; Patent specifications, procedure for obtainingpatents, patent filing through PCT, rights of a patentee; compulsory licensing, patent infringements and Remedies.

UNIT- II

Copyright: Protection and related rights; copyright piracy; Fair use doctrine, Role of creative commons in academics.
Designs: Introduction to the law on industrial designs; registration and piracy. Locarno classification and Hague Agreement. Protection of IC layouts.

UNIT – III

Trademark: Features, Importance, Registration, Trademark infringement and passing off.
Plant variety protection Act and criteria of their protection. Protection of Geographical indications of goods and undisclosed information.

UNIT-IV

Patenting in biotechnology: Biotechnology patents, disclosure requirementsand depository considerations. Ethical issues in biotechnology patenting, Material transfer Agreement and it significance in biotechnology patenting.
WIPO and its role in IPR management. International conventions/treaties – Paris convention, Berne Convention, Budapest treaty, UPOV.
Information sources in patent literature search.

REFERENCE BOOKS:

1. **U.S. Biotechnology Patent Law, 2024–2025 Edition**
2. **Patenting Life: Tales from the Front Lines of Intellectual Property and the New Biology (2025)**
3. Intellectual Property Rights and Patents in Biotechnology (2025)
4. Biotechnology Law and Policy (2025)
5. Intellectual Property Issues in Life Sciences: Disputes and Controversies (2025)

BIOT-DSC-17 Maj-604-IntellectualPropertyRightsand Biotechnology

Practical:

1. Drafting of Independent and dependant claims in a patent specification.

2. Patent literature search using InPASS.

3. Trademark searching in IPIndia.

4. PCT patent searching using PATENTSCOPE.

5. Case studies in context to patenting in Biotechnology.

BIOT-DSC-18 Maj-605- Genomics and Proteomics

Objectives: To equip students with skills to understand and learn cutting edge technologies like DNA sequencing and gene expression profiling to explore gene-protein relationships and to foster interdisciplinary thinking.

Course Duration

45 hours of lectures for Theory + 30 hours of practical
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)

Credits : 4 credits (Each credit is equal to 25 marks

External Theory	Internal Theory	External practical	Internal practical
68	07	22	03

Theory Paper: The exam of this course will be of 3 hours duration.
Instructions for paper setters and candidates:

i. The syllabus of this paper has been divided into FOUR units.
ii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. The students are required to attempt one question from each unit. **Compulsory question will be of 12 marks. Rest all questions will be of 14 marks each**

Unit I

Introduction to Genomics, DNA sequencing methods – manual & automated: Maxam & Gilbert and Sanger'smethod. Pyrosequencing, Genome Sequencing: Shotgun & Hierarchical (clone contig) methods, Computertoolsforsequencingprojects:Genome sequenceassembly software.

Unit II

Managing and Distributing Genome Data: Web based servers and softwares for genome analysis: ENSEMBL, VISTA, UCSC Genome Browser, NCBI genome. Selected Model Organisms 'Genomes and Databases.

Unit III

Introduction to protein structure, Chemical properties of proteins. Physical interactions that determine theproperty of proteins. Short-range interactions, electrostatic forces, van der waal interactions, hydrogen bonds,Hydrophobic interactions. Determination of sizes (Sedimentation analysis, gel filteration, SDS-PAGE); NativePAGE,Determinationof covalentstructures –Edman degradation.

Unit IV

Introduction to Proteomics, Analysis of proteomes. 2D-PAGE. Sample preparation, solubilization, reduction, resolution. Reproducibility of 2D-PAGE. Mass spectrometry based methods forprotein identification. Denovo sequencing using mass spectrometricdata.

REFERENCE BOOKS:

1. **Introduction to Genomics (Fourth Edition)** by Arthur Lesk (2025)
2. **The Genetic Book of the Dead: A Darwinian Reverie** by Richard Dawkins (2025)
3. **Comparative Genomics** (RECOMB-CG 2025 Proceedings)
4. Activity-Based Proteomics: Methods and Protocols (2025)
5. Proteomics Playbook 2025

Genomics and Proteomics Practical

1. Use of SNP databases at NCBI and other sites
2. Use of OMIM database
3. Detection of Open Reading Frames using OR Ffinder
4. Proteomics 2D PAGE database
5. Softwares for Protein localization.
6. Hydropathyplots
7. SDS-PAGE

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BOTANY

NEP 2020 COLLEGES SCIENCE TEMPLATE: STUDENTS OPTING FOR SINGLE MAJOR AND SINGLE MINOR (Table 1*)
and STUDENTS OPTING FOR MULTI-DISCIPLINARY/ INTER-DISCIPLINARY (Table 2*)
B.Sc., B.Sc. HONOURS AND B.Sc. HONOURS WITH RESEARCH IN BOTANY

	Table 1*				
Sem	**Level**	**Course Code**	**Course Name**	**Credits (Th+Pr+Tut)**	**Total Credits**
V	L-300	(BOT-DSC-9 Maj-501)	Plant Metabolism	3+1=4	24
		(BOT-DSC-10 Maj-502)	Economic Botany	3+1=4	
		(BOT-DSC-11 Maj-503)	Plant Pathology	3+1=4	
		(BOT-DSC-12 Maj-504)	Plant Breeding	3+1=4	
		(BOT-DSC-13 Maj-505)	Plant Resource Utilization	3+1=4	
		(BOT-INT/COM-506)	Internship/Community Work	4	
VI	L-300	(BOT-DSC-14 Maj/Min-601)	Plant Physiology	3+1=4 (x2)	24
		(BOT-DSC-15 Maj-602)	Anatomy of Angiosperms	3+1=4	
		(BOT-DSC-16 Maj-603)	Medicinal Botany	3+1=4	
		(BOT-DSC-17 Maj-604)	Plant Biotechnology	3+1=4	
		(BOT-DSC-18 Maj-605)	Bryophytes	3+1=4	
	72 (Maj)+32(Min)+9(MDC)+8(AEC)+6(VAC)+9(SEC)+4(INT/COM)+4(PBI/HCP)				144

BOTANY

(SEMESTER – V)

Course Title: PLANT METABOLISM **Course Code: BOT-DSC-9 Maj-501**

Learning Outcome: The main objective of this paper is to make students aware about various functions and metabolism of plants. It will introduce students to highly coordinated cellular functions related to carbon and nitrogen assimilation, concepts of metabolism- carbohydrates, nitrogen and lipids. Practical skills will be imparted to the student through critically designed practical related to the subject.

Max. Marks: 100 **Course Credits** = 3+1=4

1. Theory: 75 Written Exam: 67 Internal Assessment: 8
2. Practical: 25 Written Exam: 22 Internal Assessment: 3

Periods per week: 1. Theory- Four of 45 minutes duration (Three hours)

1. Practical- One of two hours duration (Two hours)

Instructions for Paper Setters:

Question Paper will have four units. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY** question consisting of 15 short answers type questions comprising 8 MCQ and 7 fill in the blanks of one mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

Instruction Aids: Teaching methodology includes series of lectures, making use of charts, transparencies, LCD, Models, slides, practical demonstrations, extension lectures from experts, field visits, discussions, quiz competitions etc.

UNIT – I **12 Hours**

Carbon Oxidation: Respiration –Glycolysis, fate of pyruvate, oxidative pentose phosphate pathway, Tri Carboxylic Acid cycle, mitochondrial electron transport, oxidative phosphorylation, ATP-Synthesis: Mechanism of ATP synthesis, substrate level phosphorylation, oxidative and photo- phosphorylation, chemiosmotic hypothesis

UNIT – II **11 Hours**

Carbon Assimilation: Photosynthesis- Photosynthetic pigments, role of photosynthetic pigments (chlorophylls and accessory pigments),action and absorption spectra, concept of two photosystems, antenna molecules and reaction centres, photochemical reactions, cyclic and non cyclic photophosphorylation, photosynthetic electron transport, PSI, PSII, Carbon assimilation pathways (Calvin Cycle (C3), Hatch and Slack cycle (C4), Crassulacean Acid Metabolism –CAM), Photorespiration.

UNIT – III **11 Hours**

Nitrogen Metabolism: Biological nitrogen fixation, Reduction of nitrogen to ammonia, *Nif* genes, nitrate assimilation, importance of nitrate reductase and its regulation, regulation of nitrogenase, ammonia assimilation.

UNIT – IV **11 Hours**

Lipid Metabolism: Structure and importance of lipids, saturated and unsaturated fatty acids, β (beta) oxidation of fatty acids, glyoxylate cycle, gluconeogenesis and its role in 53or active5353 of lipids during seed germination.

Suggested Readings:

1. Hopkins, W.G. and Huner, A. (2008). Introduction to Plant Physiology. John Wiley and Sons.U.S.A. 4th edition.
2. Taiz, L., Zeiger, E., Møller, I.M. and Murphy, A (2015). Plant Physiology and Development.Sinauer Associates Inc. USA. 6th edition.
3. Harborne, J.B. (1973). Phytochemical Methods. John Wiley & Sons. New York.
4. C. Sahoo (2018). Outlines of Plant Metabolism, Kalyani Publishers, New Delhi.
5. S, K. Gupta (2017). Plant Metabolism, Rastogi Publication, Meerut.
6. Bajracharya D. (1999). Experiments in Plant Physiology-A Laboratory Manual. Narosa Publishing House, New Delhi.
7. Harborne, J.B. (1973). Phytochemical Methods. John Wiley & Sons. New York

Suggested Laboratory Exercises:

1. To test the presence of starch in plant material using iodine.
2. To test the presence of reducing sugars in plant using Fehling's Reagent
3. To study the effect of light intensity and CO_2 concentration on rate of photosynthesis using an aquatic plant.
4. To demonstrate that chlorophyll is necessary for photosynthesis.
5. To demonstrate that light is necessary for photosynthesis.
6. To demonstrate that CO_2 is necessary for photosynthesis.
7. To demonstrate evolution of oxygen during photosynthesis in an aquatic plant.
8. To demonstrate anaerobic respiration on germinating seeds.
9. To demonstrate aerobic respiration using flask method.

GUIDELINES FOR PRACTICAL EXAMINATION

Time: 3 Hrs **Max. Marks: 25**
Practical (Written Exam): 22 **Int. Assessment: 3**

1. Write material required, procedure and precautions for plant metabolism experiment (A). Perform the experiment, record observations, calculations, if any, and results and show the experiment to the Examiner. (8)
2. Write the material required, procedure and precautions for plant metabolism experiment (B). Perform the experiment, record observations, calculations, if any, and results and show the experiment to the Examiner. (8)
3. Viva voce (3)
4. Practical Note Book. (3)

BOTANY
(SEMESTER – V)

Course Title: ECONOMIC BOTANY **Course Code: BOT-DST-10 Maj-502)**

Learning Outcome: The main objective of the paper is aimed to give insight into plant wealth of commercial and economic importance. This paper gives a sound basis of economic botany so that students can venture into fields like Forestry, Agriculture, crop production, Horticulture and conservation Biology etc.

Max. Marks: 100		**Course Credits**= 3+1=4
1. Theory: 75	Written Exam: 67	Internal Assessment: 8
2. Practical: 25	Written Exam: 22	Internal Assessment: 3

Periods per week: 1. Theory- Four of 45 minutes duration (Three hours)
2. Practical- One of two hours duration (Two hours)

Instructions for Paper Setters:

Question Paper will have four units. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY** question consisting of 15 questions having 8 MCQ and 7 fill in the blanks of one mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

Instruction Aids: Teachers can use lecture, lecture-cum-demonstration, project, assignments, problem solving, inductive and deductive methods and can use L.C.D., L.E.D., and multimedia projector in addition to black/white board.

UNIT-I **12 Hours**

Crop production: Area of cultivation, soil requirement, cultivation practices and high yielding varieties of :

i) Cereals (Rice and Maize)
ii) Fibres (Cotton)
iii) Vegetables (Potato)

UNIT-II **11 Hours**

Crop production: Area of cultivation, soil requirement, cultivation practices and high yielding varieties of :

i) Fruits (Mango, Grapes, Lemon)
ii) Sugar-yielding plants (Sugarcane)
iii) Oil-yielding plants (Groundnut, Mustard)

UNIT-III **11 Hours**

Elementary knowledge of the following plants (Botanical names, families, parts used and economic importance) :

i) Wheat, Maize, Rice (Food).
ii) Teak, Shisham, Deodar, Sal (Timbers).
iii) Cotton, Jute, Coir, Flax (Fibres).
iv) Fennel, Coriander, Turmeric, Ginger, Mint, Clove (Spices and Condiments).

UNIT-IV **11 hours**

Elementary knowledge of the following plants (Botanical names, families, parts used and economic importance):

i) Bamboo, Eucalyptus (Pulp plants).
ii) Tea and Coffee (Beverages).
iii) Bajra, Jowar, Ragi (Millets)

Suggested Readings:

1. Kochhar, S.L. (2011). Economic Botany in the Tropics. Macmillan India.
2. Hill, A.F. (1952). Economic Botany. McGraw Hill Book Co.
3. Pandey, B.P. (2010). Economic Botany. S. Chand & Co.
4. Simpson B. and Ogorzaly, M. Economic Botany: Plants in our World (3rd edition). McGraw Hill Publisher, 2000.
5. Singh, V., Pande, P.C. and Jain, D.K. Economic Botany, Rastogi Publications, Meerut, 2012.

Suggested laboratory exercises:

1. Identification and morphology of economically important part/s of crop plants mentioned below: Cereals (wheat, rice, maize); Fibres (cotton); Vegetables (potato); Fruits (mango, grapes, lemon); sugar-yielding plants (sugarcane) and oil-yielding plants (groundnut and mustard).
2. 2.Identification and morphology of economically important part/s of crop plants mentioned below: Spices and condiments (Fennel, Coriander, Turmeric, Ginger, Mint, Clove), Pulp plant (Bamboo, Eucalyptus), Millets (Bajra, Ragi)
3. Preparephoto album of atleast 10 economically important plants mentioned in syllabus.

GUIDELINES FOR PRACTICAL EXAMINATION

Time: 3 Hrs **Max Marks: 25**
Practical: 22 **Internal Assessment: 03**

1. Identify and write illustrated morphological notes on specimens A (from spices and condiments)&B (Cereals/oil yielding) giving their economic importance. (7)
2. Identify and write illustrated morphological notes on specimens C (from vegetables/ fruits/ Fibres/ Millets) & D (from pulp plants/ sugar yielding) giving their economic importance. (7)
3. Practical note-book (3)
4. Viva-voce (3)
5. Photo album (2)

BOTANY
(SEMESTER – V)

Course Title: PLANT PATHOLOGY Course Code: BOT-DST-11 Maj-503)

Learning Outcome: The main objective of the paper is to impart knowledge on various aspects of Plant Pathology like Symptomatology, Defence mechanisms, Host Parasite interactionsand Disease forecasting. It also aims to study the Etiology, Epidemiology and Control of different plant diseases caused by Fungi and other micro-organisms.

Max. Marks: 100Course Credits = 3+1=4

1.	Theory: 75	Written Exam: 67	Internal Assessment: 8
2.	Practical: 25	Written Exam: 22	Internal Assessment: 3

Periods per week: 1. Theory- Four of 45 minutes duration (Three hours)
3. Practical- One of two hours duration (Two hours)

Instructions for Paper Setters:

Question Paper will have four units. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY** question consisting of 15 short answers type questions comprising 8 MCQ and 7 fill in the blanks of one mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

Instruction Aids: Teaching methodology includes series of lectures, making use of charts, transparencies, LCD, Models, slides, practical demonstrations, extension lectures from experts, field visits, discussions, quiz competitions etc.

UNIT – I **12 Hours**

Concept and general symptoms of plant diseases.
Brief Classification of plant diseases on the basis of –
i) causal agencies,
ii) spread and severity of infection,
iii) the part of host infection
iv) the perpetuation and transmission.
Host-parasite interactions.
Dissemination /Transmission of plant pathogens.

UNIT – II **11 Hours**

Disease forecasting and assessment.
Defense Mechanism in Plants (Morphological or structural and Biochemical)
Control of Plant Diseases(Prophylactic, Therapeutic, Immunization measures).

UNIT – III **11 Hours**

Etiology, epidemiology and control of following diseases:
i) Paddy: Bacterial Blight
ii) Wheat: Tundu Disease
iii) Sugarcane: Red Rot
iv) Ergot of Rye
v) Apple: Apple Scab
vi) Damping off disease of seedlings

UNIT – IV **11 Hours**

Diseases caused by plant viruses (TMV) MLOs and Spiroplasma (Little leaf disease ofbrinjal, sandal spike) Viroids: General symptoms and Principles of Control.

Suggested Readings

1. Bilgrami, K.S. and Dube, H.C. A Text Book of Modern Plant Pathology, Vikas Publishing House, New Delhi, 1990, pp. 344
2. Sharma, P.D. The Fungi and Allied Organisms. Ist Ed. Alpha Science International Ltd., 300 2005.
3. Singh, R.S. Plant Diseases. 8 th Ed. Oxford & IBH, New Delhi. 721 pp. 200
4. Agrios, G.N. Plant Pathology. Elsevier Academic Press, 2005, pp.922
5. Mehrotra, R.S. and Aggarwal, A. Plant Pathology, Publ., Tata McGraw Hill, 2003, pp. 846.
6. Sharma, P. D. Plant Pathology, Rastogi Publ., Meerut, 1998.
7. B. K. Mishra. Mycology and Phytopathology, Kalyani Publishers, New Delhi.2017

Suggested laboratory exercises:

1. Study of plant diseases mentioned below with the help of slides/ photographs/ specimens- Paddy: Bacterial Blight, Wheat: Tundu Disease ,Sugarcane: Red Rot, Ergot of Rye Apple: Apple Scab, Damping off disease of seedlings.
2. Study of diseases caused by plant viruses (TMV) MLOs and Spiroplasma (Little leaf disease of brinjal, sandal spike).
3. Project report on various plant diseases in everyday life.

GUIDELINES FOR PRACTICAL EXAMINATION

Time: 3 Hrs **Max. Marks: 25**
Practical (Written Exam): 22 **Internal Assessment: 3**

1. Identify and write illustrated morphological note on specimen A and B. (08)
2. Identify the photograph or slide of specimen C and D giving at least two reasons. (06)
3. Viva voce (03)
4. Practical Notebook (03)
5. Project report (02)

BOTANY
(SEMESTER – V)

Course Title: PLANT BREEDING **Course Code: BOT-DSC-12**
Maj-504

Learning Outcome: The objective of this paper is to learn the significance of plant breeding and its milestone in the field of agriculture. The students will be able to distinguish the breeding method for self and cross pollinated crops

Max. Marks: 100 **Course Credits** = 3+1=4

1.	Theory: 75	Written Exam: 67	Internal Assessment: 8
2.	Practical: 25	Written Exam: 22	Internal Assessment: 3

Periods per week: 1. Theory- Four of 45 minutes duration (Three hours)
4. Practical- One of two hours duration (Two hours)

Instructions for Paper Setters:

Question Paper will have four units. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY** question consisting of 15 short answers type questions comprising 8 MCQ and 7 fill in the blanks of one mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

Instruction Aids: Teaching methodology includes delivery of lectures by the teachers using black board and many other tools like Visual Charts, Over Head projector etc. The students also make group discussions. In practical the students perform experiments pertaining to the syllabus using different equipment.

UNIT – I **12 Hours**

Principles and objectives of Plant Breeding, important conventional methods of breeding of self-pollinated and cross- pollinated crops,
Brief account of Male Sterility: Its types and uses, Male sterile lines.

UNIT – II **11 Hours**

Plant genetic resources: introduction and acclimatization; selection methods: in self-pollinated and cross-pollinated crops. Hybridisation in self-pollinated and cross-pollinated crops.

UNIT – III **11 Hours**

Concept, mechanism, example of polygenic inheritance including kernel colour in wheat. Monogenic verses polygenic inheritance. History, genetic basis of inbreeding depression and heterosis: Applications.

UNIT – IV **11 Hours**

Mutations, Polyploidy, Distant 58or active5858n and Transgenic plants, Types and its role in crop improvement.

Suggested Readings:

1. Singh, B.D. Plant Breeding Principles and Methods. Kalyani Publishers, Ludhiana. 2009.
2. Poehlman, J.M., Sleper, D.A., Breeding Field Crops, AVI. Publ., U.S.A, 1995.
3. Hayward, M.D., Bosemark, M.O., Romagosa, I., Plant Breeding: Principles and Prospects, Chapman and Hall, U.K, 1993.
4. Allard RW. 1981. Principles of Plant Breeding. John Wiley & Sons.
5. Chopra VL. 2004. Plant Breeding. Oxford & IBH.
6. Gupta SK. 2005. Practical Plant Breeding. Agribios.
7. Jain HK and Kharakwal MC. 2004.Plant Breeding and–Mendelian to Molecular Approach, Narosa Publications, New Delhi.
8. Roy D. 2003. Plant Breeding, Analysis and Exploitation of Variation. Narosa Publication House.
9. Sharma JR. 2001. Principles and Practice of Plant Breeding. Tata McGraw-Hill.

Suggested laboratory exercises:

1. Hybridisation studies in self and cross- pollinated crops.
2. To study various methods of vegetative propagation(Grafting/ Layering/ Cutting).
3. To study floral biology of wheat or rice.
4. To study the viability and germination of pollen grains
5. Prepare a report on chemical mutagens used in crop plants.

GUIDELINES FOR PRACTICAL EXAMINATION

Time- 3 Hrs **Max Marks: 25**
Practical: 22 **Internal Assessment: 03**

1. To write floral biology of crop plant (Wheat/ Rice). (07)
2. To write illustrated note on the method of vegetative propagation (Cutting/ Layering/ Grafting). (06)
3. Project Report (03)
4. Practical Note Book (03)
5. Viva –voce (03)

BOTANY
(SEMESTER – V)

Course Title: PLANT RESOURCE UTILIZATION **Course Code: (BOT-DST-13 Maj-505)**

Learning Outcome: The objective of the paper is to impart knowledge about various economic aspects of plants like history of agriculture, cultivation and uses of vegetables, fruits, legumes, millets. It will also 60or active60 student with types of fibres, petroplants.It also deals with the Genetic resources and their conservation.

Max. Marks: 100 **Course Credits** = 3+1=4

1.	Theory: 75	Written Exam: 67	Internal Assessment: 8
2.	Practical: 25	Written Exam: 22	Internal Assessment: 3

Periods per week: 1. Theory- Four of 45 minutes duration (Three hours)
5. Practical- One of two hours duration (Two hours)

Instructions for Paper Setters:

Question Paper will have four units. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY** question consisting of 15 short answers type questions comprising 8 MCQ and 7 fill in the blanks of one mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

Instruction Aids: Teaching methodology includes delivery of lectures by the teachers using black board and many other tools like Visual Charts, Over Head projector etc. The students also make group discussions. In practical the students perform experiments pertaining to the syllabus using different equipment.

UNIT – I **12 Hours**

History of Agriculture- Ancient and modern agriculture.
World Centers of Primary diversity of domesticated plants: The Indo – Burmese Centre, Plant Introductions and Secondary Centers.
Cultivation and uses of following vegetables and fruits–Spinach, Tomato, Brinjal,Citrus, Apple, Banana.

UNIT – II **11 Hours**

Cultivation and economic importance of legumes: (Pigeon pea and Chick pea)
Fibres: Classification, uses, type of fibres –soft fibres, hard fibres, surface fibres, brush fibres and braiding fibres.
Millets: General account,importance and types of millets

UNIT – III **11 Hours**

General account of petroplants.
General account of Gums and their uses.
Insecticides of plant origin; Source, part used and economic importance of Pyrethrum and Rotenone.

UNIT – IV **11 Hours**

Wood and its Uses: Wood as fuel, construction material (Plywood, Veeners, Pilings), wood containers (cooperage) and wood distillation.
General account of tannins and dyes
Genetic Resources and their conservation.

Suggested Readings:

1. Kocchar, S.L. Economic Botany of the Tropics, 2nd edition, Macmillan India Ltd., Delhi, 1998.
2. Sharma, O.P., Hill's Economic Botany (Late Dr. A.F. Hill, adapted by O.P. Sharma), Tata McGraw Hill Co. Ltd., New Delhi, 1996.
3. Swaminathan, M.S. and Kocchar, S.L., (Eds.). Plants and Society, MacMillan Publications Ltd., London, 1989
4. Chrispeels, M.J. and Sadava, D. Plants, Food and People, W.H. Freeman and Co., San Francisco, 1977
5. Dastur, J.F. Medicinal Plants of India and Pakistan.3 rd Ed. Meyerbooks, 1985
6. Simpson, B.B. and Conner-Ogorzaly. E.M. Economic Botany- Plants in Our World. McGraw-Hill Book Company, New York, 1986.
7. Randhawa, M.S. History of Agriculture.Vol. II, ICAR, New Delhi, 1982.

Suggested laboratory exercises:

1. Morphology and uses of legumes mentioned in theory syllabus.
2. Morphology and uses of fruits and vegetables pertaining to theory syllabus.
3. To study various types of fibres (surface, soft, hard, brush, braiding) through charts/ specimens/ photographs.
4. To study nutritional value of millets and prepare a project report on the same.
5. The students should be taken to a recognized Botanical Garden / Plant Nursery to show different plants of economic importance.

GUIDELINES FOR PRACTICAL EXAMINATION

Time- 3 Hrs **Max Marks: 25**
Practical: 22 **Internal Assessment: 03**

1. Identify and write illustrated morphological note on specimen A (fruit / vegetable) and specimen B (legumes). (8)
2. Identify and write an illustrated note on specimen Cand D(from fibres)(6)
3. Project Report (2)
4. Practical Note Book (3)
5. Viva –voce (3)

BOTANY
(SEMESTER – VI)

Course Title: PLANT PHYSIOLOGY **Course Code: (BOT-DSC-14 Maj/Min-601)**

Learning Outcome: The basic aim of this paper is to familiarize the students with various concepts of Plant physiology topics like plant water relations, mineral nutrition, nutrient uptake, translocation in phloem, plant growth regulators, physiology of flowering etc. Apart from the theoretical overview, the course is so designed that it imparts practical understanding of majority of physiological processes in plants. In practicals, students will be familiarize with the various experiments pertaining to theory syllabus.

Max. Marks: 100 **Course Credits** = 3+1=4

1.	Theory: 75	Written Exam: 67	Internal Assessment: 8
2.	Practical: 25	Written Exam: 22	Internal Assessment: 3

Periods per week: 1. Theory- Four of 45 minutes duration (Three hours)
6. Practical- One of two hours duration

Instructions for Paper Setters:

Question Paper will have four units. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY**question consisting of 15 short answers type questions comprising 8 MCQ and 7 fill in the blanks of one mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

Instruction Aids: Teaching methodology includes series of lectures, making use of charts, transparencies, LCD, Models, slides, practical demonstrations, extension lectures from experts, field visits, discussions, quiz competitions etc.

UNIT – I **12 Hours**

Plant Water Relations: Importance of water to plants, 62or active62, diffusion, osmosis, plasmolysis and deplasmolysis, water potential and its components, absorption of water, active and passive mechanism of water absorption; pathway of water movement (apoplast, symplast, transmembrane), ascent of sapand theories to explain ascent of sap; transpiration and mechanism of stomatal movement, mechanism of transpiration, antitranspirants.

UNIT – II **11 Hours**

Mineral nutrition: Hydroponics and its importance; essential macro-and micro elements, essentiality criteria, deficiency symptoms and their role; mineral uptake; mechanism of mineral uptake (passive absorption, active absorption, modern concepts).

Translocation in the phloem: Pressure–Flow hypothesis, Phloem loading and unloading; Source– sink relationship.

UNIT – III **11 Hours**

Proteins: Classification, role and structure (primary, secondary and tertiary) of proteins.
Basics of enzymology: Discovery and nomenclature; classification, structure, properties, factors affecting enzyme activity, mechanism of enzyme action.

UNIT – IV **11 Hours**

Growth and development: Definitions; phases of growth and development; plant movements; the concept of photoperiodism, florigen concept, vernalization, seed dormancy, physiological role of plant hormones– auxins, gibberellins, cytokinins, abscisic acid and ethylene, general account of Brassinosteroids and Jasmonic acid.

Suggested Readings:

1. Bhatia, K.N. Plant Physiology- A Modern Treatise, Trueman Book Co. Jalandhar, 2015.
2. Bhojwani, S.S. Plant Tissue Culture: applications and Limitations. Elsevier Science Publishers, New York, USA, 1990.
3. Hopkins, W.G. Introduction to Plant Physiology. John Wiley & Sons, Inc. New York, U.S.A., 1995.
4. Salisbury, F.B. and Ross, C.W. Plant Physiology (4th Edition) Wadsworth Publishing Co. California, USA, 1992.
5. Srivastava, H.N. Plant Physiology, Biochemistry & Bio-technology. Pradeep Publication, Jalandhar, 2008.
6. Srivastava, H.S. Plant Physiology, Bio-chemistry & Bio-technology, Rastogi Publications, Meerut, 2008.
7. Srivastava, H.S. and Shankar, N. Plant Physiology and Bio-chemistry, Rastogi Publications, Meerut, 2012.
8. Verma, B. Plant Physiology (1st Edition), Athena Academic, St. John Street, London, UK, 2007.
9. Taiz, L. and Zeiger, E. Plant Physiology (5th edition), Sinauer Associates, Inc., Publishers, Massachusetts, USA, 2010.
10. Bajracharya D. (1999). Experiments in Plant Physiology-A Laboratory Manual. Narosa Publishing House, New Delhi.

Suggested laboratory exercises: Plant Physiology:

1. To determine osmotic pressure of cell sap by plasmolytic method.
2. To demonstrate 63or active63 pressure using: i) Imbibition pressure apparatus. Ii) Plaster of Paris cone.
3. To demonstrate osmosis through potato osmoscope.
4. To demonstrate the measurement of transpiration using simple 63or active.
5. To demonstrate transpiration pull.
6. To study the effect of light intensity, and wind velocity on the rate of transpiration using Ganong's 63or active.
7. To compare the rate of transpiration from the two sides of a leaf using:
   i) Vaseline method ii) Cobalt chloride method
8. To demonstrate the mechanism of opening and closing of stomata.
9. To demonstrate the path of ascent of sap.
10. To test the presence of proteins in plant material using Biuret test/Xanthoproteic test.
11. To test the presence of amino acids in plant material using Ninhydrin reagent.
12. To demonstrate phototropism.
13. To demonstrate geotropism using clinostat.

GUIDELINES FOR PRACTICAL EXAMINATION

Time: 3 Hrs **Max. Marks: 25**
Practical (Written Exam): 22 **Internal Assessment: 3**

1. Write material required, procedure and precautions for physiology Experiment (A). Perform the experiment, record observations, calculations, if any, and results and show the experiment to the Examiner. (8)

2. Write the material required, procedure and precautions for physiology experiment (B). Perform the experiment, record observations, calculations, if any, and results and show the experiment to the Examiner. (8)

3. Viva voce (3)

4. Practical Note Book. (3)

BOTANY
(SEMESTER – VI)

Course Title: ANATOMY OF ANGIOSPERMS Course Code: (BOT-DSC-15 Maj-602)

Learning Outcome: The basic aim of this paper is to familiarize the students with internal structure of various plant parts, their growth, pattern and abnormality in structure development. The vast range of variation found in angiosperms provides a platform to students for acquiring basic knowledge of flowering plants which makes foundation of applied branches like horticulture, floriculture, olericulture and arboriculture.

Max. Marks: 100 **Course Credits** = 3+1=4

1. Theory: 75 Written Exam: 67 Internal Assessment: 8
2. Practical: 25 Written Exam: 22 Internal Assessment: 3

Periods per week: 1. Theory- Four of 45 minutes duration (Three hours)
7. Practical- One of two hours duration (Two hours)

Instructions for Paper Setters:

Question Paper will have four units. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY** question consisting of 15 short answers type questions comprising 8 MCQ and 7 fill in the blanks of one mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

Instruction Aids: Teaching methodology includes series of lectures, making use of charts, transparencies, LCD, Models, slides, practical demonstrations, extension lectures from experts, field visits, discussions, quiz competitions etc.

UNIT – I **12 Hours**

Tissue systems: Meristematic tissue, Classification of meristems, Permanent tissues (Simple and Complex), Secretory (special) tissues, Apical meristems: Shoot apex and organization of shoot apex (Apical cell theory, Histogen theory, Tunica Corpus theory, cyto-histological zonation) , Root apex and Organization of root apex (Apical cell theory, Histogen theory, Korper-Kappe theory); Quiescentcentre; Root cap.

UNIT – II **11 Hours**

Root: Internal structure of primary dicot root (*Helianthus*), Internal structure of primary monocot root (*Zea mays*), Secondary growth in dicot root (*Helianthus*), differences between dicot and monocot root, Anatomy of modified storage and reproductive root (*Ipomeabatatas*), Anatomy of respiratory root or pneumatophores.

UNIT – III **11 Hours**

Stem: Internal structure of primary dicot stem (*Helianthus*), Internal structure of primary monocot stem (*Zea mays*), Secondary growth in dicot stem (*Helianthus*), differences between dicot and monocot stem, Anomalous stem structure in *Boerhaavia*, *Nyctanthes*, *Mirabilis Dracaena*.

UNIT – IV **11 Hours**

Leaf: Internal structure of dicot leaf (*Mangiferaindica*), Internal structure of monocot leaf (*Zea mays*), Differences between monocot and dicot leaf,Kranz anatomy

Suggested Readings:

1. Singh, Pandey and Jain (2017). Anatomy of Angiosperms, Rastogi Publication, Meerut.
2. Eames and Mc Daniels (1986). An introduction to plant anatomy, Tata Mc Grow Hills, New Delhi
3. Esau, K. (1977). Anatomy of Seed Plants. John Wiley & Sons, Inc., Delhi.
4. M. S. Tayal (2012) Rajpal and Sons, New Delhi
5. B. K. Mishra (2017). Anatomy of Angiosperms, Kalynai Publishers, New Delhi.
6. B. P. Pandey (2017) Plant Anatomy, S. Chand Publication, New Delhi.

Suggested laboratory exercises:

1. Preparation of permanent slides to showT.S. of:
   i) dicot root (*Helianthus*)
   ii) monocot root (*Zea mays*)
   iii) dicot stem (*Helianthus*)
   iv) monocot stem (*Zea mays*)
   v) dicot leaf (*Mangiferaindica*)
   vi) monocot leaf (*Zea mays*)
2. Preparation of permanent slides to show anomalous internal structure of:
   i) 1.*Boerhaavia*
   ii) 2.*Nyctanthes*
   iii) 3.*Mirabilis*
   iv) 4.*Dracaena*
3. Study through permanent slides/ photographs/ charts of T.S. of:
   i) pneumatophore
   ii) modified storage root / reproductive root of *Ipomeabatatas*

GUIDELINES FOR PRACTICAL EXAMINATION

Time: 3 Hrs **Max. Marks: 25**
Practical (Written Exam): 22 **Internal Assessment: 3**

1. Cut T.S. of specimen A (Root/Stem/Leaf). Make its permanent slide. Identify, draw its well labeled diagram and show it to examiner. 06
2. Cut T.S. of specimen B (Anomalous stem). Make its permanent slide. Identify, draw its well labeled diagram and show it to examiner. 06
3. Identify slides C and D giving at least two reasons. 04
4. Practical Note Book 03
5. Viva –voce 03

BOTANY
(SEMESTER – VI)

Course Title: MEDICINAL BOTANY Course Code: (BOT-DSC-16 Maj-603)

Learning Outcome: The basic objective of this paper is aimed to give an insight into plants medicinal wealth. This will 67or active67 the students with the properties and uses of medicinal plants in their locality. After study this paper the students will understand utility of plants as medicine.

Max. Marks: 100 **Course Credits** = 3+1=4

1. Theory: 75 Written Exam: 67 Internal Assessment: 8
2. Practical: 25 Written Exam: 22 Internal Assessment: 3

Periods per week: 1. Theory- Four of 45 minutes duration (Three hours)
8. Practical- One of two hours duration (Two hours)

Instructions for Paper Setters:

Question Paper will have four units. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY** question consisting of 15 short answers type questions comprising 8 MCQ and 7 fill in the blanks of one mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

Instruction Aids: Teaching methodology includes series of lectures, making use of charts, transparencies, LCD, Models, slides, practical demonstrations, extension lectures from experts, field visits, discussions, quiz competitions etc.

UNIT-I **12 Hours**

Medicinal plants: history, importance and prospects. Medicinal Plants – past and present status in world and India. History and classification of drugs, general study of plants as source of drugs. Antibiotics- History, sources, production, properties and types of antibiotics: Penicillin, Streptomycin, Tetracycline, Erythromycin, Chloramphenicol

UNIT-II **11 Hours**

Elementary knowledge of the following plants (Botanical names, families, parts used and economic importance):

i) Drugs obtained from roots and other underground parts (Liquorice, Ashwagandha)
ii) Drugs obtained from bark (Quinine, Cinnamon)
iii) Drugs obtained from stem/ wood (Sandal wood, Ephedrine)

UNIT-III **11 Hours**

Elementary knowledge of the following plants (Botanical names, families, parts used and economic importance) :

i) Drugs obtained from leaves (Tulsi, Aloe)
ii) Drugs obtained from flowers (Chamomile, Clove)
iii) Drugs obtained from Fruits/ seeds (Opium, Isabgol)

UNIT-IV **11 Hours**

1. Elementary knowledge of Plants used as lotions /ointment (Pot marigold/Neem).
2. Elementary knowledge of plants used as poultice (Castor, garlic)
3. National Medicinal Plant Board and State Medicinal Plant Boards – objectives and functions. Other organizational initiatives for promotion of Medicinal plants at National and International levels.
4. Threats to medicinal plants: over-exploitation, habitat loss, climate change.

Suggested Readings:

1. Indian Medicinal Plants by P.C. Trivedi (2009).
2. Handbook of MAPs by S.K. Bhattacharjee (2009).
3. Panda H., Handbook of Ayurvedic Medicines, National Institute of Industrial Research, Delhi
4. CSIR–Cultivation and Utilization of Medicinal Plants Pandey B.P. 1978. Economic Botany. S. Chand and Company LTD. Ram Nagar, New Delhi. 1-534.
5. Raphael Ikan, Natural Products: A Lab Guide, Academic Press, 1991, 2nd edition
6. DuttAshwin, An Introduction to Medicinal Plants, Adhyayan Publishers, and distributers, 2009, 1st edition
7. Akerele, O., Heywood, V. and Synge, H. (1991). The Conservation of Medicinal Plants. Cambridge University Press.
8. Kapoor, L.D. (2001). Handbook of Ayurvedic medicinal plants. Boca Raton, FL: CRC Press. 9. Saroya, A.S. (2017). Ethnobotany. ICAR publication.
9. Sharma, R.(2003). Medicinal Plants of India-An Encyclopaedia. Delhi: DayaPublishing House.
10. Sharma, R. (2013) Agro Techniques of Medicinal Plants. Daya Publishing House,Delhi.
11. Thakur, R.S., H.S. Puri, and Husain, A.(1989). Major medicinal plants of India.Central Institute of Medicinal and Aromatic Plants, Lucknow, India.

Suggested laboratory exercises:

1. Identification and morphology of medicinally important part/s of plants mentioned below: Liquorice, Ashwagandha, Tulsi, Cinnamon, Clove, Tulsi, Neem, Pot marigold, Garlic, Castor.
2. To study various types of antibiotics.
3. Visit to plant nursery / medicinal plant garden.
4. Prepare Photo album of atleast10 local medicinal plants with brief description.

GUIDELINES FOR PRACTICAL EXAMINATION

Time: 3 Hrs **Max Marks: 25**
Practical: 22 **Internal Assessment: 03**

1. Identify and write illustrated morphological notes on specimens A, B & C giving their economic importance. (9)
2. Write a note on antibiotic (announced by examiner) (5)
3. Photo album (2)
4. Practical note-book (3)
5. Viva-voce (3)

BOTANY
(SEMESTER – VI)

Course Title: PLANT BIOTECHNOLOGY Course Code: (BOT-DSC-17 Maj-604)

Learning Outcome: The objective of the paper is to impart knowledge about various aspects of Plant Biotechnology like Concept, Scope, Tools, Tissue Culture Techniques, Protoplast Culture and Somatic Hybridization, Anther and Pollen Culture, Recombinant DNA Technology, Biosensors and Application of Biotechnology in Medicines , Agriculture, Transgenic plants, Industry and Pollution control etc.

Max. Marks: 100 **Course Credits** = 3+1=4

1.	Theory: 75	Written Exam: 67	Internal Assessment: 8
2.	Practical: 25	Written Exam: 22	Internal Assessment: 3

Periods per week: 1. Theory- Four of 45 minutes duration (Three hours)
9. Practical- One of two hours duration

Instructions for Paper Setters:

Question Paper will have four units. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY** question consisting of 15 short answers type questions comprising 8 MCQ and 7 fill in the blanks of one mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

Instruction Aids: Teaching methodology includes delivery of lectures by the teachers using black board and many other tools like Visual Charts, Over Head projector etc. The students also make group discussions. In practical the students perform experiments pertaining to the syllabus using different equipment.

UNIT – I **12 Hours**

Scope of Biotechnology
Tissue Culture Techniques: Introduction to the techniques of plant tissue culture, Nutritional requirements, Somaclonal variation, Somatic embryogenesis, Embryo culture, Germplasm storage including Cryopreservation.

UNIT – II **11 Hours**

Protoplast Culture and Somatic Hybridization: Isolation of protoplasts, culture and fusion methods, production of hybrids and cybrids.
Anther and Pollen Culture: Culture procedures for the induction of haploids, significance of haploids.

UNIT – III **11 Hours**

Recombinant DNA Technology: Isolation and purification of DNA from plant cells, DNA engineering through cutting and joining DNA molecules, Restriction Endonucleases, Ligases.
Application of Biotechnology in Medicines, Agriculture, Transgenic production and role in Human Welfare.

UNIT – IV **11 Hours**

Applications of Microbes in (a) Agriculture with special reference to Biological N2 fixation, (b) Industry Fermentation (Ethanol, Curd) (c) Biogas Production.
Biosensors: Principles of detection and Applications.

Suggested Readings:

1. Bhojwani, S. S. and Razdan, M. K. Plant Tissue Culture – Theory and Practice. Elsevier Science Pub. Co. Inc., New York, pp. 471, 1983.
2. Old, R.W. and Primrose, S.B. Principles of Gene Manipulation: An Introduction to Genetic Engineering. Blackwell Scientific Publications, Oxford, pp. 403, 1985.
3. Street, H. E. Plant Tissue and Cell Culture. Blackwell Scientific Publications, London, pp. 614, 1973.
4. Bajaj, Y.P.S. Biotechnology in Agriculture and Forestry Plant Protoplasts and Genetic Engineering I., Springer Verlag, Berlin. 1989.
5. Singh, B.D. Biotechnology: Expanding Horizon. Kalyani Publishers, New Delhi.860 pp. 2007.
6. Smith, R.H. Plant Tissue Culture, Technique and Experiments.2 nd Ed. Academic Press, New York.231pp. 2000.

Suggested laboratory exercises:

1. Composition and preparation of various culture media: (a) Knop's medium, (b) Murashige & Skoog's medium.
2. Inoculation of Plant Materials, leaf, root, stem
3. Demonstration and/ or experimentation to study callus induction, organogenesis, plant regeneration, anther culture, embryo culture, protoplast isolation and fusion in vitro.
4. Visit to university/regional laboratories for demonstration of modern instruments employed in recent biotechnological techniques and prepare report of visit.

GUIDELINES FOR PRACTICAL EXAMINATION

Time- 3 Hrs **Max Marks: 25**
Practical: 22 **Internal Assessment: 03**

1. Write the procedure and composition for the preparation of culture medium. (07)
2. Write a note on Inoculation of given plant material/ embryo culture/ anther culture. (07)
3. Laboratory Visit Report (02)
4. Practical Note Book (03)
5. Viva –voce (03)

BOTANY
(SEMESTER – VI)

Course Title: BRYOPHYTES **Course Code: (BOT-DSC-18 Maj-605)**

Learning Outcome: The focus of this paper is tounderstand theunique position of Bryophytes as land-colonizing plants, their structural diversity, and ecological importance. The students will acquire elaborate knowledge about Bryophytes and which can be used to develop skill for identification, understanding economic importance and applications of bryophytes.

Max. Marks: 100 **Course Credits** = 3+1=4

1.	Theory: 75	Written Exam: 67	Internal Assessment: 8
2.	Practical: 25	Written Exam: 22	Internal Assessment: 3

Periods per week: 1. Theory- Four of 45 minutes duration (Three hours)
10. Practical- One of two hours duration (Two hours)

Instructions for Paper Setters:

Question Paper will have four units. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY** question consisting of 15 short answers type questions comprising 8 MCQ and 7 fill in the blanks of one mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

Instruction Aids: Teaching methodology includes series of lectures, making use of charts, transparencies, LCD, Models, slides, practical demonstrations, extension lectures from experts, field visits, discussions, quiz competitions etc.

UNIT-I **12 Hours**

General characteristics and origin of Bryophytes, Fossil bryophytes (a brief account). Ecology of bryophytes: Habitat, growth forms. Economic importance of bryophytes.

UNIT-II **11 Hours**

Systematic position,Structure,reproduction and life cycle (excluding developmental details) of the following groups:

i) Marchantiales; Marchantiaceae-*Marchantia*
ii) Jungarmanniales- *Porella*
iii) Anthocerotales- *Notothylas*
iv) Polytrichales- *Polytrichum*

UNIT-III **11 Hours**

1. Structure and development of sex organs in bryophytes.
2. Water relation: Drought tolerance: Desiccation and Rehydration: Ectohydric, Endohydric and Mixohydric Mosses.
3. Experimental Studies: Effect of various factors on protonemal differentiation and bud formation.

UNIT-IV **11 Hours**

1. Apogamy and Apospory.
2. Chemistry of bryophytes- a brief account of distribution of various organic compounds in different groups of bryophytes.
3. Cytological studies in bryophytes- chromosome numbers in major groups of bryophytes and polyploidy in brief.

Suggested Readings:

1. Puri, P. (1981). Bryophytes: Morphology, Growth and differentiation. Atma Ram and Sons, Delhi and Lucknow.
2. Cavers, F. (1911). TheInter- relationship of Bryophyta. New Phytology Reprint No. 4:1-203.
3. Chopra, R.S. (1976). Inter- relationship of Indian Bryophytes. The Chronica Botanica, New Delhi.
4. Parihar, N.S. (1991).An Introduction to Embryophyta :BryophytaVol II – Central Book Depot, Allahabad.
5. Pandey B.P. (2007). Botany for Degree Students. Published by S. Chand & Company Ltd.
6. Vashistha, B.R., Sinha, A.K. and Adarsh Kumar (2014) Botany for Degree Students, Published by S. Chand & Company Ltd.
7. Vanderpoorten, A. and Goffinet, B. (2009). Introduction to Bryophytes. Cambridge University Press, UK.

Suggested LaboratoryExercises

1. Study of morphology of various genera mentioned in theory syllabus.
2. To study permanent slides/ photographs of m- chromosomes and sex chromosomes.
3. Preparation of permanent stained slide of :
   TS thallus of *Marchantia*
   T.S. of aerial shoot of *Polyrtichum*.
4. Study of permanent slides -
   Marchantia
      i) V.S. antheridiophore
      ii) V.S. archegoniophore
      iii) L.S. mature sporophyte
   Porella
      i) Mature sporogonium
   Notothylas
      i) L.S. of mature sporogonium
   Polytrichum
      i) L.S. of capsule

GUIDELINES FOR PRACTICAL EXAMINATION

Time: 3 Hrs **Max Marks: 25**
Practical: 22 **Internal Assessment: 03**

1. Cut T.S. stained and make a permanent mount of specimen A. Identify and draw its 72or act diagram and show it to examiner. (06)
2. Identify and write illustrated morphological note on specimen B. (06)
3. Identify the slide C and D giving at least one reason each. (04)
4. Practical file (03)
5. Viva voce (03)

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CHEMISTRY

Scheme of Teaching and Examination

Paper	Code of the Major Paper	Name of Major paper	Credit (T+P)	Contact Hours	Marks (T+P)
Semester 5					
A-9	CHM-DSC-9-MAJ – 501	Inorganic Chemistry – 1	3+1	45+30	75+25
A-10	CHM-DSC-10-MAJ – 502	Organic Chemistry – 1	3+1	45+30	75+25
A-11	CHM-DSC-11-MAJ – 503	Physical Chemistry – 1	3+1	45+30	75+25
A-12	CHM-DSC-12-MAJ – 504	Analytical Chemistry – 1	3+1	45+30	75+25
A-13	CHM-DSC-13-MAJ – 505	Environment Chemistry -1	3+1	45+30	75+25
	CHM-INT-1	Internship	4	-	100
Semester 6					
Paper	**Code of the Major paper**	**Name of Major paper**	**Credit (T+P)**	**Contact Hours**	**Marks (T+P)**
A-14	CHM-DSC-14-MAJ – 601	Inorganic Chemistry – 2	3+1	45+30	75+25
A-15	CHM-DSC-15-MAJ – 602	Organic Chemistry – 2	3+1	45+30	75+25
A-16	CHM-DSC-16-MAJ – 603	Physical Chemistry – 2	3+1	45+30	75+25
A-17	CHM-DSC-17-MAJ – 604	Analytical Chemistry – 2	3+1	45+30	75+25
A-18	CHM-DSC-18-MAJ – 605	Environment Chemistry -2	3+1	45+30	75+25
Paper	**Code of the Minor paper**	**Name of Minor paper**	**Credit (T+P)**	**Contact Hours**	**Marks (T+P)**
MINOR	CHM-DSC-6-MIN-601	Inorganic Chemistry – 1	3+1	45+30	75+25

Semester	**5**
Code of Paper	**CHM-DSC-9-MAJ-501 (Table 1)**
Title of Paper	**Inorganic Chemistry-1**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100 (75 Theory + 25 Practical)**
Marks distribution of Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of Practical paper	**End semester = 22** **Internal Marks = 03**

Applicable to the students

1. Those who have opted for chemistry as a Major subject in Table 1

LEARNING OUTCOMES SPECIFIC TO DISCIPLINE

The students will learn the following-

1. Important aspects of transition metals
2. Chemistry of compounds of f-block elements
3. Chemistry of coordination compounds

GENERIC LEARNING OUTCOMES

1. **The students will be able to learn Problem Solving, Practical Laboratory Skills, Data Management and Evaluation,** Ethical and Professional Responsibility, **Application based Knowledge.**
2. These generic outcomes will equip students with a broad set of transferable skills, making them capable of conducting rigorous scientific analysis while working safely, ethically, and effectively in collaborative environments.

Semester – 5
Syllabus for Theory Paper
CHM-DSC-9-MAJ-501 (Table 1)
Inorganic Chemistry-1

End semester = 67 **Credits – 03**

Internal Marks = 08 **Examination Time allowed – 3 Hours**

UNIT I: Chemistry of d-Block Elements (15 Hrs.)

First Transition Series

Position in the periodic table, Electronic configuration, Physical and Chemical properties – Atomic radii, covalent radii, density, melting and boiling points, oxidation states, ionization energies, reducing properties, electronegativity, electron affinity, metallic behavior, magnetic properties, colour, complex formation, interstitial compounds, catalytic properties, alloy formation. Relative stability of their oxidation states, coordination number and geometry. Chemistry of compounds and complexes of Sc, Ti, V, Cr, Mn, Fe, Co, Ni, Cu, and Zn.

Second and Third Transition Series

General characteristics, comparative treatment with their 3*d*-analogues in respect of ionic radii, oxidation states, magnetic 75or acti, spectral properties and stereochemistry. Important compounds of Zr, Nb, Ta, Mo, Ru, Rh, Pd, W, Re, Os, Ir, Pt, Au, Hg, Aqueous chemistry. Isopoly and heteropoly. Metal-metal bonding.

UNIT II: Chemistry of f-Block Elements (15 Hrs.)

Lanthanide Elements

Occurrence and isolation, electronic structure, oxidation states and ionic radii, lanthanide contraction, complex formation, colour and spectral properties, magnetic properties, lanthanide complexes and compounds (oxides, hydroxides, nitrides, etc) and their uses.

Actinide Elements

General features and chemistry of actinides, comparison with lanthanides, extraction of Th from Monazite sand, extraction of U, chemistry of separation of Np, Pu and Am from U, Transuranic elements. Later or heavier actinides.

UNIT III: Chemistry of Coordination Compounds (15 Hrs.)

Coordination Chemistry – 1

Definitions and terminology, Types of ligands, Werner's coordination theory and its experimental verification, effective atomic number concept, chelates, nomenclature of coordination compounds, isomerism in coordination compounds, Stereochemistry of coordination compounds, Valence bond theory of transition metal complexes. Properties of Coordination compounds, i.e. magnetic properties, colours (Qualitative approach only), and use of coordination compounds.

Books/Literature suggested

1. Cotton, F.A., Wilkinson, G., Gaus, P.L., Basic Inorganic Chemistry; 2nd edition, Pubs: John Wiley and Sons, 1995.
2. Lee, J.D., Concise Inorganic Chemistry; 4th edition, Pubs: Chapman & Hall Ltd., 1991.
3. Shriver, D.E., Atkins, P.W., Inorganic Chemistry; 4th edition, Pubs: Oxford University Press, 2006.
4. Douglas, B., Medaniel, D., Atenander, J., Concepts and Models of Inorganic Chemistry; 3rd edition, Pubs: John Wiley and Sons Inc., 1994,
5. Porterfield, W.W., Wesky, A., Inorganic Chemistry; Pubs: Addison-Wesley Publishing Company,1984.
6. Miessler, G.L., Tarr, D.A., Inorganic Chemistry; 3rd edition, Pubs: Pearson Education Inc., 2004.
7. Puri, B.R., Sharma, L.R., Kalia, K.K., Principles of Inorganic Chemistry; 30th edition, Pubs: Milestones Publisher, 2006-07

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit** (of **19 marks**) will consist of two types of questions: **9 multiple-choice questions** of one mark each and **5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:** the **compulsory question** and one question each from the **UNIT-1, UNIT-2 and UNIT-3**.
2. The use of scientific calculator is not allowed.

Semester – 5
Syllabus for Practical Paper
CHM-DSC-9-MAJ-501 (Table 1)
Inorganic Chemistry-1

End semester = 22 **Credit – 01**

Internal Marks = 03 **Examination Time allowed – 3 Hours**

I. QUALITATIVE ANALYSIS

Semimicro Analysis, cation analysis, separation and identification of ions from groups I, II, III, IV, V and VI. Anion analysis (4 ions).

Books/Literature suggested

1. Svehla, Vogel's Qualitative Inorganic Analysis (revised); 7th edition Pubs: Orient Longman, 1996.
2. Bassett, J., Denney, R.C., Jeffery, G.H., Mendham, J., Vogel's Textbook of Quantitative Inorganic Analysis (revised); 4th edition, 1978.
3. Palmer, W.G., Experimental Inorganic Chemistry; 1st edition, Pubs: Cambridge, 1954.
4. Brauer, Handbook of Preparative Inorganic Chemistry; 2nd edition, Pubs: Academic Press, 1963, Vol. I, & 1965, Vol. II

Instruction to Examiners

Four ions with no interference (anions such as PO_4^{3-}, BO_3^{-3} and similar anions like Cl^-, Br^-, I^- etc. and cations from the same group) may not be given.

Note: Practical examination will be of three hours duration divided into two parts (First, half hour for written and second half will be the next two and half hours for performance).

1. Candidate will be examined based on any two experiments given by the examiner. Each experiment carries 08 marks (3 written + 5 performance).
2. Viva-Voce: 04 marks.
3. Note Book: 02 marks.
4. Internal Assessment: 03 marks

Semester	**5**
Code of Paper	**CHM-DSC-10-MAJ-502 (Table 1)**
Title of Paper	**Organic Chemistry-1**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100 (75 Theory + 25 Practical)**
Marks distribution of Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of Practical paper	**End semester = 22** **Internal Marks = 03**

Applicable to the students

1. Those who have opted for chemistry as a Major subject in Table 1.

LEARNING OUTCOMES SPECIFIC TO DISCIPLINE

The students will learn the following-

1. Important aspects of chemistry of carbonyl compounds and a few related name reactions
2. Chemistry of carboxylic acid and its derivatives
3. Chemistry of organometallic compounds
4. Formation and use of enolates in organic synthesis

GENERIC LEARNING OUTCOMES

1. **The students will be able to learn Problem Solving, Practical Laboratory Skills, Data Management, Evaluation,** estimation and determination through a**pplication based organic chemistry techniques and concepts.**
2. These generic outcomes will equip students with a broad set of transferable skills, making them capable of conducting rigorous scientific analysis while working safely, ethically, and effectively in collaborative environments.

Semester – 5
Syllabus for Theory Paper
CHM-DSC-10-MAJ-502 (Table 1)
Organic Chemistry-1

End semester = 67 **Credits – 03**

Internal Marks = 08 **Examination Time allowed – 3 Hours**

Unit I: Carbonyl Compounds (15 Hrs)

Aldehydes and Ketones

Nomenclature and structure of the carbonyl group. Physical properties of aldehydes & ketones. Synthesis of aldehydes and ketones with particular reference to the synthesis of aldehydes from acid chorides. Synthesis of aldehydes and ketones using 1,3-dithianes, synthesis of ketones from nitriles and from carboxylic acids.

Mechanism of nucleophilic additions to carbonyl group with particular emphasis on Benzoin, Aldol, Perkin and Knoevenagel condensations. Condensation with ammonia and its derivatives, Wittig reaction. Use of acetals as protecting group. Oxidation of aldehydes, Baeyer-Villiger oxidation of ketones, Cannizzaro reaction, MPV, Clemmensen, Wolff-Kishner, $LiAlH_4$ and $NaBH_4$ reductions.

Unit II: Carboxylic Acids and Derivatives (15 Hrs)

Carboxylic Acids

Nomenclature, structure and bonding, physical properties, acidity of carboxylic acids, effects of substitutions on acid strength. Preparations of carboxylic acids. Reactions of carboxylic acids. Mechanism of decarboxylation. Methods for formation of halo acids.

Carboxylic Acid Derivatives

Structure and nomenclature of acid chlorides, esters, amides and acid anhydrides. Relative stability & reactivity of acyl derivatives. Physical properties, interconversion of acid derivatives by nucleophilic acyl substitution. Preparation of carboxylic acid derivatives, chemical reactions. Mechanisms of esterification and hydrolysis (acidic and basic).

Unit III: Organometallic Compounds and Enolates (15 Hrs)

Organic Synthesis via Enolates

Acidity of α-hydrogens, alkylation of diethyl malonate and ethyl acetoacetate. Synthesis of ethyl acetoacetate: the Claisen condensation. Keto-enol tautomerism of ethyl acetoacetate.

Organometallic Compounds:

Organomagnesium Compounds: The Grignard reagents: Formation, structure and chemical reactions. Organozinc Compounds: Formation and Chemical Reactions. Organolithium Compounds: Formation and Chemical Reactions.

Books/Literature suggested

1. Morrison, R.T., Boyd, R.N., Organic Chemistry; 6thedition, Pubs: Prentice-Hall,1992.
2. Wade Jr., L.G., Singh, M.S., Organic Chemistry; 6thedition, Pubs: Pearson Education, 2008.
3. Mukherji, S.M., Singh, S.P., Kapoor, R.P., Organic Chemistry; Pubs: New Age International, 1985, Vol. I.II, III.
4. Carey,F.A., Organic Chemistry; 4th edition, Pubs: McGraw-Hill, 2000.
5. Solomons, T.W., Fundamentals of Organic Chemistry; 5thedition, Pubs: John Wiley & Sons,1997.

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit** (of **19 marks**) will consist of two types of questions: **9 multiple-choice questions** of one mark each and **5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:** the **compulsory question** and one question each from the **UNIT-1, UNIT-2 and UNIT-3**.
2. The use of scientific calculators is not allowed.

Semester – 5
Syllabus for Practical Paper
CHM-DSC-10-MAJ-502 (Table 1)
Organic Chemistry-1

End semester = 22 **Credit – 01**
Internal Marks = 03 **Contact Hours – 2/week**
Examination Time allowed **3 Hours**

I Preparation, TLC monitoring and purification via column chromatography for a minimum of 5 of the following compounds:

a) Methyl Orange
b) Para-Aminoazobenzene
c) Aspirin
d) Picric acid
e) Acetanilide
f) Para-Bromoacetanilide
g) Phenyl azo-β-naphthol
h) 7-Hydroxy-4-methylcoumarin

II Determination of the per cent purity of the given carbonyl compound via the hydrazone formation method.

III Estimation of aniline in the given solution by acetylation/bromination.

Books/Literature suggested

1. Harwoor, L.M., Moody, J., Experimental Organic Chemistry; 1st edition, Pubs: Blackwell Scientific Publications, 1989.
2. Forniss, B.S., Rogers, V., Vogel's Text Book of Practical Organic Chemistry; Pubs: Dorling Kindhsky Pvt. Ltd., 1989.
3. Bansal, R.K., Laboratory Manual of Organic Chemistry; 3rd edition, Pubs: Wiley Eastern Limited, 1994.
4. Furniss, B.S., Hannaford, A.J., Rogers, V., Smith, P.W.G., Tatchell, A.R., Vogel's Textbook of Practical Organic Chemistry; 4th edition, Pubs: Longman group, 1978.
5. Aggarwal, O.P., Advanced Practical Organic Chemistry, 30th edition, 2013, Goel Publishing House.

General Instructions to the Examiners:

Note: Practical examination will be of three hours duration divided into two parts (First, half hour for written and second half will be the next two and half hours for performance).

1. Candidate will be examined based on any two experiments given by the examiner. Each experiment carries 08 marks (3 written + 5 performance).
2. Viva-Voce: 04 marks.
3. Note Book: 02 marks.
4. Internal Assessment: 03 marks

Semester	**5**
Code of Paper	**CHM-DSC-11-MAJ-503 (Table 1)**
Title of Paper	**Physical Chemistry-1**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100 (75 Theory + 25 Practical)**
Marks distribution of Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of Practical paper	**End semester = 22** **Internal Marks = 03**

Applicable to the students

1. Those who have opted for chemistry as a Major subject in Table 1.

LEARNING OUTCOMES SPECIFIC TO DISCIPLINE

The students will learn the following-

1. **Solutions:** Quantify molar mass and vapor pressure changes using Raoult's Law, colligative properties, and van't Hoff factors for electrolytes.
2. **Electrochemistry:** Analyze ionic conductance and mobility to construct electrochemical cells and predict reaction spontaneity via the Nernst equation.
3. **Photochemistry:** Evaluate light-matter interactions and quantum yield while mapping excited-state pathways (Jablonski diagrams) for carbonyl and alkene reactions.

GENERIC LEARNING OUTCOMES

The students will be able to learn,

1. Quantitative Reasoning: Mastery of mathematical models to solve physical chemistry problems.
2. Experimental Proficiency: Understanding the setup and logic behind classical laboratory techniques (conductometry, osmometry, etc.).
3. Molecular Analysis: Ability to relate macroscopic observations (like color or pressure) to microscopic molecular behavior.
4. Analytical Thinking: Connecting theoretical laws (like Nernst or Raoult) to real-world industrial and biological applications.

Semester – 5
Syllabus for Theory Paper
CHM-DSC-11-MAJ-503 (Table 1)
Physical Chemistry-1

End semester = 67 **Credits – 03**

Internal Marks = 08 **Examination Time allowed – 3 Hours**

UNIT – I Solutions (15 Hrs.)

Solutions and Colligative Properties:

Dilute Solutions and Colligative Properties: Ideal and non-ideal solutions, methods of expressing concentrations of solutions, activity and activity coefficient. Dilute solution, colligative properties, Raoult's law, relative lowering of vapour pressure, molecular weight determination. Osmosis, law of osmotic pressure and its measurement, determination of molecular weight from osmotic pressure. Elevation of boiling point and depression of freezing point, Thermodynamic derivation of relation between molecular weight and elevation in boiling point and depression of freezing point. Experimental methods for determining various colligative properties. Abnormal molar mass, degree of dissociation and association of solutes.

UNIT – II Electrochemistry (15 Hrs.)

Electrochemistry – I:

Electrical transport –Conduction in metals and in electrolyte solutions, specific conductance and equivalent conductance, measurement of equivalent conductance, variation of equivalent and specific conductance with dilution. Migration of ions and Kohlrausch Law, Arrhennius theory of electrolyte dissociation and its limitations, weak and strong electrolytes, Ostwald's dilution law, its uses and limitations. Debye-Huckel-Onsager's equation for strong electrolytes (elementary treatment only). Transport number

Electrochemistry-II:

Types of reversible electrodes – gas metal – ion, metal –insoluble salt – anion and redox electrodes. Electrode reactions, Nernst equation, derivation of cell E.M.F. and single electrode potential, standard hydrogen electrode – reference electrodes – standard electrode potential, sign conventions, electrochemical series and its significance.

UNIT – III Photochemistry (15 Hrs.)

Photochemistry:

Interaction of radiation with matter, difference between thermal and photochemical processes. Laws of Photochemistry: Grothus – Drapper law, Stark – Einstein law, Jablonski diagram depicting various processes occurring in the excited state. Qualitative description of fluorescence, phosphorescence, non-radiative processes (internal conversion, intersystem crossing), quantum yield, photosensitized reactions – energy transfer processes (simple examples). Photochemistry of carbonyl compounds and alkenes.

Books/Literature suggested

1. Atkins, P., Paula, J.de, Atkins Physical Chemistry; 8th edition, Pubs: Oxford University Press, 2008.
2. Puri, B.R., Sharma, L.R., Pathania, M.S., Principles of Physical Chemistry; 43rd edition, Pubs: Vishal Publishing Co., 2008.
3. Barrow, G.M., Physical Chemistry; 6th edition, Pubs: McGraw Hill Inc, 1996.
4. Rao, C.N.R., University General Chemistry; Pubs: Macmillan India, 1985.
5. Berry, R.S., Rice, S.A., Ross, J., Physical Chemistry; 2nd edition, Pubs: Oxford University Press, 2000.
6. Albert, R.A., Silbey, R.J., Physical Chemistry; 1st edition, Pubs: John Wiley & Sons Inc., 1992.
7. Dogra, S.K., Dogra, S., Physical Chemistry Through Problems; Pubs:Wiley Eastern Limited, 1991.
8. Levine, I.N., Physical Chemistry; 5th edition, Pubs: Tata McGraw Hill Publishing Co. Ltd., 2002.
9. Moore, W. J., Basic Physical Chemistry; Pubs: Prentice Hall of India Pvt. Ltd, 1983.
10. Metz, C.R., Theory and Problems of Physical Chemistry; Schaum's outline series, 2nd edition, Pubs: McGraw-Hall Book company, 1989.

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit** (of **19 marks**) will consist of two types of questions: **9 multiple-choice questions** of one mark each and **5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:** the **compulsory question** and one question each from the **UNIT-1, UNIT-2 and UNIT-3**.
2. The use of scientific calculators is not allowed.

Semester – 5
Syllabus for Practical Paper
CHM-DSC-11-MAJ-503 (Table 1)
Physical Chemistry-1

End semester = 22 **Credit – 01**

Internal Marks = 03 **Contact Hours – 2/week**

Examination Time allowed **3 Hours**

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11. Conductometry

1. To determine the Cell Constant of a Conductivity Cell
2. Study of conductometric titration of a strong acid (HCl) against a strong base (NaOH)
3. Study of conductometric titration of a weak acid (CH3COOH) against a strong base (NaOH)
4. Study of conductometric titration of a mixture of strong acid (HCl) and weak acid (CH3COOH) against a strong base (NaOH).
5. To determine the solubility and solubility product of a sparingly soluble electrolyte conductometricaly.
6. To study the saponification of ethyl acetate conductometricaly.
7. To determine the ionization constant of a weak acid conductometricaly
8. Determination of hydrolysis constant of aniline hydrochloride conductometricaly.

Books/Literature suggested

1. Khosla, B.D., Garg, V.C., Gulati, A., Senior Practical Physical Chemistry; 11th edition, Pubs: R.Chand& Co., New Delhi, 2002.
2. Das, R.C., Behra, B., Experimental Physical Chemistry; Pubs: Tata McGraw-Hill Publishing Co. Ltd., 1983.
3. Levitt, B.P., Findlay's Practical Physical Chemistry; 8th edition, Pubs: Longman Group Ltd., London & New York, 1978.

General Instructions to the Examiners:

Note: Practical examination will be of three hours duration divided into two parts (First, half hour for written and second half will be the next two and half hours for performance).

1. Candidate will be examined based on any two experiments given by the examiner. Each experiment carries 08 marks (3 written + 5 performance).
2. Viva-Voce: 04 marks.
3. Note Book: 02 marks.
4. Internal Assessment: 03 marks

Semester	**5**
Code of Paper	**CHM-DSC-12-MAJ-504 (Table 1)**
Title of Paper	**Analytical Chemistry-1**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100 (75 Theory + 25 Practical)**
Marks distribution of Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of Practical paper	**End semester = 22** **Internal Marks = 03**
Applicable to the students 1. Those who have opted for chemistry as a Major subject in Table 1	
LEARNING OUTCOMES SPECIFIC TO DISCIPLINE The students will learn the following- 1. Important aspects of analytical methods 2. Introdction to separation techniques 3. Concepts of various chromatographic techniques 4. Appliation of chromatography in various fields	
GENERIC LEARNING OUTCOMES 1. **The students will be able to learn Problem Solving, Practical Laboratory Skills, Data Management and Evaluation,**Ethical and Professional Responsibility, **Application based Knowledge.** 2. These generic outcomes will equip students with a broad set of transferable skills, making them capable of conducting rigorous scientific analysis while working safely, ethically, and effectively in collaborative environments.	

Semester – 5
Syllabus for Theory Paper
CHM-DSC-12-MAJ-504 (Table 1)
Analytical Chemistry-1

End semester = 67 **Credits – 03**

Internal Marks = 08 **Examination Time allowed – 3 Hours**

UNIT – I Analytical and statistical Methods (15 Hrs.)

Introduction to Analytical and Instrumental Methods

Analytical chemistry, Chemical analysis, Instrumental methods, Analytical methods on the basis of sample size, Sampling, Sampling statistic, Classification of analytical techniques, Classical methods, Classification of instrumental methods, Factors affecting the choice of analytical methods, Interferences, Sensitivity and detection limits, Summary.

Errors, evaluation and statistics

Types and sources of errors, Determinate errors. Indeterminate errors, Accuracy, Absolute and relative error, Precision, Minimisation of errors, Significant figures, Methods for reporting analytical data, Statistical evaluation of data, Mean, median and standard deviation, Reliability of results, Rejection of results, The Q-test, Confidence interval, Tests of significance, Student's t-test, Paired t-test, Chi-square and F test, Fisher's Z-distribution, Relations existing between different statistics, Comparison of the means of two samples, Analysis of variance, Replicate determinations, Correlation, Regression, Uses of statistics, Important relations.

UNIT – II Separation Techniques – 1 (15 Hrs.)

Chromatography

Introduction, Definitions, Theory, Classification of chromatographic methods.

Partition chromatography

Solvent system, Procedure, Theory of partition chromatography, Column efficiency, Applications, Liquid-liquid partition chromatography, Reverse phase partition or extraction chromatography, Applications of RPC.

Paper chromatography

Principle, Types and theory of paper chromatography, Rf, Rx and RG values, Techniques of paper chromatography, Two dimensional paper chromatography, Quantitative estimations, Sources of error, Applications.

Column (adsorption) chromatography

Introduction, Principles of adsorption chromatography, Experimental requirements, Identification of the compounds, Applications of column chromatography and Chiral chromatography.

UNIT – III Separation Techniques – 2 (15 Hrs.)

Thin layer chromatography (TLC)

Theory of TLC, Quantitative evaluation, types of TLC, Two-dimensional TLC, Applications of TLC, Analysis of food stuffs by TLC, High performace Thin Layer Chromatography (HPTLC).

High performance liquid chromatography (HPLC) and Gas chromatography (GC)

Introduction, Components of HPLC, Characteristic features of HPLC, Instrumentation for HPLC, Working with HPLC, Detectors, Derivatisation technique in HPLC, Applications of HPLC, Principle and theory of gas chromatography, Instrumentation, Working, Evaluation of gas chromatogram, Identification of chromatogram, Plate theory for GC, Gas-solid chromatography, Applications of GC, Preparative GC, Applications, Comparison with HPLC.

Ion exchange chromatography (IEC)

Ion-exchangers, Cation exchange resins, Ion-exchange equilibria, Anion exchange resins, Factors affecting ion exchange equilibria, Instrumentation, Technique for ion exchange, Liquid ion exchanger, Applications of IEC.

Books/Literature suggested

1. Vogel's Textbook of Quantitative Chemical Analysis (revised by Jeffery, Bassett, Mendham and Denney), 5th Edn., Pubs: ELBS (1989).
2. Christian G.D., Analytical Chemistry, Pubs: John Wiley & Sons Inc., New York (1994).
3. Fifield F.W. and Kealey D., Principles and Practice of Analytical Chemistry, 5th Edn., Blackwell Science (2000).
4. Skoog D.A., D.M.West, F.J.Holler, S.R.Crouch, Fundamentals of Analytical chemistry, Pubs: Brooks/Cole. (2004).
5. Skoog D.A., Principles of Instrumental Analysis, 3rd edition, Pubs: Holt-Saunders International edition (1985).
6. Bassett, J., Denney, R.C., Jeffery, G.H., Mendham, J., Vogel's Textbook of Quantitative Inorganic Analysis (revised); 4th edition, Pubs: Orient Longman, 1978.
7. Willard H.H., Merritt L.L. Jr, Dean J.A., Settle F.A. Jr., Instrumental Methods of Analysis, 7th edn., Pubs: Wadsworth Publishing Company, California (1988)
8. **Kevin Robards, P. E. Jackson, Paul A. Haddad, Principles and Practice of Modern Chromatographic Methods.**
9. **H. Kaur, Instrumental Methods of Analysis.**

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit** (of **19 marks**) will consist of two types of questions: **9 multiple-choice questions** of one mark each and **5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:** the **compulsory question** and one question each from the **UNIT-1, UNIT-2 and UNIT-3**.
2. The use of scientific calculators is not allowed.

Semester – 5
Syllabus for Practical Paper
CHM-DSC-12-MAJ-503 (Table 1)
Analytical Chemistry-1

End semester = 22 **Credit – 01**

Internal Marks = 03 **Contact Hours – 2/week**

Examination Time allowed **3 Hours**

I. Thin Layer Chromatography:

a) Separation of nitrobenzene and aniline and Rf calculation
b) Separation of *m*-di-chlorobenzene and p-anisidine
c) Separation of benzaldehyde and nitrobenzene using $KmnO_4$ solution as visualizing reagent

II. Paper Chromatography:

a) Separation of amino acids
b) Separation of carbohydrate molecules
c) Separation of Ni (II) and Co (II)
d) Separation of Fe (III) and Al (III)

III. Column Chromatography:

a) Separation of green plant pigments
b) Separation of mixture of dyes
g) Separation of ferrocene and acetylferrocene

Books/Literature suggested

1. Vogel, A.I., Tatchell, A.R., Furnis, B.S., Hannaford, A.J. & Smith, P.W.G., *Text book of Practical Organic Chemistry*, Prentice-Hall, 5th edition,1996.
2. *Mendham, J., A. I. Vogel's Quantitative Chemical Analysis 6th Ed., Pearson, 2009.*
3. *Universty Practical Chemistry, P. C. Kamboj.*

General Instructions to the Examiners:

Note: Practical examination will be of three hours duration divided into two parts (First, half hour for written and second half will be the next two and half hours for performance).

1. Candidate will be examined based on any two experiments given by the examiner. Each experiment carries 08 marks (3 written + 5 performance).
2. Viva-Voce: 04 marks.
3. Note Book: 02 marks.
4. Internal Assessment: 03 marks

Semester	**5**
Code of Paper	**CHM-DSC-13-MAJ-505 (Table 1)**
Title of Paper	**Environmental Chemistry-1**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100 (75 Theory + 25 Practical)**
Marks distribution of Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of Practical paper	**End semester = 22** **Internal Marks = 03**

Applicable to the students

1. Those who have opted for chemistry as a Major subject in Table 1.

LEARNING OUTCOMES SPECIFIC TO DISCIPLINE

The students will learn the following-

1. Important aspects of environmental chemistry
2. Chemicals involved in air pollution
3. Chemistry of soil
4. Waste management

GENERIC LEARNING OUTCOMES

1. **The students will be able to learn Problem Solving, Practical Laboratory Skills, Data Management and Evaluation,**Ethical and Professional Responsibility, **Application based Knowledge.**
2. These generic outcomes will equip students with a broad set of transferable skills, making them capable of conducting rigorous scientific analysis while working safely, ethically, and effectively in collaborative environments.

Semester – 5
Syllabus for Theory Paper
CHM-DSC-13-MAJ-505 (Table 1)
Environmental Chemistry-1

End semester = 67 **Credits – 03**

Internal Marks = 08 **Examination Time allowed – 3 Hours**

UNIT – 1 Fundamentals of Environmental Chemistry (15 Hrs.)

Fundamentals of Environmental Chemistry

Concept and scope of environment chemistry, Segment of environment: Atmosphere; hydrosphere; lithosphere; biosphere and anthrosphere, Biodistribution of elements, The Natural Cycles of Environment: Carbon; oxygen; nitrogen; sulphur and phosphorus cycles, Environmental pollution, Natural and manmade disasters

UNIT – 2 Air and Air Pollution (15 Hrs.)

Atmospheric Chemistry and Air Pollution

Composition of atmosphere, Major regions of atmosphere (the vertical temperatures), Heat budget of earth atmospheric system, Evolution of atmosphere, Formation and depletion of ozone layer (O_3): Chapman cycle; Role of CFCs and NO_x, Air pollutants: Sources and reactions of SO_2; NO_x; and CO, Mechanism of photochemical smog (Los Angeles type) versus Sulfurous smog (London type), Chemistry of acid rain and its impact on monuments, Greenhouse effect.

UNIT – 3 Soil Chemistry (15 Hrs.)

Soil Chemistry and Waste management

Introduction, Characteristics of soil, composition of soil organic and inorganic components of soil, water and air in soil, macronutrients and micronutrients, Sources of soil pollution: Industrial pollution; urban and domestic waste; radioactive pollutants; fertilizers; pesticides;plastics;metals; acid rain, Prevention and control of soil pollution, Waste management :Classification of waste a (Hazardous, Biomedical, e-waste) ; Principle of 3Rs (Reduce, reuse, recycle) ; composting and land fills.

Books/Literature suggested

1. S.E. Manahan, Environmental Chemistry, Lewis Publishers.
2. A.K. Dey, Environmental Chemistry, Wiley Eastern.
3. V.K. Ahlluwalia, Environmental Chemistry, Ane Books Pvt. Ltd.
4. S.M. Khopkar, Environment Pollution Analysis, Wiley Eastern Ltd.
5. Text book of Environment chemistry, Balram Pani, I.K. International Pub. House, New Delhi.
6. Principles of Environment Chemistry, Games.E.Girard, Jones & Bartlet India.
7. Environmental Chemistry, H. Kaur, Pragati Prakashan.

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit** (of **19 marks**) will consist of two types of questions: **9 multiple-choice questions** of one mark each and **5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:** the **compulsory question** and one question each from the **UNIT-1, UNIT-2 and UNIT-3**.
2. The use of scientific calculators is not allowed.

Semester – 5
Syllabus for Practical Paper
CHM-DSC-13-MAJ – 505
Environmental Chemistry-1

End semester = 22 **Credit – 01**

Internal Marks = 03 **Contact Hours – 2/week**

Examination Time allowed **3 Hours**

Soil Analysis

1. Determination of soil pH.
2. Identification of Nitrogen, Phosphorus and Potassium in soil sample.
3. Determination of percentage of $CaCO_3$ in soil sample.
4. Determination of moisture content of soil sample.
5. Determination of soil alkalinity.
6. Determination of organic matter in soil sample.

Literature/Books suggested

1. A laboratory Mannual for Environmental Chemistry, R.Gopalan, A.Anand, R.W. Sugumar, I.K.International Publ. House, New Delhi.
2. Experimental Ichthyology, S.K.Garg, A.Bhatnagar, A.Johal, CBS Publishers New Delhi.
3. Soil and Water Analysis: A Practical Mannual, Dr. R.K.Singh, Dr. .K.Yadav, Dr. R.P.Singh,R.K.Jalaj, Dr.R.Kumar. Iterative International Publisher

General Instructions to the Examiners:

Note: Practical examination will be of three hours duration divided into two parts (First, half hour for written and second half will be the next two and half hours for performance).

1. Candidate will be examined based on any two experiments given by the examiner. Each experiment carries 08 marks (3 written + 5 performance).
2. Viva-Voce: 04 marks.
3. Note Book: 02 marks.
4. Internal Assessment: 03 marks

Semester	**5**
Code of Paper	**CHM-INT-1 (Table 1)**
Title of Paper	**Internship**
Total Credit	**04**
Total Marks	**100**

Applicable to the students

12. Those who have opted for chemistry as a Major subject in Table 1

LEARNING OUTCOMES SPECIFIC TO DISCIPLINE

The internship course is designed to provide undergraduate Chemistry students with experiential learning through **industrial exposure, academic visits, training programs, and participation in scientific events**. The course aims to bridge theoretical knowledge with real-world applications in chemical industries, research institutions, and academic platforms.

Semester – 5
Instructions for Internship
CHM-INT-1 (Table 1)
Internship / Field-Based Learning Course

Total Marks = 100 **Credits: 4**

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1. Course Description

The internship course is designed to provide undergraduate Chemistry students with experiential learning through **industrial exposure, academic visits, training programs, and participation in scientific events**. The course aims to bridge theoretical knowledge with real-world applications in chemical industries, research institutions, and academic platforms.

2. Course Components

2.1 Project/Training:

Each student shall undertake **at least one** industrial visit, hands-on training, or research-based activity in any one of the following institutions or facilities:

(i) University/College Research Laboratories
(ii) Analytical Instrumentation Unit at Industry/Laboratory
(iii) Manufacturing or Processing Units in Industry
(iv) National Defence Laboratories, Metrological Laboratories, Forensic Science Labs.
(v) Environmental Monitoring Agencies, or Pollution Control Boards
(vi) Drinking Water or Wastewater Treatment Facilities or similar facilities
(vii) Quality Control or Testing Laboratories

Any other activity deemed appropriate by the Department/Institute.

The student shall spend a **minimum of 50-60 hours(7-10 days)** at the selected activity site. Winter or autumn vacations may be utilized for this purpose. Upon completion, the student must obtain a certificate or official letter from the host industry or academic institution confirming their participation and duration of engagement.

2.2 Internship Report:

Each student shall submit a **detailed internship report** of approximately **20-30 pages**, including relevant photographs. The report shall include, but not be limited to, the following aspects:

(i) Description of instrumentation, machines, or facilities used
(ii) Details of processes involved
(iii) Manpower, Infrastructure, and Safety and waste management practices
(iv) Observations made during the internship
(v) Learning outcomes and skills acquired
(vi) Any other detail

The student shall submit **three hard copies** (soft binding) of the report along with one **soft copy in PDF format** for uploading on the website. The report shall be uploaded on the Institute/College website after evaluation.

2.3 Additional Academic Activities:

Additional activities shall include participation in conferences, seminars, workshops, or activities of academic clubs organized by (i) the parent Department or Institute, or other recognized academic or research institutions.

The student shall submit a **summary report of up to 5 pages** covering such activities with relevant photographs. The summary may also include relevant academic programmes attended by the student during the **first and second years** of the course, if any.

3. Assessment Method

The assessment for this component shall be based on the following:

(i) Internship Report – 50 ***Marks***
(ii) Oral Presentation – 20 ***Marks***
(iii) Additional Academic Activities – 20 ***Marks***
(iv) ***Internal assessment***- 10 ***Marks***

(a) The internship report shall be evaluated by **two examiners** appointed by the College, one of whom shall be from a discipline other than that of the student.
(b) The internship report shall also be evaluated by an **external examiner** based on the soft copy submitted.
(c) All students shall present their internship work in a **one-day academic event**, such as a symposium, in the presence of internal examiners and fellow students. The evaluation of the presentation shall be based on the following criteria:
   i. Quality of oral presentation, clarity, innovation, understanding of the topic, and responses to questions – **60%**
   ii. Attendance, participation in discussions, and engagement during presentations of other students – **40%**

3. Teacher Involvement

A faculty member may be assigned mentorship responsibility for a group of 15–20 students for the purpose of internship. The faculty mentor will guide the students in selecting the site for the visit, coordinating with industry or academic institutions, preparing the report and presentation, arranging examiners for evaluation, and organizing related events. For undertaking these responsibilities, a workload of 4 hours/week may be allotted to the concerned faculty member.

Semester	**6**
Code of Paper	**CHM-DSC-14-MAJ-601 (Table 1)**
Title of Paper	**Inorganic Chemistry-2**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100 (75 Theory + 25 Practical)**
Marks distribution of Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of Practical paper	**End semester = 22** **Internal Marks = 03**

Applicable to the students

1. Those who have opted for chemistry as a Major subject in Table 1.

LEARNING OUTCOMES SPECIFIC TO DISCIPLINE

The students will learn the following-

1. Important aspects of coordination chemistry
2. Chemistry of organometallic compounds
3. Chemistry of carboxylic acids, fats, detergents and heterocyclic compounds
4. Concepts of quantum chemistry

GENERIC LEARNING OUTCOMES

1. **The students will be able to learn Problem Solving, Practical Laboratory Skills, Data Management and Evaluation,**Ethical and Professional Responsibility, **Application based Knowledge.**
2. These generic outcomes will equip students with a broad set of transferable skills, making them capable of conducting rigorous scientific analysis while working safely, ethically, and effectively in collaborative environments.

Semester – 6
Syllabus for Theory Paper
CHM-DSC-14-MAJ–601 (Table 1)
Inorganic Chemistry-2

End semester = 67 **Credits – 03**

Internal Marks = 08 **Examination Time allowed – 3 Hours**

UNIT I: Chemistry of Coordination Compounds **(15 Hrs.)**

Crystal field theory

Limitations of valence bond theory, Crystal field theory, CFSE in weak and strong fields, pairing energies, and factors affecting the magnitude of 10 Dq (Δo, Δt). The splitting of d-orbitals in different fields (octahedral, tetrahedral, tetragonally distorted octahedral, square planar, trigonal bipyramidal), Introduction to Orgel diagrams, Consequences and applications of orbital splitting, crystal field stabilization energy, magnetic properties, measurement of 10 Dq (Δo), Factors affecting the extent of splitting and spectrochemical series, colour of transition metal complexes. A brief outline of thermodynamic and Kinetic stability of metal complexes and factors affecting the stability, substitution reactions of square planar complexes.

UNIT II: Organometallic Chemistry **(15 Hrs.)**

Organometallic compounds

Definition, nomenclature and types of organometallic compounds. Classification of ligands, Hapticity, EAN rule, Nomenclature of organometallic compounds, Bonding in organometallic compounds, Preparation, properties, bonding and applications of alkyls and aryls of Li, Al , Hg, Sn and Ti. A brief account of metal – olefin complexes, alkyne complexes, homogeneous hydrogenation, Metal carbonyls – Preparation, bonding and reactions, Structures and the nature of bonding in bi, tri, tetranuclear metal carbonyls.

UNIT III: Bioinorganic Chemistry and Inorganic Polymers (15 Hrs.)

Bioinorganic Chemistry:

Essential and trace elements in biological processes, metalloporphyrins with special reference to haemoglobin and myoglobin. Cooperativity effect, Bohr effect, Biological role of alkali and alkaline earth metal ions. Nitrogen fixation.

Inorganic polymers

Silicones and Polyphosphazenes – Nomenclature, Preparation, Bonding, and Applications.

Books/Literature suggested

1. Cotton, F.A., Wilkinson, G., Gaus, P.L., Basic Inorganic Chemistry; 2nd edition, Pubs: John Wiley and Sons, 1995.
2. Lee, J.D., Concise Inorganic Chemistry; 4th edition, Pubs: Chapman & Hall Ltd., 1991.
3. Shriver, D.E., Atkins, P.W., Inorganic Chemistry; 4th edition, Pubs: Oxford University Press, 2006.
4. Douglas, B., Medaniel, D., Atenander, J., Concepts and Models of Inorganic Chemistry; 3rd edition, Pubs: John Wiley and Sons Inc., 1994,
5. Porterfield, W.W., Wesky, A., Inorganic Chemistry; Pubs: Addison-Wesley Publishing Company,1984.
6. Miessler, G.L., Tarr, D.A., Inorganic Chemistry; 3rd edition, Pubs: Pearson Education Inc., 2004.
7. Puri, B.R., Sharma, L.R., Kalia, K.K., Principles of Inorganic Chemistry; 30th edition, Pubs: Milestones Publisher, 2006-07

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit** (of **19 marks**) will consist of two types of questions: **9 multiple-choice questions** of one mark each and **5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:** the **compulsory question** and one question each from the **UNIT-1, UNIT-2 and UNIT-3**.
2. The use of scientific calculators is not allowed.

Semester – 6
Syllabus for Practical Paper
CHM-DSC-14-601 (Table 1)
Inorganic Chemistry – 2

End semester = 22 **Credit – 01**

Internal Marks = 03 **Contact Hours – 2/week**

Examination Time allowed **3 Hours**

I. Complexometric titrations using EDTA

a) Determination of Calcium by EDTA titration.
b) Determination of Magnesium by EDTA titration.
c) Determination of total hardness of water by EDTA.
d) Determination of Zinc/Nickel.
e) Determination of temporary and permanent hardness of water by EDTA.

II. Synthesis and analysis

a) Preparation of sodium trioxalatoferrate(III), $Na_3[Fe(C_2O_4)_3]$ and determination of its composition bypermaganometry.
b) Preparation of copper tetraammine complex $[Cu(NH_3)_4]SO_4$.
c) Preparation of cis-and trans-bisoxalatodiaqua chromate(III) ion.

Books/Literature suggested

1. Svehla, Vogel's Qualitative Inorganic Analysis (revised); 7th edition Pubs: Orient Longman, 1996.
2. Bassett, J., Denney, R.C., Jeffery, G.H., Mendham, J., Vogel's Textbook of Quantitative Inorganic Analysis (revised); 4th edition, 1978.
3. Palmer, W.G., Experimental Inorganic Chemistry; 1st edition, Pubs: Cambridge, 1954.
4. Brauer, Handbook of Preparative Inorganic Chemistry; 2nd edition, Pubs: Academic Press, 1963, Vol. I, & 1965, Vol. II

General Instructions to the Examiners:

Note: Practical examination will be of three hours duration divided into two parts (First, half hour for written and second half will be the next two and half hours for performance).

1. Candidate will be examined based on any two experiments given by the examiner. Each experiment carries 08 marks (3 written + 5 performance).
2. Viva-Voce: 04 marks.
3. Note Book: 02 marks.
4. Internal Assessment: 03 marks

Semester – 6

Syllabus for Theory Paper
CHM-DSC-15-MAJ-602 (Table 1)
Organic Chemistry-2

End semester = 67 **Credits – 03**

Internal Marks = 08 **Examination Time allowed-3 Hours**

Semester	**6**
Code of Paper	**CHM-DSC-15-MAJ-602 (Table 1)**
Title of Paper	**Organic Chemistry-2**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100 (75 Theory + 25 Practical)**
Marks distribution of Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of Practical paper	**End semester = 22** **Internal Marks = 03**

Applicable to the students

1. Those who have opted for chemistry as a Major subject in Table 1.

LEARNING OUTCOMES SPECIFIC TO DISCIPLINE

The students will learn the following-

1. Important aspects of mechanism and concepts of nitrogen containing organic compounds
2. Chemistry of heterocyclic compounds
3. Organic spectroscopic techniques for structure elucidation, such as ^{1}H NMR, IR & UV
4. Concepts of quantum chemistry

GENERIC LEARNING OUTCOMES

1. **The students will be able to learn Problem Solving, Practical Laboratory Skills, Data Management, Evaluation,** estimation and determination through a**pplication based organic chemistry techniques and concepts.**
2. These generic outcomes will equip students with a broad set of transferable skills, making them capable of conducting rigorous scientific analysis while working safely, ethically, and effectively in collaborative environments.

Semester – 6
Syllabus for Theory Paper
CHM-DSC-15-MAJ – 602 (Table 1)
Organic Chemistry-2

End semester = 67 **Credits – 03**

Internal Marks = 08 **Examination Time allowed – 3 Hours**

Unit 1: Heterocyclic and nitrogen containing organic compounds (15 Hrs)

Organic Compounds of Nitrogen

Preparation of nitroalkanes and nitroarenes. Chemical reactions of nitroalkanes. Mechanisms of nucleophilic substitution in nitroarenes and their reductions in acidic, neutral and alkaline media. Structure and nomenclature of amines, physical properties. Stereochemistry of amines, Separation of a mixture of primary, secondary and tertiary amines. Structural features affecting basicity of amines. Amine salts as phase-transfer catalysis.

Heterocyclic Compounds

Introduction: Molecular Orbital picture and aromatic character of pyrrole, furan, thiophene, pyridine. Methods of synthesis and chemical reactions with particular emphasis on the mechanism of electrophilic substitution. Mechanism of nucleophilic substitution reactions in pyridine derivatives. Comparision of basicity of pyridine, piperidine and pyrrole.

Unit 2: Organic Spectroscopy – 1 (15 Hrs)

UV-Vis absorption spectroscopy

Ultraviolet (UV) absorption spectroscopy – Absorption laws (Beer – Lambert Law), molar absorptivity, presentation and analysis of UV spectra, types of electronic transitions, effect of conjugation. Concept of chromophore and auxochrome. Bathochromic, hypsochromic, hyperchromic and hypochromic shifts. UV spectra of conjugated enes and enones. Woodward Fieser Rules and their applications in calculating maximum values of conjugated alkenes (cyclic as well as acyclic) and conjugated carbonyl compounds.

IR spectroscopy

Infrared (IR) absorption spectroscopy: Molecular vibrations, Hooke's law, selection rules, intensity and position of IR bands, measurement of IR spectrum, fingerprint region, characteristic absorptions of various functional groups and interpretation of IR spectra of simple organic compounds.

Unit 3: Organic Spectroscopy – 2 (15 Hrs)

NMR Spectroscopy

Nuclear magnetic resonance (NMR) spectroscopy. Proton magnetic resonance (^{1}H NMR) spectroscopy, nuclear shielding and deshielding, chemical shift and molecular structure, spin-spin splitting and coupling constants, area of signals, interpretation of PMR spectra of simple aliphatic and aromatic organic molecules.

Problems pertaining to the structure elucidation of simple organic compounds using a combination of UV, IR and PMR spectroscopic data.

Books/Literature suggested

1. William Kemp, Organic Spectroscopy, Bloomsbury Publishing, Ist edition, 2022.
2. P.S. Kalsi, Spectroscopy of Organic Compounds, New Age International Publishers.
3. T.L. Gilchrist, Heterocyclic Chemistry, Pearson Education, India, 3rd edition.
4. R.R. Gupta, M. Kumar, V. Gupta, Heterocyclic Chemistry, Springer Publications, Ist edition,1998.

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit** (of **19 marks**) will consist of two types of questions: **9 multiple-choice questions** of one mark each and **5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:** the **compulsory question** and one question each from the **UNIT-1, UNIT-2 and UNIT-3**.
2. The use of scientific calculators is not allowed.

Semester – 6
Syllabus for Practical Paper
CHM-DSC-15-MAJ – 602 (Table 1)
Organic Chemistry-2

End semester = 22 **Credit – 01**

Internal Marks = 03 **Contact Hours – 2/week**

Examination Time allowed **3 Hours**

I. Separation of mixture of two solid organic compounds and characterization of the components by formation of their derivatives.

II. Structure determination of organic compounds using spectroscopic data (UV, IR & NMR)

III. Tests to distinguish between the following
a. Aliphatic amines and aromatic amines
b. 1° , 2° and 3° amines

Books/Literature suggested

1. William Kemp, Organic Spectroscopy, Bloomsbury Publishing, Ist edition,2022,
2. P.S. Kalsi, Spectroscopy of Organic Compounds by, New Age International Publishers.
3. Forniss, B.S., Rogers, V., Vogel's Text Book of Practical Organic Chemistry; Pubs: Dorling Kindhsky Pvt. Ltd., 1989.
4. Aggarwal, O.P., Advanced Practical Organic Chemistry, 30th edition, 2013, Goel Publishing House.
5. Harwoor, L.M., Moody, J. Experimental Organic Chemistry;1st edition, Pubs: Blackwell Scientific Publications,1989.

General Instructions to the Examiners:

Note: Practical examination will be of three hours duration divided into two parts (First, half hour for written and second half will be the next two and half hours for performance).

1. Candidate will be examined based on any two experiments given by the examiner. Each experiment carries 08 marks (3 written + 5 performance).
2. Viva-Voce: 04 marks.
3. Note Book: 02 marks.
4. Internal Assessment: 03 marks

Semester	**6**
Code of Paper	**CHM-DSC-16-MAJ-603 (Table 1)**
Title of Paper	**Physical Chemistry-2**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100 (75 Theory + 25 Practical)**
Marks distribution of Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of Practical paper	**End semester = 22** **Internal Marks = 03**

Applicable to the students

1. Those who have opted for chemistry as a Major subject in Table 1.

LEARNING OUTCOMES SPECIFIC TO DISCIPLINE

The students will learn the following-

1. **Quantum Mechanics:** Master the wave-particle duality and Schrödinger equations to model particle behavior in boxes and the hydrogen atom.
2. **Molecular Theory:** Construct Molecular Orbitals via LCAO to analyze bonding/antibonding states and calculate coefficients forhybrid orbitals.
3. **Spectroscopy:** Apply selection rules and energy level transitions to determine bond lengths, force constants, and molecular geometry using IR, Rotational, and Electronic spectra.

GENERIC LEARNING OUTCOMES

1. **Problem Solving, Practical,** Mathematical Modeling: Translate abstract quantum postulates and spectroscopic data into concrete molecular properties like bond length and energy.
2. Instrumental Logic: Understand the physical principles governing how spectrometers and electrochemical cells function as diagnostic tools.
3. Critical Analysis: Evaluate the limitations of classical physics models compared to quantum mechanical and thermodynamic interpretations.
4. Scientific Synthesis: Integrate wave mechanics, thermodynamics, and light-matter interactions to explain the fundamental stability of chemical systems.

Semester – 6
Syllabus for Theory Paper
CHM-DSC-16-MAJ – 603
Physical Chemistry-2

End semester = 67 **Credits – 03**

Internal Marks = 08 **Examination Time allowed - 3 Hours**

UNIT – I Quantum Mechanics 1 (15 Hrs.)

Quantum Mechanics – I:

Black-body radiation, Planck's radiation law, photoelectric effect, heat capacity of solids, Bohr's model of hydrogen atom (no derivation) and its defects, Compton effect. De Broglie hypothesis, the Heisenberg's uncertainty principle, Sinusoidal wave equation, Hamiltonian operator, Schrodinger wave equation and its importance, physical interpretation of the wave function, postulates of quantum mechanics, particle in a one dimensional box. Schrodinger wave equation for H-atom, separation into three equations (without derivation).

UNIT – II Quantum Mechanics 2 (15 Hrs.)

Elementary Quantum Mechanics – II:

Molecular orbital theory, basic ideas – criteria for forming M.O. from A.O., construction of M.O.'s by LCAO – H_2^+ ion. Calculation of energy levels from wave functions, physical picture of bonding and antibonding wave functions, concept of σ, σ∗, π, π* orbitals and their characteristics. Hybrid orbitals – sp, sp2; calculation of coefficients of A.O.'s used in these hybrid orbitals.

UNIT – III Physical Spectroscopy (15 Hrs.)

Spectroscopy:

Electromagnetic radiation, regions of the spectrum, basic features of different spectrometers, statement of the Born-Oppenheimer approximation, degrees of freedom. Rotational Spectrum: Diatomic molecules. Energy levels of a rigid rotor (semi – classical principles), selection rules, spectral intensity, determination of bond length, qualitative description of non-rigid rotor, isotope effect.

Vibrational Spectrum: Infrared Spectrum : Energy levels of simple harmonic oscillator, selection rules, pure vibrational spectrum intensity, determination of force constant and qualitative relation of force constant and bond energies, effect of anharmonic motion and isotope on the spectrum,

Electronic Spectrum: Concept of potential energy curves for bonding and antibonding molecular orbitals, qualitative description of selection rules and Franck- Condon principle. Qualitative description of σ, π – and n M.O., their energy levels and the respective transitions.

Literature/Book Suggested

1. Atkins, P., Paula, J.de, Atkins Physical Chemistry; 8th edition, Pubs: Oxford University Press, 2008.
2. Puri, B.R., Sharma, L.R., Pathania, M.S., Principles of Physical Chemistry; 43rd edition, Pubs: Vishal Publishing Co., 2008.
3. Barrow, G.M., Physical Chemistry; 6th edition, Pubs: McGraw Hill Inc, 1996.
4. Rao, C.N.R., University General Chemistry; Pubs: Macmillan India, 1985.
5. Berry, R.S., Rice, S.A., Ross, J., Physical Chemistry; 2nd edition, Pubs: Oxford University Press, 2000.
6. Albert, R.A., Silbey, R.J., Physical Chemistry; 1st edition, Pubs: John Wiley & Sons Inc., 1992.
7. Dogra, S.K., Dogra, S., Physical Chemistry Through Problems; Pubs:Wiley Eastern Limited, 1991.
8. Levine, I.N., Physical Chemistry; 5th edition, Pubs: Tata McGraw Hill Publishing Co. Ltd., 2002.
9. Moore, W. J., Basic Physical Chemistry; Pubs: Prentice Hall of India Pvt. Ltd, 1983.
10. Metz, C.R., Theory and Problems of Physical Chemistry; Schaum's outline series, 2nd edition, Pubs: McGraw-Hall Book company, 1989.

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit** (of **19 marks**) will consist of two types of questions: **9 multiple-choice questions** of one mark each and **5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:** the **compulsory question** and one question each from the **UNIT-1, UNIT-2 and UNIT-3**.
2. The use of scientific calculators is not allowed.

Semester – 6
Syllabus for Practical Paper
CHM-DSC-16-MAJ – 603
Physical Chemistry-2

End semester = 22 **Credit – 01**

Internal Marks = 03 **Contact Hours – 2/week**

Examination Time allowed **3 Hours**

I. Refractometry:

1. Determination of molar refractivity of ethyl acetate, methyl acetate, ethylene chloride and chloroform and calculation of the atomic refractivities of the C, H and Cl.
2. Measurement of the average electronic polarizabilities of some of the common solvents refractometrically.
3. To find the composition of binary mixtures refractometrically.

II. Polarimetry:

1. To study the inversion of cane sugar by optical rotation measurement.
2. To determine the specific and molecular rotations of optically active substances.

III. Phase Equilibria:

1. To study the distribution of iodine between water and CCl4.
2. To study the distribution of benzoic acid between benzene and water.

IV. Colorimetry:

1. Verify Lambert-Beer's law and determine the concentration of CuSO4/KmnO4/K2Cr2O7 in a solution of unknown concentration.

Books/Literature suggested

1. Khosla, B.D., Garg, V.C., Gulati, A., Senior Practical Physical Chemistry; 11th edition, Pubs: R.Chand& Co., New Delhi, 2002.
2. Das, R.C., Behra, B., Experimental Physical Chemistry; Pubs: Tata McGraw-Hill Publishing Co. Ltd., 1983.
3. Levitt, B.P., Findlay's Practical Physical Chemistry; 8th edition, Pubs: Longman Group Ltd., London & New York, 1978.

General Instructions to the Examiners:

Note: Practical examination will be of three hours duration divided into two parts (First, half hour for written and second half will be the next two and half hours for performance).

1. Candidate will be examined based on any two experiments given by the examiner. Each experiment carries 08 marks (3 written + 5 performance).
2. Viva-Voce: 04 marks.
3. Note Book: 02 marks.
4. Internal Assessment: 03 marks

Semester	**6**
Code of Paper	**CHM-DSC-17-MAJ-604 (Table 1)**
Title of Paper	**Analytical Chemistry-2**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100 (75 Theory + 25 Practical)**
Marks distribution of Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of Practical paper	**End semester = 22** **Internal Marks = 03**
Applicable to the students 1. Those who have opted for chemistry as a Major subject in Table 1.	
LEARNING OUTCOMES SPECIFIC TO DISCIPLINE The students will learn the following- 1. Basic principles and aspects of flame absorption spectroscopy. 2. Chemistry of electroanalytical and thermogravimetric methods of analysis. 3. Concepts of various extraction techniques for isolation and purification.	
GENERIC LEARNING OUTCOMES 1. **The students will be able to learn Problem Solving, Practical Laboratory Skills, Data Management and Evaluation,** Ethical and Professional Responsibility, **Application based Knowledge.** 2. These generic outcomes will equip students with a broad set of transferable skills, making them capable of conducting rigorous scientific analysis while working safely, ethically, and effectively in collaborative environments.	

Semester – 6
Syllabus for Theory Paper
CHM-DSC-17-MAJ – 604
Analytical Chemistry-2

End semester = 67 **Credits – 03**

Internal Marks = 08 **Examination Time allowed - 3 Hours**

UNIT–I: Optical methods of analysis-II (15 Hours)

Flame Atomic Absorption and Emission Spectrometry:

Basic principles of instrumentation (choice of source, monochromator, detector, choice of flame and Burner designs.

Techniques of atomization and sample introduction; Method of background correction, sources of chemical interferences and their method of removal. Techniques for the quantitative estimation of trace level of metal ions from water samples.

UNIT–II: Thermal methods of analysis (15 Hours)

Thermogravimetry:

Theory of thermogravimetry (TG), basic principle of instrumentation, Techniques for quantitative estimation of Ca and Mg from their mixture.

Electroanalytical methods: Classification of electroanalytical methods, basic principle of pH metric, potentiometric and conductometric titrations. Techniques used for the determination of equivalence points, Techniques used for the determination of pKa values.

UNIT–III: Separation techniques (15 Hours)

Solvent extraction: Classification, principle and efficiency of the technique, Mechanism of extraction: Extraction by 110or activ and chelation.

Technique of extraction: batch, continuous and counter current extractions. Qualitative and quantitative aspects of solvent extraction: extraction of metal ions from aqueous solution, extraction of organic species from the aqueous and non-aqueous media

Literature/Book Suggested

1. Willard, H.H.; Merritt, Jr, L L; Dean, J A; Settle, Jr, F A, Instrumental Methods of Analysis, 7th Ed. Wardsworth Publishing
2. Mendham, J., A. I. Vogel's Quantitative Chemical Analysis 6th Ed., Pearson, 2009.
3. Christian, G.D. Analytical Chemistry, 6th Ed. John Wiley Company, Belmont, California, USA, 1988.
4. Khopkar, S.M. Basic Concepts of Analytical Chemistry. New Age International Pvt Ltd Publishers.
5. Harris, D.C.: Quantitative Chemical Analysis, 9th Ed. New York, W.H. Freeman & Co, New York, 2007.
6. Skoog, D.A., Holler F.J., Crouch S.R., Principles of Instrumental Analysis, 7th Ed., Cengage Publisher, 2018.
7. Mikes, O. Laboratory Hand Book of Chromatographic Allied Methods, London, D. van Nostrand Company Ltd., 1979.
8. Ditts, R.V. Analytical Chemistry: Methods of separation, van Nostrand, 1974.

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit** (of **19 marks**) will consist of two types of questions: **9 multiple-choice questions** of one mark each and **5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:** the **compulsory question** and one question each from the **UNIT-1, UNIT-2 and UNIT-3**.
2. The use of scientific calculators is not allowed.

Semester – 6
Syllabus for Practical Paper
CHM-DSC-16-MAJ – 603
Physical Chemistry-2

End semester = 22 **Credit – 01**

Internal Marks = 03 **Contact Hours – 2/week**

Examination Time allowed **3 Hours**

I. Solvent Extractions:

To separate a mixture of Ni^{2+}& Fe^{2+} by complexation with DMG and extracting the Ni^{2+}-DMG complex in chloroform, and to determine its concentration using a spectrophotometer.

II. Real sample analysis:

(i) Determine the pH of the given aerated drinks fruit juices, shampoos and soaps.

(ii) Determination of Na, Ca, Li in cola drinks and fruit juices.

(iii) Analysis of soil to determine pH, calcium, magnesium, phosphate, nitrate

(iv) To analyze moisture content, water holding capacity, texture and relative salinity of soil samples

III. Spectrophotometric Analysis:

(i) Determination of *p*Ka values of indicator using spectrophotometer.

(ii) Determination of dissolved oxygen in water

(iii) Determine the composition of the Ferric-salicylate/ ferric-thiocyanate complex by Job's method.

(iv) To determine the concentration of Na^{+}& K^{+} using flame photometer.

*The experiments may be modified according to the availability of reagents, apparatus, instruments and safety features

Books/Literature suggested

1. Vogel, A.I., Tatchell, A.R., Furnis, B.S., Hannaford, A.J. & Smith, P.W.G., Text book of Practical Organic Chemistry, Prentice-Hall, 5th edition,1996.
2. Mendham, J., A. I. Vogel's Quantitative Chemical Analysis 6th Ed., Pearson, 2009.
3. P. C. Kamboj, University Practical Chemistry, 2nd Ed., Vishal Publishing Co, 2018.

General Instructions to the Examiners:

Note: Practical examination will be of three hours duration divided into two parts (First, half hour for written and second half will be the next two and half hours for performance).

1. Candidate will be examined based on any two experiments given by the examiner. Each experiment carries 08 marks (3 written + 5 performance).
2. Viva-Voce: 04 marks.
3. Note Book: 02 marks.
4. Internal Assessment: 03 marks

Semester	**6**
Code of Paper	**CHM-DSC-18-MAJ-605 (Table 1)**
Title of Paper	**Environmental Chemistry-2**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100 (75 Theory + 25 Practical)**
Marks distribution of Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of Practical paper	**End semester = 22** **Internal Marks = 03**
Applicable to the students 1. Those who have opted for chemistry as a Major subject in Table 1.	
LEARNING OUTCOMES SPECIFIC TO DISCIPLINE The students will learn the following- 1. Important aspects of water pollution 2. Details of Environmental Toxicology 3. Concepts of Green Chemistry	
GENERIC LEARNING OUTCOMES 1. **The students will be able to learn Problem Solving, Practical Laboratory Skills, Data Management and Evaluation,**Ethical and Professional Responsibility, **Application based Knowledge.** 2. These generic outcomes will equip students with a broad set of transferable skills, making them capable of conducting rigorous scientific analysis while working safely, ethically, and effectively in collaborative environments.	

Semester – 6
Syllabus for Theory Paper
CHM-DSC-18-MAJ – 605
Environmental Chemistry-2

End semester = 67 **Credits – 03**

Internal Marks = 08 **Examination Time allowed - 3 Hours**

UNIT – 1 Water Chemistry (15 Hrs.)

Aquatic Chemistry and Water Pollution

Chemical composition of water bodies (Ground water, river water, sea water, wet lands), Hydrological cycle, Water quality parameters: Physicochemical parameters(pH, turbidity, hardness) ; Dissolved oxygen(DO) ; Biological oxygen demand (BOD) ; Chemical oxygen demand (COD), Water pollution: Organic water pollutants; Inorganic water pollutants, Eutrophication, Water treatment: primary; Secondary (aerobic/115or activ) and tertiary treatment (reverse osmosis, ion exchange and adsorption)

UNIT – 2 Environmental toxicology (15 Hrs.)

Environmental toxicology

Toxic chemicals in environment, Biochemical effects of: As; Cd; Pb and Hg, Carcinogens, Biodegradability, Principles of Biodegradation: Biooxidation; bioreduction, Environmental episodes: Bhoal gas tragedy, Chernobyl disaster

UNIT – 3 Green Chemistry (15 Hrs.)

Green Chemistry and Sustainability

The 12 Principles of green chemistry with real world examples (e.g. Atom economy), Use of supercritical CO_2, ionic liquids and biocatalysts in chemical synthesis, Renewable Energy: Chemistry of Hydrogen fuel cell, Photovoltaic cells, biofuels (Bioethanol, biodiesel)

Literature/Books Suggested

1. S.E. Manahan, Environmental Chemistry, Lewis Publishers.
2. A.K. Dey, Environmental Chemistry, Wiley Eastern.
3. V.K. Ahlluwalia, Environmental Chemistry, Ane Books Pvt. Ltd.
4. S.M. Khopkar, Environment Pollution Analysis, Wiley Eastern Ltd.
5. Text book of Environment chemistry, Balram Pani, I.K. International Pub. House, New Delhi.
6. Principles of Environment Chemistry, Games.E.Girard, Jones & Bartlet India.
7. Environmental Chemistry, H. Kaur, Pragati Prakashan

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit** (of **19 marks**) will consist of two types of questions: **9 multiple-choice questions** of one mark each and **5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:** the **compulsory question** and one question each from the **UNIT-1, UNIT-2 and UNIT-3**.
2. The use of scientific calculators is not allowed.

Semester – 6
Syllabus for Practical Paper
CHM-DSC-18-MAJ – 605
Environmental Chemistry-2

End semester = 22 **Credit – 01**

Internal Marks = 03 **Contact Hours – 2/week**

Examination Time allowed **3 Hours**

Water Analysis

1. Determination of total dissolved solids (TDS).
2. Determination of water salinity.
3. Determination of alkalinity.
4. Determination of dissolved oxygen.
5. Determination of chlorides.
6. Determination of total hardness.
7. Determination of calcium.

Literature/Books suggested

1. A laboratory Mannual for Environmental Chemistry, R.Gopalan, A.Anand, R.W. Sugumar, I.K.International Publ. House, New Delhi.
2. Experimental Ichthyology, S.K.Garg, A.Bhatnagar, A.Johal, CBS Publishers New Delhi.
3. Soil and Water Analysis: A Practical Mannual, Dr. R.K.Singh, Dr. K.Yadav, Dr. R.P.Singh,R.K.Jalaj, Dr.R.Kumar. Iterative International Publisher

General Instructions to the Examiners:

Note: Practical examination will be of three hours duration divided into two parts (First, half hour for written and second half will be the next two and half hours for performance).

1. Candidate will be examined based on any two experiments given by the examiner. Each experiment carries 08 marks (3 written + 5 performance).
2. Viva-Voce: 04 marks.
3. Note Book: 02 marks.
4. Internal Assessment: 03 marks

Semester	**6**
Code of Paper	**CHM-DSC-6-MIN-601 (Table 1)**
Title of Paper	**Inorganic Chemistry**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100 (75 Theory + 25 Practical)**
Marks distribution of Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of Practical paper	**End semester = 22** **Internal Marks = 03**
Applicable to the students 1. Those who have opted for chemistry as a Major subject in Table 1.	
LEARNING OUTCOMES SPECIFIC TO DISCIPLINE The students will learn the following- 1. Important aspects of transition metals 2. Chemistry of compounds of f-block elements 3. Chemistry of coordination compounds	
GENERIC LEARNING OUTCOMES 1. **The students will be able to learn Problem Solving, Practical Laboratory Skills, Data Management and Evaluation,** Ethical and Professional Responsibility, **Application based Knowledge.** 2. These generic outcomes will equip students with a broad set of transferable skills, making them capable of conducting rigorous scientific analysis while working safely, ethically, and effectively in collaborative environments.	

Semester – 6
Syllabus for Theory Paper
CHM-DSC-6-MIN-601 (Table 1)
Inorganic Chemistry

End semester = 67 **Credits – 03**

Internal Marks = 08 **Examination Time allowed – 3 Hours**

UNIT I: Chemistry of d-Block Elements (15 Hrs.)

First Transition Series

Position in the periodic table, Electronic configuration, Physical and Chemical properties – Atomic radii, covalent radii, density, melting and boiling points, oxidation states, ionization energies, reducing properties, electronegativity, electron affinity, metallic behavior, magnetic properties, colour, complex formation, interstitial compounds, catalytic properties, alloy formation. Relative stability of their oxidation states, coordination number and geometry. Chemistry of compounds and complexes of Sc, Ti, V, Cr, Mn, Fe, Co, Ni, Cu, and Zn.

Second and Third Transition Series

General characteristics, comparative treatment with their 3*d*-analogues in respect of ionic radii, oxidation states, magnetic 119or acti, spectral properties and stereochemistry. Important compounds of Zr, Nb, Ta, Mo, Ru, Rh, Pd, W, Re, Os, Ir, Pt, Au, Hg, Aqueous chemistry. Isopoly and heteropoly. Metal-metal bonding.

UNIT II: Chemistry of f-Block Elements (15 Hrs.)

Lanthanide Elements

Occurrence and isolation, electronic structure, oxidation states and ionic radii, lanthanide contraction, complex formation, colour and spectral properties, magnetic properties, lanthanide complexes and compounds (oxides, hydroxides, nitrides, etc) and their uses.

Actinide Elements

General features and chemistry of actinides, comparison with lanthanides, extraction of Th from Monazite sand, extraction of U, chemistry of separation of Np, Pu and Am from U, Transuranic elements. Later or heavier actinides.

UNIT III: Chemistry of Coordination Compounds (15 Hrs.)

Coordination Chemistry – 1

Definitions and terminology, Types of ligands, Werner's coordination theory and its experimental verification, effective atomic number concept, chelates, nomenclature of coordination compounds, isomerism in coordination compounds, Stereochemistry of coordination compounds, Valence bond theory of transition metal complexes. Properties of Coordination compounds, i.e. magnetic properties, colours (Qualitative approach only), and use of coordination compounds.

Books/Literature suggested

1. Cotton, F.A., Wilkinson, G., Gaus, P.L., Basic Inorganic Chemistry; 2nd edition, Pubs: John Wiley and Sons, 1995.
2. Lee, J.D., Concise Inorganic Chemistry; 4th edition, Pubs: Chapman & Hall Ltd., 1991.
3. Shriver, D.E., Atkins, P.W., Inorganic Chemistry; 4th edition, Pubs: Oxford University Press, 2006.
4. Douglas, B., Medaniel, D., Atenander, J., Concepts and Models of Inorganic Chemistry; 3rd edition, Pubs: John Wiley and Sons Inc., 1994,
5. Porterfield, W.W., Wesky, A., Inorganic Chemistry; Pubs: Addison-Wesley Publishing Company,1984.
6. Miessler, G.L., Tarr, D.A., Inorganic Chemistry; 3rd edition, Pubs: Pearson Education Inc., 2004.
7. Puri, B.R., Sharma, L.R., Kalia, K.K., Principles of Inorganic Chemistry; 30th edition, Pubs: Milestones Publisher, 2006-07

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit** (of **19 marks**) will consist of two types of questions: **9 multiple-choice questions** of one mark each and **5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:** the **compulsory question** and one question each from the **UNIT-1, UNIT-2 and UNIT-3**.
2. The use of scientific calculators is not allowed.

Semester – 6
Syllabus for Theory Paper
CHM-DSC-6-MIN-601 (Table 1)
Inorganic Chemistry

End semester = 22 **Credit – 01**

Internal Marks = 03 **Examination Time allowed – 3 Hours**

1. QUALITATIVE ANALYSIS

Semimicro Analysis, cation analysis, separation and identification of ions from groups I, II, III, IV, V and VI. Anion analysis (4 ions).

Books/Literature suggested

1. Svehla, Vogel's Qualitative Inorganic Analysis (revised); 7th edition Pubs: Orient Longman, 1996.
2. Bassett, J., Denney, R.C., Jeffery, G.H., Mendham, J., Vogel's Textbook of Quantitative Inorganic Analysis (revised); 4th edition, 1978.
3. Palmer, W.G., Experimental Inorganic Chemistry; 1st edition, Pubs: Cambridge, 1954.
4. Brauer, Handbook of Preparative Inorganic Chemistry; 2nd edition, Pubs: Academic Press, 1963, Vol. I, & 1965, Vol. II

General Instructions to the Examiners:

Four ions with no interference (anions such as PO_4^{3-}, BO_3^{-3} and similar anions like Cl^-, Br^-, I^- etc. and cations from the same group) may not be given.

Note: Practical examination will be of three hours duration divided into two parts (First, half hour for written and second half will be the next two and half hours for performance).

1. Candidate will be examined based on any two experiments given by the examiner. Each experiment carries 08 marks (3 written + 5 performance).
2. Viva-Voce: 04 marks.
3. Note Book: 02 marks.
4. Internal Assessment: 03 marks

BSc III Sem 5 and Sem 6
CHEMISTRY
as
(MAJOR and MINOR)

DOUBLE MAJOR AND SINGLE MINOR
(TABLE 2)

Paper	Code of the Major Paper	Name of Major paper	Credit (T+P)	Contact Hours	Marks (T+P)
Semester 5					
A-6	CHM-DSC-6-MAJ – 501	Inorganic Chemistry	3+1	45+30	75+25
A-7	CHM-DSC-7-MAJ – 502	Organic Chemistry	3+1	45+30	75+25
	CHM-INT-1	Internship	4	-	100
Paper	**Code of the Minor paper**	**Name of Minor paper**	**Credit (T+P)**	**Contact Hours**	**Marks (T+P)**
Minor	CHM-DSC-6-MIN-501	Inorganic Chemistry	3+1	45+30	75+25
Semester 6					
Paper	**Code of the Major paper**	**Name of Major paper**	**Credit (T+P)**	**Contact Hours**	**Marks (T+P)**
A-8	CHM-DSC-8-MAJ – 601	Physical Chemistry	3+1	45+30	75+25
A-9	CHM-DSC-9-MAJ – 602	General Chemistry	3+1	45+30	75+25
Paper	**Code of the Minor paper**	**Name of Minor paper**	**Credit (T+P)**	**Contact Hours**	**Marks (T+P)**
Minor	CHM-DSC-7-MIN-601	Organic Chemistry	3+1	45+30	75+25
Minor	CHM-DSC-8-MIN-602	Physical Chemistry	3+1	45+30	75+25

Semester	**5**
Code of Paper	**CHM-DSC-6-MAJ-501 (Table 2)**
Title of Paper	**Inorganic Chemistry**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100 (75 Theory + 25 Practical)**
Marks distribution of Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of Practical paper	**End semester = 22** **Internal Marks = 03**
Applicable to the students 1. Those who have opted for chemistry as a Major subject in Table 2.	
LEARNING OUTCOMES SPECIFIC TO DISCIPLINE The students will learn the following- 1. Important aspects of transition metals 2. Chemistry of compounds of f-block elements 3. Chemistry of coordination compounds	
GENERIC LEARNING OUTCOMES 1. **The students will be able to learn Problem Solving, Practical Laboratory Skills, Data Management and Evaluation,** Ethical and Professional Responsibility, **Application based Knowledge.** 2. These generic outcomes will equip students with a broad set of transferable skills, making them capable of conducting rigorous scientific analysis while working safely, ethically, and effectively in collaborative environments.	

Semester – 5
Syllabus for Theory Paper
CHM-DSC-6-MAJ-501 (Table 2)
Inorganic Chemistry

End semester = 67 **Credits – 03**

Internal Marks = 08 **Examination Time allowed – 3 Hours**

UNIT I: Chemistry of d-Block Elements (15 Hrs.)

First Transition Series

Position in the periodic table, Electronic configuration, Physical and Chemical properties – Atomic radii, covalent radii, density, melting and boiling points, oxidation states, ionization energies, reducing properties, electronegativity, electron affinity, metallic behavior, magnetic properties, colour, complex formation, interstitial compounds, catalytic properties, alloy formation. Relative stability of their oxidation states, coordination number and geometry. Chemistry of compounds and complexes of Sc, Ti, V, Cr, Mn, Fe, Co, Ni, Cu, and Zn.

Second and Third Transition Series

General characteristics, comparative treatment with their 3*d*-analogues in respect of ionic radii, oxidation states, magnetic 124or acti, spectral properties and stereochemistry. Important compounds of Zr, Nb, Ta, Mo, Ru, Rh, Pd, W, Re, Os, Ir, Pt, Au, Hg, Aqueous chemistry. Isopoly and heteropoly. Metal-metal bonding.

UNIT II: Chemistry of f-Block Elements (15 Hrs.)

Lanthanide Elements

Occurrence and isolation, electronic structure, oxidation states and ionic radii, lanthanide contraction, complex formation, colour and spectral properties, magnetic properties, lanthanide complexes and compounds (oxides, hydroxides, nitrides, etc) and their uses.

Actinide Elements

General features and chemistry of actinides, comparison with lanthanides, extraction of Th from Monazite sand, extraction of U, chemistry of separation of Np, Pu and Am from U, Transuranic elements. Later or heavier actinides.

UNIT III: Chemistry of Coordination Compounds (15 Hrs.)

Coordination Chemistry – 1

Definitions and terminology, Types of ligands, Werner's coordination theory and its experimental verification, effective atomic number concept, chelates, nomenclature of coordination compounds, isomerism in coordination compounds, Stereochemistry of coordination compounds, Valence bond theory of transition metal complexes. Properties of Coordination compounds, i.e. magnetic properties, colours (Qualitative approach only), and use of coordination compounds.

Books/Literature suggested

1. Cotton, F.A., Wilkinson, G., Gaus, P.L., Basic Inorganic Chemistry; 2nd edition, Pubs: John Wiley and Sons, 1995.
2. Lee, J.D., Concise Inorganic Chemistry; 4th edition, Pubs: Chapman & Hall Ltd., 1991.
3. Shriver, D.E., Atkins, P.W., Inorganic Chemistry; 4th edition, Pubs: Oxford University Press, 2006.
4. Douglas, B., Medaniel, D., Atenander, J., Concepts and Models of Inorganic Chemistry; 3rd edition, Pubs: John Wiley and Sons Inc., 1994,
5. Porterfield, W.W., Wesky, A., Inorganic Chemistry; Pubs: Addison-Wesley Publishing Company,1984.
6. Miessler, G.L., Tarr, D.A., Inorganic Chemistry; 3rd edition, Pubs: Pearson Education Inc., 2004.
7. Puri, B.R., Sharma, L.R., Kalia, K.K., Principles of Inorganic Chemistry; 30th edition, Pubs: Milestones Publisher, 2006-07

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit** (of **19 marks**) will consist of two types of questions: **9 multiple-choice questions** of one mark each and **5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:** the **compulsory question** and one question each from the **UNIT-1, UNIT-2 and UNIT-3**.
2. The use of scientific calculators is not allowed.

Semester – 5

Syllabus for Theory Paper
CHM-DSC-6-MAJ-501 (Table 2)
Inorganic Chemistry

End semester = 22 **Credit – 01**

Internal Marks = 03 **Examination Time allowed – 3 Hours**

2. QUALITATIVE ANALYSIS

Semimicro Analysis, cation analysis, separation and identification of ions from groups I, II, III, IV, V and VI. Anion analysis (4 ions).

Instruction to Examiners:

Four ions with no interference (anions such as PO_4^{3-}, BO_3^{-3} and similar anions like Cl^-, Br^-, I^- etc. and cations from the same group) may not be given.

Books/Literature suggested

1. Svehla, Vogel's Qualitative Inorganic Analysis (revised); 7th edition Pubs: Orient Longman, 1996.
2. Bassett, J., Denney, R.C., Jeffery, G.H., Mendham, J., Vogel's Textbook of Quantitative Inorganic Analysis (revised); 4th edition, 1978.
3. Palmer, W.G., Experimental Inorganic Chemistry; 1st edition, Pubs: Cambridge, 1954.
4. Brauer, Handbook of Preparative Inorganic Chemistry; 2nd edition, Pubs: Academic Press, 1963, Vol. I, & 1965, Vol. II.

General Instructions to the Examiners:

Note: Practical examination will be of three hours duration divided into two parts (First, half hour for written and second half will be the next two and half hours for performance).

1. Candidate will be examined based on any two experiments given by the examiner. Each experiment carries 08 marks (3 written + 5 performance).
2. Viva-Voce: 04 marks.
3. Note Book: 02 marks.
4. Internal Assessment: 03 marks

Semester	**5**
Code of Paper	**CHM-DSC-7-MAJ-502 (Table 2)**
Title of Paper	**Organic Chemistry**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100 (75 Theory + 25 Practical)**
Marks distribution of Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of Practical paper	**End semester = 22** **Internal Marks = 03**

Applicable to the students

1. Those who have opted for chemistry as a Major subject in Table 2.

LEARNING OUTCOMES SPECIFIC TO DISCIPLINE

The students will learn the following-

1. Important aspects of chemistry of carbonyl compounds and a few related name reactions
2. Chemistry of carboxylic acid and its derivatives
3. Chemistry of organometallic compounds
4. Formation and use of enolates in organic synthesis

GENERIC LEARNING OUTCOMES

1. **The students will be able to learn Problem Solving, Practical Laboratory Skills, Data Management, Evaluation,** estimation and determination through a**pplication based organic chemistry techniques and concepts.**
2. These generic outcomes will equip students with a broad set of transferable skills, making them capable of conducting rigorous scientific analysis while working safely, ethically, and effectively in collaborative environments.

Semester – 5
Syllabus for Theory Paper
CHM-DSC-7-MAJ-502 (Table 2)
Organic Chemistry

End semester = 67 **Credits – 03**

Internal Marks = 08 **Examination Time allowed – 3 Hours**

Unit I: Carbonyl Compounds (15 Hrs)

Aldehydes and Ketones

Nomenclature and structure of the carbonyl group. Physical properties of aldehydes & ketones. Synthesis of aldehydes and ketones with particular reference to the synthesis of aldehydes from acid chorides. Synthesis of aldehydes and ketones using 1,3-dithianes, synthesis of ketones from nitriles and from carboxylic acids.

Mechanism of nucleophilic additions to carbonyl group with particular emphasis on Benzoin, Aldol, Perkin and Knoevenagel condensations. Condensation with ammonia and its derivatives, Wittig reaction. Use of acetals as protecting group. Oxidation of aldehydes, Baeyer-Villiger oxidation of ketones, Cannizzaro reaction, MPV, Clemmensen, Wolff-Kishner, $LiAlH_4$ and $NaBH_4$ reductions.

Unit II: Carboxylic Acids and Derivatives (15 Hrs)

Carboxylic Acids

Nomenclature, structure and bonding, physical properties, acidity of carboxylic acids, effects of substitutions on acid strength. Preparations of carboxylic acids. Reactions of carboxylic acids. Mechanism of decarboxylation. Methods for formation of halo acids.

Carboxylic Acid Derivatives

Structure and nomenclature of acid chlorides, esters, amides and acid anhydrides. Relative stability & reactivity of acyl derivatives. Physical properties, interconversion of acid derivatives by nucleophilic acyl substitution. Preparation of carboxylic acid derivatives, chemical reactions. Mechanisms of esterification and hydrolysis (acidic and basic).

Unit III: Organometallic Compounds and Enolates (15 Hrs)

Organic Synthesis via Enolates

Acidity of α-hydrogens, alkylation of diethyl malonate and ethyl acetoacetate. Synthesis of ethyl acetoacetate: the Claisen condensation. Keto-enol tautomerism of ethyl acetoacetate.

Organometallic Compounds:

Organomagnesium Compounds: The Grignard reagents: Formation, structure and chemical reactions. Organozinc Compounds: Formation and Chemical Reactions. Organolithium Compounds: Formation and Chemical Reactions.

Books/Literature suggested

1. Morrison, R.T., Boyd, R.N., Organic Chemistry; 6thedition, Pubs: Prentice-Hall,1992.
2. Wade Jr., L.G., Singh, M.S., Organic Chemistry; 6thedition, Pubs: Pearson Education, 2008.
3. Mukherji, S.M., Singh, S.P., Kapoor, R.P., Organic Chemistry; Pubs: New Age International, 1985, Vol. I.II, III.
4. Carey,F.A., Organic Chemistry; 4th edition, Pubs: McGraw-Hill, 2000.
5. Solomons, T.W., Fundamentals of Organic Chemistry; 5thedition, Pubs: John Wiley & Sons,1997.

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit** (of **19 marks**) will consist of two types of questions: **9 multiple-choice questions** of one mark each and **5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:** the **compulsory question** and one question each from the **UNIT-1, UNIT-2 and UNIT-3**.
2. The use of scientific calculators is not allowed.

Semester – 5
Syllabus for Practical Paper
CHM-DSC-7-MAJ-502 (Table 2)
Organic Chemistry-1

End semester = 22 **Credit – 01**

Internal Marks = 03 **Contact Hours – 2/week**

Examination Time allowed **3 Hours**

I Preparation, TLC monitoring and purification via column chromatography for a minimum of 5 of the following compounds:

i) Methyl Orange
j) Para-Aminoazobenzene
k) Aspirin
l) Picric acid
m) Acetanilide
n) Para-Bromoacetanilide
o) Phenyl azo-β-naphthol
p) 7-Hydroxy-4-methylcoumarin

II Determination of the per cent purity of the given carbonyl compound via the hydrazone formation method.

III Estimation of aniline in the given solution by acetylation/bromination.

Books/Literature suggested

1. Harwoor, L.M., Moody, J., Experimental Organic Chemistry; 1st edition, Pubs: Blackwell Scientific Publications, 1989.
2. Forniss, B.S., Rogers, V., Vogel's Text Book of Practical Organic Chemistry; Pubs: Dorling Kindhsky Pvt. Ltd., 1989.
3. Bansal, R.K., Laboratory Manual of Organic Chemistry; 3rd edition, Pubs: Wiley Eastern Limited, 1994.
4. Furniss, B.S., Hannaford, A.J., Rogers, V., Smith, P.W.G., Tatchell, A.R., Vogel's Textbook of Practical Organic Chemistry; 4th edition, Pubs: Longman group, 1978.
5. Aggarwal, O.P., Advanced Practical Organic Chemistry, 30th edition, 2013, Goel Publishing House.

General Instructions to the Examiners:

Note: Practical examination will be of three hours duration divided into two parts (First, half hour for written and second half will be the next two and half hours for performance).

1. Candidate will be examined based on any two experiments given by the examiner. Each experiment carries 08 marks (3 written + 5 performance).
2. Viva-Voce: 04 marks.
3. Note Book: 02 marks.
4. Internal Assessment: 03 marks

Semester – 5

Instructions for Internship

CHM-INT-1 (Table 1)

Internship / Field-Based Learning Course

Total Marks = 100 **Credits: 4**

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1. Course Description

The internship course is designed to provide undergraduate Chemistry students with experiential learning through **industrial exposure, academic visits, training programs, and participation in scientific events**. The course aims to bridge theoretical knowledge with real-world applications in chemical industries, research institutions, and academic platforms.

2. Course Components

2.1 Project/Training:

Each student shall undertake **at least one** industrial visit, hands-on training, or research-based activity in any one of the following institutions or facilities:

(viii) University/College Department Research Laboratories
(ix) Analytical Instrumentation Unit at Industry/Laboratory
(x) Manufacturing or Processing Units in Industry
(xi) National Defence Laboratories, Metrological Laboratories, Forensic Science Labs.
(xii) Environmental Monitoring Agencies, or Pollution Control Boards
(xiii) Drinking Water or Wastewater Treatment Facilities or similar facilities
(xiv) Quality Control or Testing Laboratories
Any other activity deemed appropriate by the Department/Institute.

The student shall spend a **minimum of 50-60 hours(7-10 days)** at the selected activity site. Winter or autumn vacations may be utilized for this purpose. Upon completion, the student must obtain a certificate or official letter from the host industry or academic institution confirming their participation and duration of engagement.

2.2 Internship Report:

Each student shall submit a **detailed internship report** of approximately **20-30 pages**, including relevant photographs. The report shall include, but not be limited to, the following aspects:

(vii) Description of instrumentation, machines, or facilities used
(viii) Details of processes involved
(ix) Manpower, Infrastructure, and Safety and waste management practices
(x) Observations made during the internship
(xi) Learning outcomes and skills acquired
(xii) Any other detail

The student shall submit **three hard copies** (soft binding) of the report along with one **soft copy in PDF format** for uploading on the website. The report shall be uploaded on the Institute/College website after evaluation.

2.3 Additional Academic Activities:

Additional activities shall include participation in conferences, seminars, workshops, or activities of academic clubs organized by (i) the parent Department or Institute, or other recognized academic or research institutions. The student shall submit a **summary report of up to 5 pages** covering such activities with relevant photographs. The summary may also include relevant academic programmes attended by the student during the **first and second years** of the course, if any.

3. **Assessment Method**

The assessment for this component shall be based on the following:

(v) Internship Report – 50 ***Marks***
(vi) Oral Presentation – 20 ***Marks***
(vii) Additional Academic Activities – 20 ***Marks***
(viii) ***Internal assessment***- 10 ***Marks***

(a) The internship report shall be evaluated by **two examiners** appointed by the College, one of whom shall be from a discipline other than that of the student.
(b) The internship report shall also be evaluated by an **external examiner** based on the soft copy submitted.
(c) All students shall present their internship work in a **one-day academic event**, such as a symposium, in the presence of internal examiners and fellow students. The evaluation of the presentation shall be based on the following criteria:
   i. Quality of oral presentation, clarity, innovation, understanding of the topic, and responses to questions – **60%**
   ii. Attendance, participation in discussions, and engagement during presentations of other students – **40%**

4. **Teacher Involvement**

A faculty member may be assigned mentorship responsibility for a group of 15–20 students for the purpose of internship. The faculty mentor will guide the students in selecting the site for the visit, coordinating with industry or academic institutions, preparing the report and presentation, arranging examiners for evaluation, and organizing related events. For undertaking these responsibilities, a workload of 4 hours/week may be allotted to the concerned faculty member.

Semester	**5**
Code of Paper	**CHM-DSC-6-MIN-501 (Table 2)**
Title of Paper	**Inorganic Chemistry**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100 (75 Theory + 25 Practical)**
Marks distribution of Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of Practical paper	**End semester = 22** **Internal Marks = 03**

Applicable to the students

1. Those who have opted for chemistry as a Major subject in Table 2.

LEARNING OUTCOMES SPECIFIC TO DISCIPLINE

The students will learn the following-

1. Important aspects of transition metals
2. Chemistry of compounds of f-block elements
3. Chemistry of coordination compounds

GENERIC LEARNING OUTCOMES

1. **The students will be able to learn Problem Solving, Practical Laboratory Skills, Data Management and Evaluation,** Ethical and Professional Responsibility, **Application based Knowledge.**
2. These generic outcomes will equip students with a broad set of transferable skills, making them capable of conducting rigorous scientific analysis while working safely, ethically, and effectively in collaborative environments.

Semester – 5
Syllabus for Theory Paper
CHM-DSC-6-MIN-501 (Table 2)
Inorganic Chemistry

End semester = 67 **Credits – 03**

Internal Marks = 08 **Examination Time allowed – 3 Hours**

UNIT I: Chemistry of d-Block Elements (15 Hrs.)

First Transition Series

Position in the periodic table, Electronic configuration, Physical and Chemical properties – Atomic radii, covalent radii, density, melting and boiling points, oxidation states, ionization energies, reducing properties, electronegativity, electron affinity, metallic behavior, magnetic properties, colour, complex formation, interstitial compounds, catalytic properties, alloy formation. Relative stability of their oxidation states, coordination number and geometry. Chemistry of compounds and complexes of Sc, Ti, V, Cr, Mn, Fe, Co, Ni, Cu, and Zn.

Second and Third Transition Series

General characteristics, comparative treatment with their 3*d*-analogues in respect of ionic radii, oxidation states, magnetic 134or acti, spectral properties and stereochemistry. Important compounds of Zr, Nb, Ta, Mo, Ru, Rh, Pd, W, Re, Os, Ir, Pt, Au, Hg, Aqueous chemistry. Isopoly and heteropoly. Metal-metal bonding.

UNIT II: Chemistry of f-Block Elements (15 Hrs.)

Lanthanide Elements

Occurrence and isolation, electronic structure, oxidation states and ionic radii, lanthanide contraction, complex formation, colour and spectral properties, magnetic properties, lanthanide complexes and compounds (oxides, hydroxides, nitrides, etc) and their uses.

Actinide Elements

General features and chemistry of actinides, comparison with lanthanides, extraction of Th from Monazite sand, extraction of U, chemistry of separation of Np, Pu and Am from U, Transuranic elements. Later or heavier actinides.

UNIT III: Chemistry of Coordination Compounds (15 Hrs.)

Coordination Chemistry – 1

Definitions and terminology, Types of ligands, Werner's coordination theory and its experimental verification, effective atomic number concept, chelates, nomenclature of coordination compounds, isomerism in coordination compounds, Stereochemistry of coordination compounds, Valence bond theory of transition metal complexes. Properties of Coordination compounds, i.e. magnetic properties, colours (Qualitative approach only), and use of coordination compounds.

Books/Literature suggested

1. Cotton, F.A., Wilkinson, G., Gaus, P.L., Basic Inorganic Chemistry; 2nd edition, Pubs: John Wiley and Sons, 1995.
2. Lee, J.D., Concise Inorganic Chemistry; 4th edition, Pubs: Chapman & Hall Ltd., 1991.
3. Shriver, D.E., Atkins, P.W., Inorganic Chemistry; 4th edition, Pubs: Oxford University Press, 2006.
4. Douglas, B., Medaniel, D., Atenander, J., Concepts and Models of Inorganic Chemistry; 3rd edition, Pubs: John Wiley and Sons Inc., 1994,
5. Porterfield, W.W., Wesky, A., Inorganic Chemistry; Pubs: Addison-Wesley Publishing Company,1984.
6. Miessler, G.L., Tarr, D.A., Inorganic Chemistry; 3rd edition, Pubs: Pearson Education Inc., 2004.
7. Puri, B.R., Sharma, L.R., Kalia, K.K., Principles of Inorganic Chemistry; 30th edition, Pubs: Milestones Publisher, 2006-07

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit** (of **19 marks**) will consist of two types of questions: **9 multiple-choice questions** of one mark each and **5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:** the **compulsory question** and one question each from the **UNIT-1, UNIT-2 and UNIT-3**.
2. The use of scientific calculators is not allowed.

Semester – 5
Syllabus for Theory Paper
CHM-DSC-6-MIN-501 (Table 2)
Inorganic Chemistry

End semester = 22 **Credit – 01**

Internal Marks = 03 **Examination Time allowed – 3 Hours**

I. QUALITATIVE ANALYSIS

Semimicro Analysis, cation analysis, separation and identification of ions from groups I, II, III, IV, V and VI. Anion analysis (4 ions).

Books/Literature suggested

1. Svehla, Vogel's Qualitative Inorganic Analysis (revised); 7th edition Pubs: Orient Longman, 1996.
2. Bassett, J., Denney, R.C., Jeffery, G.H., Mendham, J., Vogel's Textbook of Quantitative Inorganic Analysis (revised); 4th edition, 1978.
3. Palmer, W.G., Experimental Inorganic Chemistry; 1st edition, Pubs: Cambridge, 1954.
4. Brauer, Handbook of Preparative Inorganic Chemistry; 2nd edition, Pubs: Academic Press, 1963, Vol. I, & 1965, Vol. II.

General Instructions to the Examiners:

Instruction to Examiners:

Four ions with no interference (anions such as PO_4^{3-}, BO_3^{-3} and similar anions like Cl^-, Br^-, I^- etc. and cations from the same group) may not be given.

Note: Practical examination will be of three hours duration divided into two parts (First, half hour for written and second half will be the next two and half hours for performance).

1. Candidate will be examined based on any two experiments given by the examiner. Each experiment carries 08 marks (3 written + 5 performance).
2. Viva-Voce: 04 marks.
3. Note Book: 02 marks.
4. Internal Assessment: 03 marks

Semester	**6**
Code of Paper	**CHM-DSC-8-MAJ-601 (Table 2)**
Title of Paper	**Physical Chemistry**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100 (75 Theory + 25 Practical)**
Marks distribution of Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of Practical paper	**End semester = 22** **Internal Marks = 03**

Applicable to the students

1. Those who have opted for chemistry as a Major subject in Table 2.

LEARNING OUTCOMES SPECIFIC TO DISCIPLINE

The students will learn the following-

1. **Solutions:** Quantify molar mass and vapor pressure changes using Raoult's Law, colligative properties, and van't Hoff factors for electrolytes.
2. **Electrochemistry:** Analyze ionic conductance and mobility to construct electrochemical cells and predict reaction spontaneity via the Nernst equation.
3. **Photochemistry:** Evaluate light-matter interactions and quantum yield while mapping excited-state pathways (Jablonski diagrams) for carbonyl and alkene reactions.

GENERIC LEARNING OUTCOMES

The students will be able to learn.

1. Quantitative Reasoning: Mastery of mathematical models to solve physical chemistry problems.
2. Experimental Proficiency: Understanding the setup and logic behind classical laboratory techniques (conductometry, osmometry, etc.).
3. Molecular Analysis: Ability to relate macroscopic observations (like color or pressure) to microscopic molecular behavior.
4. Analytical Thinking: Connecting theoretical laws (like Nernst or Raoult) to real-world industrial and biological applications.

Semester – 6
Syllabus for Theory Paper
CHM-DSC-8-MAJ-601 (Table 2)
Physical Chemistry

End semester = 67 **Credits – 03**

Internal Marks = 08 **Examination Time allowed – 3 Hours**

UNIT – I Solutions (15 Hrs.)

Solutions and Colligative Properties:

Dilute Solutions and Colligative Properties: Ideal and non-ideal solutions, methods of expressing concentrations of solutions, activity and activity coefficient. Dilute solution, colligative properties, Raoult's law, relative lowering of vapour pressure, molecular weight determination. Osmosis, law of osmotic pressure and its measurement, determination of molecular weight from osmotic pressure. Elevation of boiling point and depression of freezing point, Thermodynamic derivation of relation between molecular weight and elevation in boiling point and depression of freezing point. Experimental methods for determining various colligative properties. Abnormal molar mass, degree of dissociation and association of solutes.

UNIT – II Electrochemistry (15 Hrs.)

Electrochemistry – I:

Electrical transport –Conduction in metals and in electrolyte solutions, specific conductance and equivalent conductance, measurement of equivalent conductance, variation of equivalent and specific conductance with dilution. Migration of ions and Kohlrausch Law, Arrhennius theory of electrolyte dissociation and its limitations, weak and strong electrolytes, Ostwald's dilution law, its uses and limitations. Debye-Huckel-Onsager's equation for strong electrolytes (elementary treatment only). Transport number

Electrochemistry-II:

Types of reversible electrodes – gas metal – ion, metal –insoluble salt – anion and redox electrodes. Electrode reactions, Nernst equation, derivation of cell E.M.F. and single electrode potential, standard hydrogen electrode – reference electrodes – standard electrode potential, sign conventions, electrochemical series and its significance.

UNIT – III Photochemistry (15 Hrs.)

Photochemistry:

Interaction of radiation with matter, difference between thermal and photochemical processes. Laws of Photochemistry: Grothus – Drapper law, Stark – Einstein law, Jablonski diagram depicting various processes occurring in the excited state. Qualitative description of fluorescence, phosphorescence, non-radiative processes (internal conversion, intersystem crossing), quantum yield, photosensitized reactions – energy transfer processes (simple examples). Photochemistry of carbonyl compounds and alkenes.

Books/Literature suggested

1. Atkins, P., Paula, J.de, Atkins Physical Chemistry; 8th edition, Pubs: Oxford University Press, 2008.
2. Puri, B.R., Sharma, L.R., Pathania, M.S., Principles of Physical Chemistry; 43rd edition, Pubs: Vishal Publishing Co., 2008.
3. Barrow, G.M., Physical Chemistry; 6th edition, Pubs: McGraw Hill Inc, 1996.
4. Rao, C.N.R., University General Chemistry; Pubs: Macmillan India, 1985.
5. Berry, R.S., Rice, S.A., Ross, J., Physical Chemistry; 2nd edition, Pubs: Oxford University Press, 2000.
6. Albert, R.A., Silbey, R.J., Physical Chemistry; 1st edition, Pubs: John Wiley & Sons Inc., 1992.
7. Dogra, S.K., Dogra, S., Physical Chemistry Through Problems; Pubs:Wiley Eastern Limited, 1991.
8. Levine, I.N., Physical Chemistry; 5th edition, Pubs: Tata McGraw Hill Publishing Co. Ltd., 2002.
9. Moore, W. J., Basic Physical Chemistry; Pubs: Prentice Hall of India Pvt. Ltd, 1983.
10. Metz, C.R., Theory and Problems of Physical Chemistry; Schaum's outline series, 2nd edition, Pubs: McGraw-Hall Book company, 1989.

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit** (of **19 marks**) will consist of two types of questions: **9 multiple-choice questions** of one mark each and **5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:** the **compulsory question** and one question each from the **UNIT-1, UNIT-2 and UNIT-3**.
2. The use of scientific calculators is not allowed.

Semester – 5
Syllabus for Practical Paper
CHM-DSC-8-MAJ-601 (Table 2)
Physical Chemistry

End semester = 22 **Credit – 01**

Internal Marks = 03 **Contact Hours – 2/week**

Examination Time allowed **3 Hours**

I. Conductometry

1. To determine the Cell Constant of a Conductivity Cell
2. Study of conductometric titration of a strong acid (HCl) against a strong base (NaOH)
3. Study of conductometric titration of a weak acid (CH3COOH) against a strong base (NaOH)
4. Study of conductometric titration of a mixture of strong acid (HCl) and weak acid (CH3COOH) against a strong base (NaOH).
5. To determine the solubility and solubility product of a sparingly soluble electrolyte conductometricaly.
6. To study the saponification of ethyl acetate conductometricaly.
7. To determine the ionization constant of a weak acid conductometricaly
8. Determination of hydrolysis constant of aniline hydrochloride conductometricaly.

Books/Literature suggested

1. Khosla, B.D., Garg, V.C., Gulati, A., Senior Practical Physical Chemistry; 11th edition, Pubs: R.Chand& Co., New Delhi, 2002.
2. Das, R.C., Behra, B., Experimental Physical Chemistry; Pubs: Tata McGraw-Hill Publishing Co. Ltd., 1983.
3. Levitt, B.P., Findlay's Practical Physical Chemistry; 8th edition, Pubs: Longman Group Ltd., London & New York, 1978.

General Instructions to the Examiners:

Note: Practical examination will be of three hours duration divided into two parts (First, half hour for written and second half will be the next two and half hours for performance).

1. Candidate will be examined based on any two experiments given by the examiner. Each experiment carries 08 marks (3 written + 5 performance).
2. Viva-Voce: 04 marks.
3. Note Book: 02 marks.
4. Internal Assessment: 03 marks

Semester	**6**
Code of Paper	**CHM-DSC-9-MAJ-602 (Table 2)**
Title of Paper	**General Chemistry-6**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100 (75 Theory + 25 Practical)**
Marks distribution of Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of Practical paper	**End semester = 22** **Internal Marks = 03**

Applicable to the students

1. Those who have opted for chemistry as a Major subject in Table 2.

LEARNING OUTCOMES SPECIFIC TO DISCIPLINE

The students will learn the following-

1. Important aspects of coordination chemistry
2. Chemistry of organometallic compounds
3. Chemistry of carboxylic acids, fats, detergents and heterocyclic compounds
4. Concepts of chemical equilibria, acid bases, buffers.

GENERIC LEARNING OUTCOMES

1. **The students will be able to learn Problem Solving, Practical Laboratory Skills, Data Management and Evaluation,**Ethical and Professional Responsibility, **Application based Knowledge.**
2. These generic outcomes will equip students with a broad set of transferable skills, making them capable of conducting rigorous scientific analysis while working safely, ethically, and effectively in collaborative environments.

Semester – 6
Syllabus for Theory Paper
CHM-DSC-9-MAJ-602 (Table 2)
General Chemistry-6

End semester = 67 **Credits – 03**

Internal Marks = 08 **Examination Time allowed-3 Hours**

UNIT I: Inorganic Chemistry (15 Hrs.)

Coordination Chemistry – II

Limitations of valence bond theory, Crystal field theory, CFSE in weak and strong fields, pairing energies, factors affecting the magnitude of 10 Dq (Δo, Δt). The splitting of d-orbitals in different fields (octahedral, tetrahedral, tetragonally distorted octahedral, square planar, trigonal bipyramidal), Introduction to Orgel diagrams, Consequences and applications of orbital splitting, crystal field stabilization energy, magnetic properties, measurement of 10 Dq (Δo), Factors affecting the extent of splitting and spectrochemical series, colour of transition metal complexes.

Organometallic compounds

Definition, nomenclature and types of organometallic compounds. Classification of ligands, Hapticity, EAN rule, Nomenclature of organometallic compounds, Bonding in organometallic compounds, Preparation, properties, bonding and applications of alkyls and aryls of Li, Al , Hg, Sn and Ti. A brief account of metal – olefin complexes, alkyne complexes, homogeneous hydrogenation, Metal carbonyls – Preparation, bonding and reactions, Structures and the nature of bonding in bi, tri, tetranuclear metal carbonyls.

UNIT II: Organic Chemistry (15 Hrs.)

Substituted carboxylic acids:

Hydroxyl acids: Malic, tartaric and citric acids (structural features only). Methods of formation and chemical reactions of unsaturated monocarboxylic acids. Dicarboxylic acids: Methods of formation and effects of heat and hydrating agents.

Fats, Oils and Detergents:

Natural fats, edible and industrial oils of vegetable origin, common fatty acids, glycerides, hydrogenation of unsaturated oils.Saponification value, iodine value, acid value.Soaps and detergents (classification and mechanism of action); alkyl and aryl sulphonates.

Fused Heterocyclic compounds:

Introduction to condensed-five and six-membered heterocycles. Preparation and reactions of indole, quinoline and isoquinoline.Mechanism of electrophilic substitution reactions of indole, quinoline and isoquinoline.

UNIT III: Physical (15 Hrs.)

Ionic Equilibria:

Strong, moderate and weak electrolytes, degree of ionization, factors affecting degree of ionization, ionization constant and ionic product of water. Ionization of weak acids and bases, pH scale, common ion effect; dissociation constants of mono-, di-and triprotic acids (exact treatment). Salt hydrolysis-calculation of hydrolysis constant, degree of hydrolysis and pH for different salts. Buffer solutions; derivation of Henderson equation and its applications; buffer capacity, buffer range, buffer action and applications of buffers in analytical chemistry and biochemical processes in the human body. Solubility and solubility product of sparingly soluble salts – applications of solubility product principle. Qualitative treatment of acid – base titration curves (calculation of pH at various stages). Theory of acid–base indicators; selection of indicators and their limitations.

Books/Literature suggested

1. Cotton, F.A., Wilkinson, G., Gaus, P.L., Basic Inorganic Chemistry; 2nd edition, Pubs: John Wiley and Sons, 1995.
2. Lee, J.D., Concise Inorganic Chemistry; 4th edition, Pubs: Chapman & Hall Ltd., 1991.
3. Shriver, D.E., Atkins, P.W., Inorganic Chemistry; 4th edition, Pubs: Oxford University Press, 2006.
4. Douglas, B., Medaniel, D., Atenander, J., Concepts and Models of Inorganic Chemistry; 3rd edition, Pubs: John Wiley and Sons Inc., 1994,
5. Porterfield, W.W., Wesky, A., Inorganic Chemistry; Pubs: Addison-Wesley Publishing Company,1984.
6. Miessler, G.L., Tarr, D.A., Inorganic Chemistry; 3rd edition, Pubs: Pearson Education Inc., 2004.
7. Puri, B.R., Sharma, L.R., Kalia, K.K., Principles of Inorganic Chemistry; 30th edition, Pubs: Milestones Publisher, 2006-07
8. Atkins, P., Paula, J. de, Atkins Physical Chemistry; 8th edition, Pubs: Oxford University Press, 2008.
9. Puri, B.R., Sharma, L.R., Pathania, M.S., Principles of Physical Chemistry; 43rd edition, Pubs: Vishal Publishing Co., 2008.
10. Barrow, G.M., Physical Chemistry; 6th edition, Pubs: McGraw-Hill Inc, 1996.
11. Berry, R.S., Rice, S.A., Ross, J., Physical Chemistry; 2nd edition, Pubs: Oxford University Press, 2000.
12. Albert, R.A., Silbey, R.J., Physical Chemistry; 1st edition, Pubs: John Wiley & Sons Inc., 1992.
13. Dogra, S.K., Dogra, S., Physical Chemistry Through Problems; Pubs: Wiley Eastern Limited, 1991.

14. Levine, I.N., Physical Chemistry; 5th edition, Pubs: Tata McGraw-Hill Publishing Co. Ltd., 2002.
15. Moore, W. J., Basic Physical Chemistry; Pubs: Prentice Hall of India Pvt. Ltd, 1983.
16. Metz, C.R., Theory and Problems of Physical Chemistry; Schaum's outline series, 2nd edition, Pubs: McGraw-Hall Book Company, 1989.
17. Morrison, R.T., Boyd, R.N., Organic Chemistry; 6th edition, Pubs: Prentice-Hall, 1992.
18. Solomons, T.W., Fryhle, C.B., Organic Chemistry; 9th edition, Pubs: Wiley India, 2007.
19. Wade Jr., L.G., Singh, M.S., Organic Chemistry; 6th edition, Pubs: Pearson Education, 2008.
20. Mukherji, S.M., Singh, S.P., Kapoor, R.P., Organic Chemistry; Pubs: New Age International, 1985.
21. Carey, F.A., Sundberg, R.J., Advanced Organic Chemistry Part B: Reactions and Synthesis; 5th edition, Pubs: Springer, 2007

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit** (of **19 marks**) will consist of two types of questions: **9 multiple-choice questions** of one mark each and **5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:** the **compulsory question** and one question each from the **UNIT-1, UNIT-2 and UNIT-3**.
2. The use of scientific calculators is not allowed.

Semester – 6
Syllabus for Practical Paper
CHM-DSC-9-MAJ-602 (Table 2)
General Chemistry-6

End semester = 22 **Credit – 01**

Internal Marks = 03 **Contact Hours – 2/week**

Examination Time allowed **3 Hours**

I. INORGANIC CHEMISTRY

1. Complexometric titrations using EDTA
   a. Determination of Calcium by EDTA titration.
   b. Determination of Magnesium by EDTA titration.
   c. Determination of total hardness of water by EDTA.
   d. Determination of zinc/Nickel.
   e. Determination of temporary and permanent hardness of water by EDTA.

II. ORGANIC CHEMISTRY

1. Determination of the following for any type of oil, such as coconut oil, mustard oil, sunflower oil, etc.
   a. Acid value
   b. Iodine value
   c. Saponification value
2. Estimation of Citric Acid in Lemon Juice
3. Complexometric Estimation of Citric Acid (Calcium Salt Method)
4. Volumetric determination of Equivalent Weight of Citric Acid
5. To study optical rotation of Tartaric acid.

III. PHYSICAL CHEMISTRY

1. To prepare buffer solutions using 0.2M acetic acid and 0.2M sodium acetate with help of Henderson equation and determine the mole ratio of solution of unknown pH using pH-meter.
2. Determination of hydrolysis constant of aniline hydrochloride / ammonium chloride by using pH meter.
3. To determine the concentration of strong acid (HCl) by titration against standard solution of 0.1N NaOH solution with the help of pH meter.

Books/Literature suggested

1. Khosla, B.D., Garg, V.C., Gulati, A., Senior Practical Physical Chemistry; 11th edition, Pubs: R.Chand& Co., New Delhi, 2002.
2. Das, R.C., Behra, B., Experimental Physical Chemistry; Pubs: Tata McGraw-Hill Publishing Co. Ltd., 1983.
3. Levitt, B.P., Findlay's Practical Physical Chemistry; 8th edition, Pubs: Longman Group Ltd., London & New York, 1978.
4. Svehla, Vogel's Qualitative Inorganic Analysis (revised); 7th edition Pubs: Orient Longman, 1996.
5. Bassett, J., Denney, R.C., Jeffery, G.H., Mendham, J., Vogel's Textbook of Quantitative Inorganic Analysis (revised); 4th edition, 1978.
6. Palmer, W.G., Experimental Inorganic Chemistry; 1st edition, Pubs: Cambridge, 1954.
7. Brauer, Handbook of Preparative Inorganic Chemistry; 2nd edition, Pubs: Academic Press, 1963, Vol. I, & 1965, Vol. II

General Instructions to the Examiners:

Note: Practical examination will be of three hours duration divided into two parts (First, half hour for written and second half will be the next two and half hours for performance).

1. Candidate will be examined based on any two experiments given by the examiner. Each experiment carries 08 marks (3 written + 5 performance).
2. Viva-Voce: 04 marks.
3. Note Book: 02 marks.
4. Internal Assessment: 03 marks

Semester	**6**
Code of Paper	**CHM-DSC-7-MIN-601 (Table 2)**
Title of Paper	**Organic Chemistry**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100 (75 Theory + 25 Practical)**
Marks distribution of Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of Practical paper	**End semester = 22** **Internal Marks = 03**

Applicable to the students

1. Those who have opted for chemistry as a Major subject in Table 2

LEARNING OUTCOMES SPECIFIC TO DISCIPLINE

The students will learn the following-

1. Important aspects of chemistry of carbonyl compounds and a few related name reactions
2. Chemistry of carboxylic acid and its derivatives
3. Chemistry of organometallic compounds
4. Formation and use of enolates in organic synthesis

GENERIC LEARNING OUTCOMES

1. **The students will be able to learn Problem Solving, Practical Laboratory Skills, Data Management, Evaluation,** estimation and determination through a**pplication based organic chemistry techniques and concepts.**
2. These generic outcomes will equip students with a broad set of transferable skills, making them capable of conducting rigorous scientific analysis while working safely, ethically, and effectively in collaborative environments.

Semester – 5
Syllabus for Theory Paper
CHM-DSC-7-MIN-601 (Table 2)
Organic Chemistry

End semester = 67 **Credits – 03**

Internal Marks = 08 **Examination Time allowed – 3 Hours**

Unit I: Carbonyl Compounds (15 Hrs)

Aldehydes and Ketones

Nomenclature and structure of the carbonyl group. Physical properties of aldehydes & ketones. Synthesis of aldehydes and ketones with particular reference to the synthesis of aldehydes from acid chorides. Synthesis of aldehydes and ketones using 1,3-dithianes, synthesis of ketones from nitriles and from carboxylic acids.

Mechanism of nucleophilic additions to carbonyl group with particular emphasis on Benzoin, Aldol, Perkin and Knoevenagel condensations. Condensation with ammonia and its derivatives, Wittig reaction. Use of acetals as protecting group. Oxidation of aldehydes, Baeyer-Villiger oxidation of ketones, Cannizzaro reaction, MPV, Clemmensen, Wolff-Kishner, $LiAlH_4$ and $NaBH_4$ reductions.

Unit II: Carboxylic Acids and Derivatives (15 Hrs)

Carboxylic Acids

Nomenclature, structure and bonding, physical properties, acidity of carboxylic acids, effects of substitutions on acid strength. Preparations of carboxylic acids. Reactions of carboxylic acids. Mechanism of decarboxylation. Methods for formation of halo acids.

Carboxylic Acid Derivatives

Structure and nomenclature of acid chlorides, esters, amides and acid anhydrides. Relative stability & reactivity of acyl derivatives. Physical properties, interconversion of acid derivatives by nucleophilic acyl substitution. Preparation of carboxylic acid derivatives, chemical reactions. Mechanisms of esterification and hydrolysis (acidic and basic).

Unit III: Organometallic Compounds and Enolates (15 Hrs)

Organic Synthesis via Enolates

Acidity of α-hydrogens, alkylation of diethyl malonate and ethyl acetoacetate. Synthesis of ethyl acetoacetate: the Claisen condensation. Keto-enol tautomerism of ethyl acetoacetate.

Organometallic Compounds:

Organomagnesium Compounds: The Grignard reagents: Formation, structure and chemical reactions. Organozinc Compounds: Formation and Chemical Reactions. Organolithium Compounds: Formation and Chemical Reactions.

Books/Literature suggested

1. Morrison, R.T., Boyd, R.N., Organic Chemistry; 6thedition, Pubs: Prentice-Hall,1992.
2. Wade Jr., L.G., Singh, M.S., Organic Chemistry; 6thedition, Pubs: Pearson Education, 2008.
3. Mukherji, S.M., Singh, S.P., Kapoor, R.P., Organic Chemistry; Pubs: New Age International, 1985, Vol. I.II, III.
4. Carey,F.A., Organic Chemistry; 4th edition, Pubs: McGraw-Hill, 2000.
5. Solomons, T.W., Fundamentals of Organic Chemistry; 5thedition, Pubs: John Wiley & Sons,1997.

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit** (of **19 marks**) will consist of two types of questions: **9 multiple-choice questions** of one mark each and **5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:** the **compulsory question** and one question each from the **UNIT-1, UNIT-2 and UNIT-3**.
2. The use of scientific calculators is not allowed.

Semester – 6
Syllabus for Practical Paper
CHM-DSC-7-MIN-601 (Table 2)
Organic Chemistry-1

End semester = 22 **Credit – 01**

Internal Marks = 03 **Contact Hours – 2/week**

Examination Time allowed **3 Hours**

I Preparation, TLC monitoring and purification via column chromatography for a minimum of 5 of the following compounds:

q) Methyl Orange
r) Para-Aminoazobenzene
s) Aspirin
t) Picric acid
u) Acetanilide
v) Para-Bromoacetanilide
w) Phenyl azo-β-naphthol
x) 7-Hydroxy-4-methylcoumarin

II Determination of the per cent purity of the given carbonyl compound via the hydrazone formation method.

III Estimation of aniline in the given solution by acetylation/bromination.

Books/Literature suggested

1. Harwoor, L.M., Moody, J., Experimental Organic Chemistry; 1st edition, Pubs: Blackwell Scientific Publications, 1989.
2. Forniss, B.S., Rogers, V., Vogel's Text Book of Practical Organic Chemistry; Pubs: Dorling Kindhsky Pvt. Ltd., 1989.
3. Bansal, R.K., Laboratory Manual of Organic Chemistry; 3rd edition, Pubs: Wiley Eastern Limited, 1994.
4. Furniss, B.S., Hannaford, A.J., Rogers, V., Smith, P.W.G., Tatchell, A.R., Vogel's Textbook of Practical Organic Chemistry; 4th edition, Pubs: Longman group, 1978.
5. Aggarwal, O.P., Advanced Practical Organic Chemistry, 30th edition, 2013, Goel Publishing House.

General Instructions to the Examiners:

Note: Practical examination will be of three hours duration divided into two parts (First, half hour for written and second half will be the next two and half hours for performance).

1. Candidate will be examined based on any two experiments given by the examiner. Each experiment carries 08 marks (3 written + 5 performance).
2. Viva-Voce: 04 marks.
3. Note Book: 02 marks.
4. Internal Assessment: 03 marks

Semester	**6**
Code of Paper	**CHM-DSC-8-MIN-602 (Table 2)**
Title of Paper	**Physical Chemistry**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100 (75 Theory + 25 Practical)**
Marks distribution of Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of Practical paper	**End semester = 22** **Internal Marks = 03**

Applicable to the students

1. Those who have opted for chemistry as a Major subject in Table 2.

LEARNING OUTCOMES SPECIFIC TO DISCIPLINE

The students will learn the following-

1. **Solutions:** Quantify molar mass and vapor pressure changes using Raoult's Law, colligative properties, and van't Hoff factors for electrolytes.
2. **Electrochemistry:** Analyze ionic conductance and mobility to construct electrochemical cells and predict reaction spontaneity via the Nernst equation.
3. **Photochemistry:** Evaluate light-matter interactions and quantum yield while mapping excited-state pathways (Jablonski diagrams) for carbonyl and alkene reactions.

GENERIC LEARNING OUTCOMES

The students will be able to learn

1. Quantitative Reasoning: Mastery of mathematical models to solve physical chemistry problems.
2. Experimental Proficiency: Understanding the setup and logic behind classical laboratory techniques (conductometry, osmometry, etc.).
3. Molecular Analysis: Ability to relate macroscopic observations (like color or pressure) to microscopic molecular behavior.
4. Analytical Thinking: Connecting theoretical laws (like Nernst or Raoult) to real-world industrial and biological applications

Semester – 6
Syllabus for Theory Paper
CHM-DSC-8-MIN-602 (Table 2)
Physical Chemistry

End semester = 67 **Credits – 03**

Internal Marks = 08 **Examination Time allowed – 3 Hours**

UNIT – I Solutions (15 Hrs.)

Solutions and Colligative Properties:

Dilute Solutions and Colligative Properties: Ideal and non-ideal solutions, methods of expressing concentrations of solutions, activity and activity coefficient. Dilute solution, colligative properties, Raoult's law, relative lowering of vapour pressure, molecular weight determination. Osmosis, law of osmotic pressure and its measurement, determination of molecular weight from osmotic pressure. Elevation of boiling point and depression of freezing point, Thermodynamic derivation of relation between molecular weight and elevation in boiling point and depression of freezing point. Experimental methods for determining various colligative properties. Abnormal molar mass, degree of dissociation and association of solutes.

UNIT – II Electrochemistry (15 Hrs.)

Electrochemistry – I:

Electrical transport –Conduction in metals and in electrolyte solutions, specific conductance and equivalent conductance, measurement of equivalent conductance, variation of equivalent and specific conductance with dilution. Migration of ions and Kohlrausch Law, Arrhennius theory of electrolyte dissociation and its limitations, weak and strong electrolytes, Ostwald's dilution law, its uses and limitations. Debye-Huckel-Onsager's equation for strong electrolytes (elementary treatment only). Transport number

Electrochemistry-II:

Types of reversible electrodes – gas metal – ion, metal –insoluble salt – anion and redox electrodes. Electrode reactions, Nernst equation, derivation of cell E.M.F. and single electrode potential, standard hydrogen electrode – reference electrodes – standard electrode potential, sign conventions, electrochemical series and its significance.

UNIT – III Photochemistry (15 Hrs.)

Photochemistry:

Interaction of radiation with matter, difference between thermal and photochemical processes. Laws of Photochemistry: Grothus – Drapper law, Stark – Einstein law, Jablonski diagram depicting various processes occurring in the excited state. Qualitative description of fluorescence, phosphorescence, non-radiative processes (internal conversion, intersystem crossing), quantum yield, photosensitized reactions – energy transfer processes (simple examples). Photochemistry of carbonyl compounds and alkenes.

Books/Literature suggested

1. Atkins, P., Paula, J.de, Atkins Physical Chemistry; 8th edition, Pubs: Oxford University Press, 2008.
2. Puri, B.R., Sharma, L.R., Pathania, M.S., Principles of Physical Chemistry; 43rd edition, Pubs: Vishal Publishing Co., 2008.
3. Barrow, G.M., Physical Chemistry; 6th edition, Pubs: McGraw Hill Inc, 1996.
4. Rao, C.N.R., University General Chemistry; Pubs: Macmillan India, 1985.
5. Berry, R.S., Rice, S.A., Ross, J., Physical Chemistry; 2nd edition, Pubs: Oxford University Press, 2000.
6. Albert, R.A., Silbey, R.J., Physical Chemistry; 1st edition, Pubs: John Wiley & Sons Inc., 1992.
7. Dogra, S.K., Dogra, S., Physical Chemistry Through Problems; Pubs:Wiley Eastern Limited, 1991.
8. Levine, I.N., Physical Chemistry; 5th edition, Pubs: Tata McGraw Hill Publishing Co. Ltd., 2002.
9. Moore, W. J., Basic Physical Chemistry; Pubs: Prentice Hall of India Pvt. Ltd, 1983.
10. Metz, C.R., Theory and Problems of Physical Chemistry; Schaum's outline series, 2nd edition, Pubs: McGraw-Hall Book company, 1989.

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit** (of **19 marks**) will consist of two types of questions: **9 multiple-choice questions** of one mark each and **5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:** the **compulsory question** and one question each from the **UNIT-1, UNIT-2 and UNIT-3**.
2. The use of scientific calculators is not allowed.

Semester – 6
Syllabus for Practical Paper
CHM-DSC-8-MIN-602 (Table 2)
Physical Chemistry

End semester = 22 **Credit – 01**
Internal Marks = 03 **Contact Hours – 2/week**
Examination Time allowed **3 Hours**

I. Conductometry

1. To determine the Cell Constant of a Conductivity Cell
2. Study of conductometric titration of a strong acid (HCl) against a strong base (NaOH)
3. Study of conductometric titration of a weak acid (CH_3COOH) against a strong base (NaOH)
4. Study of conductometric titration of a mixture of strong acid (HCl) and weak acid (CH_3COOH) against a strong base (NaOH).
5. To determine the solubility and solubility product of a sparingly soluble electrolyte conductometricaly.
6. To study the saponification of ethyl acetate conductometricaly.
7. To determine the ionization constant of a weak acid conductometricaly
8. Determination of hydrolysis constant of aniline hydrochloride conductometricaly.

Books/Literature suggested

1. Khosla, B.D., Garg, V.C., Gulati, A., Senior Practical Physical Chemistry; 11th edition, Pubs: R.Chand& Co., New Delhi, 2002.
2. Das, R.C., Behra, B., Experimental Physical Chemistry; Pubs: Tata McGraw-Hill Publishing Co. Ltd., 1983.
3. Levitt, B.P., Findlay's Practical Physical Chemistry; 8th edition, Pubs: Longman Group Ltd., London & New York, 1978.

General Instructions to the Examiners:

Note: Practical examination will be of three hours duration divided into two parts (First, half hour for written and second half will be the next two and half hours for performance).

1. Candidate will be examined based on any two experiments given by the examiner. Each experiment carries 08 marks (3 written + 5 performance).
2. Viva-Voce: 04 marks.
3. Note Book: 02 marks.
4. Internal Assessment: 03 marks

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COMPUTER SCIENCE & APPLICATIONS

<table>
<tr><th colspan="8">Table 1*</th><th colspan="8">Table 2*</th></tr>
<tr><th>Sem</th><th>Level</th><th>Course Code</th><th>Course Name</th><th colspan="2">Credits (Th+Pr)</th><th>Total Credits</th><th>Sem</th><th>Level</th><th>Course Code</th><th>Course Name</th><th colspan="2">Credits (Th+Pr)</th><th>Total Credits</th></tr>
<tr><td rowspan="6">V</td><td rowspan="6">L-300</td><td>CSA-DSC-5(Maj)-501</td><td>Python Programming</td><td>2+2</td><td>4</td><td rowspan="6">24</td><td rowspan="6">V</td><td rowspan="6">L-300</td><td>CSA-DSC-5(Maj/Min)-501</td><td>Python Programming</td><td>2+2</td><td>4 (x3)</td><td rowspan="6">24</td></tr>
<tr><td>CSA-DSC-5(Maj)-502</td><td>E-Commerce</td><td>4+0</td><td>4</td><td>CSA-DSC-5(Maj)-502</td><td>E-Commerce</td><td>4+0</td><td rowspan="3">4 (x2)</td></tr>
<tr><td rowspan="2">CSA-DSC-5(Maj)-503</td><td rowspan="2">Programming with JAVA</td><td rowspan="2">2+2</td><td rowspan="2">4</td><td>CSA-DSC-5(Maj)-503</td><td>Programming with JAVA</td><td>2+2</td></tr>
<tr><td rowspan="3">Internship/ Community Work</td><td rowspan="3">0+4</td><td rowspan="3">4</td></tr>
<tr><td>CSA-DSC-5(Maj)-504</td><td>Operating System with Linux</td><td>2+2</td><td>4</td></tr>
<tr><td>CSA-DSC-5(Maj)-505</td><td>Computer Organization</td><td>4+0</td><td>4</td></tr>
<tr><td>CSA-Intern-501</td><td>Internship/ Community Work</td><td>0+4</td><td>4</td><td colspan="8">Note: Students opting for Computer Sci & Applications as Major subject have to choose any 2 (two) Major out of 3 (three) given Major in 5th Sem.</td></tr>
<tr><td rowspan="6">VI</td><td rowspan="6">L-300</td><td>CSA-DSC-6(Maj/Min)-601</td><td>Introduction to Data Science</td><td>2+2</td><td>4 (x2)</td><td rowspan="6">24</td><td rowspan="6">VI</td><td rowspan="6">L-300</td><td>CSA-DSC-6(Maj/Min)-601</td><td>Introduction to Data Science</td><td>2+2</td><td>4 (x3)</td><td rowspan="6">24</td></tr>
<tr><td>CSA-DSC-6(Maj)-602</td><td>Serverside Scripting with PHP</td><td colspan="2">2+2=4</td><td>CSA-DSC-6(Maj/Min)-602</td><td>Serverside Scripting with PHP</td><td>2+2</td><td rowspan="3">4 (x3)</td></tr>
<tr><td>CSA-DSC-6(Maj)-603</td><td>Cyber Security</td><td colspan="2">4+0=4</td><td>CSA-DSC-6(Maj/Min)-603</td><td rowspan="2">Cyber Security</td><td>4+0</td></tr>
<tr><td>CSA-DSC-6(Maj)-604</td><td>Computer Networks</td><td colspan="2">4+0=4</td><td colspan="3" rowspan="2">Note: Students opting for Computer Sci & Applications as Major subject have to choose any 2 (two) Major out of 3 (three) given Major in 6th Sem. Similarly, Minor subject have to choose any 2 (two) Minor out of 3 (three) given Minor in 6th Sem.</td></tr>
<tr><td>CSA-DSC-6(Maj)-605</td><td>Management Information System</td><td colspan="2">4+0=4</td></tr>
<tr><td colspan="16">The student exiting the programme after two years securing 144 credits will be awarded BSc degree</td></tr>
</table>

Notes:

1. In the first semester student is offered three different subjects A, B & C.
2. At the end of second semester each student has the option to choose any one of the two streams namely Single Major or Multi-Disciplinary. The student choosing Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student who chooses multi-disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3-year degree) shall be eligible for 4 YEAR DEGREE B.Sc. HONOURS with research, on the condition that the Institution is a PU approved Research Centre
5. The student opting for Single Major must secure 50% credits of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for Minor in a particular subject, the student must attain atleast 12 credits in that subject(s).
6. The student exiting the programme after one year securing 48 credits will be awarded UG Certificate provided he/she secures 4 credits from summer internship/community work. Further, the student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secure 4 credits from summer internship. The student successfully securing 144 credits will be awarded BA degree.
7. Language paper chosen as MIL-1 and MIL-2 should be in the same subject.
8. Once Major and Minor papers are chosen in 3rd semester, the same will continue as Major and Minor subject in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write Dissertation of 08 credits in one of the two Major subjects.

Paper Title: Python Programming
Paper Code: CSA-DSC-5(Maj)-501

Th	Pr	Cr	
2	2	4	TheoryExternalMarks:45
			TheoryInternalMarks:5
Level: L-300			NumberofTheoryHours:30
			Practical External Marks: 50
			Practical Hours: 60

Time Duration: 3 Hrs.

Objective: The objective of the module is to enable students to understand the basic concepts of the Python programming language, including lists, tuples, classes, and dictionaries. The module will enable students to do programming using the Python library.

Learning Outcome: After the completion of the module, students will gain knowledge of:

- Students would be able to develop logic for problem-solving.
- The fundamental concepts of Python programming, syntax and semantics.
- Students will learn the basic constructs of Python programming, such as data, operations, conditions, loops, functions, etc.
- Using Lists, tuples, Dictionaries, and functions in Python.
- Creating and using Files in Python.
- Using Python modules and functions.

Note:

The Question Paper will consist of Four Units.

- The examiner will set a total of nine questions, including two questions from each unit and one compulsory short answer question covering the entire syllabus.
- Students are required to attempt one question from each unit and the compulsory question.
- All questions carry equal marks unless specified.

UNIT – I

Introduction to Python Programming: IDLE interpreter, Interactive Mode Programming, Script Mode Programming, The Role of Indentation in Python, Variables, Assigning Values to Variables, Input/Output with Print and Input Functions, Basic Data Types in Python.
Operators in Python: Arithmetic, Relational, Logical, Compound Assignment, Increment and Decrement, Ternary, Bitwise, and Comma Operators, Precedence and Order of Evaluation, Statements and Expressions, Automatic and Explicit Type Conversions.
The Conditional Statements in Python: If, If-else, If-elif-else, Python loops, for-loop, while-loop, for-else, while-else, Nested Loops, the built-in Function range(), continue and break statement.

UNIT – II

Strings: Introduction to String, String Operations, Transversal of string, Methods and Inbuilt Functions.
Lists: Definition, Operations, Traversal, Methods and Inbuilt Functions, Nested Lists, Copying Lists, List as an argument, Mutable and Immutable Data types.
Tuples: Introduction, Operations, Traversal, Methods and Inbuilt Functions, Nested Tuples
Dictionaries: Introduction, Traversal, Methods and Inbuilt Functions, Manipulating Dictionaries.

UNIT – III

Function: definition, advantages, User-defined Functions, defining a function, Function Parameters, Formal Parameters, Actual Parameters, Global and Local Variables, return values, Introduction to Anonymous Functions, passing a list, Python standard Libraries.
Classes: creating and using a class, working with classes and instances, importing classes, Python standard library.

UNIT IV

Exception Handling: exceptions and errors, try-except block, the else block, handling the Zero Division Error and File Not Found Error exception.
File Handling: Introduction, Types of Files, Opening and Closing a File, writing to a file, reading from a file, Setting offset in a file, Creating and Traversing a file.

Suggested Readings:

1. Learning Python, Mark Lutz, 5th Edition, O'Reilly Media, Inc., 2015
2. Python Programming: A Complete Guide for Beginners to Master, Python Programming Language: A Complete Guide For Beginners To Master And Become An Expert In Python Programming Language, Brian Draper, Createspace Independent Pub; 1st edition, 2016
3. Python: Python Programming For Beginners – The Comprehensive Guide To Python Programming: Computer Programming, Computer Language, Computer Science, Adam Stark, CreateSpace Independent Publishing Platform, 2016
4. Python Essential Reference, David M Beazley, 4th edition, Pearson Education Inc., 2009.
5. Python Programming: An Introduction to Computer Science, John M Zelle, 2nd Edition,2003 6. Python Crash Course, A Hands-On, Project-Based Introduction to Programming, Eric Maththes, 2nd Edition, No Starch Press, 2019

Paper Code: CSA-DSC-5(Maj)-501P
(Practical based on paper CSA-DSC-5(Maj)-501)

Time Duration: 3 Hrs.

List of Practicals:

1. Program to print a message.
2. Program to take name, age, and marks as input and display them.
3. Program to check the data type of variables using type().
4. Program to perform all arithmetic operations on two numbers.
5. Program to show relational operator results between two numbers.
6. Program to test logical operators (and, or, not).
7. Program to demonstrate compound assignment operators (+=, -=, etc.).
8. Program using the ternary operator to find the largest of two numbers.
9. Program to check whether a number is even or odd using if-else.
10. Program to find the greatest of three numbers using if-elif-else.
11. Program to print numbers from 1 to 10 using a for-loop.
12. Program to print numbers from 1 to 10 using a while loop.
13. Program to print even numbers using for and range().
14. Program to calculate the sum of the first N natural numbers using a loop.
15. Program using nested loops to print a pattern (e.g., a rectangle of stars).
16. Program to demonstrate the break statement inside a loop.
17. Program to demonstrate the continue statement inside a loop.
18. Program using for-else (e.g., searching for an item in a list).
19. Program using while-else (e.g., counter loop finishing normally).
20. Program to input a string and print its length.
21. Program to count vowels in a string.
22. Program to check if a substring exists in a string.
23. Program to reverse a string using slicing.
24. Program to demonstrate string methods (upper(), lower(), replace(), etc.).
25. Program to create a list and display all its elements.
26. Program to find the largest and smallest element in a list.
27. Program to count even and odd numbers in a list.
28. Program to demonstrate list methods (append, remove, sort, etc.).
29. Program to take a list input from the user and print it using traversal.
30. Program to demonstrate nested lists and print elements.
31. Program to create a tuple and access its elements.
32. Program to count occurrences of an element using count().
33. Program to find index of an element using index().
34. Program to create a dictionary of student details and print keys & values.
35. Program to add, update, or delete items from a dictionary.
36. Program to iterate through dictionary using for key in dict.
37. Program to check if a key exists in a dictionary.
38. Program to define a simple function and call it.
39. Program to add two numbers using a user-defined function.
40. Program to calculate the factorial of a number using a function.
41. Program to check if a number is prime using a function.
42. Program to demonstrate default arguments in a function.
43. Program to demonstrate return statement.
44. Program using a lambda function to find the square and cube.
45. Program to pass a list to a function and print its sum.

46. Program to create a simple class with attributes and methods.
47. Program to define a class Student and display details of objects.
48. Program to demonstrate constructor (__init__).
49. Program to create multiple objects of a class.
50. Program to import a class from another file and use it.
51. Program to demonstrate try and except.
52. Program to handle ZeroDivisionError.
53. Program to handle FileNotFoundError.
54. Program using try-except-else block.
55. Program to accept integer input and handle invalid inputs.
56. Program to create and write to a text file.
57. Program to read a text file and display its contents.
58. Program to append new lines to an existing file.
59. Program to count the number of lines in a file.
60. Program to use seek() to move the file pointer to a specific position.
61. Program to copy content from one file to another.
62. Program to write list elements to a file and read them back.
63. Program to read a file and count words/characters.

Paper Title: E-Commerce
Paper Code: CSA-DSC-5(Maj)-502

Th	Pr	Cr		
4	-	4	Theory External Marks :	90
Level: L-300			Theory Internal Marks :	10
			Number of Theory Hours :	60

Objective: The objective of this E-commerce course is to provide students with a comprehensive understanding of online business concepts, technologies, and strategies. It aims to develop knowledge of digital marketplaces, electronic payment systems, online marketing, and secure transaction methods. Students will learn how to design, manage, and promote e-commerce websites, understand consumer behavior in the digital environment, and explore legal, ethical, and security issues.

Learning Outcomes:
By the end of the course

- Understand key concepts, models, and frameworks of e-commerce and digital business.
- Explain online payment systems, security measures, and legal/ethical issues in electronic transactions.
- Analyze consumer behavior and apply digital marketing strategies for online business growth.
- Design, manage, and evaluate e-commerce websites and online storefronts.
- Use digital tools and platforms to create, operate, and promote e-business solutions.

Note:

- The Question Paper will consist of Four Units.
- The examiner will set a total of nine questions, including two questions from each unit and one compulsory short answer question covering the entire syllabus.
- Students are required to attempt one question from each unit and the compulsory question.
- All questions carry equal marks unless specified.

UNIT I

Introduction to E-Commerce: Definition of E-commerce, Definition of E-business, Difference between E-commerce and E-business, Features of E-commerce, Scope of E-commerce, Advantages of E-commerce, Limitations of E-commerce, Traditional commerce vs. E-commerce, Types of E-commerce: B2B – Business to Business, B2C – Business to Consumer, C2C – Consumer to Consumer, C2B – Consumer to Business, B2G – Business to Government, G2C – Government to Citizen.
Definition of EDI, Need for EDI, Traditional system vs. EDI-enabled system, Components of EDI: EDI standards.

UNIT II

Web based E-Commerce: Definition; Need for web based business, steps in setting up business on Internet: Selection and registration of domain name, Website development: Planning a website, steps for creating a website, Elements of a webpage, definition of website hosting, Server, domain, and DNS concepts, Hosting control panels (cPanel, Plesk, DirectAdmin), types of web hosting

UNIT III

E-Payment Systems & Security: Electronic payment methods, Credit/Debit cards, Net banking, UPI, Mobile wallets, COD, RTGS, NEFT, IMPS, Digital currency, Blockchain basics, E-commerce security issues, Encryption, Digital signatures, Digital certificates, SSL/TLS, Cybercrime, IT Act 2000, Intellectual property issues.

UNIT IV

E-Marketing & Customer Relationship Management: Digital marketing concepts, Web Traffic Analysis: Hits, View pages, Visits and Other web-reporting tools, Search Engine Optimisation (SEO), Social Media Optimisation (SMO) , Email marketing, Pay Per Click (PPC), Customer Relationship Management tools.

Suggested Readings:

1. Bhasker, Bharat: Electronic Commerce: Framework, Technologies and Applications; Tata McGraw-Hill.
2. Bajaj, Kamlesh& Nag, Debjani: E-Commerce-The Cutting Edge of Business; Tata McGraw- Hill.
3. Young, Margaret Levine : The Complete Reference: Internet; Tata McGraw Hill.
4. KalaKota, Ravi &Whinston, Andrew B.: Frontiers of Electronic Commerce; Addison Wesley.
5. Stallings, William: Network Security Essentials: Applications & Standards; Pearson Education.

Paper Title: Programming with JAVA
Paper Code: CSA-DSC-5(Maj)-503

Th	Pr	Cr	
2	2	4	Theory External Marks : 45
			Theory Internal Marks : 05
Level: L- 300			Number of Theory Hours : 30
			Practical External Marks: 50
			Practical Hours: 60

Time Duration: 3 Hrs.

Course Objective: This course aims to provide students with knowledge of all the programming concepts of the Java programming language.

Learning Outcomes: On successful completion of the course, students will be able to:

- Describe the meaning of the Object-Oriented Paradigm, and create class hierarchies using the Object-Oriented Design process.
- Design and implement Java programs for complex problems

NOTE:

- The Question Paper will consist of Four Units.
- Examiner will set a total of NINE questions comprising TWO questions from each Unit and ONE compulsory question of short answer type covering the whole syllabus.
- The students are required to attempt ONE question from each Unit and the Compulsory question.
- All questions carry equal marks unless specified.

UNIT – I

Basic Principles of Object Oriented Programming: Introduction and Features of Java, Creating the Java program, Compiling and Running the programs, Java Virtual Machine, Java Bytecode
Basic Language Elements: Tokens, Identifiers, Keywords, Literals, Comments, Primitive Data types, Operators, Assignments; Input/output in Java: Basics, I/O Classes, Reading Console Input, Decision and Loop Control Statements, break, continue

UNIT – II

Class and Object in Java: Defining a Class, Creating Objects, Accessing Class Members, Constructors, Methods Overloading, Static Members, Abstract Class and methods; Final classes, variables and Methods; Finalizer Methods
Array & String in Java: Defining an Array, Initializing & Accessing Array, Multi–Dimensional Arrays, Defining Strings, Operation on Array and String, Managing Strings using String and StringBuffer Classes

UNIT – III

Inheritance: Benefits of Inheritance, Types of Inheritance in Java, Extending a Class, Defining a subclass, Subclass constructor, Multilevel inheritance, Hierarchical inheritance, Overriding Methods, Visibility Control

Packages & Interfaces: Basic Concepts of Package and Interface, Organizing Classes and Interfaces in Packages, Defining and naming a Package, Importing Packages, CLASSPATH Setting for Packages, Access Protection in Packages, Standard Packages

UNIT IV

Exception Handling in Java: Basics of Exception, Types of Exception, Use of try, catch, finally, throw, throws in Exception Handling, Catching more than one Exception, In-built and User Defined Exceptions, Checked and Un-Checked Exceptions,

Applet in Java: Applet Basics, Applet Architecture, Applet Life Cycle, Applet Tag, Parameters to Applet, Embedding Applets in a Web page, Creating Simple Applets

Multithreaded Programming: Creating a single and multiple threads using different ways, thread priorities, synchronization.

Suggested Readings:

1. Programming with JAVA – E Balgurusamy
2. The Complete Reference – JAVA Herbert Schildt.
3. "JAVA 2 UNLEASHED, Tech Media Publications, New Delhi.
4. Patrick Naughton and Herbert Schlitz,"JAVA-2 Complete Reference" ,TMH, New Delhi.

Paper Code: CSA-DSC-5(Maj)-503P
(Practical based on CSA-DSC-5(Maj)-503)

Time Duration: 3 Hrs.

List of Practicals:

1. Program to print a simple message.
2. Program to show the use of tokens (identifiers, keywords, literals).
3. Program to demonstrate primitive data types (int, float, double, char, 164or act).
4. Program to demonstrate arithmetic operators (+, −, *, /, %).
5. Program to demonstrate relational and logical operators.
6. Program to demonstrate assignment operators.
7. Program to read input using Scanner.
8. Program to demonstrate break and continue in loops.
9. Program to check even/odd using if-else.
10. Program to print numbers from 1 to 10 using a loop.
11. Program to create a simple class and object.
12. Program to access class members (variables & methods).
13. Program to demonstrate constructors.
14. Program to demonstrate constructor overloading.
15. Program to demonstrate method overloading.
16. Program to demonstrate static variables and static methods.
17. Program to demonstrate final variable, final method, final class.
18. Program to demonstrate abstract class and abstract method.
19. Program to demonstrate the finalize() method.
20. Program to create and print a 1D array.

21. Program to find the sum of array elements.
22. Program to find the largest element in an array.
23. Program to create and print a 2D array.
24. Program to demonstrate simple String operations (length, concat, compare).
25. Program to reverse a string.
26. Program to show StringBuffer operations (append, insert, delete).
27. Program to demonstrate single inheritance.
28. Program to demonstrate multilevel inheritance.
29. Program to demonstrate hierarchical inheritance.
30. Program to demonstrate method overriding.
31. Program to demonstrate super keyword.
32. Program to demonstrate visibility (public, private, protected).
33. Program to create a user-defined package.
34. Program to import a package and use its class.
35. Program to demonstrate multiple inheritance using interfaces.
36. Program to demonstrate try-catch.
37. Program to demonstrate multiple catch blocks.
38. Program to demonstrate finally block.
39. Program to throw an exception using throw.
40. Program to create a user-defined exception.
41. Program to demonstrate checked exception (FileNotFound).
42. Program to demonstrate unchecked exception (ArithmeticException).
43. Program to create a simple Applet and display text.
44. Program to demonstrate applet lifecycle (init, start, paint).
45. Program to pass parameters to an Applet using the APPLET tag.
46. Program to create a thread using the Thread class.
47. Program to create a thread using the Runnable interface.
48. Program to demonstrate multiple threads running together.
49. Program to demonstrate thread priorities.
50. Program to demonstrate synchronization.

Paper Title: Operating System with Linux

Paper Code: CSA-DSC-5(Maj)-504

Th	Pr	Cr	TheoryExternalMarks:45
2	2	4	TheoryInternalMarks:5
Level: L- 300			NumberofTheoryHours:30
			Practical External Marks: 50
			Practical External hours: 60

TimeDuration:3Hours.

Objective: The objective of the module is to make the student understand the basic concepts of anoperating system mainly the functionsof an operating system. The module introduces Linux operating system structure, concepts and commands. The objective of this course is to introduce students to the fundamental concepts of Linux and its role as an open-source operating system.

LearningOutcomes:

- Onsuccessfulcompletionofthecourse,studentswillbeabletounderstand:
- Operatingsystemfeatures,typesandfunctions
- Use Linux commands for file management, process handling, user and group management
- Write simple shell scripts for automation of tasks
- The hands-on practice can be done using any Linux distribution preferably Ubuntu Desktop Edition

Note:

TheQuestionPaperwillconsistofFourUnits.

- Examiner will set total of **NINE**questions comprising **TWO**questions from each Unit and **ONE**compulsory question of short answer type covering whole syllabi.
- The students are required to attempt **ONE**question from each Unit and the Compulsory question.
- Allquestionscarryequalmarksunlessspecified.

UNIT-I

Operating Systems (OS): Introduction, its needs and services, Functions of an operating system Types of OS: Single-User, Multi-user, batch system, Multiprogramming, Multiprocessing and Real time Operating Systems, Parallel systems, Distributed systems- peer to peer system, client-server system. Operating system structure- Monolithic, Layered, Microkernel, Modular, Hybrid. Concept and purpose of virtualization.

UNIT–II

Operating System Functions, Linux, Features of Linux, Basic Command Syntax, Login , Logout, Shutdown commands, Files and directories: File Naming Conventions, Hierarchical file system: Concept of Root Directory, Home Directory, parent and Child Directory, System Directories in Linux, Absolute and Relative File path, Types of Files in Linux, File and Directory permissions and chmod command (symbolic and octal mode), File Ownership, Pipes, Filters, Input, output and Error Redirection.

UNIT-III

Vi Editor: Editing Modes, Save and Quit commands in vi, Cursor Movement Commands in vi, Copy and paste text, Delete Text commands in vi
Basic Linux File and Directory Handling commands with options: pwd, ls, cd, mkdir, rmdir, cd, cat, cp, rm, mv, wc, more, less, head, tail, clear, man, grep, tr, sort, cut, cmp, 167or a , diff, find, umask, chown
Process handling commands: foreground and background processes, sleep, fg, bg, kill, ps, nice, jobs, top
Basic Commands: date, banner, who, who am I, cal, bc, prompt, path
Using Wildcard Characters (*, ? and []) with file commands, Environment Variables

UNIT-IV

User and Group Management: Commands to create, delete, modify users and groups in Linux. Passwords in Linux: passwd command, Structure of passwd file, sudo and su command, Aliases, File compression: tar and gzip, BASH Shell Scripting Programming Constructs: Create, save and run shell script file, Variables, echo and read commands, expr, command substitution, test command, Decision making constructs (if-else, if-elseif, case-esac), Loops (while, for, until), tput clear, tput cup. Basic shell programs.

SuggestedReadings:

1. Peterson,J.L.&Silberschatz,A.,OperatingSystemConcept,AddisonWesley,reading.
2. John Goerzen: Linux Programming Bible, IDG Books, New Delhi
3. Ryan Turne: Linux: The Ultimate Beginner's Guide to Learn Linux Operating System, Command Line and Linux Programming Step by Step
4. David Clinton: Ubuntu Linux Bible, Christop, Publisher: Wiley, Edition: 10th
5. Tim Parker: Linux Unleashed, SAMS, Edition 3rd
6. Andrew Mallett Mokhtar Ebrahim: Mastering Linux Shell Scripting – Second Edition: A practical guide to Linux command-line, Bash scripting, and Shell programming, Packt, Edition 2nd
7. Richard Blum, Christine Bresnahan: Linux Command Line and Shell Scripting Bible, Wiley, Edition 4th

Paper Code: CSA-DSC-5(Maj)-504P
(Practical based on paper CSA-DSC-5(Maj)-504)

Time Duration: 3 Hrs. **Number of Practical Hours: 60**

Note: Paper will be set at the time of examination. Due weight-age may be given to the practical notebook and Assignments in evaluation.

Suggested List of Practical Assignments

Write shell scripts to

1. Display message "Hello World"
2. Display date and this month's calendar and shell used.
3. Display environment variables.
4. Change the values of environment variables.
5. Define, set, unset and display a variable
6. Make a variable read only Variable.
7. Assign values to variables using read command.
8. Accept a string to be searched in a given file.
9. Perform simple arithmetic operation.
10. Determines whether the file exits in home directory.
11. Accept user input for a scriptname and determine the permissions of the given file.
12. Input two string and assess whether one of the strings is null, both equal or not equal.
13. Createmenus using a switch case statement to show system configurations like:
    a. Currently logged user and his log name
    b. Your current shell
    c. Your home directory
    d. our operating system type
    e. Your current path setting
    f. Your current working directory
    g. Show the Currently logged number of users
    h. About your os and version, release number, kernel version
    i. output username, time and date and Whois logged on.
14. Print the sum of five numbers using while loop.
15. Calculate result of 'n' students (while, do, expr).
16. Copy a file from one directory to another
17. Find smaller of two numbers.
18. Swap values of two variables.
19. Find the greatest number.
20. Find factorial of a number
21. Print sum of n numbers.
22. Print multiplication table.

Paper Title: Computer Organization
Paper Code: CSA-DSC-5(Maj)-505

Th	Pr	Cr	TheoryExternalMarks:90
4	-	4	TheoryInternalMarks:10
Level: L-300			NumberofTheoryHours:60

Objective: The objective of the course is to help learners to understand how fundamental digital components work together to form the building blocks of computer hardware and processing systems. It provides a solid foundation in how digital systems are structured and operate at the hardware level. The course aims to develop skills in working with number systems, performing conversions and arithmetic, and applying 169or act algebra to simplify logic expressions. It introduces the design and analysis of combinational circuits as well as sequential circuits.

Learning Outcomes:
By the end of the course, students will be able to:

- Understand and perform operations with different number systems.
- Simplify and apply Boolean expressions to design digital circuits.
- Design and analyze combinational circuits like adders, multiplexers, and decoders.
- Understand and use flip-flops in sequential circuits such as counters and registers.

Note:

- The Question Paper will consist of Four Units.
- The examiner will set a total of nine questions, including two questions from each unit and one compulsory short answer question covering the entire syllabus.
- Students are required to attempt one question from each unit and the compulsory question.
- All questions carry equal marks unless specified.

UNIT – I

Overview of Computer Organization: Definition of computer organization, basic block diagram of computer; Introduction to Memory Unit.

Data Representation and Arithmetic: Number Systems and its conversions – Binary, Decimal, Octal and Hexadecimal; Signed Number Representation – sign-magnitude, 1's complement and 2's complement; fixed and floating-point representation, Two's Complement representation; Binary Arithmetic Operations: Addition, Subtraction, Multiplication and Division; Introduction to Computer Codes:BCD, Excess-3 Code, Gray Code

UNIT – II

Digital Logic and Gates: Boolean algebra – Boolean Laws, duality principle, De-Morgan's Theorem, simplification of Boolean algebra; Logic Gates: AND, OR, NOT; Universal Gates: NAND and NOR gates; XOR and XNOR Gates, Characteristics of Logic Gates (Fan-in, Fan-out, Power dissipation, Propagation Delay), Minterms and maxterms, standard and canonical form, Karnaugh map and minimization techniques, Don't care condition.

UNIT – III

Combinational Logic Circuits: Adders – Half and Full Adder, Full adder using two half adders, Subtractor – Half and Full Subtractor, n-bit parallel adder, n-bit binary adder, binary adder-subtractor, Multiplexer, Multiplexer tree, Demultiplexer, Applications of Multiplexer and Demultiplexers, Encoder, Decoder, Chip cascading – Design of 4 x 16, 5 x 32 decoder using 2 x 4 and 3 x 8 decoders.

UNIT – IV

Flip-flops : Latch, SR-FF, D-FF, JK-FF, T-FF, Race-Around Condition, Master-Slave JK-FF, Applications of flip-flops.

Sequential Circuit Design: Registers, Shift Registers: SISO, SIPO, PISO, PIPO; Bi-directional shift register, applications of shift registers; Counters: Asynchronous counters, mod counters, up-down counter, synchronous counters, design of synchronous counter.

Suggested Readings:

1. Jain, R P : Modern Digital Electronics, Fourth Edition, Tata McGraw Hill Education Private Limited, New Delhi, 2010.
2. Mano, M. Morris: Computer System Architecture, Third Edition, Pearson Education India, 2007.
3. Hayes, John P. : Computer Architecture and Organization, Third Edition, McGraw Hill Education, 2017.
4. Chaudhuri, P. Pal : Computer Organization and Design, 3rd Ed, Prentice Hall of India Learning Pvt. Ltd., 2008.

Paper Title: Introduction to Data Science
Paper Code: CSA-DSC-6(Maj)-601

Th	Pr	Cr	
2	2	4	Theory External Marks : 45
			Theory Internal Marks : 05
Level: L- 300			Number of Theory Hours : 30
			Practical External Marks: 50
			Practical Hours: 60

Time Duration: 3 Hrs.

Course Objective: The objective of this course is to introduce students to the fundamental concepts of Data Science, including data handling, preprocessing, visualization, and basic machine learning techniques. It aims to develop the ability to analyze data, draw meaningful insights, and understand the tools and technologies used in modern data-driven applications. The course prepares students to apply data science methods in real-life problem-solving.

Learning Outcomes: On successful completion of the course, students will be able to:

- Students will understand the basic concepts, scope, and life cycle of data science.
- Students will learn to preprocess, clean, and manage data using appropriate techniques.
- Students will be able to apply basic machine learning algorithms for regression, classification, and clustering.
- Students will develop skills in data visualization and exploratory data analysis.
- Students will gain awareness of real-world applications, tools, ethical issues, and emerging trends in data science.

NOTE:

- The Question Paper will consist of Four Units.
- Examiner will set a total of NINE questions comprising TWO questions from each Unit and ONE compulsory question of short answer type covering the whole syllabus.
- The students are required to attempt ONE question from each Unit and the Compulsory question.
- All questions carry equal marks unless specified.

UNIT I

Introduction to Data Science Meaning and importance of data science, types of data, data science life cycle, data collection methods, basics of Python/R, simple data operations, introduction to data analysis, and basic data visualization.

UNIT II

Data Preprocessing and Data Management Data cleaning, handling missing data, removing errors, data transformation, feature creation, reducing data size, basics of databases, simple SQL commands, basics of NoSQL, and introduction to data warehousing.

UNIT III

Machine Learning Fundamentals Introduction to machine learning, supervised and unsupervised learning, basic regression methods, simple classification methods, basic clustering, model checking methods, train-test split, cross-validation, and overfitting/underfitting concepts.

UNIT IV

Applications, Tools, and Emerging Trends Basics of big data, introduction to Hadoop and Spark, introduction to Power BI, introduction to Tableau, applications of data science in various fields, data privacy, ethics, and case studies (Retail – Product Demand Prediction and Healthcare – Early Disease Detection)

Reference Books:

1. V.K. Jain – Data Science and Analytics, Publisher: BPB Publications
2. Anil Maheshwari – Data Analytics Made Accessible, Publisher: McGraw Hill Education (India Edition Available)
3. G. K. Gupta – Introduction to Data Mining with Case Studies, Publisher: PHI Learning (Prentice Hall of India)
4. U. Dinesh Kumar – Business Analytics: The Science of Data-Driven Decision Making, Publisher: Wiley India
5. S. Chandrasekaran& M. Umamaheswari – Data Mining and Warehousing, Publisher: PHI Learning
6. Saurabh Mukherjee – Machine Learning, Publisher: McGraw Hill Education (India Edition)
7. Seema Acharya &SubhasiniChellappan – Big Data and Analytics, Publisher: Wiley India

Paper Code: CSA-DSC-6(Maj)-601P
(Practical based on paper CSA-DSC-6(Maj)-601)

Time Duration: 3 Hrs.

List of practicals:

1. Load a CSV file using Python and display the first 5 rows.
2. Identify the type of data (numerical, categorical, ordinal) for each column in a given dataset.
3. Plot a simple bar chart for categorical data using Matplotlib.
4. Plot a line chart to show the trend of any numeric column.
5. Write Python code to calculate mean, median, mode, variance, and standard deviation for a numeric column.
6. Perform basic string operations (split, replace, uppercase, count).
7. Detect and display missing values in a dataset.
8. Replace missing values using mean/median/mode.
9. Identify duplicate rows and remove them.
10. Write SQL commands for:
    Creating a table; Inserting records; Selecting data
11. Write SQL queries to filter data using WHERE, LIKE, ORDER BY.
12. Perform a JOIN operation between two SQL tables.
13. Create a basic Power BI dashboard using a dataset
14. Case Study: Predict product demand using a simple regression model.

Paper Title: Serverside Scripting with PHP
Paper Code: CSA-DSC-6(Maj)-602

Th	Pr	Cr	Theory External Marks : 45
2	2	4	Theory Internal Marks : 05
Level: L-300			Number of Theory Hours : 30
			Practical External Marks : 50
			Number of Practical Hours: 60

Objective: The objective of the course is to enable students to understand how dynamic web applications are created by using PHP to process user input, manage server-side logic, and interact with databases. The course aims to develop skills in writing PHP scripts, handling forms, managing sessions, and implementing authentication, while also focusing on secure coding practices. Students learn to generate dynamic content, connect PHP with MySQL, and build functional, data-driven web pages.

Learning Outcomes:
By the end of the course, students will be able to:

- Understand the basics of server-side scripting and PHP execution.
- Write PHP scripts to process user input and generate dynamic web pages.
- Handle forms using GET and POST methods.
- Connect PHP applications to databases for data storage and retrieval.
- Use sessions and cookies for user login and state management.
- Apply basic security practices to prevent common web vulnerabilities.

Note:

- The Question Paper will consist of Four Units.
- The examiner will set a total of nine questions, including two questions from each unit and one compulsory short answer question covering the entire syllabus.
- Students are required to attempt one question from each unit and the compulsory question.
- All questions carry equal marks unless specified.

UNIT – I

Introduction to Server-Side Scripting and PHP: Client-side vs server-side scripting, Need of server-side scripting in dynamic web development, Overview of web servers, Role of PHP in web development, PHP Installation (XAMPP/WAMP).

PHP Basics: PHP Syntax and structure, PHP Variables, constants and data types, Embedded PHP in HTML, Operators, expressions, operator precedence, comments, echo and Print statement.

Control structures: Conditional statements: if, if-else, nested-if, switch; looping structures: for, while, do-while, foreach; break and continue.

UNIT – II

Functions: Built-in PHP functions, User defined functions: function definition, formal parameter vs actual parameter, variable scope, file inclusion Statements

Arrays: Indexed, Associative and Multidimensional Arrays; Creating Arrays, The array() construct; Accessing Array elements, Iterating with each() and foreach(), Iterative functions: current(), next(), prev(), reset(), end().

Strings: String declaration and syntax, string operators, concatenation and assignment, The heredoc syntax, common string functions : string cleanup functions, case functions, string replacement, finding characters and substrings.

UNIT – III

Working with Forms and User Input: Super global variables, Creating forms using HTML, GET and POST methods, Retrieving form data in PHP, Validating user input, Handling file uploads in PHP.

Database Connectivity: Connecting to MySQL, MySQL Functions, Making MySQL queries, executing queries, fetching data sets, creating MySQL database with PHP; Integrating forms and Databases: Basic form submission to a database, editing data with an HTML form.

UNIT – IV

Sessions and Cookies: Understanding sessions, starting and ending sessions, working with session variables, encoding and decoding session variables, Introduction to cookies, creating and retrieving cookies, modifying and deleting cookies, cookie expiration and lifetime.

Error and Exceptional Handling : Types of PHP Errors (notice, warning, parse error), PHP Exceptions: try, catch and finally blocks, throwing an exception, creating custom exceptions, multiple catch blocks.

Suggested Readings:

1. PHP6 and MySQL Bible, Steve Suehring, Wiley India edition, 2015 reprint
2. PHP: The Complete Reference, “Steven Holzner”, Tata McGraw Hill
3. PHP6, Apache, MySQL Web development, Timothy Boronczyk, Wiley India edition
4. Programming PHP, RasmusLerdorf, Kevin Tetroi (O’Reilly, ISBN 1565926102).
5. PHP, MySQL, and JavaScript: A Step-By-Step Guide to Creating Dynamic Websites by Robin Nixon O’Reilly Media
6. Core PHP Programming. Leon Atkinson (Prentice Hall, ISBN 0130463469).

Paper Code: CSA-DSC-6(Maj)-602P
(Practical based on paper CSA-DSC-6(Maj)-602)

Time Duration : 3 Hrs
List of Practical

1. Write a PHP code to print your name, course, and current date & time.
2. Write a PHP code demonstrating variables and constants.
3. Write a PHP code to perform arithmetic operations (addition, subtraction, multiplication, division).
4. Create a PHP code to check whether a number is even or odd.
5. Write a PHP code to find the greatest number of any three given numbers using nested-if.
6. Write a PHP code to print the day of the week using a switch-case statement.
7. Write a PHP code to print multiplication table using loops.
8. Create indexed, associative, and multidimensional arrays and display their elements.
9. Write a PHP code to demonstrate various loop iterative functions.
10. Write a PHP code to demonstrate string cleanup functions
11. Write a PHP code to demonstrate string case functions.
12. Write a user-defined function to find the largest number in an array.
13. Write a PHP code to demonstrate file inclusion function – include()
14. Write a PHP code to demonstrate file inclusion function – require()
15. Create a form (using GET/POST) and display submitted user data on another page.
16. Write a PHP code to connect to a MySQL database
17. Create a registration form and insert the form data into a MySQL table.
18. Fetch and display all records from a database table in an HTML table.
19. Create a file upload form that allows users to upload an image with size/type validation.
20. Write a PHP code to create a cookie with a name, value and expiration time.
21. Write a PHP code that sets a cookie based on user input.
22. Write a PHP code to retrieve and display a session variable.
23. Demonstrate exception handling during database connection using try–catch.
24. Write a PHP code to Handle different types of exceptions using multiple catch blocks.
25. Write a PHP code to Create and throw a user-defined exception class.

Paper Title: Cyber Security
Paper Code: CSA-DSC-6(Maj)-603

Th	Pr	Cr	Theory External Marks : 90
4	--	4	Theory Internal Marks : 10
Level: L- 300			Number of Theory Hours : 60

Time Duration: 3 Hrs.

Course Objective: This course aims to make students understand the Cyber Security and its implications.

Learning Outcomes: On successful completion of the course, students will be able to:

- Understand the architecture and regulatory aspects of cyberspace and the key terminologies in cyber security.
- Identify and assess web-based attacks, network vulnerabilities and exploit various tools.
- Recognize and classify different types of cybercrimes and understand legal frameworks.
- Grasp data privacy principles including compliance with regulations.

NOTE:

- The Question Paper will consist of Four Units.
- Examiner will set a total of NINE questions comprising TWO questions from each Unit and ONE compulsory question of short answer type covering the whole syllabus.
- The students are required to attempt ONE question from each Unit and the Compulsory question.
- All questions carry equal marks unless specified.

UNIT-I

Introduction to Cyber security: Understanding Cyberspace: Meaning of cyberspace and its components like Computers, networks, websites and apps and how they work together, Internet vs World Wide Web, How data moves on the internet

Architecture of cyberspace: Users, devices, servers, networks, Internet infrastructure for data transfer: routers, ISPs, data centers, Internet Society and bodies involved in governance, Regulations of cyberspace and its needs

Cyber Security: Concept and need, Online threats, Basic security goals: confidentiality, integrity, availability (CIA Triad), Issues and challenges in cyberspace.

Cyber security terminologies: Attack, Attack vector, Attack surface; Threat, Vulnerability, Risk; Exploit and exploitation; Hacker types(white/black/grey hat); Controls and security measures; Authentication(passwords, biometrics); Access Control and Cryptography.

UNIT-II

Web and Browser Attacks: Browser based attacks, Web Attacks Targeting Users(phishing pages, fake forms etc.), Methods to steal user or website data, Email-based attacks (spam, attachments, spoofing)

Network Vulnerabilities: Overview of vulnerability, scanning and why is it done, open port / service Identification, Detecting services and versions, Traffic Probes, Vulnerability Probes, Examples of common vulnerabilities.

Tools for Vulnerability Scanning (Concepts Only): Open VAS, Metasploit, Netcat and Socat, Network Sniffers and Injection tools.

UNIT-III

Cyber-crimes and their types: Classification of cyber-crimes: Crimes targeting computers and mobiles, Crimes against women and children, Financial and online frauds, Social engineering attacks, Malware and ransom-ware attacks, Zero day and zero click attacks.

Cybercriminals modus-operandi: Simple understanding of planning and executing attacks, Hiding identity and maintaining anonymity, Typical methods used by criminals.

Reporting of cybercrimes: Filing complaints, Importance of evidence, Using national cybercrime portals, Remedial and mitigation measures.

Legal perspective of cybercrime: IT Act 2000 and its amendments (overview), Cyber offences covered under the Act, Punishments for cybercrimes

Agencies covering Cyber crimes in India: CERT-In, Cyber Crime Cells, Other national and state-level security organizations

UNIT-IV

Understanding Data: Data, meta-data, big data; Personal and non-personal data. Data protection and its importance.

Data Privacy and Data Security: Difference between data privacy and data security, Big data security: issues and challenges Principles of data protection, India's Personal Data Protection Bill and its compliance

International Data Protection Regulations: General Data Protection Regulations(GDPR), 2016, Personal Information Protection and Electronic Documents Act (PIPEDA), Basic comparison with Indian context

Security Tools and Threats: Firewalls and Packet Filtering, Password Cracking approaches (Basic explanation), Key loggers and Spyware, Virus vs Worms, Trojan and backdoors, Steganography, DoS and DdoS attack, SQL injection, Buffer Overflow attacks and attacks on wireless Networks.

Suggested Readings:

1. Cyber security and Cyber war: What Everyone Needs to Know" by P.W. Singer and Allan Friedman (2014)
2. Cyber Security Understanding Cyber Crimes, Computer Forensics and Legal Perspectives by SumitBelapure and Nina Godbole, Wiley India Pvt. Ltd.(2011)
3. Security in the Digital Age: Social Media Security Threats and Vulnerabilities by Henry A. Oliver, Create Space Independent Publishing Platform (2015).
4. Cybercrime and Digital Forensics: An Introduction" by Marjie T. Britz (3rd Edition, 2019)
5. Security and Privacy in Communication Networks" by Nosseir, Ekici, and Cavallaro (2021)
6. Information Security Governance, Guidance for Information Security Managers by W. KragBrothy, 1st Edition, Wiley Publication.(2009)
7. Auditing IT Infrastructures for Compliance by Martin Weiss, Michael G. Solomon, 2nd Edition, Jones Bartlett Learning.(2015)

Paper Title: ComputerNetworks
Paper Code: CSA-DSC-6(Maj)-604

Th	Pr	Cr	Theory External Marks: 90
2	2	4	Theory Internal Marks: 10
Level: L-300			Number of Theory Hours: 60

TimeDuration:3Hrs.

Objective: To provide students with a strong foundation in computer networking concepts, architectures, and protocols, and to equip them with practical skills in designing, analyzing, and securing networks.

Learning Outcomes:

- Explain the fundamental concepts, types of networks, and reference models (OSI & TCP/IP).
- Analyze transmission media, switching, and multiplexing techniques.
- Apply error detection, flow control, and medium access protocols in data communication.
- Demonstrate understanding of routing algorithms, transport protocols, and application layer services.
- Evaluate network security mechanisms including firewalls, cryptography, and intrusion prevention.

Note:

- TheQuestionPaperwillconsistofFourUnits.
- Examinerwillsettotalof**NINE**questionscomprising**TWO**questionsfromeach Unit and **ONE**compulsory question ofshort answer type covering whole syllabi.
- Thestudentsarerequiredtoattempt**ONE**questionfromeachUnitandthe Compulsory question.
- Allquestionscarryequalmarksunlessspecified.

UNIT-I

ComputerNetwork: Types of Networks: LAN, MAN, WAN Network Topologies: Bus, Star, Ring, Mesh, Tree, Hybrid, Modes of communication: Simplex, half duplex, full duplex, Uses of Computer Networks, OSI Reference Model, TCP/ Ipreferencemodel, Comparison of OSI with TCP/IP model.

Physical Layer: Transmission media: Wired transmissions (Twisted pair, Coaxial cable, Fiber optics), Wireless Transmission (Radio, Microwave, and Infrared), Switching: (Circuit Switching, Message Switching, Packet Switching), Multiplexing: (Frequency Division, Time Division, Wave Length Division).

UNIT–II

DataLinkLayer: Framing,Types of addresses: (unicast, multicast, and broadcast),Address Resolution Protocol (ARP),ErrorsDetectionandCorrectionCode:Check sum, CRC, Hamming code, Data Link Protocols for noisy and noiseless channels, Sliding Window Protocol: Stop and Wait ARQ, Go-back-N ARQ, Selective Repeat ARQ.

Medium Access Sub-Layer: Introduction to Static and Dynamic channel allocation, Random-access protocols ALOHA, CSMA, CSMA/CD, CSMA/CA, IEEE standards 802.3, 802.11, Repeaters, Hubs, Bridges, Switches, Routers, and Gateways.

UNIT–III

Network Layer: Design Issues, network layer addressing, network layer datagram, IP addressed Classes. Sub netting-Sub network, Subnet mask, Routing Algorithm: Shortest Path Routing, Flooding,Distance Vector Routing,Link State Routing, Broadcast and Multicast routing, Routing Protocols: RIP, OSPF, BGP. Congestion control: Principles of Congestion Control, Congestion prevention policies, Leaky bucket and token bucketalgorithms.
Transport Layer: Design issues, Elements of transport protocols: Addressing, Connection establishment and release, Flow control and buffering, Introduction to TCP/UDP protocols.

UNIT–IV

Application Layer: Domain Name system (DNS), DNS name space, DNS Servers, World Wide Web, HTTP, e-mail: Architecture and Services, Message Component, Multipurpose Internet Mail Extensions (MIME), Simple Mail Transfer Protocol (SMTP), Post OfficeProtocol (POP), Remote Login and File Transfer Protocol.

Network Security: Fundamentals of Network Security, Sniffing and Snooping, Firewalls, Intrusion Detection and Prevention, Introduction to Cryptography, Substitution Ciphers, Transposition Ciphers, Symmetric-Key Algorithms, Public-Key Algorithms (RSA),Digital Signatures.

REFERENCES:

1. AndrewS.Tanenbaum,“ComputerNetworks”,PearsonPublication.
2. BehrouzA.Forouzan,“DataCommunicationandNetworking”,TataMcGrawHill.
3. TheodoreS.Rappaport,“WirelessCommunication:PrincipleandPractices”,PearsonPublication.
4. CharlieKaufman,RadioPerlman,MikeSpeciner,“NetworkSecurity”,PHI

Paper Title: Management Information System
Paper Code: CSA-DSC-6(Maj)-605

Th	Pr	Cr	Theory External Marks : 90
4	--	4	Theory Internal Marks : 10
Level: L- 300			Number of Theory Hours : 60

Objective: The objective of this course is to provide students with a basic understanding of Management Information Systems and their role in supporting business operations and decision-making. It aims to introduce different types of information systems, explain how MIS is developed and used in organizations, and familiarize students with modern technologies and security issues related to MIS. The course helps students understand how information systems improve efficiency, productivity, and managerial decision-making.

Learning Outcomes:
By the end of the course

- Students will understand the basic concepts, components, and importance of Management Information Systems in organizations.
- Students will be able to identify and differentiate various types of information systems used at different managerial levels.
- Students will learn to analyze and design simple MIS using system development processes and tools.
- Students will understand the role of MIS in major business functions such as finance, marketing, HR, and operations.
- Students will gain awareness of emerging technologies and security issues related to information systems.

Note:

- The Question Paper will consist of Four Units.
- The examiner will set a total of nine questions, including two questions from each unit and one compulsory short answer question covering the entire syllabus.
- Students are required to attempt one question from each unit and the compulsory question.
- All questions carry equal marks unless specified.

UNIT I

Introduction to MIS Meaning and definition of MIS, nature and scope of MIS, role and importance of MIS, system and subsystem concepts, elements of a system, types of systems (open, closed, physical, abstract), data and information, quality of information, sources of information, components of information systems, MIS and decision making, types of decisions (structured, semi-structured, unstructured)

UNIT II

Types of Information Systems Classification of information systems, Transaction Processing System (TPS), Management Information System (MIS), Decision Support System (DSS), Executive Support System (ESS), Office Automation System (OAS), Knowledge Management System (KMS), Marketing Information System, Finance and Accounting Information System, Human Resource Information System, Production and Inventory Information System.

UNIT III

MIS Development Process System Development Life Cycle (SDLC), planning, analysis, design, implementation, maintenance, feasibility studies (technical, economic, operational, legal), tools of system analysis (Data Flow Diagram), MIS design process, input and output design, testing and evaluation of MIS.

UNIT IV

Emerging Trends, Security, and Applications Cloud computing, big data analytics, AI and machine learning in MIS, ERP systems (SAP, Oracle), information security issues, cybersecurity threats, ethical issues in MIS.

Suggested Readings:

1. Ralph Stair & George Reynolds – Principles of Information Systems, Publisher: Cengage Learning
2. Gordon B. Davis & Margrethe H. Olson – Management Information Systems: Conceptual Foundations, Structure and Development, Publisher: McGraw Hill Education
3. Turban, Rainer & Potter – Introduction to Information Systems, Publisher: John Wiley & Sons
4. S. Sadagopan – Management Information Systems, Publisher: Prentice Hall of India (PHI Learning)
5. H. James – Management Information Systems for the Information Age, Publisher: McGraw Hill Education

Paper Title: Internship/Community Work
Paper Code: CSA- Intern-1

Th	Pr	Cr	
0	2	2	Practical External Marks: 50
Level:			Number of Hours for Internship/Community Work: 60

- The Project will involve development of application/system software in industrial, commercial or scientific environment. The Project can be any application or Web Site using Database connectivity.
- Community work
    - May involve developing projects which may uplift the life of unprivileged or add to the betterment of society in large.
    - May also include educating people about the use of technology safely by adopting a 60-hour well-defined plan. However, it will be compulsory to show the outcome in the form of feedback from trained people. For training purposes, a tie up can be made with schools. Same study material and awareness campaign can be used for three to four different groups of minimum 20 people in each group.

Evaluation

A student will have to submit project report of Internship/Community Work which will be evaluated based on Project Report (10 marks), Presentation (10 marks), Viva-Voce (10 marks), and execution of the Project/Feedback of the Community work (20 marks).

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ENVIRONMENT CONSERVATION

SEMESTER-V

MINOR COURSE
BIODIVERSITY

Course Code: MINC5 **Credits=4 (T=3+P=1)**

Total Hours = 60 (Theory=45+Practical=15)
Total Marks = 100 (Theory=75 [Final =60; Internal Assessment =15] +Practical=25)

Course Outcomes

1. Students will be able to understand the concept, levels, and importance of biodiversity.
2. Students will be able to analyse threats to biodiversity and explore conservation strategies.
3. Students will gain knowledge of Indian biodiversity, protected areas, and endemic species.
4. Students will understand global and national biodiversity conservation initiatives and legislations.
5. Students will develop skills to document, conserve, and manage biodiversity resources.

THEORY

Unit–I (15 Hrs)

Introduction to Biodiversity

1.1 Concept and definition

1.2 Levels of biodiversity – genetic, species, ecosystem diversity

1.3 Values of biodiversity – ecological, economic, social, cultural, ethical

1.4 Biodiversity hotspots in India and the world

Unit–II (15 Hrs)

Threats to Biodiversity

2.1 Causes of biodiversity loss – habitat destruction, deforestation, overexploitation, pollution, climate change, invasive species

2.2 Endangered and endemic species

2.3 Ecosystem services and biodiversity

2.4 Extinction – natural and anthropogenic causes

Unit-III (15 Hrs)

Conservation of Biodiversity

3.1 In-situ conservation – protected areas, biosphere reserves, national parks, wildlife sanctuaries

3.2 Ex-situ conservation – botanical gardens, zoos, seed banks, gene banks, tissue culture

3.3 Role of local communities, indigenous knowledge, and NGOs in biodiversity conservation

3.4 Case studies on successful conservation practices

Unit–IV (15 Hrs)

Biodiversity Management

4.1 Indian biodiversity – flora, fauna, agro-biodiversity, marine biodiversity

4.2 Biodiversity and sustainable development

4.3 People's Biodiversity Registers (PBRs) and community participation

4.4 International and national institutions for biodiversity conservation (UNEP, WWF, BNHS, Botanical Survey of India, Zoological Survey of India)

4.5 Biodiversity management and climate resilience

PRACTICALS (15 Hrs)

1. Preparation of a checklist of flora and fauna of the local area.
2. Visit to a biodiversity-rich site (forest, wetland, grassland, or national park).
3. Identification and documentation of local medicinal plants.
4. Survey of invasive alien species in the area.
5. Preparation of a People's Biodiversity Register (PBR) for a selected village or locality.
6. Case study preparation on a critically endangered or endemic species.

SUGGESTED READINGS

1. Odum, E.P. and Barrett, G.W. 2008. Fundamentals of Ecology. Thomson Brooks, Australia.
2. Groom, M.J., Meffe, G.K., Carroll, C.R. 2006. Principles of Conservation Biology. Sinauer Associates.
3. Krishnamurthy, K.V. 2003. Textbook of Biodiversity. Science Publishers, USA.
4. Sharma, P.D. 2018. Ecology and Environment. Rastogi Publication, Meerut.
5. Singh, J.S., Singh, S.P., Gupta, S.R. 2014. Ecology, Environment and Resource Conservation. Anamaya Publishers, New Delhi.
6. Heywood, V.H. and Watson, R.T. 1995. Global Biodiversity Assessment. Cambridge University Press.

Instructions for paper setter for Theory Paper :

There will be nine questions in all. Question # 1 shall be compulsory comprising of four short answer questions covering whole of the syllabus. There shall be two questions each from Unit I to IV. The candidate shall be required to attempt compulsory question, and one question each from Unit I to IV. Each question shall be of 12 marks.

SEMESTER- VI MINOR COURSE

ENVIRONMENTAL LAW, POLICY AND SUSTAINABLE DEVELOPMENT

Course Code: MINC6 **Credits=4 (T=3+P=1)**

Total Hours = 60 (Theory=45+Practical=15)
Total Marks = 100 (Theory=75 [Final =60; Internal Assessment =15] +Practical=25)

Course Outcomes

1. Students will be able to understand the Environment laws and policy in context with India
2. Students will be able to analyse latest changes and situations with respect to environmental laws and policy in the country.
3. Students will understand global environmental legislation.
4. Students will be equipped with the current concepts on sustainability and lifestyle.

THEORY

Unit-I (15 Hrs)

Philosophy and Concept

1.1 Environmental Law and Policy – Definition, Objectives
1.2 Core Principles: Precautionary principle,
1.3 Core Principles: Polluter Pays principle
1.4 Core Principles: Sustainable Development
1.5 Sustainable Development: Definition and concepts of sustainable development

Unit-II (15 Hrs)

National Laws

2.1 National Environment Policy
2.2 The Water and Air Acts & rules with amendments
2.3 The Environment (Protection) Act (EPA) 1986 with amendments
2.4 Forest Conservation Act 1980 with amendments
2.5 Wildlife Protection Act with amendments
2.6 National Green Tribunal Act, 2010 with amendments

Unit-III (15 Hrs)

International laws and Treaties

3.1 Principles of International Law and International treaties.
3.2 Earth Summit 1992
3.3 Convention on Biodiversity
3.3. Montreal Protocol
3.4 Convention on Climate Change
3.5 Kyoto Protocol

UNIT-IV (15 Hrs)
International Initiative
4.1 Millennium Development Goals
4.2 Sustainable Development Goals; Hurdles to sustainability

4.3 Cyber Waste Management

4.4 Environmental Management Systems; Life cycle assessment

4.5 Role of NGOs in Environment Protection

4.6 Role of Individual in Environment Protection

PRACTICALS (15Hrs)

1. Case studies on Environment laws in India
2. Case studies on International Environment Legislations
3. Visit to an NGO and prepare a report on NGO formation and its working
4. Project Report based on the syllabus

SUGGESTED READINGS:

1. P. Leela Krishnan (editor) — Environmental Law and Practice (multiple editions). Practical focus on compliance, litigation strategy and regulatory procedures.
2. Indian Law Institute — The Environment Protection Act and Rules: A Commentary. Handy for practitioners dealing with central statutes and rules.
3. R. Rangarajan & S. R. Bhatt — Environmental Law: Cases and Materials (various publishers). Case-oriented resource for litigation practice.
4. S. C. Shastri & V. S. Mani (eds.) — Landmark Judgments in Environmental Law. Collection and analysis of major Supreme Court .and High Court decisions.
5. A. Nagaraj (editor) — Annotated Environmental Protection Act and Rules. Useful for quick statutory interpretation with citations to cases and notifications.
6. NGT Reports and Compilation — National Green Tribunal: Selected Decisions. Collections of important NGT orders (various publishers/NGT website).
7. P. Chaturvedi — Air Pollution Law and Control in India. Focused on legal treatment of air quality regulation.
8. R. R. Nighute — Water Law in India: Statutes, Policies and Jurisprudence. Coverage of water rights, pollution control and tribunal practice.
9. Kritika Banerjee / M. K. Ranjitsinh (ed.) — Forest and Biodiversity Law (various editors/publishers). Treatments of forest conservation, Wildlife Protection Act and biodiversity legislation.

Interdisciplinary and policy-oriented works

1. Shyam Divan & Armin Rosencranz — Protecting the Environment: A Study of Indian Environmental Law and Policy (essay collections). Deeper policy analysis and historical development.
2. Sunita Narain — Environmental debates and policy essays (various titles). Accessible discussions linking law, policy and environmental justice.
3. Jayanta Bandyopadhyay / Smita Nakhooda (eds.) — Climate policy and law analyses (research-institute reports). Useful for policy-context chapters.

Instructions for paper setter for Theory Paper :

There will be nine questions in all. Question # 1 shall be compulsory comprising of four short answer questions covering whole of the syllabus. There shall be two questions each from Unit I to IV. The candidate shall be required to attempt compulsory question, and one question each from Unit I to IV. Each question shall be of 12 marks.

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INDUSTRIAL CHEMISTRY

as

(MAJOR and MINOR)

DOUBLE MAJOR AND SINGLE MINOR

(TABLE 2)

Paper	Code of the Paper	Name of paper Major Paper	Credit (T+P)	Contact Hours	Marks (T+P)
Semester 5					
A-6	ICH-DSC-6-MAJ-501	Material and Energy Balance	3+1	45+30	75+25
A-7	ICH-DSC-7-MAJ-502	Chemical Process Economics	3+1	45+30	75+25
	ICH-INT-1	Internship	4	-	100
Paper	**Code of the Minor paper**	**Name of Minor paper**	**Credit (T+P)**	**Contact Hours**	**Marks (T+P)**
Minor	ICH-DSC-6-MIN-501	Material and Energy Balance	3+1	45+30	75+25
Semester 6					
Paper	**Code of the Major paper**	**Name of Major paper**	**Credit (T+P)**	**Contact Hours**	**Marks (T+P)**
A-8	ICH-DSC-8-MAJ-601	Pharmaceuticals	3+1	45+30	75+25
A-9	ICH-DSC-9-MAJ-602	Heavy and Fine chemicals	3+1	45+30	75+25
Paper	**Code of the Minor paper**	**Name of Minor paper**	**Credit (T+P)**	**Contact Hours**	**Marks (T+P)**
Minor	ICH-DSC-7-MIN-601	Chemical Process Economics	3+1	45+30	75+25
Minor	ICH-DSC-8-MIN-602	Pharmaceuticals	3+1	45+30	75+25

Semester	**5**
Code of Paper	**ICH-DSC-6-MAJ-501**
Title of Paper	**Material and Energy Balance**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100 (75 Theory + 25 Practical)**
Marks distribution of the Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of the Practical paper	**End semester = 22** **Internal Marks = 03**
Contact Hours	**Theory: 3 Hrs. Per week** **Practical: 2 Hrs. Per week**

Applicable to the students

1. **Those who have opted for Industrial chemistry as a Major subject in Table 2.**
2. **Those who have opted for Industrial chemistry as a Minor subject in Table 1 or Table 2.**

LEARNING OUTCOMES SPECIFIC TO DISCIPLINE

By studying this course, the students will be able to:

- Explain what are the basic dimensions and units.
- The student shall be able to carry out the important material and energy balance calculations.
- Explain the theory behind various types of purification and pre-treatment techniques.
- The student will be able to explain safety of chemical handling through the comprehensive understanding of laboratory equipment and glassware.
- Emphasis is given to developing practical skills that will aid the student to set up accurate and reproducible, result-oriented experiments in higher classes.

GENERIC OUTCOMES

The third year of the course is designed to heighten the chemical temperament amongst the students. This is done by teaching them how to collect samples/data/evidence, carry out data analysis, evaluation and draw logical conclusions. They are also encouraged to work and improve upon their communication skills, through proper listening, plenty of hands-on laboratory exposure, group discussions and individual expression.

For practical work, groups shall be constituted, with student strength not exceeding 20 students per group.

Semester – 5
Syllabus for Theory Paper
ICH-DSC-6-MAJ-501

Material and Energy Balance

End semester = 67 **Credits – 03**

Internal Marks = 08 **Contact Hours – 3/week**

Examination Time allowed **3 Hours**

UNIT I: Dimension and Energy Balance **(15 Hrs.)**

Dimensions and units: Basic Chemical Calculations-Atomic weight, molecular weight, equivalent weight, mole, mole fractions.

Material balance: The concept of material Balance, Open and closed systems, Steady state and Unsteady state systems, Multiple component systems, Accounting for Chemical Reactions in material balances, Material balance for batch and semi batch process Concept of limiting reactant conversion, yield (Qualitative treatment only).

Energy Balance: Heat capacity of pure gases and gaseous mixtures at constant pressures. Sensible heat changes in liquids, enthalpy changes (Qualitative treatment only).

UNIT II: Purification Techniques **(15 Hrs.)**

Distillation: Introduction, Batch and continuous distillation, Separation of azeotropes, Plate columns and packed columns

Absorption: Introduction, Equipments-packed columns, spray Columns, bubble columns, packed bubble columns, mechanically agitated contractors.

Evaporation: Introduction, Equipments short tube (standard) Evaporator, forced circulation evaporators, falling film evaporators, climbing film (upward flow) evaporators, wiped (agitated) film evaporator.

Filtration: Introduction, filter media and filter aids, equipments – plate and frame filter press, nutch filter, rotary drum filter, sparkler filter, candle filter, bag filter, centrifuge.

UNIT III: Pre-treatment Techniques (15 Hrs.)

Drying: Introduction, free moisture, equipments-tray dryer, rotary dryer, flash dryer, fluidized bed dryer, drum dryer, spray dryer.

Extraction: Introduction, selection of solvent.
Fuel: Types of fuels – advantages and disadvantages of combustion of fuels, calorific value, specifications for fuel oil.

Boilers: Types of boilers and their functioning.

Books/Literature suggested

1. Vogel's Qualitative Inorganic Analysis (revised); 7th Longman, 1996. Edition Pubs: Orient
2. Furniss, B.S., Hannaford, A.J., Smith, P.W.G., Tatchell, A.R., Vogel's Textbook of Practical Organic Chemistry; 5th edition, Pubs: Dorling Kindersley (India) Pvt. Ltd. Delhi., 2006
3. Bassett, J., Denney, R.C., Jeffery, G.H., Mendham, J., Vogel's Textbook of Quantitative th Inorganic Analysis (revised); 4 edition, 1978. St
4. Palmer, W.G., Experimental Inorganic Chemistry; 1st edition, Pubs: Cambridge, 1954.
5. RD Bansal, R.K., Laboratory Manual of Organic chemistry; 1994. Limited, edition, Pubs: Wiley Eastern
6. Khosla, B.D., Garg, V.C., Gulati, A., Senior Practical Physical Chemistry; 11 Pubs: R. Chand & Co., New Delhi, 2002. Edition
7. Das, R.C., Behra, B., Experimental Physical Chemistry; Pubs: Tata McGraw Hill Publishing Co. Ltd., 1998.
8. Levitt, B.P., Findlays Practical Physical Chemistry; 8th Ltd., London & New York, 1978. Edition, Pubs: Longman Group

Instructions for the examiner

4. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
5. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts equally distributed within the syllabus of the respective unit.
6. The **compulsory unit** (of **19 marks**) will consist of two types of questions: **9 multiple-choice questions** of one mark each and **5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:** the **compulsory question** and one question each from the **UNIT-1, UNIT-2 and UNIT-3**.
2. The use of scientific calculators is not allowed.

Semester – 5

Syllabus for Practical Paper
ICH-DSC-6-MAJ-501

Material and Energy Balance

End semester = 22 **Credit – 01**

Internal Marks = 03 **Contact Hours – 2/week**

Examination Time allowed **3 Hours**

Techniques used in Chemistry lab

1. Fractional Crystallization (Copper Sulphate-Mohr Salt Solution)
2. Distillation: Simple, Vacuum, Fractional Distillation (1Expt.Each)
3. Filtration: Simple Filtration, Vacuum Filtration
4. Determination of Elevation and depression in b.p/m.p of liquids.
5. Determination of Vant Hoff Factor, Degree of Dissociation in strong electrolyte.
6. Determine the concentration of sugar in a solution refractometrically.
7. Determine specific rotation of a given compound.
8. Determine the percentages of two active substances in a solution with polarimeter.
9. Identify whether the given compound is dextro-rotatory or laevo-rotatory.
10. Power point presentations.

Books/Literature suggested

1. Vogel's Qualitative Inorganic Analysis (revised); 7th Longman, 1996. Edition Pubs: Orient
2. Furniss, B.S., Hannaford, A.J., Smith, P.W.G., Tatchell, A.R., Vogel's Textbook of Practical Organic Chemistry; 5th edition, Pubs: Dorling Kindersley (India) Pvt. Ltd. Delhi., 2006
3. Bassett, J., Denney, R.C., Jeffery, G.H., Mendham, J., Vogel's Textbook of Quantitative th Inorganic Analysis (revised); 4 edition, 1978.
4. Palmer, W.G., Experimental Inorganic Chemistry; 1st edition, Pubs: Cambridge, 1954.
5. RD Bansal, R.K., Laboratory Manual of Organic chemistry; 1994. Limited, edition, Pubs: Wiley Eastern .
6. Khosla, B.D., Garg, V.C., Gulati, A., Senior Practical Physical Chemistry; 11 Pubs: R. Chand & Co., New Delhi, 2002. Edition
7. Das, R.C., Behra, B., Experimental Physical Chemistry; Pubs: Tata McGraw Hill Publishing Co. Ltd., 1998.
8. Levitt, B.P., Findlays Practical Physical Chemistry; 8th Ltd., London & New York, 1978. Edition, Pubs: Longman Group

General Instructions to the Examiners:

Note: Practical examination will be of three hours duration divided into two parts (i.e. half hour for written and next two and half hours for performance).

1. Candidate will be examined based on any two experiments given by the examiner. Each experiment carries 08 marks (3 written + 5 performance).
2. Viva-Voce: 04 marks.
3. Note Book: 02 marks.

Semester	**5**
Code of Paper	**ICH-DSC-7-MAJ-502**
Title of Paper	**Chemical Process Economics**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100 (75 Theory + 25 Practical)**
Marks distribution of the Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of the Practical paper	**End semester = 22** **Internal Marks = 03**
Contact Hours	**Theory: 3 Hrs. Per week** **Practical: 2 Hrs. Per week**
Applicable to the students **1. Those who have opted for Industrial chemistry as a Major subject in Table 2.** **2. Those who have opted for Industrial chemistry as a Minor subject in Table 1 or Table 2.**	
LEARNING OUTCOMES SPECIFIC TO DISCIPLINE By studying this course, the students will be able to: • The student shall be able to differentiate between marketing and scientific management. • Explain the theory behind various types of costings involved in an industry. • The student will be able to explain safety of chemical handling through the comprehensive understanding of laboratory equipment and glassware. • Emphasis is given to developing practical skills that will aid the student to set up accurate and reproducible, result-oriented experiments in higher classes.	
GENERIC OUTCOMES The third year of the course is designed to heighten the chemical temperament amongst the students. This is done by teaching them how to collect samples/data/evidence, carry out data analysis, evaluation and draw logical conclusions. They are also encouraged to work and improve upon their communication skills, through proper listening, plenty of hands-on laboratory exposure, group discussions and individual expression.	

For practical work, groups shall be constituted, with student strength not exceeding 20 students per group.

Semester – 5
Syllabus for Theory Paper
ICH-DSC-7-MAJ-502
Chemical Process Economics

End semester = 67 **Credits – 03**

Internal Marks = 08 **Contact Hours – 3/week**

Examination Time allowed **3 Hours**

UNIT I: Costing and Marketing Management **(15 Hrs.)**

Cost Estimation: Factors involved in project cost estimation, Methods employed for the estimation of capital investment.

Variation of cost with capacity break-even Point. Optimum batch sizes, production, scheduling etc.

Marketing management: Some aspects of marketing, pricing policy. Profitability criteria, Economics of selecting alternatives.

Unit II: Scientific Management (15 Hrs)

Scientific Management:
Concept of scientific management in Industry.
Functions of management decision making, Planning organizing directing and control.
Location of industry, concept and importance.

Human resource Management: Management of human resources-selection Incentives, welfare and safety.

Unit III: Sampling, Packaging and IPR (15 Hrs)

Sampling: Sampling procedures, sampling of bulk materials. Techniques of sampling solids, liquids and gases collecting and processing of data.

Pharmaceutical packaging: Introduction package, Selection of packaging materials ancillary materials packaging machinery quality control of packaging materials.

Intellectual Property Rights: Basic knowledge of IPR, patents and trademarks.

Books/Literature suggested

1. Economics of Chemical Industry, Hempel. E. H
2. Shreve R.N. Brink. J.A., Chemical Process Industries, International student edition, Pubs: McGraw Hill Book Co. New York, 1960
3. Industrial Engineering and Management Science, Banga T.R., Agarwal N.K
4. Business Organization and Management, Bhushan Y.K.
5. Skoog D.A., Principles of Instrumental Analysis, 3rd Saunders International edition,1985. Edition, Pubs: Holt
6. Bassett, J., Denney, R.C., Jeffery, G.H., Mendham, J., Vogel's Textbook of Quantitative th Inorganic Analysis (revised); 4 edition, Pubs: Orient Longman,1978.
7. Willard H.H., Merritt L.L. Jr, Dean J.A., Settle F.A. Jr., Instrumental Methods of Analysis, 7th edn., Pubs: Wadsworth Publishing Company, California,1988.
8. Patents & Technology: *Indian Patent Law* by Kalyan C. Kankanala.
9. General IPR & Business: *Intellectual Property Rights: Protection and Management* by Nithyananda K. V.

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit** (of **19 marks**) will consist of two types of questions: **9 multiple-choice questions** of one mark each and **5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:** the **compulsory question** and one question each from the **UNIT-1, UNIT-2 and UNIT-3**.
2. The use of scientific calculators is not allowed.

Semester – 5

Syllabus for Practical Paper

ICH-DSC-7-MAJ-502

Chemical Process Economics

End semester = 22 **Credit – 01**

Internal Marks = 03 **Contact Hours – 2/week**

Examination Time allowed **3 Hours**

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1. Literature survey;
   Searching Patents with given keywords,
   name of molecule
   structure of compound, etc.
2. Demonstration and compilation of various pharmaceutical packaging materials with documentation of their properties.
3. Quality control tests of some packaging materials,aluminium strips, cartons, glass bottles, vials, syringes, etc.
4. Power point presentations

Books/Literature suggested

1. Skoog D.A., Principles of Instrumental Analysis, 3rd Saunders International edition,1985. Edition, Pubs: Holt
2. Bassett, J., Denney, R.C., Jeffery, G.H., Mendham, J., Vogel's Textbook of Quantitative th Inorganic Analysis (revised); 4 edition, Pubs: Orient Longman,1978.
3. Willard H.H., Merritt L.L. Jr, Dean J.A., Settle F.A. Jr., Instrumental Methods of Analysis, 7th edn., Pubs: Wadsworth Publishing Company, California,1988.
4. Block J, Beale J(Eds.), Wilson and Gisvold's Textbook of Organic and Pharmaceutical Chemistry, Pubs: Lippincott Williams & Wilkins, 2003.
5. Patents & Technology: Indian Patent Law by Kalyan C. Kankanala.
6. General IPR & Business: Intellectual Property Rights: Protection and Management by Nithyananda K. V.

General Instructions to the Examiners:

Note: Practical examination will be of three hours duration divided into two parts (i.e. half hour for written and next two and half hours for performance).

1. Candidate will be examined based on any two experiments given by the examiner. Each experiment carries 08 marks (3 written + 5 performance).
2. Viva-Voce: 04 marks.
3. Note Book: 02 marks.

Semester	**5**
Code of Paper	**ICH-INT-1 (Table 2)**
Title of Paper	**Internship**
Total Credit	**04**
Total Marks	**100**

Applicable to the students

1. Those who have opted for Industrial chemistry as a Major subject in Table 2

LEARNING OUTCOMES SPECIFIC TO DISCIPLINE

The internship course is designed to provide undergraduate Industrial Chemistry students with experiential learning through **industrial exposure, academic visits, training programs and participation in scientific events**. The course aims to bridge theoretical knowledge with real-world applications in chemical industries, research institutions, and academic platforms.

Semester – 5

Instructions for Internship
ICH-INT-1 (Table 2)
Internship / Field-Based Learning Course

Total Marks = 100 **Credits: 4**

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1. Course Description

The internship course is designed to provide undergraduate Industrial Chemistry students with experiential learning through **industrial exposure, academic visits, training programs, and participation in scientific events**. The course aims to bridge theoretical knowledge with real-world applications in chemical industries, research institutions, and academic platforms.

2. Course Components

2.1 Project/Training:

Each student shall undertake **one of the following** industrial hands-on training, or research-based activity in any one of the following institutions or facilities:

I. Training at a local industry.
II. Training in College research Laboratory.

The student shall spend a **minimum of 50-60 hours(7-10 days)** at the selected activity site. Winter or summer vacations may be utilized for this purpose. Upon completion, the student must obtain a certificate or official letter from the host industry or academic institution confirming their participation and duration of engagement.

2.2 Internship Report:

Each student shall submit a **detailed internship report** of approximately **20-30 pages**, including relevant photographs. The report shall include, but not be limited to, the following aspects:

I. Description of Raw materials used
II. Steps involved in manufacturing
III. Finished product
IV. Quality control measures involved
V. Manpower, Infrastructure, and Safety and waste management practices
VI. Observations made during the internship
VII. Learning outcomes and skills acquired

The student shall submit **two hard copies** (soft binding) of the report along with one **soft copy in PDF format** for uploading on the website. The report shall be uploaded on the Institute/College website after evaluation.

2.3 Additional Academic Activities:

Additional activities shall include participation in conferences, seminars, workshops, or activities of academic clubs organized by (i) the parent Department or Institute, or other recognized academic or research institutions.

The student shall submit a **summary report of up to 5 pages** covering such activities with relevant photographs. The summary may also include relevant academic programmes attended by the student during the **first and second years** of the course, if any.

3. **Assessment Method**

The assessment for this component shall be based on the following:

(i) Internship Report with certificate – 50 ***Marks***

(ii) Oral Presentation – 20 ***Marks***

(iii) Additional Academic Activities – 20 ***Marks***

(iv) ***Internal assessment***- 10 ***Marks***

(a) The internship report shall be evaluated by **two examiners** appointed by the College, one of whom shall be from a discipline other than that of the student.

(b) All students shall present their internship work in a **one-day academic event**, in the presence of internal examiners and fellow students. The evaluation of the presentation shall be based on the following criteria:

i. Quality of oral presentation, clarity, innovation, understanding of the topic, and responses to questions – **60%**

ii. Attendance, participation in discussions, and engagement during presentations of other students – **40%**

3. Teacher Involvement

A faculty member may be assigned mentorship responsibility for a group of 15 students for the purpose of internship. The faculty mentor will guide the students in preparing the report and presentation, arranging examiners for evaluation and organizing related events.For undertaking these responsibilities, a workload of 4 hours/week may be allotted to the concerned faculty member.

Semester	**5**
Code of Paper	**ICH-DSC-6-MIN-501**
Title of Paper	**Material and Energy Balance**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100 (75 Theory + 25 Practical)**
Marks distribution of the Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of the Practical paper	**End semester = 22** **Internal Marks = 03**
Contact Hours	**Theory: 3 Hrs. Per week** **Practical: 2 Hrs. Per week**

Applicable to the students

1. **Those who have opted for Industrial chemistry as a Major subject in Table 2.**
2. **Those who have opted for Industrial chemistry as a Minor subject in Table 1 or Table 2.**

LEARNING OUTCOMES SPECIFIC TO DISCIPLINE

By studying this course, the students will be able to:

- Explain what are the basic dimensions and units.
- The student shall be able to carry out the important material and energy balance calculations.
- Explain the theory behind various types of purification and pre-treatment techniques.
- The student will be able to explain safety of chemical handling through the comprehensive understanding of laboratory equipment and glassware.
- Emphasis is given to developing practical skills that will aid the student to set up accurate and reproducible, result-oriented experiments in higher classes.

GENERIC OUTCOMES

The third year of the course is designed to heighten the chemical temperament amongst the students. This is done by teaching them how to collect samples/data/evidence, carry out data analysis, evaluation and draw logical conclusions. They are also encouraged to work and improve upon their communication skills, through proper listening, plenty of hands-on laboratory exposure, group discussions and individual expression.

For practical work, groups shall be constituted, with student strength not exceeding 20 students per group.

Semester – 5
Syllabus for Theory Paper
ICH-DSC-6-MIN-501

Material and Energy Balance

End semester = 67 **Credits – 03**

Internal Marks = 08 **Contact Hours – 3/week**

Examination Time allowed **3 Hours**

UNIT I: Dimension and Energy Balance **(15 Hrs.)**

Dimensions and units: Basic Chemical Calculations-Atomic weight, molecular weight, equivalent weight, mole, mole fractions.

Material balance: The concept of material Balance, Open and closed systems, Steady state and Unsteady state systems, Multiple component systems, Accounting for Chemical Reactions in material balances, Material balance for batch and semi batch process Concept of limiting reactant conversion, yield (Qualitative treatment only).

Energy Balance: Heat capacity of pure gases and gaseous mixtures at constant pressures. Sensible heat changes in liquids, enthalpy changes (Qualitative treatment only).

UNIT II: Purification Techniques **(15 Hrs.)**

Distillation: Introduction, Batch and continuous distillation, Separation of azeotropes, Plate columns and packed columns

Absorption: Introduction, Equipments-packed columns, spray Columns, bubble columns, packed bubble columns, mechanically agitated contractors.

Evaporation: Introduction, Equipments short tube (standard) Evaporator, forced circulation evaporators, falling film evaporators, climbing film (upward flow) evaporators, wiped (agitated) film evaporator.

Filtration: Introduction, filter media and filter aids, equipments – plate and frame filter press, nutch filter, rotary drum filter, sparkler filter, candle filter, bag filter, centrifuge.

UNIT III: Pre-treatment Techniques (15 Hrs.)

Drying: Introduction, free moisture, equipments-tray dryer, rotary dryer, flash dryer, fluidized bed dryer, drum dryer, spray dryer.

Extraction: Introduction, selection of solvent.
Fuel: Types of fuels – advantages and disadvantages of combustion of fuels, calorific value, specifications for fuel oil.

Boilers: Types of boilers and their functioning.

Books/Literature suggested

1. Vogel's Qualitative Inorganic Analysis (revised); 7th Longman, 1996. Edition Pubs: Orient
2. Furniss, B.S., Hannaford, A.J., Smith, P.W.G., Tatchell, A.R., Vogel's Textbook of Practical Organic Chemistry; 5th edition, Pubs: Dorling Kindersley (India) Pvt. Ltd. Delhi., 2006
3. Bassett, J., Denney, R.C., Jeffery, G.H., Mendham, J., Vogel's Textbook of Quantitative th Inorganic Analysis (revised); 4 edition, 1978. St
4. Palmer, W.G., Experimental Inorganic Chemistry; 1st edition, Pubs: Cambridge, 1954.
5. RD Bansal, R.K., Laboratory Manual of Organic chemistry; 1994. Limited, edition, Pubs: Wiley Eastern
6. Khosla, B.D., Garg, V.C., Gulati, A., Senior Practical Physical Chemistry; 11 Pubs: R. Chand & Co., New Delhi, 2002. Edition
7. Das, R.C., Behra, B., Experimental Physical Chemistry; Pubs: Tata McGraw Hill Publishing Co. Ltd., 1998.
8. Levitt, B.P., Findlays Practical Physical Chemistry; 8th Ltd., London & New York, 1978. Edition, Pubs: Longman Group

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit** (of **19 marks**) will consist of two types of questions: **9 multiple-choice questions** of one mark each and **5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:** the **compulsory question** and one question each from the **UNIT-1, UNIT-2 and UNIT-3**.
2. The use of scientific calculators is not allowed.

Semester – 5

Syllabus for Practical Paper

ICH-DSC-6-MIN-501

Material and Energy Balance

End semester = 22 **Credit – 01**

Internal Marks = 03 **Contact Hours – 2/week**

Examination Time allowed **3 Hours**

Techniques used in Chemistry lab

1. Fractional Crystallization (Copper Sulphate-Mohr Salt Solution)
2. Distillation: Simple, Vacuum, Fractional Distillation (1Expt.Each)
3. Filtration: Simple Filtration, Vacuum Filtration
4. Determination of Elevation and depression in b.p/m.p of liquids.
5. Determination of Vant Hoff Factor, Degree of Dissociation in strong electrolyte.
6. Determine the concentration of sugar in a solution refractometrically.
7. Determine specific rotation of a given compound.
8. Determine the percentages of two active substances in a solution with polarimeter.
9. Identify whether the given compound is dextro-rotatory or laevo-rotatory.
10. Power point presentations.

Books/Literature suggested

1. Vogel's Qualitative Inorganic Analysis (revised); 7th Longman, 1996. Edition Pubs: Orient
2. Furniss, B.S., Hannaford, A.J., Smith, P.W.G., Tatchell, A.R., Vogel's Textbook of Practical Organic Chemistry; 5th edition, Pubs: Dorling Kindersley (India) Pvt. Ltd. Delhi., 2006
3. Bassett, J., Denney, R.C., Jeffery, G.H., Mendham, J., Vogel's Textbook of Quantitative Inorganic Analysis (revised); 4 edition, 1978.
4. Palmer, W.G., Experimental Inorganic Chemistry; 1st edition, Pubs: Cambridge, 1954.
5. RD Bansal, R.K., Laboratory Manual of Organic chemistry; 1994. Limited, edition, Pubs: Wiley Eastern .
6. Khosla, B.D., Garg, V.C., Gulati, A., Senior Practical Physical Chemistry; 11 Pubs: R. Chand & Co., New Delhi, 2002. Edition
7. Das, R.C., Behra, B., Experimental Physical Chemistry; Pubs: Tata McGraw Hill Publishing Co. Ltd., 1998.
8. Levitt, B.P., Findlays Practical Physical Chemistry; 8th Ltd., London & New York, 1978. Edition, Pubs: Longman Group

General Instructions to the Examiners:

Note: Practical examination will be of three hours duration divided into two parts (i.e. half hour for written and next two and half hours for performance).

1. Candidate will be examined based on any two experiments given by the examiner. Each experiment carries 08 marks (3 written + 5 performance).
2. Viva-Voce: 04 marks.
3. Note Book: 02 marks.

Semester	**6**
Code of Paper	**ICH-DSC-8-MAJ-601**
Title of Paper	**Pharmaceuticals**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100 (75 Theory + 25 Practical)**
Marks distribution of the Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of the Practical paper	**End semester = 22** **Internal Marks = 03**
Contact Hours	**Theory: 3 Hrs. Per week** **Practical: 2 Hrs. Per week**

Applicable to the students

1. **Those who have opted for Industrial chemistry as a Major subject in Table 2.**
2. **Those who have opted for Industrial chemistry as a Minor subject in Table 1 or Table 2.**

LEARNING OUTCOMES SPECIFIC TO DISCIPLINE

By studying this course, the students will be able to:

- Explain what are the different types of pharmacopoeia.
- Explain the theory behind various types of excipients, phytochemicals and crude drugs.
- The students will be able to explain the different types of isolation processes and chemical constituents of plants.
- Emphasis is given to developing practical skills that will aid the student to set up accurate and reproducible, result-oriented experiments in higher classes.

GENERIC OUTCOMES

The third year of the course is designed to heighten the chemical temperament amongst the students. This is done by teaching them how to collect samples/data/evidence, carry out data analysis, evaluation and draw logical conclusions. They are also encouraged to work and improve upon their communication skills, through proper listening, plenty of hands-on laboratory exposure, group discussions and individual expression.

For practical work, groups shall be constituted, with student strength not exceeding 20 students per group.

Semester – 6
Syllabus for Theory Paper
ICH-DSC-8-MAJ-601
Pharmaceuticals

End semester = 67 **Credits – 03**
Internal Marks = 08 **Contact Hours – 3/week**
Examination Time allowed **3 Hours**

UNIT – I Indian Pharmacopoeia (15 Hrs.)

Historical background and development of Pharmaceutical Industry in India.

Pharmacopoeias: Development of India Pharmacopoeia and Introduction to B.P., U.S.P., E.P., N.F. and other important pharmacopoeias.

Tools related to Chemistry: CHEMDRAW, Mercury. Use and management of chemical data (Scoupus, Web of Science)

Scientific ethics: Plagiarism, Scientific integrity, data integrity.

Excipients: Various types of pharmaceutical excipients; their Chemistry, process of manufacture and quality specifications- Glidants, lubricants, diluents, preservatives, antioxidants, emulsifying agents, coating agents, binders, colouring agents, flavouring agents, gelatin and other additives, sorbitol, mannitol, viscosity builders, etc.

UNIT – II Regulatory authority and Analysis (15 Hrs.)

Pharmaceutical packaging: FDA, Important schedules and some legal aspects of drugs

Phytochemicals: Introduction to plant classification and Crude drugs cultivation collection preparation for the market and storage of medicinal plants.

Crude Drugs: Evaluation of crude drugs-Moisture content Extractive Value, volatile oil content, foreign organic matter.

Quantitative analysis: Quantitative microscopic exercises including of starch leaf content (Palisade ratio stomatal number and index vein islet number and vein termination number) crude fibre content.
Introduction to chromatographic method for identification of crude drugs.

UNIT – III Plant analysis and Drugs (15 Hrs.)

Chemical constitution of plants: including carbohydrates, aminoacids, proteins fats waxes, volatile oils, terpenoids, 203or acti, saponins, flavonoids, tannins, glycosides, alkaloids.

Isolation Procedures: Various isolation procedures 203 or active ingradients with example for alkaloid e.g., vinca alkaloids, reserpine, sapogenin.

Pharmaceutical quality control (other than the Analytical methods covered under core subject) Sterility testing, Pyrogenic testing glass testing bulk density of powder etc. Effluent Handling

Classification of various types of drugs with examples.
Sulphadrugs-Sulphaguandine, sulphamethoxazele

Antimicrobial-chloramphenicol, tetracycline, Penicillin- G and their monographs.
NSAID-salicyclicacid and its derivatives, Ibuprofen, mefenamic acid.

Books/Literature suggested

1. Ramstead E, Modren Pharmacognosy, Pubs:McGraw-Hill,1959.
2. Lemke T. L., Williams D.A., Foye's Principles of Medicinal Chemistry, Pubs: Lippincott Williams & Wilkins, 2007.
3. Lednicer D, Strategies for Organic Drug Synthesis and Design, Second Edition, Pubs: Wiely Interscience, 2008.
4. Block J, Beale J(Eds.), Wilson and Gisvold's Textbook of Organic and Pharmaceutical Chemistry, Pubs: Lippincott Williams & Wilkins, 2003.
5. Indian Pharmacopoeia Commission, Indian Pharmacopoeia 2007, 3 Vols. With One Supplement (Addendum 2008).
6. The British Pharmacopoeia 2010.

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit** (of **19 marks**) will consist of two types of questions: **9 multiple-choice questions** of one mark each and **5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:** the **compulsory question** and one question each from the **UNIT-1, UNIT-2 and UNIT-3**.
2. The use of scientific calculators is not allowed.

Semester – 6

Syllabus for Practical Paper

ICH-DSC-8-MAJ-601

Pharmaceuticals

End semester = 22 **Credit – 01**

Internal Marks = 03 **Contact Hours – 2/week**

Examination Time allowed **3 Hours**

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1. Study of monographs of antimicrobial and NSAID.
2. Complete I.P. monograph of three drugs representing variety of testing methods.
3. Report Write compound data and chemical procedures for drugs.
4. Limit tests for chlorine, heavy metals, arsenic Etc. of two representative bulk drugs.
5. Active ingredient analysis of few types of formulations.
6. Representing different methods of analysis Acidimetry, Alkalimetry, nonaqueous complexometry, potentiometry etc.
7. Determination of sulphate ash, loss on drying and other tests of bulk drugs.
8. Power point Presentations

Books/Literature suggested

1. Vogel's Qualitative Inorganic Analysis (revised); 7th Longman, 1996. Edition Pubs: Orient
2. Furniss, B.S., Hannaford, A.J., Smith, P.W.G., Tatchell, A.R., Vogel's Textbook of Practical Organic Chemistry; 5th edition, Pubs: Dorling Kindersley (India) Pvt. Ltd. Delhi., 2006
3. Bassett, J., Denney, R.C., Jeffery, G.H., Mendham, J., Vogel's Textbook of Quantitative Inorganic Analysis (revised); 4 edition, 1978.
4. Palmer, W.G., Experimental Inorganic Chemistry; 1st edition, Pubs: Cambridge, 1954.
5. RD Bansal, R.K., Laboratory Manual of Organic chemistry; 1994. Limited, edition, Pubs: Wiley Eastern .
6. Khosla, B.D., Garg, V.C., Gulati, A., Senior Practical Physical Chemistry; 11 Pubs: R. Chand & Co., New Delhi, 2002. Edition
7. Das, R.C., Behra, B., Experimental Physical Chemistry; Pubs: Tata McGraw Hill Publishing Co. Ltd., 1998.
8. Levitt, B.P., Findlays Practical Physical Chemistry; 8th Ltd., London & New York, 1978. Edition, Pubs: Longman Group

General Instructions to the Examiners:

Note: Practical examination will be of three hours duration divided into two parts (i.e. half hour for written and next two and half hours for performance).

1. Candidate will be examined based on any two experiments given by the examiner. Each experiment carries 08 marks (3 written + 5 performance).
2. Viva-Voce: 04 marks.
3. Note Book: 02 marks.

Semester	**6**
Code of Paper	**ICH-DSC-9-MAJ-602**
Title of Paper	**Heavy and Fine Chemicals**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100 (75 Theory + 25 Practical)**
Marks distribution of the Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of the Practical paper	**End semester = 22** **Internal Marks = 03**
Contact Hours	**Theory: 3 Hrs. Per week** **Practical: 2 Hrs. Per week**

Applicable to the students

1. **Those who have opted for Industrial chemistry as a Major subject in Table 2.**
2. **Those who have opted for Industrial chemistry as a Minor subject in Table 1 or Table 2.**

LEARNING OUTCOMES SPECIFIC TO DISCIPLINE

By studying this course, the students will be able to:

- Explain what are the different types of manufacturing parameters involved in heavy chemical production.
- Explain the theory behind various types of fine chemicals such as, laboratory reagents, analytical reagents and chromatographic materials.
- The students will be able to explain the different types of dyes, oils and colouring agents.
- Emphasis is given to developing practical skills that will aid the student to set up accurate and reproducible, result-oriented experiments in higher classes.

GENERIC OUTCOMES

The third year of the course is designed to heighten the chemical temperament amongst the students. This is done by teaching them how to collect samples/data/evidence, carry out data analysis, evaluation and draw logical conclusions. They are also encouraged to work and improve upon their communication skills, through proper listening, plenty of hands-on laboratory exposure, group discussions and individual expression.

For practical work, groups shall be constituted, with student strength not exceeding 20 students per group.

Semester – 6
Syllabus for Theory Paper
ICH-DSC-9-MAJ-602
Heavy and Fine Chemicals

End semester = 67 **Credits – 03**

Internal Marks = 08 **Contact Hours – 3/week**

Examination Time allowed **3 Hours**

UNIT I: Manufacturing parameters-1 (15 Hrs.)

Manufacture of the following with reference to consumption Pattern, Raw materials, Production process, Major engineering aspects, Special material of construction, Quality control, Hazards, safety and Effluent management:

Synthetic nitrogen products: Ammonia, nitric acid ammonium nitrate and ammonium sulphate.

Chlor-alkali industrial products: Caustic soda Chlorine.

Phosphorus chemicals: Phosphorus, phosphoric acid ammonium phosphate, superphosphate, triple superphosphate.

Industrial carbon-carbon blocks, manufacture of graphite and carbon.

Lime, gypsum, Silicon, calcium carbide, silicon carbide

Flourine, bromine, Iodine, hydrobromic acid, Interhalogen compounds

UNIT II: Manufacturing parameters-2 (15 Hrs.)

Manufacture of the following with reference to Raw materials, Flow chart, Effluent management, Kinetics and Uses of:

Aluminium alkoxides, titanium tetra chloride, and titanates, titanium dioxide.

Applications and uses of zeolites as catalyst. Their use in isomerization and dehydration / dehydroxylation.

Reagents-Laboratory chemicals from heavy chemical

Analytical reagents-Sodium carbonate, sodium bicarbonate, potassium dichromate, oxalic acid perchloric acid, Common solutions-Fehling solution, Karl Fisher reagent.

Chromatographic materials and HPLC solvents-Coating material, pre-coating of plates, Spectroscopy grade chemicals methanol, ethanol, potassium bromide, carbon tetrachloride nujol, chloroform.

UNIT III: Dyes, Oils and Colouring Agents (15 Hrs.)

Dyes: Indigo vat dyes, reactive dyes

Essential oils: General, organic 207or ac, camphor, citral, citronellol, menthol.

Surfactants and emulsifying agents-PEG, Tweens, Spans.

Colouring agents: Manufacture of some natural colours and synthetic colours flavouring agents-Fragrances and Food additives.

Books/Literature suggested

1. Shreve R.N. Brink. J.A., Chemical Process Industries, International student edition, Pubs: McGraw Hill Book Co.New York, 1st Edn 960.
2. Groggins P.M., Unit Processing Organic Synthesis, 5edition, International Student edition, Pubs: McGraw-Hill Book Co., New York, 1998.
3. Dryden's outlines of Chemical Technology, edited and revised by Gopala Rao M. and Marshall S, Pubs: East-West Press, New Delhi, 2004.

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit** (of **19 marks**) will consist of two types of questions: **9 multiple-choice questions** of one mark each and **5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:** the **compulsory question** and one question each from the **UNIT-1, UNIT-2 and UNIT-3**.
2. The use of scientific calculators is not allowed.

Semester – 6
Syllabus for Practical Paper
ICH-DSC-9-MAJ-602
Heavy and Fine Chemicals

End semester = 22 **Credit – 01**

Internal Marks = 03 **Contact Hours – 2/week**

Examination Time allowed **3 Hours**

1. Preparation of synthetic zeolites.
2. Preparation using zeolites.
3. Preparation of aluminium isopropoxide and reactions using the same.
4. Synthesis of few fine chemicals –for example,
5. Amyl acetate, 209or ac chemicals Paracetamol, sulphanilamide.
6. Extraction and Purification of lemon grass oil to obtain citral.
7. Resolution of tartaric acid and-phenyl ethyl amine.
8. Isolation of some natural products, like tartaric acid citric acid, etc.
9. Power Point Presentaions

Books/Literature suggested

1. A Comprehensive Guide to Physical Chemistry Experiments and Viva Questions, Neelam Seedher, 2nd edition, Notion Press, India.
2. Vogel's Qualitative Inorganic Analysis (revised); 7th Longman, 1996. Edition Pubs: Orient
3. Furniss, B.S., Hannaford, A.J., Smith, P.W.G., Tatchell, A.R., Vogel's Textbook of Practical Organic Chemistry; 5th edition, Pubs: Dorling Kindersley (India) Pvt. Ltd. Delhi., 2006
4. Bassett, J., Denney, R.C., Jeffery, G.H., Mendham, J., Vogel's Textbook of Quantitative th Inorganic Analysis (revised); 4 edition, 1978. St
5. Khosla, B.D., Garg, V.C., Gulati, A., Senior Practical Physical Chemistry; 11 Pubs: R. Chand & Co., New Delhi, 2002. Edition
6. Das, R.C., Behra, B., Experimental Physical Chemistry; Pubs: Tata McGraw Hill Publishing Co. Ltd., 1998.
7. Levitt, B.P., Findlays Practical Physical Chemistry; 8th Ltd., London & New York, 1978. Edition, Pubs: Longman Group

General Instructions to the Examiners:

Note: Practical examination will be of three hours duration divided into two parts (i.e. half hour for written and next two and half hours for performance).

1. Candidate will be examined based on any two experiments given by the examiner. Each experiment carries 08 marks (3 written + 5 performance).
2. Viva-Voce: 04 marks.
3. Note Book: 02 marks.

Semester	**6**
Code of Paper	**ICH-DSC-7-MIN-601**
Title of Paper	**Chemical Process Economics**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100 (75 Theory + 25 Practical)**
Marks distribution of the Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of the Practical paper	**End semester = 22** **Internal Marks = 03**
Contact Hours	**Theory: 3 Hrs. Per week** **Practical: 2 Hrs. Per week**

Applicable to the students

1. **Those who have opted for Industrial chemistry as a Major subject in Table 2.**
2. **Those who have opted for Industrial chemistry as a Minor subject in Table 1 or Table 2.**

LEARNING OUTCOMES SPECIFIC TO DISCIPLINE

By studying this course, the students will be able to:

- The student shall be able to differentiate between marketing and scientific management.
- Explain the theory behind various types of costings involved in an industry.
- The student will be able to explain safety of chemical handling through the comprehensive understanding of laboratory equipment and glassware.
- Emphasis is given to developing practical skills that will aid the student to set up accurate and reproducible, result-oriented experiments in higher classes.

GENERIC OUTCOMES

The third year of the course is designed to heighten the chemical temperament amongst the students. This is done by teaching them how to collect samples/data/evidence, carry out data analysis, evaluation and draw logical conclusions. They are also encouraged to work and improve upon their communication skills, through proper listening, plenty of hands-on laboratory exposure, group discussions and individual expression.

For practical work, groups shall be constituted, with student strength not exceeding 20 students per group.

Semester – 5
Syllabus for Theory Paper
ICH-DSC-7-MIN-601

Chemical Process Economics

End semester = 67 **Credits – 03**

Internal Marks = 08 **Contact Hours – 3/week**

Examination Time allowed **3 Hours**

UNIT I: Costing and Marketing Management **(15 Hrs.)**

Cost Estimation: Factors involved in project cost estimation, Methods employed for the estimation of capital investment.

Variation of cost with capacity break-even Point. Optimum batch sizes, production, scheduling etc.

Marketing management: Some aspects of marketing, pricing policy. Profitability criteria, Economics of selecting alternatives.

Unit II: Scientific Management (15 Hrs)

Scientific Management:
Concept of scientific management in Industry.
Functions of management decision making, Planning organizing directing and control.
Location of industry, concept and importance.

Human resource Management: Management of human resources-selection Incentives, welfare and safety.

Unit III: Sampling, Packaging and IPR (15 Hrs)

Sampling: Sampling procedures, sampling of bulk materials. Techniques of sampling solids, liquids and gases collecting and processing of data.

Pharmaceutical packaging: Introduction package, Selection of packaging materials ancillary materials packaging machinery quality control of packaging materials.

Intellectual Property Rights: Basic knowledge of IPR, patents and trademarks.

Books/Literature suggested

1. Economics of Chemical Industry, Hempel. E. H
2. Shreve R.N. Brink. J.A., Chemical Process Industries, International student edition, Pubs: McGraw Hill Book Co. New York, 1960

3. Industrial Engineering and Management Science, Banga T.R., Agarwal N.K
4. Business Organization and Management, Bhushan Y.K.
5. Skoog D.A., Principles of Instrumental Analysis, 3rd Saunders International edition,1985. Edition, Pubs: Holt
6. Bassett, J., Denney, R.C., Jeffery, G.H., Mendham, J., Vogel's Textbook of Quantitative th Inorganic Analysis (revised); 4 edition, Pubs: Orient Longman,1978.
7. Willard H.H., Merritt L.L. Jr, Dean J.A., Settle F.A. Jr., Instrumental Methods of Analysis, 7th edn., Pubs: Wadsworth Publishing Company, California,1988.
8. Patents & Technology: *Indian Patent Law* by Kalyan C. Kankanala.
9. General IPR & Business: *Intellectual Property Rights: Protection and Management* by Nithyananda K. V.

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit** (of **19 marks**) will consist of two types of questions: **9 multiple-choice questions** of one mark each and **5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:** the **compulsory question** and one question each from the **UNIT-1, UNIT-2 and UNIT-3**.
2. The use of scientific calculators is not allowed.

Semester – 5
Syllabus for Practical Paper
ICH-DSC-7-MIN-601
Chemical Process Economics

End semester = 22 **Credit – 01**
Internal Marks = 03 **Contact Hours – 2/week**
Examination Time allowed **3 Hours**

1. Literature survey;
   a. Searching Patents with given keywords,
   b. name of molecule
   c. structure of compound, etc.
2. Demonstration and compilation of various pharmaceutical packaging materials with documentation of their properties.
3. Quality control tests of some packaging materials,
   a. aluminium strips,
   b. cartons,
   c. glass bottles,
   d. vials,
   e. syringes, etc.
4. Power point presentations

Books/Literature suggested

1. Skoog D.A., Principles of Instrumental Analysis, 3rd Saunders International edition,1985. Edition, Pubs: Holt
2. Bassett, J., Denney, R.C., Jeffery, G.H., Mendham, J., Vogel's Textbook of Quantitative th Inorganic Analysis (revised); 4 edition, Pubs: Orient Longman,1978.
3. Willard H.H., Merritt L.L. Jr, Dean J.A., Settle F.A. Jr., Instrumental Methods of Analysis, 7th edn., Pubs: Wadsworth Publishing Company, California,1988.
4. Block J, Beale J(Eds.), Wilson and Gisvold's Textbook of Organic and Pharmaceutical Chemistry, Pubs: Lippincott Williams & Wilkins, 2003.
5. Patents & Technology: Indian Patent Law by Kalyan C. Kankanala.
6. General IPR & Business: Intellectual Property Rights: Protection and Management by Nithyananda K. V.

General Instructions to the Examiners:

Note: Practical examination will be of three hours duration divided into two parts (i.e. half hour for written and next two and half hours for performance).

1. Candidate will be examined based on any two experiments given by the examiner. Each experiment carries 08 marks (3 written + 5 performance).
2. Viva-Voce: 04 marks.
3. Note Book: 02 marks.

Semester	**6**
Code of Paper	**ICH-DSC-8-MIN-602**
Title of Paper	**Pharmaceuticals**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100 (75 Theory + 25 Practical)**
Marks distribution of the Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of the Practical paper	**End semester = 22** **Internal Marks = 03**
Contact Hours	**Theory: 3 Hrs. Per week** **Practical: 2 Hrs. Per week**

Applicable to the students

3. **Those who have opted for Industrial chemistry as a Major subject in Table 2.**
4. **Those who have opted for Industrial chemistry as a Minor subject in Table 1 or Table 2.**

LEARNING OUTCOMES SPECIFIC TO DISCIPLINE

By studying this course, the students will be able to:

- Explain what are the different types of pharmacopoeia.
- Explain the theory behind various types of excipients, phytochemicals and crude drugs.
- The students will be able to explain the different types of isolation processes and chemical constituents of plants.
- Emphasis is given to developing practical skills that will aid the student to set up accurate and reproducible, result-oriented experiments in higher classes.

GENERIC OUTCOMES

The third year of the course is designed to heighten the chemical temperament amongst the students. This is done by teaching them how to collect samples/data/evidence, carry out data analysis, evaluation and draw logical conclusions. They are also encouraged to work and improve upon their communication skills, through proper listening, plenty of hands-on laboratory exposure, group discussions and individual expression.

For practical work, groups shall be constituted, with student strength not exceeding 20 students per group.

Semester – 6
Syllabus for Theory Paper
ICH-DSC-8-MIN-602
Pharmaceuticals

End semester = 67 **Credits – 03**

Internal Marks = 08 **Contact Hours – 3/week**

Examination Time allowed **3 Hours**

UNIT – I Indian Pharmacopoeia (15 Hrs.)

Historical background and development of Pharmaceutical Industry in India.

Pharmacopoeias: Development of India Pharmacopoeia and Introduction to B.P., U.S.P., E.P., N.F. and other important pharmacopoeias.

Tools related to Chemistry: CHEMDRAW, Mercury. Use and management of chemical data (Scoupus, Web of Science)

Scientific ethics: Plagiarism, Scientific integrity, data integrity.

Excipients: Various types of pharmaceutical excipients; their Chemistry, process of manufacture and quality specifications- Glidants, lubricants, diluents, preservatives, antioxidants, emulsifying agents, coating agents, binders, colouring agents, flavouring agents, gelatin and other additives, sorbitol, mannitol, viscosity builders, etc.

UNIT – II Regulatory authority and Analysis (15 Hrs.)

Pharmaceutical packaging: FDA, Important schedules and some legal aspects of drugs

Phytochemicals: Introduction to plant classification and Crude drugs cultivation collection preparation for the market and storage of medicinal plants.

Crude Drugs: Evaluation of crude drugs-Moisture content Extractive Value, volatile oil content, foreign organic matter.

Quantitative analysis: Quantitative microscopic exercises including of starch leaf content (Palisade ratio stomatal number and index vein islet number and vein termination number) crude fibre content.
Introduction to chromatographic method for identification of crude drugs.

UNIT – III Plant analysis and Drugs (15 Hrs.)

Chemical constitution of plants: including carbohydrates, aminoacids, proteins fats waxes, volatile oils, terpenoids, 215or acti, saponins, flavonoids, tannins, glycosides, alkaloids.
Isolation Procedures: Various isolation procedures 215 or active ingradients with example for alkaloid e.g., vinca alkaloids, reserpine, sapogenin.

Pharmaceutical quality control (other than the Analytical methods covered under core subject) Sterility testing, Pyrogenic testing glass testing bulk density of powder etc. Effluent Handling

Classification of various types of drugs with examples.

Sulphadrugs-Sulphaguandine, sulphamethoxazele
Antimicrobial-chloramphenicol, tetracycline, Penicillin- G and their monographs.
NSAID-salicyclicacid and its derivatives, Ibuprofen, mefenamic acid.

Books/Literature suggested

1. Ramstead E, Modren Pharmacognosy, Pubs:McGraw-Hill,1959.
2. Lemke T. L., Williams D.A., Foye's Principles of Medicinal Chemistry, Pubs: Lippincott Williams & Wilkins, 2007.
3. Lednicer D, Strategies for Organic Drug Synthesis and Design, Second Edition, Pubs: Wiely Interscience, 2008.
4. Block **J,** Beale J(Eds.), Wilson and Gisvold's Textbook of Organic and Pharmaceutical Chemistry, Pubs: Lippincott Williams & Wilkins, 2003.
5. Indian Pharmacopoeia Commission, Indian Pharmacopoeia 2007, 3 Vols. With One Supplement (Addendum 2008).
6. The British Pharmacopoeia 2010.

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit** (of **19 marks**) will consist of two types of questions: **9 multiple-choice questions** of one mark each and **5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:** the **compulsory question** and one question each from the **UNIT-1, UNIT-2 and UNIT-3**.
2. The use of scientific calculators is not allowed.

Semester – 6
Syllabus for Practical Paper
ICH-DSC-8-MIN-602
Pharmaceuticals

End semester = 22 **Credit – 01**
Internal Marks = 03 **Contact Hours – 2/week**
Examination Time allowed **3 Hours**

1. Study of monographs of antimicrobial and NSAID.
2. Complete I.P. monograph of three drugs representing variety of testing methods.
3. Report Write compound data and chemical procedures for drugs.
4. Limit tests for chlorine, heavy metals, arsenic Etc. of two representative bulk drugs.
5. Active ingredient analysis of few types of formulations.
6. Representing different methods of analysis Acidimetry, Alkalimetry, nonaqueous complexometry, potentiometry etc.
7. Determination of sulphate ash, loss on drying and other tests of bulk drugs.
8. Power point Presentations

Books/Literature suggested

1. Vogel's Qualitative Inorganic Analysis (revised); 7th Longman, 1996. Edition Pubs: Orient
2. Furniss, B.S., Hannaford, A.J., Smith, P.W.G., Tatchell, A.R., Vogel's Textbook of Practical Organic Chemistry; 5th edition, Pubs: Dorling Kindersley (India) Pvt. Ltd. Delhi., 2006
3. Bassett, J., Denney, R.C., Jeffery, G.H., Mendham, J., Vogel's Textbook of Quantitative Inorganic Analysis (revised); 4 edition, 1978.
4. Palmer, W.G., Experimental Inorganic Chemistry; 1st edition, Pubs: Cambridge, 1954.
5. RD Bansal, R.K., Laboratory Manual of Organic chemistry; 1994. Limited, edition, Pubs: Wiley Eastern .
6. Khosla, B.D., Garg, V.C., Gulati, A., Senior Practical Physical Chemistry; 11 Pubs: R. Chand & Co., New Delhi, 2002. Edition
7. Das, R.C., Behra, B., Experimental Physical Chemistry; Pubs: Tata McGraw Hill Publishing Co. Ltd., 1998.
8. Levitt, B.P., Findlays Practical Physical Chemistry; 8th Ltd., London & New York, 1978. Edition, Pubs: Longman Group

General Instructions to the Examiners:

Note: Practical examination will be of three hours duration divided into two parts (i.e. half hour for written and next two and half hours for performance).

1. Candidate will be examined based on any two experiments given by the examiner. Each experiment carries 08 marks (3 written + 5 performance).
2. Viva-Voce: 04 marks.
3. Note Book: 02 marks.

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MATHEMATICS

AS PER NEP (2020) Syllabus of **Mathematics** for UG Classes

Table-1 (SCIENCE TEMPLETE)

Sem.	Level	MAJOR (4 Cr)MINOR (4 Cr)		MDC(3 Cr) Any paper in each of the first three semesters	VAC(2Cr) Required three papers, one for each in 1st, 2nd, & 4th semesters	SEC(3Cr) Required three papers, one for each of the first three semesters	AEC (2Cr)	PBI / HCP (2Cr)
I	100	• MAT-DSC1 (MAJ/MIN)-101: Algebra and Trigonometry	• MAT-DSC1 (MAJ/MIN)-101: Algebra and Trigonometry	• MAT-MDC-101:BasicMathematics • MAT-MDC-102:Calculus • MAT-MDC-103:Algebra	• MAT-VAC-101:Vedic Mathematics • MAT-VAC-102:Vedic Algebra • MAT-VAC-103:Theory of Numbers • MAT-VAC-104:Theory of Numbers with html	• MAT-SEC-101:Maths Based Practicals • MAT-SEC-102:Maths Based Programming And Practicals • MAT-SEC-103:Latex And Html • MAT-SEC-104:Quantitative Aptitude • MAT-SEC-105:Discrete Mathematics	Eng -1	Pbi-1/ HCP-1
II	100	• MAT-DSC2 (MAJ/MIN)-201: Calculus-I	• MAT-DSC2 (MAJ/MIN)-201: Calculus-I				Eng -2	Pbi-2/ HCP-2
III	200	• MAT-DSC3 (MAJ/MIN)-301: Theory of Equations • MAT-DSC3 (MAJ)-302: Differential Equations-I	• MAT-DSC3 (MAJ/MIN)-301: Theory of Equations • MAT-DSC3 (MAJ)-302: Differential Equations-I				Eng-3/ MIL-1	
IV	200	• MAT-DSC4 (MAJ/MIN)-401: Calculus-II • MAT-DSC4 (MAJ)-402: Discrete Mathematics and Graph Theory • MAT-DSC4 (MAJ)-403: Coordinate Geometry-I • MAT-DSC4 (MAJ)-404: Numerical Analysis	• MAT-DSC4 (MAJ/MIN)-401: Calculus-II				Eng-4/ MIL-2	
V	300	• MAT-DSC5 (MAJ/MIN)-501: Linear Algebra • MAT-DSC5 (MAJ)-502: Sequence and Series of real numbers • MAT-DSC5 (MAJ)-503: Differential Equations-II • MAT-DSC5 (MAJ)-504: Solid Geometry • MAT-DSC5 (MAJ)-505: Statics			Internship / community work			
VI	300	• MAT-DSC6 (MAJ/MIN)-602: Analysis • MAT-DSC6 (MAJ)-603: Linear Programming • MAT-DSC6 (MAJ)-604: Probability Theory • MAT-DSC6 (MAJ)-605: Dynamics • MAT-DSC6 (MAJ)-606: Modern Algebra	MAT-DSC6 (MAJ/MIN)-601: Analytical Geometry					

Table-2 (SCIENCE TEMPLET)

MULTI-DISCIPLINARY / INTER-DISCIPLINARY**. BSc, BSc HONOURS AND BSc HONOURS WITH RESEARCH**

Sem.	Level	MAJOR (4 Cr)	MINOR (4 Cr)	MDC(3 Cr) Any paper in each of the first three semesters	VAC(2Cr) Required three papers, one for each in 1st, 2nd, & 4th semesters	SEC(3Cr) Required three papers, one for each of the first three semesters	AEC (2Cr)	PBI / HCP (2Cr)
I	100	• MAT-DSC1 (MAJ/MIN)-101: Algebra and Trigonometry	• MAT-DSC1 (MAJ/MIN)-101: Algebra and Trigonometry	• MAT-MDC-101:BasicMathematics • MAT-MDC-102:Calculus • MAT-MDC-103:Algebra	• MAT-VAC-101:Vedic Mathematics • MAT-VAC-102:Vedic Algebra • MAT-VAC-103:Theory of Numbers • MAT-VAC-104:Theory of Numbers with html	• MAT-SEC-101:Maths Based Practicals • MAT-SEC-102:Maths Based Programming And Practicals • MAT-SEC-103:Latex And Html • MAT-SEC-104:Quantitative Aptitude • MAT-SEC-105:Discrete Mathematics	Eng -1	Pbi-1/ HCP-1
II	100	• MAT-DSC2 (MAJ/MIN)-201: Calculus-I	• MAT-DSC2 (MAJ/MIN)-201: Calculus-I				Eng -2	Pbi-2/ HCP-2
III	200	• MAT-DSC3 (MAJ/MIN)-301: Theory of Equations	• MAT-DSC3 (MAJ/MIN)-301: Theory of Equations • MAT-DSC3 (MAJ)-302: Differential Equations-I				Eng-3/ MIL-1	
IV	200	• MAT-DSC4 (MAJ/MIN)-401: Calculus-II • MAT-DSC4 (MAJ)-402: Differential Equations-I	• MAT-DSC4 (MAJ/MIN)-401: Calculus-II				Eng-4/ MIL-2	
V	300	• MAT-DSC5 (MAJ/MIN)-501: Linear Algebra • MAT-DSC5 (MAJ)-502: Sequence and Series of real numbers	• MAT-DSC5 (MAJ/MIN)-501: Linear Algebra		Internship / community work			
VI	300	• MAT-DSC6 (MAJ/MIN)-601: Analytical Geometry • MAT-DSC6 (MAJ)-602: Analysis	• MAT-DSC6 (MAJ/MIN)-601: Analytical Geometry • MAT-DSC6 (MAJ)-602: Analysis					

Notes.

1. Members of the Board of Studies unanimously decided not to offer mathematics in Table-3 of Arts stream.
2. Practicals (wherever included in the syllabi) may be conducted into small groups(say 15 – 20 students)to facilitate the smooth conduct of practical sessions.
3. Scientific Calculator is allowed in the Numerical Analysis Paper.
   IMPORTANT NOTE: All papers in VAC/SEC/MDC would be floated in each of the first three semesters. However, the students are not allowed to repeat any paper.

Semester - V

MAT-DSC5 (MAJ/MIN)-501: LINEAR ALGEBRA

Credits = 4

Teaching hours = 6 hrs / week including tutorials /practicals etc. in a semester of 15 weeks.

Exam Time: 3 hours

Max. Marks: 100 (External: 80 + Internal Assessment: 20)

Note:
1. The syllabus has been split into Four Units. The examiner will set in total nine questions including a compulsory question. The compulsory question (QuestionNo.1) will contain 4 parts covering the entire syllabus. Two questions will be set from each Unit.
2. A student will be required to attempt five questions in all, selecting one question from each Unit and the compulsory question. Each question will carry equal marks.
3. The 20 marks of internal assessment has following division

Mid Semester Exam = 10 marks,
Assignment / Project work = 5 marks,
Classroom participation = 3 marks**,**
Attendance = 2 marks for more than 85%and 1mark for more than 75% attendance.

Objective. This syllabus aims to introduce students to the foundational concepts and structures of linear algebra, with a focus on vector spaces, linear transformations, matrices, and inner product spaces. The course is designed to equip students with the theoretical understanding and practical skills necessary to apply these concepts in various mathematical fields.

Learning Outcomes:

1. Understand and apply the concepts of vector spaces, subspaces, and their properties, including basis, dimension, and direct sums.
2. Analyze and solve problems related to linear transformations, including the use of the Rank-Nullity theorem and invertible transformations.
3. Use matrix representations to solve problems involving linear transformations, characteristic roots, and change of basis.
4. Apply advanced techniques like the Cayley-Hamilton theorem, orthogonal projections, and the Gram-Schmidt process to diagonalize matrices and work with inner product spaces.

Unit-I

Definition of a Group, Ring, Field and examples.Definition and Examples of Vector Spaces, Subspaces, Algebra of subspaces, Linear span, Linear dependence and independence of vectors, Basis and dimension of a vector space and its subspaces. Direct sums and complements.

Unit-II

Linear transformations, kernel and range of linear transformation Rank and Nullity of a linear transformation, Rank-Nullity theorem, invariant sub space, Vector space of linear transformations, invertible linear transformations.

Unit-III

Linear transformations and matrices, Matrix representation of a linear transformation relative to an ordered basis. Matrix of an inverse operator, Change of basis. Characteristic roots and characteristic vectors, Algebraic and Geometric multiplicity of characteristic roots.

Unit-IV

Cayley-Hamilton theorem, Diagonalizable operators and matrices. Minimal polynomial of a linear operator (matrix). Inner product and inner product spaces, Orthogonal sets, Orthogonal projections, The Gram Schmidt process.

References:

1. **K. Hoffman and R. Kunze**: *Linear Algebra*, 2nd Edition, Prentice Hall, New Jersey, 1971.
2. **P. B. Bhattacharya, S. K. Jain, and S. R. Nagpaul**: *First Course in Linear Algebra* (Wiley Eastern, Delhi).
3. **J. Gilbert and L. Gilbert**: *Linear Algebra and Matrix Theory* (Academic Press).
4. **J. Gilbert**: *Linear Algebra and Matrix Theory* (Academic Press).
5. **I. N. Herstein**: *Topics in Algebra* (Delhi, Vikas).
6. **V. Bist and V. Sahai**: *Linear Algebra* (Narosa, Delhi).

Semester - V

*MAT-DSC5 (MAJ)-502: SEQUENCE AND SERIES OF REAL NUMBERS

Credits = 4

Teaching hours = 6 hrs / week including tutorials /practicals etc. in a semester of 15 weeks.

Exam Time: 3 hours

Max. Marks: 100 (External: 80 + Internal Assessment: 20)

Note: 1. The syllabus has been split into Four Units. The examiner will set in total nine questions including a compulsory question. The compulsory question (QuestionNo.1) will contain 4 parts covering the entire syllabus. Two questions will be set from each Unit.

2. A student will be required to attempt five questions in all, selecting one question from each Unit and the compulsory question. Each question will carry equal marks.

3. The 20 marks of internal assessment has following division

Mid Semester Exam = 10 marks,

Assignment / Project work = 5 marks,

Classroom participation = 3 marks**,**

Attendance = 2 marks for more than 85%and 1mark for more than 75% attendance.

Objective: The objective of this course is to develop a comprehensive understanding of sequences and series, emphasizing their convergence properties and criteria. Students will learn to analyze and test the convergence of sequences and series using various methods, including Cauchy's Criterion, Condensation Test, Integral Test, and Root Test etc. The course will also cover practical applications of convergence, preparing students for advanced mathematical studies and research.

Learning Outcomes: By the end of this course, students will be able to:

1. Understand the fundamental properties of Sequences and Series
2. Analyze convergence of sequences and series by using established tests including Cauchy's Condensation Test, Integral Test, and Root Test, to determine the behavior of series.
3. Develop skills for advanced mathematical proofs and Problem-Solving

Unit-I

Definition of a real sequence, Bounds of a sequence, Convergent, Divergent and Oscillatory sequences. Limit of a sequence, Algebra of limits, sandwich theorem or squeeze principal, application of squeeze principal, Monotonic Sequences, Theorems on monotonic sequences, applications of monotonic sequence in numerical questions, Cauchy's theorems on limits. Cauchy's first theorem on limits, Cauchy's seconds theorem on limits. Application of Cauchy's theorems on limits.

*For students of BA / BSc who follow either template-1 or opted mathematics as a major subject in template-2.

Unit-II

Subsequences of a real sequence, subsequential limit (cluster point), peak point, Bolzano-Weierstrass Theorem, Cauchy's sequence, Cauchy's general principal of convergence.Application of Cauchy's sequence in numerical questions of real sequences.

Sequential continuity and uniform continuity of functions of single real variable. Numerical questions related to sequential continuity and uniform continuity of function.

Unit-III

Infinite Series, convergence of series, convergence of geometric series,Series of non-negative terms. Convergences of series of non-negative terms, P-Test. Comparison tests, Cauchy's Condensation test, Cauchy's Integral test, Cauchy's Root test.

Unit-IV

Ratio tests: Kummer's Test, D'Alembert's test, Raabe's test, De Morgan and Bertrand's test, Gauss Test, Logarithmic test. Alternating series. Leibnitz's theorem. Absolute and conditional convergence, Rearrangement of absolutely convergent series, Riemann's rearrangement theorem.

References:

1. **D. Soma Sundaramand B. Choudhary**:A First Course in Mathematical Analysis, Narosa PublishingHouse, New Delhi 1997.
2. **P.K. Jain and S.K. Kaushik**:AnIntroductiontoRealAnalysis,S.Chand&Co.,NewDelhi2000.
3. **MurrayR.Spiegel**:TheoryandProblemsofAdvancedCalculus,SchaumPublishing Co.,NewYork.
4. **S.C.Malik**:MathematicalAnalysis,WileyEasternLtd.,NewDelhi.
5. **O.E.Stanaitis**:AnIntroductiontoSequences,SeriesandImproperIntegrals,Holden–Dey,Inc.,SanFrancisco,California.
6. **Earl D.Rainville**:InfiniteSeries,TheMacmillanCompany,NewYork.
7. **N. Piskunov**:DifferentialandIntegralCalculus,PeacePublishers,Moscow.

Semester - V

*MAT-DSC5 (MAJ)-503: DIFFERENTIAL EQUATION-II

Credits = 4

Teaching hours = 6 hrs / week including tutorials /practicals etc. in a semester of 15 weeks.

Exam Time: 3 hours

Max. Marks: 100 (External: 80 + Internal Assessment: 20)

Note: 1. The syllabus has been split into Four Units. The examiner will set in total nine questions including a compulsory question. The compulsory question (QuestionNo.1) will contain 4 parts covering the entire syllabus. Two questions will be set from each Unit.

2. A student will be required to attempt five questions in all, selecting one question from each Unit and the compulsory question. Each question will carry equal marks.

3. The 20 marks of internal assessment has following division

Mid Semester Exam = 10 marks,

Assignment / Project work = 5 marks,

Classroom participation = 3 marks**,**

Attendance = 2 marks for more than 85%and 1mark for more than 75% attendance.

Objective. The objective of this course is to introduce students to special types of differential equations, particularly Bessel and Legendre equations, and solving these using power series methods. The course will explore the properties of these solutions, such as recurrence relations and orthogonality. Additionally, students will learn partial differential equations. An important technique of solving differential equations by using Laplace and Inverse Laplace transformations will be discussed.

Learning Outcomes: By the end of this course, students will be able to:

1. Identify and solve Bessel and Legendre differential equations using power series methods.
2. Analyze the properties of Bessel and Legendre functions, including recurrence relations and orthogonality.
3. Understand the formulation and solution of partial differential equations.
4. Apply Laplace and Inverse Laplace transformations to solve ordinary differential equations.

Unit-I

Series solution of differential equations, Power Series and Frobenius method, Bessel and Legendre equations. Bessel functions of First and Second kind. Legendre function. Generating function. Equation reducible to Bessel's equation and its solution.

*Only for those students of BA/BSc who follow template-1.

Unit-II

Recurrence relation and orthogonality of Bessel and Legendre function. Partial Differential Equations: Origin of first order Partial Differential Equations, Linear Equation of first order, Integral surfaces passing through a given curve, surfaces orthogonal to a given system of surfaces.

Unit-III

Laplace Transformation-Linearity of the Laplace transformation. Existence theorem for Laplace transformations, Shifting Theorems, Laplace transforms of derivatives and integrals, Multiplication of t^n, division by t.

Unit-IV

Inverse Laplace Transforms-Linearity property, shifting properties, Change of Scale Property. Inverse Laplace transforms of derivatives and integrals, Convolution theorem. (without proof) Applications of Laplace Transforms- solution of integral equation, Solution of differential equations with constant coefficients, Solution of differential equations with variable coefficients,

References:

1. **Erwin Kreyszig**: Advanced Engineering Mathematics, John Wiley & Sons, New York, 1999.
2. **D. A. Murray**: Introductory Course on Differential Equations, Orient Longmen, (India) 1967.
3. **A. R. Forsyth**: A Treatise on Differential Equations, Macmillan and Co. Ltd., London.
4. **Sneddon, I. N.**: Elements of Partial Differential Equations, McGraw Hill, 1957.
5. **J. D. Murray & M.R. Spiegel**:Schaum Series, Laplace Transforms.

Semester - V

*MAT-DSC5 (MAJ)-504: SOLID GEOMETRY

Credits = 4

Teaching hours = 6 hrs / week including tutorials /practicals etc. in a semester of 15 weeks.

Exam Time: 3 hours

Max. Marks: 100 (External: 80 + Internal Assessment: 20)

Note: 1. The syllabus has been split into Four Units. The examiner will set in total nine questions including a compulsory question. The compulsory question (QuestionNo.1) will contain 4 parts covering the entire syllabus. Two questions will be set from each Unit.

2. A student will be required to attempt five questions in all, selecting one question from each Unit and the compulsory question. Each question will carry equal marks.

3. The 20 marks of internal assessment has following division

Mid Semester Exam = 10 marks,

Assignment / Project work = 5 marks,

Classroom participation = 3 marks**,**

Attendance = 2 marks for more than 85%and 1mark for more than 75% attendance.

Objective: The main goal of this course is to deliver basics of the solid geometry. The course introduces the concepts of plane, sphere, cone and cylinder. Methods will be taught to write various kinds of equations of these objects embedded in multiple ways in the 3-dimensional space.

Learning Outcomes:

The students will be able to write equations of plane, sphere, cone, cylinder and conicoids. Students will learn various properties of these objects, their tangents and normals.

Unit-I

Sphere:Section of a sphere and a plane, spheres through a given circle, intersection of a line and a sphere, tangent line,tangent plane, angle of intersection of two spheres and condition of orthogonality, power ofa point w.r.t. asphere, radicalaxis,radicalcenter,co-axialfamilyofspheres, limitingpoints.

*Only for those students of BA/BSc who follow template-1.

Unit-II

Transformationofaxes: Shifting of origin and rotation of axes.

Cylinder: Cylinder as a surface generated by a line moving parallel to a fixed line and through a fixed curve, differentkindsofcylinderssuchasrightcircular,elliptic,parabolicandhyperboliccylindersinstandardforms,envelopingcylinders.

Unit-III

Cone:Cone with a vertex at the origin as the graph of a homogeneous equation of second degree in x,y,z, cone as asurface generated by a line passing through a fixed curve and a fixed point outside the plane of the curve,Reciprocal cones, elliptic cones, ,envelopingcones.

Unit-IV

Right circular cones, right circular cone as a surface of revolution obtained byrotatingthe curve ina plane about anaxis.

Conicoid:Equationsofellipsoid,hyperboloidandparaboloidinstandardform.Reductionofseconddegreeequationinthreevariablesinstandardform.

References:

1. **P.K.Jain and Khalil Ahmad**: A Text Book of Analytical Geometry of Three Dimensions,Wiley Eastern Ltd., 1999.
2. **Shanti Narayan and P. K. Mittal**: Analytical Solid Geometry, Seventeenth Revised Edition, S. Chand & Co., New Delhi, 2006.
3. **R.J.T. Bill**: Elementary Treatise on Coordinate Geometry of Three Dimensions, Macmillan India Ltd., 1994.
4. **G. B. Thomas and R. L. Finney:** Calculus and Analytic Geometry, Addison-Wesley Publishing Company.

Semester - V

*MAT-DSC5 (MAJ)-505: STATICS

Credits = 4

Teaching hours = 6 hrs / week including tutorials /practicals etc. in a semester of 15 weeks.

Exam Time: 3 hours

Max. Marks: 100 (External: 80 + Internal Assessment: 20)

Note: 1. The syllabus has been split into Four Units. The examiner will set in total nine questions including a compulsory question. The compulsory question (QuestionNo.1) will contain 4 parts covering the entire syllabus. Two questions will be set from each Unit.

2.A student will be required to attempt five questions in all, selecting one question from each Unit and the compulsory question. Each question will carry equal marks.

3. The 20 marks of internal assessment has following division

Mid Semester Exam = 10 marks,

Assignment / Project work = 5 marks,

Classroom participation = 3 marks**,**

Attendance = 2 marks for more than 85%and 1mark for more than 75% attendance.

Objective: The objective of this course is to provide students with a fundamental understanding of forces, their effects on bodies at rest, and the conditions required for equilibrium. Key topics include the Parallelogram Law, the Triangle Law, Lami's Theorem, and Varignon's Theorem, along with the study of moments, couples, and friction.

Learning Outcomes: Upon completion of this course, students will be able to:

1. Apply fundamentallaws, such as the Parallelogram and Triangle Laws, to resolve and combine forces acting on a body.
2. Determine the resultant and equilibrium conditions for coplanar forces, including concurrent and non-concurrent force systems.
3. Analyze moments, couples, and use Varignon's Theorem to simplify complex force systems.

Unit-I

Basic notions. Composition and resolution of concurrent forces–Parallelogramlawof forces, Components of a force in given directions, Resolved parts of a force, Parallel Forces, resultant of two like (unlike) parallel forces acting at distinct points and its applications.

*Only for those students of BA / BSc who follow template-1.

Unit-II

Resultant of any number of coplanar concurrent forces, Equilibriumconditionsforcoplanarconcurrentforces, equilibrium ofabodyrestingonasmoothinclined plane. Equilibrium of three forces acting at a point – Triangle law of forces,converse of Triangle law of Forces, λ–μtheorem, Lami'stheorem.

Unit-III

Moments and Couples – Moment of a force about a point and a line, Centre of Parallel forces, theorems on moment of a couple, Equivalent couples, Varignon's theorem, generalized theoremof moments, resultant of a force and a couple, resolution of a force into a force and a couple, reduction of a system of coplanar forces to a force and a couple.

Unit-IV

Equilibrium conditions for any number of coplanar non-concurrent forces.Friction; definition and nature of friction, laws of friction, equilibriumofa particle ona roughplane, Problems on ladders, rods, spheres and circles.

References

1. **S.L.Loney**:Statics,MacmillanandCompany,London.
2. **R.S.Verma**:ATextBookonStatics,PothishalaPvt.Ltd.,Allahabad.
3. **K.R.Chaudhery and A.C. Aggarwal**:Elementsof Mechanics, Statics and Dynamics, S. Chandand Company.
4. **S. L. Loney**:TheelementsofStaticsandDynamics,5th edition, Cambridge University Press, 1947.

Semester - VI

MAT-DSC6 (MAJ/MIN)-601: ANALYTICALGEOMETRY

Credits = 4

Teaching hours = 6 hrs / week including tutorials /practicals etc. in a semester of 15 weeks.

Exam Time: 3 hours

Max. Marks: 100 (External: 80 + Internal Assessment: 20)

Note: 1. The syllabus has been split into Four Units. The examiner will set in total nine questions including a compulsory question. The compulsory question (QuestionNo.1) will contain 4 parts covering the entire syllabus. Two questions will be set from each Unit.

2.A student will be required to attempt five questions in all, selecting one question from each Unit and the compulsory question. Each question will carry equal marks.

3. The 20 marks of internal assessment has following division

Mid Semester Exam = 10 marks,

Assignment / Project work = 5 marks,

Classroom participation = 3 marks**,**

Attendance = 2 marks for more than 85%and 1mark for more than 75% attendance.

Objective: The main goal of this course is to deliver basics of plane geometry. The course introduces the concept of pair of straight lines, circles, conicsections, sphere, cone and cylinder. Methods will be taught to write equations of these objects whenever requisite information is given.

Learning Outcomes:Students will be able to write joint equation of a pair of straight lines, equations of circle and other conics, equations of tangents, normal and chord of contact in case of circle and other conics. Equations and various properties of sphere, cone, cylinder and their tangents and normal will be learnt.

.Unit-I

Change of axes: Translation and rotation of axes.

Pairof Straightlines: Equation of a straight line in polar coordinates,Joint equation of pair of straight lines (Cartesian coordinates), angle between them, Condition of parallelism and perpendicularity.

Circle: Generalequationofcircle, parametric equations,tangents,normals,chordofcontact,pairoftangentsfroma point,equationofchordintermsof mid-point.

Scope. Chapters 3, 4 and 5 of [1].

Unit-II

Conic sections : Equations of parabola, ellipse and hyperbola in Cartesian coordinate system, parametric forms of conics, equations of their tangents,normal,chordsofcontact,poleandpolar,tangentsfroma point,equationofchordintermsofmid-point, some important results and properties of conics (e.g. see [1]).

Scope. Chapters 7, 8 and 9 of [1].

Unit-III

More on conic section. Polar equation of conic, diameter and conjugate diameters ofthe conics. Special propertiesof Parabola, Hyperbola and Ellipse. Conjugate hyperbola, asymptotesto thehyperbola, rectangularhyperbola. Identification of conic in general second degree equations.

Unit-IV

Sphere, Cone, Cylinder. Sphere and its intersections with plane. Polar plane, radical plane, intersection of two spheres. Equation of a cone, intersections of cone with a plane. Enveloping cone, right circular cone. Equation of a cylinder, Enveloping cylinder, right circular cylinder.

Scope. Chapters 15, 16 and 17 of [1].

References:

1. **P.K. Jain and Khalil Ahmad**:*A Text Book of Analytical Geometry,*New Age International (P) Ltd. Publishers, London, New Delhi, Nairobi 2018.

2. **S.L. Loney**:*The Elements of Coordinate Geometry,* Macmillan andCompany,London, 2nd Edition 2007.

3. **Erwin Kreyszig**:*Advanced Engineering Mathematics, John Wiley & Sons,1999.*

4. **Gorakh Prasad and H.C. Gupta**:*Text Book on Coordinate Geometry, Pothishala Pvt. Ltd., Allahabad, 1955.*

5. **G. B. Thomas and R. L. Finney:** Calculus and Analytic Geometry, Addison-Wesley Publishing Company.

Semester - VI

*MAT-DSC6 (MAJ/MIN)-602: ANALYSIS

Credits = 4

Teaching hours = 6 hrs / week including tutorials /practicals etc. in a semester of 15 weeks.

Exam Time: 3 hours

Max. Marks: 100 (External: 80 + Internal Assessment: 20)

Note: 1. The syllabus has been split into Four Units. The examiner will set in total nine questions including a compulsory question. The compulsory question (QuestionNo.1) will contain 4 parts covering the entire syllabus. Two questions will be set from each Unit.

2.A student will be required to attempt five questions in all, selecting one question from each Unit and the compulsory question. Each question will carry equal marks.

3. The 20 marks of internal assessment has following division

Mid Semester Exam = 10 marks,

Assignment / Project work = 5 marks,

Classroom participation = 3 marks**,**

Attendance = 2 marks for more than 85%and 1mark for more than 75% attendance.

Objective: The objective of this course is to provide students with a solid foundation in advanced calculus, focusing on multivariable functions, special functions, integration, and vector calculus. Students will explore key theorems, including Euler's theorem on homogeneous functions, Taylor's theorem for multivariable functions, and the Mean Value Theorems of integral calculus. They will also study improper integrals, convergence tests, and special functions such as Beta and Gamma functions. The course further covers vector differentiation, equipping students with analytical skills applicable to physics, engineering, and advanced mathematics.

Learning Outcomes:Upon successful completion of this course, students will be able to:

1. Apply Euler's and Taylor's theorems to solve mathematical problems.
2. Differentiate between countable and uncountable sets.
3. Determine integrability conditions for functions and apply the fundamental theorem of calculus, mean value theorems, and change of variables in integral calculations.
4. Evaluate improper integrals and analyze their convergence using comparison, Abel's, and Dirichlet's tests.
5. Calculate gradients, divergence, and curl, and apply these concepts in solving problems in vector calculus.

Unit-I

Maxima and minima of functions of two variables, Taylor's theorem for functions of two variables. Jacobians. Envelopes. Evolutes.Countable and uncountable sets.

*Not for students of BA who opted mathematics as a minor subject in template-II.

Unit-II

Riemann integral, Integrability of continuous and monotonic functions, Integrability of a function having finite points of discontinuity, Properties of integrable functions. The fundamental theorem of integral calculus, change of variables, Mean value theorems of integral calculus (First Mean Value Theorem, Generalised first Mean Value Theorem, Bonet's Mean value Theorem, Weirerstrass' Mean Value Theorem).

Unit-III

Improper integrals and their convergence, Comparison tests, Absolute and conditional convergence,Abel's and Dirichlet's tests.Beta and Gamma functions.

Unit-IV

Frullani's integral,convergences Integral as a function of a parameter. Continuity, derivability and integrability of an integral of a function of a parameter. Vector differentiation, Gradient,Divergence and Curl with their properties and applications.

References:

1. **T. M. Apostol**:MathematicalAnalysis,NorosapublishingHouse,NewDelhi,1985.
2. **R.R. Goldberg**:RealAnalysis,Oxford&IBHPublishingCo.,NewDelhi,1970.
3. **S.Lang**:UndergraduateAnalysis,Springer-Verlag,NewYork,1983.
4. **D. Somasundaram and B. Choudhary**: AFirstCourseinMathematicalAnalysis,NarosaNewDelhi,1997.
5. **ShantiNarayan**:ACourseofMathematicalAnalysis,S.Chand&Co.NewDelhi.
6. **P.K. Jain and S.K. Kaushik**:AnIntroductiontoRealAnalysis,S.Chand&Co.,NewDelhi,2000.
7. **S.C.Malik and Savita Arora**:MathematicalAnalysis,2nd edition,NewAgeInternationalPublishers.
8. **G.B.Thomasand R.L. Finney**:CalculusandAnalyticGeometry(Ninthedition),PearsonPublication.

Semester - VI

*MAT-DSC6 (MAJ)-603: Linear Programming

Credits = 4

Teaching hours = 6 hrs / week including tutorials /practicals etc. in a semester of 15 weeks.

Exam Time: 3 hours

Max. Marks: 100 (External: 80 + Internal Assessment: 20)

Note: 1. The syllabus has been split into Four Units. The examiner will set in total nine questions including a compulsory question. The compulsory question (QuestionNo.1) will contain 4 parts covering the entire syllabus. Two questions will be set from each Unit.

2.A student will be required to attempt five questions in all, selecting one question from each Unit and the compulsory question. Each question will carry equal marks.

3. The 20 marks of internal assessment has following division

Mid Semester Exam = 10 marks,

Assignment / Project work = 5 marks,

Classroom participation = 3 marks**,**

Attendance = 2 marks for more than 85%and 1mark for more than 75% attendance.

Objective: This course is application oriented which introduces the concepts of linear programming to the students.

Course outcomes: After the completion of the course, the students will be able to formulate and solve linear programming problems.

UNIT-I

Introduction –General Linear programming problem (LPP) and examples, Formulation of LPP, Standard and canonical form of LPP, Convex set and its properties, Hyperplane, Open and Closed half-spaces, Feasible, Basic Feasible and Optimal Solutions, Extreme Point & graphical methods (Chapter-2, Reference [3])

UNIT-II

Simplex method, Artificial variable techniques- Two Phase Method; Big M Method, Special cases in LPP.
Finding Inverse of a matrix using Simplex method, Solving system of linear equations using Simplex method. (Chapter-3, Reference [3])

UNIT-III

Sensitivity analysis- removal of constraint, change in cost of basic or non-basic variable.Duality: Definition of the dual problem, Primal-dual relationships, Economic Interpretation of Duality, Dual simplex Method. (Chapter-4, Reference [3])

UNIT-IV

LP Formulation of Transportation Problem, Existence of Solution in Transportation Problem.
Assignment Problem, Mathematical Formulation of the Problem Test for Optimality by using Hungarian Method – Maximization Case in Assignment Problem

References:

1. **G. Hadley**: Linear Programming. Narosa Publishing House, 2002 (reprint).
2. **A. Ravindran, D. T. Phillips and James J. Solberg**: Operations Research- Principles and Practice, John Wiley & Sons, 2005.
3. **H. A. Taha**: Operations Research-An Introduction, Prentice Hall, 9th Edition, 2010.
4. **H. S. Kasana**: Introductory Operation Research: Theory and Applications, Springer Verlag ,2005.
5. **Kanti Swaroop, P.K. Gupta and Manmohan**: Operation Research, Sultan Chand and Sons, 2010.

*For students of BA / BSc who follow template-1 and opted mathematics as a major subject.

Semester - VI

*MAT-DSC6 (MAJ)-604: PROBABILITY THEORY

Credits = 4

Teaching hours = 6 hrs / week including tutorials /practicals etc. in a semester of 15 weeks.

Exam Time: 3 hours

Max. Marks: 100 (External: 80 + Internal Assessment: 20)

Note: 1. The syllabus has been split into Four Units. The examiner will set in total nine questions including a compulsory question. The compulsory question (QuestionNo.1) will contain 4 parts covering the entire syllabus. Two questions will be set from each Unit.

2.A student will be required to attempt five questions in all, selecting one question from each Unit and the compulsory question. Each question will carry equal marks.

3. The 20 marks of internal assessment has following division

Mid Semester Exam = 10 marks,

Assignment / Project work = 5 marks,

Classroom participation = 3 marks**,**

Attendance = 2 marks for more than 85%and 1mark for more than 75% attendance.

Objective: The objective of this course is tointroduce some useful topics,to the students, of probability theory, which will be helpful in solving problems in other areas of mathematics as well as in daily life. Continuous as well as discrete random variables will be studied at length. Many illustrations will be given to convince their applications in mathematics as well as in other areas.

Learning Outcomes:By the end of this course, students will be able to:

1. Compute mean, variance, moment generating functions.
2. Apply Bayes' theorem in computing probabilities.
3. Use random variable in computing probability.

Unit-I

ReviewofnotionofProbability,conditionalProbabilityandindependence,Bayes'Theorem.Random variables, probabilitydensity function, cumulative distribution function, discrete and continuous random variables, expectations, mean, variance, moment generating function, skewness and kurtosis.

Unit-II

DiscreteRandomVariables:Bernoullirandomvariable,binomialrandomvariable,negativebinomialrandom variable, geometric random variable, Poisson random variable.

*For students of BA / BSc who follow template-1 and opted mathematics as a major subject.

Unit-III

ContinuousRandomVariables:Uniformrandomvariable,exponentialrandomvariable,Betarandomvariable, Gamma random variable, Chi-square random variable, normal random variable**.**

Unit-IV

BivariateRandomVariables:Jointdistribution,jointandconditionaldistributions,ConditionalExpectations, Independent random variables, the correlation coefficient, Bivariate normal distribution.

References

Ross, S.M.:IntroductiontoProbabilityModels(Sixthedition)AcademicPress,1997.

Hogg., R.V.and Craig, A.T:Introduction to Mathematical Statistics, MacMillan, 2002.

I.Blake:AnIntroductiontoAppliedProbability,JohnWiley&Sons,1979.

J.Pitman:Probability,Narosa,1993.

P. L. Meyer:IntroductoryProbabilityandStatisticalApplications,2nd Edition,Oxford and IBH Publishing Co.

Semester - VI

*MAT-DSC6 (MAJ)-605: DYNAMICS

Credits = 4

Teaching hours = 6 hrs / week including tutorials /practicals etc. in a semester of 15 weeks.

Exam Time: 3 hours

Max. Marks: 100 (External: 80 + Internal Assessment: 20)

Note: 1. The syllabus has been split into Four Units. The examiner will set in total nine questions including a compulsory question. The compulsory question (QuestionNo.1) will contain 4 parts covering the entire syllabus. Two questions will be set from each Unit.

2.A student will be required to attempt five questions in all, selecting one question from each Unit and the compulsory question. Each question will carry equal marks.

3. The 20 marks of internal assessment has following division

Mid Semester Exam = 10 marks,

Assignment / Project work = 5 marks,

Classroom participation = 3 marks**,**

Attendance = 2 marks for more than 85%and 1mark for more than 75% attendance.

Objective: The objective of teaching Dynamics is to provide students with an understanding of the fundamental principles governing the motion of particles and rigid bodies. This includes studying linear and curvilinear motion, analyzing forces, and understanding energy transformations in dynamic systems. Key concepts include Newton's Laws of Motion, work-energy principles, impulse-momentum relationships, etc.

Learning Outcomes:By the end of this course, students will be able to:

1. Apply Newton's laws to solve problems involving motion with constant or variable acceleration.
2. Analyze simple harmonic motion and motion of particles connected by elastic strings.
3. Understand and calculate work, power, and potential energy, and evaluate work done in gravitational fields and conservative systems.
4. Analyze relative motion, including displacement, velocity, and acceleration in rotating reference frames.

Unit-I

Motion of a particle with constant acceleration, acceleration of falling bodies, motion under gravity, motion of a body projected vertically up wards: Newton's Laws of Motion, Motion of two particles connected by astring, motion along a smooth inclined plane.

*For students of BA / BSc who follow template-1 and opted mathematics as a major subject.

Unit-II

Constrainedmotion along a smooth inclined plane. Variable acceleration: Simple harmonic motion, elastic string.

Unit-III

Curvilinear motion of a particle in a plane: Definition of velocity and acceleration, projectiles, motion in a circle.Work, power, conservative fields and the potential energy, work done against gravity, potential energy of a gravitational field.

Unit-IV

Relative motion, relative displacement, velocity and acceleration, motion relative to a rotating frame of reference. Linear momentum, angular momentum, conservation of angular momentum, impulsive forces, principle of impulse and momentum, motion with respect to centre ofmass of a system of particles, collisions of elastic bodies, loss of energy during impact.

References:

1. **S.L.Loney**:AnElementaryTreatiseontheDynamicsofaParticleandofRigidbodies,CambridgeUniversityPress,1956.
2. **K.R.Chaudhery and A.C. Aggarwal**:ElementsofMechanics,StaticsandDynamics.SChandandCompany.
3. **S. L. Loney**:TheelementsofStaticsandDynamics,5th edition,Cambridge University Press, 1947.
4. **DonaldT. GreenWood**:PrinciplesofDynamics,SecondEdition,PrenticeHallofIndia.
5. **M. Ray**:ATextBookonDynamics,S.ChandandCompany-1989

Semester - VI

*MAT-DSC6 (MAJ)-606: MODREN ALGEBRA

Credits = 4

Teaching hours = 6 hrs / week including tutorials /practicals etc. in a semester of 15 weeks.

Exam Time: 3 hours

Max. Marks: 100 (External: 80 + Internal Assessment: 20)

Note: 1. The syllabus has been split into Four Units. The examiner will set in total nine questions including a compulsory question. The compulsory question (QuestionNo.1) will contain 4 parts covering the entire syllabus. Two questions will be set from each Unit.

2.A student will be required to attempt five questions in all, selecting one question from each Unit and the compulsory question. Each question will carry equal marks.

3. The 20 marks of internal assessment has following division

Mid Semester Exam = 10 marks,

Assignment / Project work = 5 marks,

Classroom participation = 3 marks**,**

Attendance = 2 marks for more than 85%and 1mark for more than 75% attendance.

Objective: The objective of this course is to introduce some foundational concepts from group theory and ring theory which include definitions of groups, subgroups, cosets, homomorphisms, class equation, rings, integral domains, ideals, quotient rings, and polynomial rings etc. Some important theorems will also be introduced. The application of these topics in other fields of mathematics will also be discussed.

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Learning Outcomes:By the end of this course, students will be able to:

1. Apply algebraic concepts in advanced mathematical contexts and related fields.
2. Analyze and construct homomorphisms and isomorphisms in group and ring theory,
3. Differentiate between rings, integral domains, and division rings, and work with ideals, quotient rings, and polynomial rings.

Unit-I

Groups, Elementary Properties of Groups, order of element of a group, cyclic group, Subgroups, Cosets and Lagrange's Theorem, Normal subgroups, Normalizer of of element in Group, Centre of group and Quotient Groups.

*For students who have opted mathematics as a major subject in table-1.

Unit-II

Group Homomorphismand Isomorphism, Theorems on Homomorphism, Permutation Groups, Caley's Theorem, cyclic Permutation, even and odd permutation, disjoint permutation, Alternating groups, Simplicity of An, n≥ 5 (without proof). Conjugate elements, conjugate class, Class equation of finite group and its applications. Similar permutation, relation between similar and conjugate permutation, number of conjugate class of S*n* is partition of *n*

Unit-III

Rings, Integral domains, unit element, Division Ring, some general properties of Rings,Characteristic of a ring, idempotent and nilpotent elements. Subrings, properties of sub ring, Centre of a ring , Ideals, properties of Ideals, principal Ideals, Product of Ideals, Modular Law.

Unit-IV

Prime and Maximal Ideals, Quotient Rings,Ring Homomorphisms, Types of Ring Homomorphism ,Theorems on Ring Homomorphism, Polynomial rings, Division algorithm in ring of polynomials, Roots of polynomials, remainder Theorem

References:

1. **S.Singh andQ.Zameerudin: *ModernAlgebra*,8th Edition,VikasPublication,NewDelhi.**
2. I.N.Herstein:*TopicsinAlgebra*,WileyEasternLtd.,NewDelhi.
3. J.A.Gallian: *ContemporaryAbstractAlgebra*,NarosaPublicationHouse,NewDelhi.
4. M.Artin:*Algebra*, Prentice Hall of India, New Delhi,1994.

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MICROBIOLOGY

NEP 2020 COLLEGES SCIENCE TEMPLATE: STUDENTS OPTING FOR SINGLE MAJOR AND SINGLE MINOR (Table 1*) and STUDENTS OPTING FOR MULTI-DISCIPLINARY/ INTER-DISCIPLINARY (Table 2*)

B.Sc., B.Sc. HONOURS AND B.Sc. HONOURS WITH RESEARCH IN MICROBIOLOGY

Table 1*						Table 2*					
Sem	**Level**	**Course Code**	**Course Name**	**Credits (Th+Pr+Tut)**	**Total Credits**	**Sem**	**Level**	**Course Code**	**Course Name**	**Credits (Th+Pr+Tut)**	**Total Credits**
V	L-300	DSC-MAJ	To be opted by the student	3+1=4	24	V	L-300	MIC-DSC-9 MIN - 501	Agro and Industrial Microbiology	3+1=4	24
		DSC-MAJ	To be opted by the student	3+1=4				DSC- MAJ	To be opted by the student	3+1=4	
		DSC-MAJ	To be opted by the student	3+1=4				DSC- MAJ	To be opted by the student	3+1=4	
		DSC-MAJ	To be opted by the student	3+1=4				DSC- MAJ	To be opted by the student	3+1=4	
		DSC- MAJ	To be opted by the student	3+1=4				DSC- MAJ	To be opted by the student	3+1=4	
		*(INT/COM-506)	Internship/CommunityWork	4				*(INT/COM-506)	Internship/Community Work	4	
VI	L-300	(DSC-14 /Min-601)	To be opted by the student	3+1=4	24	VI	L-300	MIC-DSC-14 MIN - 601	Food Microbiology	3+1=4	24
		DSC-MAJ	To be opted by the student	3+1=4				MIC-DSC-15-MIN - 602	Microbial Pathogenesis	3+1=4	
		DSC-MAJ	To be opted by the student	3+1=4				DSC- MAJ	To be opted by the student	3+1=4	
		DSC-MAJ	To be opted by the student	3+1=4				DSC- MAJ	To be opted by the student	3+1=4	
		DSC- MAJ	To be opted by the student	3+1=4				DSC- MAJ	To be opted by the student	3+1=4	
		DSC-MAJ	To be opted by the student	3+1=4				DSC- MAJ	To be opted by the student	3+1=4	
72 (Maj)+32(Min)+9(MDC)+8(AEC)+6(VAC)+9(SEC)+4(INT/COM)+4(PBI/HCP)											
The student successfully securing 144 credits will be awarded B.Sc.degree											

Notes:*The internship will consist of a 4-credit course (100 marks) comprising 30 hours (2 hours per week) of practical or field training. The internship may be completed within the parent institute or through institute/industrial training for 2–4 weeks, with a minimum requirement of 30 hours. Evaluation will be based on the project report and a detailed presentation of the internship. The internal assessment will constitute 10% of the total marks, as per university and institutional guidelines.

1. In the first semester student is offered three different subjects A, B & C.
2. At the end of second semester each student has the option to choose any one of the two streams namely Single Major or Multi-Disciplinary. The student choosing Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student who chooses multi-disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3-year degree) shall be eligible for 4 YEAR DEGREE B.Sc. HONOURS with research, on the condition that the Institution is a PU approved Research Centre
5. The student opting for Single Major must secure 50% credits of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for Minor in a particular subject, the student must attain atleast 12 credits in that subject(s).
6. The student exiting the programme after one year securing 48 credits will be awarded UG Certificate provided he/she secures 4 credits from summer internship/community work. Further, the student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secure 4 credits from summer internship. The student successfully securing 144 credits will be awarded BA degree.
7. Language paper chosen as MIL-1 and MIL-2 should be in the same subject.
8. Once Major and Minor papers are chosen in 3rd semester, the same will continue as Major and Minor subject in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write Dissertation of 08 credits in one of the two Major subjects.

SEMESTER V

MIC-DSC-9 MIN - 501: AGRO AND INDUSTRIAL MICROBIOLOGY (THEORY)

Total marks:75
Theory: 68
Internal assessment:07
Time: 3hours
Credits: 03

Instructions for paper setters and candidates:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit.
- Compulsory question will be of 12 marks. Rest all questions will be of 14 marks each

Objective And Learning outcome: On completion of this course, students will understand the role of agriculture as an industry and the importance of microbes in industrial and agro-industrial processes, including their isolation, identification, and preservation. They will be able to explain methods for improving industrial microorganisms through mutation, genetic recombination, and process optimization to enhance product yield. Learners will gain knowledge of microbial production of commercially important products such as vitamins, organic acids, antibiotics, amino acids, alcohol, and fermented foods.

UNIT I

Basic concept of agriculture as industry: Industrially important microbes, its screening, selection and identification.
Maintenance and preservation of industrially important microbial cultures.

UNIT II

Improvement programme of industrial microbes: Mutational programme, Development of auxotrophic mutants, protoplast fusion, recombinant techniques, selection pressure in maintaining the hyper producer, media formulation and process optimization of industrial and agro industrial microbes.

UNIT III

Microbes in agro industries: Introduction of primary and secondary metabolites, production of vitamin B12, alcohol, wine, cheese, citric acid, gluconic acid, antibiotics (penicillin), amino acid (lysine/glutamic acid)

UNIT IV

Emerging energy technologies in agro industries: production of vermiculture, composting, production of biofertilizers: Blue green algae, azolla, fungi, mycorrhiza (VAM), bacteria – Azospirrilum, Single cell proteins (bacterial, fungal and algal).

Reference Books:

1. Marwaha, S.S., Arora, J.K. (2003). Biotechnological strategies in agro-Processing (Asiatech publishers Inc. New Delhi, India). 40
2. Stanbury, P.F., Whitaker A. (2011) 2nd edition. Principles of fermentation technology.
3. Patel A.H. (2007). Industrial microbiology (New Age International Publishers).
4. Loessner, J. M, Jay J.M, D.A. Golden (2005). Modern food microbiology (Golden Springer).
5. Cappuccino J.G., Sherman N. (2007). Microbiology: A laboratory manual (Pearson Benjamin Cummings).

MIC-DSC-9 MIN - 501: AGRO AND INDUSTRIAL MICROBIOLOGY (PRACTICAL)

Total marks:25
Theory:22
Internal assessment:3
Time: 3 hours
Credits: 01

1. Isolation of industrially important microbes from soil/agro-waste samples
2. Screening of microbes for enzyme production (amylase, protease)
3. Glycerol stock preparation
4. Selection of auxotrophic mutants using replica plating technique
5. Fermentation for production of alcohol
6. Visit to vermicomposting unit

SEMESTER VI

MIC-DSC-14 MIN - 601: FOOD MICROBIOLOGY (THEORY)

Total marks:75
Theory: 68
Internal assessment:07
Time: 3 hours
Credits: 03

Instructions for paper setters and candidates:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit.
- Compulsory question will be of 12 marks. Rest all questions will be of 14 marks each

Objective And Learning outcome: On completion of this course, students will be able to understand the role of microorganisms in food, the factors affecting their growth, and the methods used for detection and enumeration of food-borne microbes. They will gain knowledge of the principles of food preservation and the causes of contamination and spoilage in different food products. Learners will be able to explain food adulterants, additives, fermented foods, and the concepts of probiotics and prebiotics. They will also develop an understanding of food- and water-borne diseases, microbial toxins, and their impact on food safety and public health.

UNIT – I

Food and Microorganisms: History and scope of Food microbiology, food as substrates for microbes, factors affecting growth of microorganisms, Detection of microorganisms in food: Sampling plan & procedure for microbial analysis; Qualitative methods to isolate pathogenic microorganisms; Quantitative methods for microbial enumeration: Direct enumeration, indirect estimations and standard and recommended methods; Introduction to Rapid and advanced estimation methods (Immunoassays, nucleic acid probe) for detection of pathogens.

UNIT – II

Principles of food preservation: Physical, chemical, and biological methods of preservations.
Contamination, preservation and spoilage of different kind of foods: Milk and milk products (milk, butter, yoghurt and cheese), beverages (beer, wine), meat and fish products (sausages, nuggets, frozen fish, canned fish), vegetables and fruits.

UNIT – III

Food adulterants and food additives: Major food adulterants, types and their methods of assay, food additives their function and uses, flavoring agents, coloring agents and vitamins as food additives.
Fermented foods and their production: Bakers yeast, Bread, cheese, yoghurt, tofu, miso, tempeh, sauerkraut, meat and alcoholic beverages (beer and wine)
Concept of Probiotics and prebiotics.

UNIT –IV

Food and water borne diseases: *Shigellosis*, *Salmonellosis*, *Cholera*.
Food borne intoxications: *Staphylococcal*, *Bacillus* and *Clostridium botulinum*
Exotoxins and Endotoxins, their mechanism of action

Reference Books:

1. Frazier W.C., Westhoff, D.C. (Ed). (1988). Food microbiology (McGraw-Hill).
2. Admas, M.R., Moss, M.O (2005). Food microbiology (Edition 3, Illustrated Publisher Royal Society of Chemistry).
3. SriLakshmi B. (2003) Food science (New Age International Publishers, India).
4. Jay J.M., M.J. Loessner, D.A. Golden. (2005). Modern food microbiology (Edition 7, Illustrated Publisher Springer).
5. B. Sivasankar (2004). Food processing and preservation (PHI Private Ltd, New Delhi).
6. Michael P. Doyle (1989). Food borne bacterial pathogens (Edition illustrated, Publisher Marcel Dekker).

MIC-DSC-14 MIN - 601: FOOD MICROBIOLOGY (PRACTICAL)

Total marks:25
Theory:22
Internal assessment:3
Time: 3 hours
Credits: 01

1. Isolation and identification of microorganisms in spoiled food (fungi and bacteria).
2. Inhibitory effect of low temperature on microbial growth.
3. Methylene blue reductase test (MBRT) for determination of quality of milk.
4. Identification of mushrooms by spore prints.
5. Checking the effect of pasteurization of milk by alkaline phosphatase.

SEMESTER VI

MIC-DSC- 15-MIN -602 : MICROBIAL PATHOGENESIS (THEORY)

Total marks:75
Theory: 67
Internal assessment:08
Time: 3hours
Credits: 03

Instructions for paper setters and candidates:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit.'
- Compulsory question will be of 12 marks. Rest all questions will be of 14 marks each

Objective and Learning outcome: This course introduces students to the mechanisms by which microorganisms cause disease, including host–pathogen interactions, virulence factors, immune evasion strategies, and molecular basis of pathogenicity. Emphasis is placed on medically important bacteria, viruses, fungi, and parasites, along with laboratory techniques used to study microbial pathogenesis. The student will be able to explain determinants of microbial pathogenicity and virulence, mechanisms of disease transmission and progression, Correlate virulence factors with clinical manifestations, Perform and interpret basic diagnostic and pathogenicity-related experiments, Apply biosafety and ethical principles in handling pathogens

UNIT-I

Concept of disease, infection, pathogenicity, virulence.
Commensalism, mutualism, opportunistic infections. Koch's postulates. Classification of pathogens: bacteria, viruses, fungi, parasites.
Normal human microbiota, dysbiosis, and disease association. Routes of transmission and portals of entry.

UNIT-II

Adherence, colonization, invasion, and dissemination.
Host defense mechanisms: innate and adaptive immunity (overview)
Microbial immune evasion strategies
Virulence factors: adhesins, invasins, enzymes
Toxins: exotoxins and endotoxins (mechanisms and examples)
Biofilms, quorum sensing, and pathogenicity islands

UNIT-III

Pathogenesis of major bacterial diseases: *Staphylococcus aureus, Streptococcus pyogenes, Mycobacterium tuberculosis, Salmonella, Shigella, pathogenic E. coli*
Emerging and re-emerging bacterial pathogens
Antibiotic resistance: mechanisms and clinical significance
Horizontal gene transfer and virulence evolution

UNIT-IV

General principles of viral pathogenesis: entry, replication, latency, oncogenesis
Pathogenesis of selected viral diseases: HIV, Hepatitis
Fungal pathogenesis: *Candida, Aspergillus, Cryptococcus*
Parasitic pathogenesis: *Plasmodium, Entamoeba histolytica*
Zoonotic diseases.

MIC-DSC- 15-MIN -602 : MICROBIAL PATHOGENESIS (PRACTICAL)

Total marks:25
Theory:22
Internal assessment:3
Time: 3 hours
Credits: 01

1. Sterilization techniques and aseptic handling of pathogens
2. Isolation and identification of pathogenic bacteria using selective media
3. Acid-fast staining.
4. Capsule staining)
5. Antibiotic susceptibility testing (Kirby–Bauer method)
6. Detection of biofilm formation (tube/plate method)
7. Visit to a diagnostic microbiology laboratory

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PHYSICS

Table 1*						Table 2*					
Sem	**Level**	**Course Code**	**Course Name**	**Credits (Th+Pr+Tut)**	**Total Credits**	**Sem**	**Level**	**Course Code**	**Course Name**	**Credits (Th+Pr+Tut)**	**Total Credits**
V	L-300	(PHY-DSC-9 Maj-501)	Condensed Matter Physics - II	3+1=4	24	V	L-300	(PHY-DSC-7 Maj/Min-501)	Statistical Physics and Thermodynamics	3+1=4 (x3)	24
		(PHY-DSC-10 Maj-502)	Electronics and Solid-State Devices - II	3+1=4				(PHY-DSC-8 Maj-502)	Electronics and Solid-State Devices	3+1=4 (x2)	
		(PHY-DSC-11 Maj-503)	Electromagnetic Theory	3+1=4							
		(PHY-DSC-12 Maj-504)	Mathematical Physics - I	4							
		(PHY-DSC-13 Maj-505)	Fundamentals of Computational Physics	2+2= 4							
		Internship / Community Work		4				Internship / Community Work		4	
VI	L-300	(PHY-DSC-14 Min-601)	Condensed Matter Physics	3+1=4	24	VI	L-300	(PHY-DSC-9 Maj/Min-601)	Condensed Matter Physics	3+1=4 (x3)	24
		(PHY-DSC-15 Maj-602)	Nuclear Physics	3+1= 4				(PHY-DSC-10 Maj/Min-602)	Nuclear Physics	3+1=4 (x3)	
		(PHY-DSC-16 Maj-603)	Mathematical Physics - II	4							
		(PHY-DSC-17 Maj-604)	Advanced Computational Physics	2+2= 4							
		(PHY-DSC-18 Maj-605)	Classical Mechanics	4							
		(PHY-DSC-19 Maj-606)	Atomic and Molecular Physics	3+1= 4							

Notes:

1. In the first semester student is offered three different subjects A, B & C.
2. At the end of second semester each student has the option to choose any one of the two streams namely Single Major or Multi-Disciplinary. The student choosing Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student who chooses multi-disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3-year degree) shall be eligible for 4 YEAR DEGREE B.Sc. HONOURS with research, on the condition that the Institution is a PU approved Research Centre
5. The student opting for Single Major must secure 50% credits of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for Minor in a particular subject, the student must attain atleast 12 credits in that subject(s).
6. The student exiting the programme after one year securing 48 credits will be awarded UG Certificate provided he/she secures 4 credits from summer internship/community work. Further, the student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secure 4 credits from summer internship. The student successfully securing 144 credits will be awarded BA degree.
7. Language paper chosen as MIL-1 and MIL-2 should be in the same subject.
8. Once Major and Minor papers are chosen in 3rd semester, the same will continue as Major and Minor subject in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write Dissertation of 08 credits in one of the two Major subjects.

B.Sc. Semester 5
PHY-DSC-7 Maj/Min-501
(Statistical Physics and Thermodynamics)
(for Table-2)

Maximum marks: 100
(Theory 75 marks & Practical 25 marks)
Theory 75 marks [60 marks+15 marks (internal assessment 20%)]
Practical 25 marks [20 marks+05 marks (internal assessment 20%)]
Teaching hours: Theory-3 Hrs per week
Total Credits: 3 (Theory) + 1 (Practical)

Learning Outcomes

- To understand the basic ideas of statistical physics, distribution of four distinguishable particles in two compartments, deviation from the state of maximum probability
- To study phase space and its division into elementary cells, B.E. statistics and F.D. statistics.
- To explore the statistical definition of entropy, change of entropy of a system, brief review of the terms
- To understand the Derivation of Maxwell's thermodynamical relations, Clausius Clayperon Equation and Joule-Thomson effect

Note:

- The theory paper will be of 75 marks including 15 marks for internal assessment and three-Hour duration.
- The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of 8 short answer type questions covering the whole syllabus and the students have to attempt any 6 parts of the compulsory question. Each question will have a weightage of 12 marks. The candidates will attempt five questions in all, selecting two questions each from Unit I and II and the compulsory question from unit III.
- The question paper is expected to contain problems with a weightage of 25-30%.
- Use of non-programmable calculators will be allowed in the examination centre.
- The Logarithmic table will be provided in the examination centre, if required.

UNIT-I

Basic ideas of Statistical Physics, Scope of Statistical Physics, basic ideas about probability, distribution of four distinguishable particles in two compartments of equal size. Concept of macrostates, microstates, thermodynamic probability, effects of constraints on the system, distribution of n particles in two compartments, deviation from the state of maximum probability, equilibrium state of dynamic system, Phase space and its division into elementary cells, three kinds of statistics. The basic approach in the three statistics, Maxwell-Boltzmann statistics applied to an ideal gas in equilibrium, Need of quantum statistics-B.E. statistics, derivation of Planck's law of radiation, F.D. statistics, Comparison of M.B., B.E. and F.D. statistics.

UNIT-II

Statistical definition of entropy, change of entropy of a system, additive nature of entropy, law of increase of entropy, reversible and irreversible processes with examples, Work done in a reversible process, Examples of increase of entropy in natural processes, Entropy and disorder, Entropy of a perfect gas, Equation of state of ideal gas from simple statistical consideration. Heat death of the universe, Derivation of Maxwell's thermodynamical relations, cooling produced by adiabatic stretching, Expression for (C_p-C_v), change of state and Clausius Clayperon Equation, Thermodynamical treatment of Joule-Thomson effect and applications (Qualitative).

Books Suggested:

Essential Readings:

1. "Statistical Physics and Thermodynamics", V.S. Bhatia, (Shoban Lal Nagin Chand, Jalandhar).
2. "A Treatise on Heat" Saha and Srivastava (Indian Press, Ahmedabad, 1972).

Further Readings:

1. Thermal Physics by C. Kittel & H. Kroemer, CBS Pub., 1987.
2. Thermal Physics, S.C. Garg, R.M. Bansal, and C.K. Ghosh, TMH, 2000.

PHYSICS PRACTICALS
(PHY-DSC-7 Maj/Min-501)
(for Table-2)

Max. Marks: 25 (20 + 5 internal assessment) **Total Credit: 1**

Teaching hours: 2Hrs per week

General Guidelines for Physics Practical Examinations:

1. The distribution of marks is as follows:

(i)	One full experiment requiring the student to take some data, analyze it and draw conclusions. (Candidates are expected to state their results with limits of error).	09
(ii)	One exercise based on experiment	04
(iii)	Viva-Voce	04
(iv)	Record (Practical file)	03
(v)	Internal Assessment	05

Note for Examiners:

2. The marks scored under each head must be clearly written on the answer sheet.
3. There will be one session of 3 hours duration. The paper will have two sections. In Section-A one experiment is to be allotted by the external examiner.
4. Section–B will consist of exercises which will be set by the external examiner on the spot. The length of the exercises should be such that any of these could be completed in one hour.
5. The examiner should take care that the experiment allotted to an examinee from section–A and exercise allotted from section–B are not directly related to each other.
6. Number of candidates in a group for practical examination should not exceed 12.
7. In a single group, no experiment to be allotted to more than three examinees in the group.

Note: Each student will perform at least five experiments in the whole semester.

List of Practicals:

1. To study adiabatic expansion of a gas.
2. To measure thermal expansion of crystal using interference fringes.
3. To measure thermal conductivity in poor conductor by Lee's method.
4. To measure the thermo e.m.f.
5. To study Probability distribution using coloured dice, coins.
6. To determine the heating efficiency of an electric kettle with varying voltages.
7. Total radiation law, temperature dependence of radiation.
8. To determine principal points of a lens system.
9. To determine the wavelength of the spectral lines of mercury using diffraction grating and mercury source.
10. To find the thermal conductivity of copper using Searle's conductivity apparatus.

Texts and Reference Books:

1. "B.Sc. Practical Physics by C.L. Arora, S. Chand & Co. (2014).
2. "A Laboratory Manual of Physics for Undergraduate Classes" by D.P. Khandelwal.

B.Sc. Semester 5
PHY-DSC-8 Maj-502
(Electronics and Solid-State Devices)
(for Table-2)

Maximum marks: 100
(Theory 75 marks & Practical 25 marks)
Theory 75 marks [60 marks+15 marks (internal assessment 20%)]
Practical 25 marks [20 marks+5 marks (internal assessment 20%)]
Teaching hours: Theory-3Hrs per week
Total Credits: 3 (Theory) +1 (Practical)

Learning Outcomes

- Understand the Energy band diagrams in semiconductors, Position of Fermi level in p and n semiconductors, forward and reverse biased diode, V-I characteristics, zener diode, LED.
- Learn the diode circuits, Clippers, half wave, full wave and bridge rectifiers, Bipolar Junction transistors, and Transistor biasing.
- Explore JFET, Feedback in amplifiers, operational amplifiers and their applications
- Understand the concepts of binary number system, different types of logic gates, Minterms and Maxterms.

Note:

1. The theory paper will be of 75 marks including 15 marks for internal assessment and three-Hour duration.
2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of 8 short answer type questions covering the whole syllabus and the students have to attempt any 6 parts of the compulsory question. Each question will have a weightage of 12 marks. The candidates will attempt five questions in all, selecting two questions each from Unit I and II and the compulsory question from unit III.
3. The question paper is expected to contain problems with a weightage of 25-30%.
4. Use of non-programmable calculators will be allowed in the examination centre.
5. The Logarithmic table will be provided in the examination centre, if required.

UNIT-I

Energy band diagrams in semiconductors, Direct and indirect semiconductors, Formula to calculate Position of Fermi level in p and n semiconductors (Qualitative), Barrier formation, energy band diagram of p-n junction, Formula for Depletion width, Qualitative ideas of current flow mechanism in forward and reverse biased diode, V-I characteristics, static and dynamic resistance, zener diode, LED, Diode circuits, Clipping, Rectification: half wave, full wave and bridge rectifiers, filter circuits (L filter, C filter) (Qualitative), Bipolar Junction transistors: Structure and working, Characteristics of CB and CE configurations, Load line analysis of transistors, Q-point, Transistor biasing and stabilization of operating point, fixed bias, collector to base bias, voltage divider biasing circuit.

UNIT-II

Structure, working and characteristics of JEFT (drain and transconductance curve), FET amplifier and its voltage gain, Structure and working of MOSFET, Feedback in amplifiers, voltage gain of negative feedback amplifier, advantages of negative voltage feedback, Operational amplifier (black box approach): Characteristics of ideal operational amplifier, open-loop and closed-loop gain, applications - inverting and non-inverting amplifiers, adder, subtractor, differentiator and integrator, Analog and digital circuits, binary numbers, decimal to binary conversions, AND, OR, NOT gates, NAND NOR gates as universal gates, XOR and XNOR gates, De Morgan's theorem, Simplification of logic circuit using Boolean algebra, Minterms and Maxterms

Books Suggested:

Essential Readings:

1. Electronic Devices and Circuit Theory, 7th Ed., R. Boylestad, L. Nashelsky, Prentice Hall Inc.
2. Electronic Principles, A.P. Malvino, and D.J. Bates, 7th ed. McGraw Hill
3. Solid State Electronic Devices, 6th Ed., Ben G. Streetman and S. Banerjee, Eastern Economy Edition.
4. Digital Principles and Applications, 7th Ed., A.P. Malvino, D.P. Leach and Saha, 2011, Tata McGraw Hill

Further Readings:

1. Basic Electronics, 5 th Edition, B.L. Thareja, S. Chand.
2. Basic Electronics and Linear Circuits, N.N. Bhargave, D.C. Kulshreshtha, and S.C. Gupta,Tata Mc Graw Hill.
3. Foundations of Electronics, D. Chatopadhyay, P.C. Rakshit, B. Saha, and N.N. Purkit, New Age International

PHYSICS PRACTICALS
(PHY-DSC-8 Maj-502)
(for Table-2)

Max. Marks: 25 (20+5 internal assessment) **Total Credit: 1**

Teaching hours: 2Hrs per week

General Guidelines for Physics Practical Examinations:

1. The distribution of marks is as follows:

(i)	One full experiment requiring the student to take some data, analyze it and draw conclusions. (Candidates are expected to state their results with limits of error).	09
(ii)	One exercise based on experiment	04
(iii)	Viva-Voce	04
(iv)	Record (Practical file)	03
(v)	Internal Assessment	05

Note for Examiners:

2. The marks scored under each head must be clearly written on the answer sheet.
3. There will be one session of 3 hours duration. The paper will have two sections. In Section-A one experiment is to be allotted by the external examiner.
4. Section–B will consist of exercises which will be set by the external examiner on the spot. The length of the exercises should be such that any of these could be completed in one hour.
5. The examiner should take care that the experiment allotted to an examinee from section–A and exercise allotted from section–B are not directly related to each other.
6. Number of candidates in a group for practical examination should not exceed 12.
7. In a single group, no experiment to be allotted to more than three examinees in the group.

Note: Each student will perform at least five experiments in the whole semester.

List of Practicals

1. To draw forward and reverse bias characteristics of a p-n junction diode and draw a load line.
2. To draw the characteristics of a Zener diode and LED.
3. To study the function of a diode as a clipping element.
4. To measure the efficiency and ripple factors for (a) Half-wave, (b) Full wave, and (c) Bridge rectifier circuits.
5. To study the Common Base Characteristics of a transistor (PNP or NPN).
6. To draw output and mutual characteristics of an FET and determine its parameters.
7. To set up an oscillator and study its output on CRO for different V values.
8. To determine the Hall coefficient for a given semiconductor.
9. To find the conductivity of a given semi-conductor crystal using four probe method.

Texts and Reference Books:

1. "B.Sc. Practical Physics by C.L. Arora, S. Chand& Co. (2014).
2. "A Laboratory Manual of Physics for Undergraduate Classes" by D.P. Khandelwal.
3. "Programming with C, Schaum series" by Byron Gottfried & Jitender Chhabra.

B.Sc. Semester 5
(PHY-DSC-9-Maj-501)
(Condensed Matter Physics-II)
(for Table-1)

Maximum marks: 100
(Theory 75 marks & Practical 25 marks)
Theory 75 marks [60 marks+15 marks (internal assessment 20%)]
Teaching hours: 45 Hrs. / 3 Hrs. per week
Total Credits: 3 (Theory)+1 (Practical)

Learning Outcomes

- Explain lattice dynamics and thermal properties of solids, including phonon concepts, vibrational modes, density of states, and Einstein and Debye models of specific heat.
- Analyze magnetic and dielectric properties of materials, including classical and quantum theories of magnetism, ferroelectric and piezoelectric behavior, and frequency-dependent polarization mechanisms.
- Understand and explain advanced states of matter, such as liquid crystals and superconductors, including their physical properties, theoretical models, and technological applications.
- Describe nanoscale materials and nanotechnology concepts, including size-dependent properties, fabrication and characterization techniques, carbon nanostructures, and applications across science and engineering domains.

Note:

1. The theory paper will be of 75 marks including 15 marks for internal assessment and three-Hour duration.
2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of 8 short answer type questions covering the whole syllabus and the students have to attempt any 6 parts of the compulsory question. Each question will have a weightage of 12 marks. The candidates will attempt five questions in all, selecting two questions each from Unit I and II and the compulsory question from unit III.
3. The question paper is expected to contain problems with a weightage of 25-30%.
4. Use of non-programmable calculators will be allowed in the examination centre.
5. The Logarithmic table will be provided in the examination centre, if required.

UNIT-I

Lattice Dynamics : Lattice vibrations and phonons, Scattering of photons by phonons, Dynamics of a linear chain of similar atoms and chain of two types of atoms, optical and acoustic modes, Density of modes, Einstein and Debye theories of specific heats of solids.

Magnetic classification of materials (Dia, para, ferro, ferri, antiferro), Langevin theory of dia and paramagnetism, Quantum theory, Weiss's theory of Ferromagnetism, temperature dependence, hysteresis of ferromagnetic materials.

UNIT-II

Dielectric constant & polarizability, electric susceptibility, Clausius Mosotti equation, frequency dependence, ferroelectrics and Piezoelectrics.

Liquid crystals, various types and properties. Applications.

Superconductivity: Meisner effect, London's equation and penetration depth, critical magnetic field and temperature, DC and AC Josephson effect, BCS theory (formation of cooper pairs), ground state and energy gap.

Basic ideas of materials at nanoscale: Difference from bulk material properties, Nanoparticles, introduction to fabrication and characterization techniques, Carbon Nanostructures– nanotubes, graphene. Applications of nanotechnology in various fields.

Recommended Books :

1. Introduction to Solid State Physics, C. Kittel, Wiley Eastern
2. Elements of Modern Physics,S.H. Patil, Tata McGraw Hill, 1985.
3. Solid State Physics, 6th Edition, S.O. Pillai, New Age International Publishers.
4. Fundamental of Physics, Vol. II, R.M.P. Jaiswal, S.K. Gupta and A. Rani, R. Chand and Co.

Further Readings :

1. Elements of Solid State Physics, 2nd Edition, J.P. Srivastava, Prentice Hall.
2. Elementary Solid State Physics, M. Ali Omar, Pearson.
3. Crystallography for Solid State Physics, A.R. Verma, O.N. Srivastava, Wiley Eastern.

PHYSICS PRACTICALS
(PHY-DSC-9-Maj-501)
(for Table-1)

Max. Marks: 25 (20+5 internal assessment) **Total Credit: 1**

Teaching hours: 30 Hrs. / 2 Hrs. per week

General Guidelines for Physics Practical Examinations:

1. The distribution of marks is as follows:

(i)	One full experiment requiring the student to take some data, analyze it and draw conclusions. (Candidates are expected to state their results with limits of error).	09
(ii)	One exercise based on experiment	04
(iii)	Viva-Voce	04
(iv)	Record (Practical file)	03
(v)	Internal Assessment	05

Note for Examiners:

2. The marks scored under each head must be clearly written on the answer sheet.
3. There will be one session of 3 hours duration. The paper will have two sections. In Section-A one experiment is to be allotted by the external examiner.
4. Section–B will consist of exercises which will be set by the external examiner on the spot. The length of the exercises should be such that any of these could be completed in one hour.
5. The examiner should take care that the experiment allotted to an examinee from section–A and exercise allotted from section–B are not directly related to each other.
6. Number of candidates in a group for practical examination should not exceed 12.
7. In a single group, no experiment to be allotted to more than three examinees in the group.

Note: Each student will perform at least five experiments in the whole semester.

List of Practicals:

1. To trace the B-H curves for different materials using CRO and find the magnetic parameters from these.
2. To measure the magnetic susceptibility of $FeCl_2$ solution by Quincke's method.
3. To measure heat capacity of solid at high and low temperatures.
4. To determine dipole moment of an organic molecule, Acetone.
5. To obtain the potential energy curve due to magnet-magnet interactions using air track apparatus.
6. To measure dielectric constant of barium titanate as function of temperature and frequency, and hence study its phase transition.
7. To determine dielectric constant at high frequency using Leacher Wire. To study reflected waves in Leacher Wire for different terminating loads.
8. To Study the Phase Diagram of Pb-Sn Alloy using Cooling Curve and Determine its composition.
9. To measure the transition temperature of a high-temperature superconductor.

Texts and Reference Books:

1. "B.Sc. Practical Physics by C.L. Arora, S.Chand & Co. (2014).
2. "A Laboratory Manual of Physics for Undergraduate Classes" by D.P. Khandelwal.

B.Sc. Semester 5
(PHY-DSC-10-Maj-502)
(Electronics and Solid-State Devices – II)
(for Table-1)

Maximum marks: 100
(Theory 75 marks & Practical 25 marks)
Theory 75 marks [60 marks+15 marks (internal assessment 20%)]
Teaching hours: 45 Hrs. / 3 Hrs. per week
Total Credits: 3 (Theory) + 1 (Practical)

Learning Outcomes

- Understand and explain the physical principles of semiconductor materials, crystal growth techniques, p–n junction fabrication, and the operation of modern electronic devices
- Analyze and design amplifier circuits with feedback and oscillator circuits using RC and LC networks, applying concepts such as loop gain, phase condition, Barkhausen criterion, and frequency of oscillation.
- Apply operational amplifier concepts to design and evaluate linear and nonlinear analog circuits including amplifiers, comparators, waveform generators, 555 timer circuits, and analog computation blocks.
- Explain and compare communication systems, including analog and digital modulation techniques, and analyze multiple access methods used in satellite and mobile communication systems such as FDMA, TDMA, and CDMA.

Note:

1. The theory paper will be of 75 marks including 15 marks for internal assessment and three-Hour duration.
2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of 8 short answer type questions covering the whole syllabus and the students have to attempt any 6 parts of the compulsory question. Each question will have a weightage of 12 marks. The candidates will attempt five questions in all, selecting two questions each from Unit I and II and the compulsory question from unit III.
3. The question paper is expected to contain problems with a weightage of 25-30%.
4. Use of non-programmable calculators will be allowed in the examination centre.
5. The Logarithmic table will be provided in the examination centre, if required.

Unit-I

Growth of semiconductor crystals, Direct and indirect semiconductors, Fabrication of p-n junction, Varactors.

Fundamentals of operation of BJT, FET, MOSFET and UJT. Liquid crystal display. High frequency devices: Gunn diode, IMPATT diode.

Feed-back in amplifiers, voltage gain of negative feedback amplifier, advantages of negative voltage feedback, negative current feedback circuit, emitter follower.

Theory of sinusoidal oscillations, loop gain and phase, Lead-lag RC circuit, Wein bridge oscillator. Barkhausen criterion of sustained oscillations, positive feedback amplifier, LC oscillators, Colpitts and Hartley oscillators.

Unit-II

Differential amplifiers, common mode rejection ratio, Transfer characteristics of OPAMP, inverting and noninverting configurations, open loop and close loop gain, Slew rate, Basic internal circuit of IC Opamp. Comparators with hysteresis, Window comparator, Rectangular and triangular wave generators. 555 timer based circuits.

Analogue computation – Summing amplifier, Integrators and Differentiators, Solving differential equations and simultaneous linear equations, Logarithmic and antilogarithmic amplifiers, Current- to- voltage and Voltage-to-current converters.

General communication system, Generation and detection of amplitude modulated, Single-side band, Double-side band suppressed carrier and Frequency modulated wave. ASK, PSK and FSK, Satellite and mobile communication - TDMA, FDMA, CDMA.

Recommended Books:

Essential Readings:

1. Electronic Devices and Circuit Theory, 7th Ed., R. Boylestad, L. Nashelsky, Prentice Hall Inc.
2. Electronic Principles, A.P. Malvino, and D.J. Bates, 7th ed. McGraw Hill
3. Communication Systems, S. Haykin, John Wiley and Sons.

Further Readings:

1. Basic Electronics, 5th Edition, B.L. Thareja, S. Chand.
2. Basic Electronics and Linear Circuits, N.N. Bhargave, D.C. Kulshreshtha, and S.C. Gupta, Tata Mc Graw Hill.
3. Foundations of Electronics, D. Chatopadhyay, P.C. Rakshit, B. Saha, and N.N. Purkit, New Age International.

PHYSICS PRACTICALS

(PHY-DSC-10-Maj-502)

(for Table-1)

Max. Marks: 25 (20+5 internal assessment) **Total Credit: 1**

Teaching hours: 30 Hrs. / 2 Hrs. per week

General Guidelines for Physics Practical Examinations:

1. The distribution of marks is as follows:

(i)	One full experiment requiring the student to take some data, analyze it and draw conclusions. (Candidates are expected to state their results with limits of error).	09
(ii)	One exercise based on experiment	04
(iii)	Viva-Voce	04
(iv)	Record (Practical file)	03
(v)	Internal Assessment	05

Note for Examiners:

2. The marks scored under each head must be clearly written on the answer sheet.
3. There will be one session of 3 hours duration. The paper will have two sections. In Section A, one experiment is to be allotted by the external examiner.
4. Section B will consist of exercises that will be set by the external examiner on the spot. The length of the exercises should be such that any of these could be completed in one hour.
5. The examiner should take care that the experiment allotted to an examinee from section A and the exercise allotted from section B are not directly related to each other.
6. The number of candidates in a group for practical examination should not exceed 12.
7. In a single group, no experiment is to be allotted to more than three examinees in the group.

Note: Each student will perform at least five experiments in the whole semester.

List of Practicals:

1. To draw output and mutual characteristics of an FET and determine its parameters.
2. To study MOSFET characteristics, biasing and its applications as an amplifier.
3. To study the response of an RC circuit to various input voltages (square, sine, and triangular).
4. To study Hartley/Colpitts oscillators.
5. To study solid state power supply and voltage multiplier circuits (using breadboard).
6. Design different Clipping and Clamping circuits: positive, negative and bias (using breadboard)

7. To study the gain of an amplifier at different frequencies and to find bandwidth and gain-bandwidth.
8. To verify the statement: Power dissipation in the sidebands in amplitude modulation is directly proportional to the square of the modulation.
9. To study the various aspects of frequency modulation and demodulation.
10. To measure characteristic parameters of an OPAMP and use of operational amplifier for different mathematical operations.

Texts and Reference Books:

1. "B.Sc. Practical Physics by C.L. Arora, S.Chand & Co. (2014).
2. "A Laboratory Manual of Physics for Undergraduate Classes" by D.P. Khandelwal.

B.Sc. Semester 5

(PHY-DSC-11-Maj-503)

(Electromagnetic Theory)

Maximum marks: 100

(Theory 75 marks & Practical 25 marks)

Theory 75 marks [60 marks+15 marks (internal assessment 20%)]

Practical 25 marks [20 marks+05 marks (internal assessment 20%)]

Teaching hours: Theory-45 Hrs. / 3 Hrs. per week

Total Credits: 3(Theory)+1(Practical)

Learning Outcomes

- To understand fundamental electrostatic and magnetostatic phenomena using Coulumb's Law, Biot-Savart's Law and Ampere's Law.
- To understand Maxwell' s equations and explain how they unify electricity, magnetism and optics.
- To solve boundary value problems using Laplace and Poisson's equations.
- To derive expressions for electromagnetic fields for symmetric charge and current distributions in various types of media.
- To use computational tools (C++ and GNUplot) to simulate field distributions, visualize potentials, fields and waves.

Note:

1. The theory paper will be of 75 marks including 15 marks for internal assessment and three-Hour duration.
2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of 8 short answer type questions covering the whole syllabus and the students have to attempt any 6 parts of the compulsory question. Each question will have a weightage of 12 marks. The candidates will attempt five questions in all, selecting two questions each from Unit I and II and the compulsory question from unit III.
3. The question paper is expected to contain problems with a weightage of 25-30%.
4. Use of non-programmable calculators will be allowed in the examination centre.
5. The Logarithmic table will be provided in the examination centre, if required.

UNIT-I

Fundamental concepts of vector calculus, physical significance and related problems for gradient, divergence and curl of a vector field, Electrostatics in Vacuum : Coulomb's Law, Gauss Law, Scalar potential. Laplace and Poisson's equations. Electrostatic potentials, energy and energy density of the electromagnetic field.

Multipole Expansion : Multipole expansion of the scalar potential of a charge distribution. Dipole moment, quadrupole moment. Multipole expansion of the energy of a charge distribution in an external field.

Magnetostatics: The differential equations of magnetostatics, Vector potential, Magnetic field of a localized current distribution.

Electrostatics of Dielectrics : Static fields in material media. Polarization vector macroscopic equations. Molecular polarizability and electric susceptibility. Clasusius- Mossetti relations. Models of Molecular Polarizability, Boundary value Problems: Uniqueness Theorem. Dirichlet and Neumann Boundary conditions, Green's Theorem, Method of images with examples.

UNIT-II

Time Varying Fields and Maxwell Equations : Faraday's Law of induction. Displacement current. Maxwell equations. Scalar and vector potentials. Gauge transformation, Lorentz and Coulomb gauges, General Expression for the electromagnetic fields energy, conservation of energy, Poynting's Theorem.

EM waves in various unbounded media: Wave equation, plane waves in free space and isotropic dielectrics, polarization, energy transmitted by a plane wave, Poynting's theorem for a complex vector field. Waves in conducting media, skin depth, EM waves in rare field plasma and their propagation in ionosphere.

EM waves in bounded media-Applications : Reflection and Refraction of EM waves at plane dielectrics interface, Fresnel's amplitude relations. Reflection and transmission coefficients. Polarization by reflection. Brewster's angle, Total internal reflection.

Suggested Books

1. Classical Electrodynamics : S.P. Puri, (Tata McGraw Hill, New Delhi) (2nd edition)1997, Alpha Science International Ltd., India (2011).

2. Classical Electrodynamics : J.D. Jackson, (Wiley Eastern, New Delhi) (1998).

3. Introduction to Electrodynamics: D.J. Griffiths, (Prentice Hall India, New Delhi) (2008).

4. Electromagnetic Field Theory Fundamentals: Bhag Singh Guru and H.R. Hiziroglu (Cambridge University Press) 2nd edition, (2004).

PHYSICS PRACTICALS

(PHY-DSC-11-Maj-503)

Max. Marks: 25 (20+5 internal assessment) **Total Credit: 1**

Teaching hours: 30 Hrs. / 2 Hrs. per week

General Guidelines for Physics Practical Examinations:

1. The distribution of marks is as follows:

(i) One full experiment requiring the student to discuss the theory and algorithm of the problem assigned along with designing and executing a C++ program based on it. 10

(ii) Viva-Voce 5

(iii) Record (Practical file) 5

(iv) Internal Assessment 5

Note for Examiners:

2. The marks scored under each head must be clearly written on the answer sheet.
3. There will be one session of 3 hours duration. Any one experiment is to be allotted by the external examiner.
4. Number of candidates in a group for practical examination should not exceed 12.
5. In a single group, no experiment to be allotted to more than three examinees in the group.

List of Practicals:

Design C++ programs for the following problems based on the concepts of electrodynamics.

1. Calculate Coulomb force between two point charges and graphically study the variation of this force with distance.
2. Study the motion of an electric charge in a uniform electric field and plot its trajectory.
3. Study the motion of an electric charge in a uniform magnetic field and plot its trajectory.
4. Calculate the electric potential due to a dipole on its axis and study its variation with distance.
5. Using any numerical differentiation technique, demonstrate the Faraday's Law of electromagnetic induction.
6. Study the motion of a charged particle in combined electric and magnetic fields and draw its trajectory.
7. Study graphically the EM oscillations in a LCR Circuit. Show the variations of (a) charge vs time, (b) current vs time.
8. Carry out a numerical study of the variation of skin depth with frequency.
9. Study the variation of magnetic field due to an infinite long straight wire with distance.

Suggested Books

- Classical Electrodynamics : S.P. Puri, (Tata McGraw Hill, New Delhi) (2nd edition)1997, Alpha Science International Ltd., India (2011).
- Introduction to Electrodynamics: D.J. Griffiths, (Prentice Hall India, New Delhi) (2008).
- Computational Physics: R.C. Verma, P.K. Ahluwalia and K.C. Sharma (New Age) (2005).

B.Sc. Semester 5

(PHY-DSC-12-Maj-504)

(Mathematical Physics-I)

Maximum marks: 100

Theory 100 marks [80 marks+20 marks (internal assessment 20%)]

Teaching hours: 60 Hrs. / 4 Hrs. per week

Total Credits: 4 (Theory)

Learning Outcomes

- Apply complex variable theory to analyze analytic functions using Cauchy–Riemann conditions, contour integration, residue calculus, series expansions, and dispersion relations for evaluating definite integrals.
- Understand and use generalized functions, including the Dirac delta function and its representations, along with Gamma and Beta functions, to solve problems arising in mathematical and physical applications.
- Solve differential equations and matrix problems relevant to theoretical physics using separation of variables, Frobenius method, eigenvalue–eigenvector analysis, and the Cayley–Hamilton theorem.
- Analyze and apply special functions such as Bessel, Legendre, Hermite, and Laguerre functions, including their generating functions, recurrence relations, and orthogonality properties, in physical systems and boundary-value problems.

Note:

1. The theory paper will be of 100 marks including 20 marks for internal assessment and three-Hour duration.
2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of 10 short answer type questions covering the whole syllabus and the students have to attempt any 8 parts of the compulsory question. Each question will have a weightage of 16 marks. The candidates will attempt five questions in all, selecting two questions each from Unit I and II and the compulsory question from unit III.
3. The question paper is expected to contain problems with a weightage of 25-30%.
4. Use of non-programmable calculators will be allowed in the examination centre.
5. The Logarithmic table will be provided in the examination centre, if required.

Unit-I

Complex Variables: Introduction, Cauchy Riemann conditions. Cauchy's Integral theorem, Cauchy's Integral formula, Taylor & Laurent expansion, singularities, calculus of residues, evaluation of definite integrals. Dispersion relation.

Delta and Gamma Functions: Dirac delta function. Delta sequences for one dimensional function, properties of delta function, Orthogonal function and Integral representation of Delta function. Gamma function and its various form, Weierstrass form, factorial notation and applications. Beta function. Relation with gamma function.

Unit-II

Differential Equations and Matrices: Partial differential equations of theoretical physics, separation of variables, singular points, series solutions-Frobenius method, second solution. Matrices, Cayley-Hamilton Theorem. Eigenvalues and eigenvectors.

Special Functions: Bessel function of first and second kind, Generating function, integral representation and recurrence relations for Bessel's functions of first kind, Legendre function: generating function, recurrence relations and special properties, orthogonality. Various Legendre polynomials, Associated Legendre functions, Hermite functions and Lagurerre functions. Generating function, Recurrence relations and orthogonality.

Recommended Books:

1. Mathematical methods for Physicists : G. Arfken and H.J. Weber (Academic Press, SanDiego) (sixth edition) (2005)

2. Matrices and Tensors in Physics: A.W. Joshi (Wiley Eastern, New Delhi) (2002)

3. Mathematical Physics: P.K. Chatopadhyay (Wiley Eastern, New Delhi) (2005)

4. Introduction to Mathematical Physics: C. Harper (Prentice Hall of India, New Delhi) (2004).

B.Sc. Semester 5

(PHY-DSC-13 Maj-505)

(Fundamentals of Computational Physics)

Maximum marks: 100

(Theory 50 marks & Practical 50 marks)

Theory 50 marks [40 marks+10 marks (internal assessment 20%)]

Practical 50 marks [40 marks+10 marks (internal assessment 20%)]

Teaching hours: Theory-30 Hrs. / 2 Hrs. per week

Total Credits: 2(Theory)+2(Practical)

Learning Outcomes

- Understand the fundamentals of computer systems and programming, including computer hardware, memory types, software classifications, programming languages, and the basic structure and syntax of the C++ programming language.
- Apply core C++ programming concepts such as data types, operators, variables, expressions, input–output mechanisms, preprocessor directives, and data formatting tools to develop simple programs.
- Design and implement control structures and data structures using decision-making statements, looping constructs, arrays, and structures to solve computational problems efficiently.
- Develop and test structured C++ programs and algorithms for solving numerical, logical, and matrix-based problems using file handling and modular programming principles.

Note:

1. The theory paper will be of 50 marks including 10 marks for internal assessment and three Hour duration.
2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of 8 short answer type questions covering the whole syllabus and the students have to attempt any 6 parts of the compulsory question. Each question will have a weightage of 8 marks. The candidates will attempt five questions in all, selecting two questions each from Unit I and II and the compulsory question from unit III.
3. The question paper is expected to contain problems with a weightage of 25-30%.
4. Use of non-programmable calculators will be allowed in the examination centre.
5. The Logarithmic table will be provided in the examination centre, if required.

UNIT-I

Fundamentals and history of Computer systems, basic hardware components, types of computer memory-primary memory, secondary memory, introduction to software-system software, application software and utility software, programming languages and their types-low level, high level and middle level languages, introduction to C++, data types-integer, character, floating point, Boolean, void, constants and variables; operators in C++- assignment, arithmetic, compound assignment, increment and decrement, mathematical, sizeof, bitwise, relational, logical, conditional ternary, explicit type casting and sizeof operators with suitable examples, Constants and Variables, Block, Local and Global variables. Auto, Static and External Variables. Expressions and statements in C++, I/O Statements : printf, scanf, getc, getch, getchar, getche, etc., Streams-cin and cout, ifstream, ofstream , Preprocessor : #include and #define directives; Manipulators for Data Formatting: setw, width, endl and setprecision etc.

UNIT-II

Control Statements :- If-statement. If-else Statement. Nested if Structure. Else-if Statement. Ternary Operator. Goto Statement. Switch Statement. Unconditional and Conditional Looping. While Loop. Do-while Loop. For Loop. Break and Continue Statements. Nested Loops. Switch statement, Arrays and Structures - One and Two Dimensional Arrays. Idea of Structures with suitable examples, Use of input and output data files, Introduction to Programming Language Algorithms and Structured Programming. Design of simple algorithms and computer programs based on the fundamental programming concepts learnt above. Programs: (a) based on simple arithmetic operations such as calculating area, volume or simple interest etc. (b)finding out the greatest number amongst any set of 20 numbers (c) plotting and evaluation of mathematical functions (d) sum and difference of matrices (d) product of matrices (e) finding sum of first n natural numbers etc.

Recommended Books

1. Numerical Mathematical Analysis, J.B. Scarborough (Oxford Book Co.) 5th edition, (1964).
2. A first course in Computational Physics: P.L. DeVries (Wiley) (1994).
3. Schaum's outline of Programming with C, Byron Gottfried, McGraw-Hill
4. Computer Programming in C, V. Rajaraman (Prentice Hall of India Pvt. Ltd.) (2006)
5.. Computational Physics: R.C. Verma, P.K. Ahluwalia and K.C. Sharma (New Age)(2005)
6.. Computer Applications in Physics: S. Chandra (Narosa)(2008).

PHYSICS PRACTICALS

(PHY-DSC-13-Maj-505)

Max. Marks: 50 (40+10 internal assessment) **Total Credit: 2**

Teaching hours: 60 Hrs. / 4 Hrs. per week

General Guidelines for Physics Practical Examinations:

1. The distribution of marks is as follows:

(i) One full experiment requiring the student to discuss the theory and algorithm of the problem assigned along with designing and executing a C++ program based on it.	20
(ii) Viva-Voce	10
(iii) Record (Practical file)	10
(iv) Internal Assessment	10

Note for Examiners:

2. The marks scored under each head must be clearly written on the answer sheet.
3. There will be one session of 3 hours duration. Any one experiment is to be allotted by the external examiner.
4. Number of candidates in a group for practical examination should not exceed 12.
5. In a single group, no experiment to be allotted to more than three examinees in the group.

List of Practicals:

1. To find out roots of a quadratic equation
2. Data handling: find standard deviation, mean, variance, moments etc. of at least 25 entries.
3. To generate random numbers between (i) 1 and 0, (ii) 1 and 100.
4. Perform numerical integration of a given mathematical function using Simpson's rule.
5. Perform numerical integration of a given mathematical function using Weddle's rule.
6. To find the determinant of a 3x3 matrix.
7. To find eigen values and eigen vectors of a matrix.
8. Choose a set of 10 values and find the least squared fitted curve.
9. To study superposition of waves and generate patterns like stationary waves and beats.
10. To study fourier analysis of a square wave.

Books Suggested:

1. Schaum's outline of Programming with C, Byron Gottfried, McGraw-Hill.
2. Computational Physics: R.C. Verma, P.K. Ahluwalia and K.C. Sharma (New Age) (2005).
3. Computer Applications in Physics: S. Chandra (Narosa) (2008).

B.Sc. Semester 6
(PHY-DSC-9 Maj/Min-601) for Table 2
(PHY-DSC-14 Min-601) for Table 1
(Condensed Matter Physics)

Maximum marks: 100
(Theory 75 marks & Practical 25 marks)
Theory 75 marks [60 marks+15 marks (internal assessment 20%)]
Practical 25 marks [20 marks+05 marks (internal assessment 20%)]
Teaching hours: Theory-45 Hrs. / 3 Hrs. per week
Total Credits: 3(Theory) + 1 (Practical)

Learning Outcomes

- To understand Crystal structure, Two and Three dimensional Bravais lattices, Crystal planes and Miller indices
- To study the Bragg's Law, Laue equations, Bragg's law in reciprocal lattice, Brillouin zones and band theory of solids
- To explain Lattice Dynamics, dynamics of a linear chain of similar atoms and two types of atoms, Einstein and Debye theories of specific heats of solids, Magnetic materials.
- To understand the concept of Superconductors, Meissner effect, critical magnetic field and temperature, BCS theory

Note:

1. The theory paper will be of 75 marks including 15 marks for internal assessment and three-Hour duration.
2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of 8 short answer type questions covering the whole syllabus and the students have to attempt any 6 parts of the compulsory question. Each question will have a weightage of 12 marks. The candidates will attempt five questions in all, selecting two questions each from Unit I and II and the compulsory question from unit III.
3. The question paper is expected to contain problems with a weightage of 25-30%.
4. Use of non-programmable calculators will be allowed in the examination centre.
5. The Logarithmic table will be provided in the examination centre, if required.

UNIT-I

Crystal structure: Symmetry operations for a two-dimensional crystal. Two dimensional Bravais lattices, Three dimensional Bravais lattices, Basic primitive cells, Crystal planes and Miller indices, Diamond and NaCl structure. Crystal diffraction: Bragg's Law, Laue equations, Reciprocal lattices of SC, BCC and FCC, Bragg's law in reciprocal lattice, Brillouin zones and its derivation in two dimensions, band Theory of solids, periodic potential and Bloch theorem, Kronig-Penney model, band gaps, band structures in conductors, direct and indirect semiconductors and insulators.

UNIT-II

Lattice Dynamics: Lattice vibrations and phonons, Scattering of photons by phonons, Dynamics of a linear chain of similar atoms and chain of two types of atoms, optical and acoustic modes, Einstein and Debye theories of specific heats of solids. Magnetic classification of materials (Dia, para, ferro, ferri, antiferro), Langevin theory of dia and paramagnetism, Ferromagnetic domains, hysteresis of ferromagnetic materials, Superconductivity: Meissner effect, London's equation and penetration depth, critical magnetic field and temperature, DC and AC Josephson effect, BCS theory (formation of cooper pairs)

Books Suggested:

Essential Readings:

1. Introduction to Solid State Physics, C. Kittel, Wiley Eastern
2. Elements of Modern Physics,S.H. Patil, Tata McGraw Hill, 1985.
3. Solid State Physics, 6th Edition, S.O. Pillai, New Age International Publishers.
4. Fundamental of Physics, Vol. II, R.M.P. Jaiswal, S.K. Gupta and A. Rani, R. Chand and Co.

Further Readings:

1. Elements of Solid-State Physics, 2nd Edition, J.P. Srivastava, Prentice Hall.
2. Elementary Solid-State Physics, M. Ali Omar, Pearson.
3. Crystallography for Solid State Physics, A.R. Verma, O.N. Srivastava, Wiley Eastern.

PHYSICS PRACTICALS
(PHY-DSC-9 Maj/Min-601) for Table 2
(PHY-DSC-14 Min-601) for Table 1

Max. Marks: 25 (20+5 internal assessment) **Total Credit: 1**
Teaching hours: 2Hrs per week

General Guidelines for Physics Practical Examinations:

1. The distribution of marks is as follows:

(i)	One full experiment requiring the student to take some data, analyze it and draw conclusions. (Candidates are expected to state their results with limits of error).	09
(ii)	One exercise based on experiment	04
(iii)	Viva-Voce	04
(iv)	Record (Practical file)	03
(v)	Internal Assessment	05

Note for Examiners:

2. The marks scored under each head must be clearly written on the answer sheet.
3. There will be one session of 3 hours duration. The paper will have two sections. In Section-A one experiment is to be allotted by the external examiner.
4. Section–B will consist of exercises which will be set by the external examiner on the spot. The length of the exercises should be such that any of these could be completed in one hour.
5. The examiner should take care that the experiment allotted to an examinee from section–A and exercise allotted from section–B are not directly related to each other.
6. Number of candidates in a group for practical examination should not exceed 12.
7. In a single group, no experiment to be allotted to more than three examinees in the group.

Note: Each student will perform at least five experiments in the whole semester.

List of Practicals:

1. Measurement of reverse saturation current in p-n junction diode at various temperatures and to find the approximate value of energy gap.
2. To study the stabilization of output voltage of a power supply with Zener diode.
3. To study the characteristics of a thermistor and find its parameters.
4. To measure and plot Common Emitter Characteristics of a transistor (PNP or NPN).
5. To trace the B-H curves for different materials using CRO and find the magnetic parameters from these.
6. To measure the magnetic susceptibility of $FeCl_2$ solution by Quincke's method.
7. To study the response of RC-circuit to various input voltages (square, sine and triangular).
8. To study OR, AND, NOR and NAND gates.

Texts and Reference Books:

1. "B.Sc. Practical Physics by C.L. Arora, S. Chand & Co. (2014).

2. "A Laboratory Manual of Physics for Undergraduate Classes" by D.P. Khandelwal.

B.Sc. Semester 6
(PHY-DSC-10 Maj/Min-602) for Table 2
(PHY-DSC-15 Maj-602) for Table 1

(Nuclear Physics)

Maximum marks: 100
(Theory 75 marks & Practical 25 marks)
Theory 75 marks [60 marks+15 marks (internal assessment 20%)]
Practical 25 marks [20 marks+05 marks (internal assessment 20%)]
Teaching hours: Theory-45 Hrs. / 3 Hrs. per week
Total Credits: 3(Theory) + 1 (Practical)

Learning Outcomes

- To understand general properties of Nuclei, properties of nuclear forces, Liquid drop model and Nuclear shell model,
- To study radioactive decay, Alpha decay, Beta decay, Gamma-ray emission and Nuclear Reactions
- To explain Interaction of nuclear radiation with matter, Gamma-ray interaction with matter, Detectors for nuclear radiation
- To learn Particle Physics, Concept of the quark model, origin and composition of Cosmic rays and Particle accelerators

Note:

1. The theory paper will be of 75 marks including 15 marks for internal assessment and three-Hour duration.
2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of 8 short answer type questions covering the whole syllabus and the students have to attempt any 6 parts of the compulsory question. Each question will have a weightage of 12 marks. The candidates will attempt five questions in all, selecting two questions each from Unit I and II and the compulsory question from unit III.
3. The question paper is expected to contain problems with a weightage of 25-30%.
4. Use of non-programmable calculators will be allowed in the examination centre.
5. The Logarithmic table will be provided in the examination centre, if required.

UNIT-I

General properties of Nuclei: Constituents of nucleus and their intrinsic properties, Quantitative facts about nuclear size, mass, density, binding energy and its variation with mass number, Wave mechanical properties of nucleus, parity; magnetic moment, properties of nuclear forces, Nuclear Models: Liquid drop model, semi-empirical mass formula, most stable isobar. Evidence for nuclear shell structure, Nuclear shell model

Alpha decay, energetic, alpha spectrum, Beta decay, Qualitative discussion of beta spectrum, Evidence of existence of Neutrino, Conservation of nuclear energy in Beta minus, Beta plus and Electron capture decays. Gamma-ray emission

Nuclear Reactions: Types, Concept of compound and direct (pickup and stripping) reactions, Q-value equation, Energy classification of neutrons, Nuclear fission in reactors, Reactor facilities available in India, Nuclear fusion in stars.

UNIT-II

Interaction of nuclear radiation with matter: Energy loss due to ionization (Bethe Bloch formula), Bremsstrahlung, electron-positron annihilation, Gamma-ray interaction with matter, photoelectric effect, Compton scattering, pair production (qualitative description).

Detectors for nuclear radiation: Gas-filled detectors, G.M. counter, Scintillation detector and Photomultiplier tube, Brief account of Semiconductor detectors.

Particle Physics: Particle interactions: Classification of elementary particles, Antiparticles, charge conjugation Symmetries and Conservation principles, Lepton number, baryon number, Isospin, Hypercharge, Strangeness and charm, Gell-mann Nishijima formula, Concept of the quark model, color quantum number and gluons. Origin and composition of Cosmic rays, Secondary cosmic rays, Van Allen belts. Particle accelerators: Tandem accelerator Linear accelerator, Cyclotron, Accelerator facilities available in India.

Books Suggested:

Essential Readings:

1. Concept of Modern Physics, 6 th Ed., A. Beiser, S. Mahajan and S.R. Choudhury, Tata McGraw Hill.
2. Nuclear Physics, I. Kaplan,Addition-Wesley, Publishing Company Inc.
3. Physics for Degree Students, C.L. Arora and P.S. Hemne, S. Chand & Co., 2014.

Further Readings:

1. An Introduction to Nuclear Physics, M.R. Bhiday, and V.A. Joshi, Orient Longman.
2. Concepts of Nuclear Physics, B.L. Cohen, Tata McGraw Hill 3. Fundamentals of Nuclear Physics, J. Verma, CBS.

PHYSICS PRACTICALS
(PHY-DSC-10 Maj/Min-602) for Table 2
(PHY-DSC-15 Maj-602) for Table 1

Max. Marks: 25 (20+5 internal assessment) **Total Credit: 1**
Teaching hours: 2Hrs per week

General Guidelines for Physics Practical Examinations:

1. The distribution of marks is as follows:

(i)	One full experiment requiring the student to take some data, analyze it and draw conclusions. (Candidates are expected to state their results with limits of error).	09
(ii)	One exercise based on experiment	04
(iii)	Viva-Voce	04
(iv)	Record (Practical file)	03
(v)	Internal Assessment	05

Note for Examiners:

2. The marks scored under each head must be clearly written on the answer sheet.
3. There will be one session of 3 hours duration. The paper will have two sections. In Section-A one experiment is to be allotted by the external examiner.
4. Section–B will consist of exercises which will be set by the external examiner on the spot. The length of the exercises should be such that any of these could be completed in one hour.
5. The examiner should take care that the experiment allotted to an examinee from section–A and exercise allotted from section–B are not directly related to each other.
6. Number of candidates in a group for practical examination should not exceed 12.
7. In a single group, no experiment to be allotted to more than three examinees in the group.

Note: Each student will perform at least five experiments in the whole semester.

List of Practicals:

1. To draw the Plateau of a G.M. Counter and find its dead time.
2. To study the statistical fluctuations using G.M. Counter.
3. Verify the inverse square law for gamma ray using G.M. Counter.
4. To study the absorption of beta-particles using G.M. Counter. Also determine the absorption co-efficient (for aluminium) from it.
5. To find the half life period of a given radioactive substance using a G.M. Counter.
6. Verification of Rutherford scattering experiment-mechanical analogue.
7. To study a full wave bridge rectifier using a step down transformer with several output tappings.
8. To study the reduction in the ripples in the rectified output with L, C, LC and π-filters.
9. To study the alpha spectrum from natural sources Th and U.
10. To determine the gamma-ray absorption coefficient for Pb, Sn and Fe elements and Pb-Sn alloy.

Texts and Reference Books:

1. "B.Sc. Practical Physics by C.L. Arora, S. Chand & Co. (2014).
2. "A Laboratory Manual of Physics for Undergraduate Classes" by D.P. Khandelwal.

B.Sc. Semester 6

(PHY-DSC-16-Maj-603)

(Mathematical Physics - II)

Maximum marks: 100

Theory 100 marks [80 marks+20 marks (internal assessment 20%)]

Teaching hours: 60 Hrs. / 4 Hrs. per week

Total Credits: 4 (Theory)

Learning Outcomes

- Understand and apply group theory concepts including finite and continuous groups, representations, characters, and symmetry operations to analyze physical systems, with emphasis on rotation and unitary groups.
- Apply Fourier and Laplace transform techniques to solve physical problems involving series expansions, integral transforms, differential equations, and momentum-space representations.
- Analyze and solve integral equations and tensor problems, including Green's function methods, tensor algebra, covariant differentiation, and applications in non-Cartesian coordinate systems.
- Implement numerical methods and simulation techniques for solving differentiation, integration, and differential equations, performing data fitting, and modeling physical systems using Monte Carlo methods.

Note:

1. The theory paper will be of 100 marks including 20 marks for internal assessment and three-Hour duration.
2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of 10 short answer type questions covering the whole syllabus and the students have to attempt any 8 parts of the compulsory question. Each question will have a weightage of 16 marks. The candidates will attempt five questions in all, selecting two questions each from Unit I and II and the compulsory question from unit III.
3. The question paper is expected to contain problems with a weightage of 25-30%.
4. Use of non-programmable calculators will be allowed in the examination centre.
5. The Logarithmic table will be provided in the examination centre, if required.

Unit-I

Group Theory: Basic definitions, Multiplication table, conjugate elements and classes. Subgroups, Direct product of groups, Isomorphism and Homomorphism. Permutation groups, Definition of representation and its properties. Reducible and irreducible presentation. Schurs' Lemmas (Statement only), Orthogonal theorem, Characters of a representation. Example of C4v.Topological groups and Lie groups, Three dimensional rotation group. Unitary groups: SU(2), O(3), Applications of group theory.

Fourier Series and Integral Transforms: Fourier series, General properties, Advantages and applications, Gibbs phenomenon. Development of the Fourier integral, Inversion theorem, Fourier transform, Fourier transforms of derivatives, Momentum representation. Laplace transforms, Laplace transforms of derivatives, Properties of Laplace transform, Faltung theorem, Inverse Laplace transformation.

Unit-II

Integral Equations: Definitions and classifications, Neumann series, Separable kernels, Hilbert-Schmidt theory. Green's function in one dimension.

Tensors: Introduction, definitions, contraction, direct product, Quotient rule. Pseudo tensors, Levi-Civita symbol, irreducible tensors, Non-Cartesian tensors - metric tensor. Christoffel symbols, Covariant differentiation.

Elementary Numerical Analysis: Numerical differentiation, Numerical integration by Simpson and Weddle's rules. Numerical solution of differential equations by Euler and Runge-Kutta Method, Linear and non-linear least square fitting, generation of random numbers, Monte-Carlo technique, integration, simulations.

Recommended Books:

1. Mathematical methods for Physicists: G. Arfken and H.J. Weber (Academic Press, San Diego) (sixth edition) (2005).
2. Group Theory for Physicists: A.W. Joshi (Wiley Eastern, New Delhi) (2005).
3. Numerical Mathematical Analysis, J.B. Scarborough (Oxford Book Co. Kolkata) (1961).
4. A First Course in Computational Physics: P.L. Devries (Wiley, New York) (1994).

B.Sc. Semester 6

(PHY-DSC-17-Maj-604)

(Advanced Computational Physics)

Maximum marks: 100

(Theory 50 marks & Practical 50 marks)

Theory 50 marks [40 marks+10 marks (internal assessment 20%)]

Practical 50 marks [40 marks+10 marks (internal assessment 20%)]

Teaching hours: Theory-30 Hrs. / 2 Hrs. per week

Total Credits: 2(Theory)+2(Practical)

Learning Outcomes

1. To understand the fundamental principles of object oriented programming and the advantages of OOP over procedural programming.
2. To design and implement classes and objects using appropriate access specifiers.
3. To learn effective use of functions and apply pointers, references, constructors and destructors in OOP.
4. To learn some fundamental numerical methods and utilise them to solve real world problems and model them using OOP.

Note:

1. The theory paper will be of 50 marks including 10 marks for internal assessment and three Hour duration.
2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of 8 short answer type questions covering the whole syllabus and the students have to attempt any 6 parts of the compulsory question. Each question will have a weightage of 8 marks. The candidates will attempt five questions in all, selecting two questions each from Unit I and II and the compulsory question from unit III.
3. The question paper is expected to contain problems with a weightage of 25-30%.
4. Use of non-programmable calculators will be allowed in the examination centre.
5. The Logarithmic table will be provided in the examination centre, if required.

UNIT-I

C++ Programming Language

Functions :- Standard Library Functions and User-defined Functions. Void Functions and Functions returning Values. Function Prototypes. Function Call by Value and by Reference. Recursion. Idea of Function Overloading, friend functions and friend classes, virtual functions, scope resolution operator, Procedural vs Object Oriented Programming, Idea of Classes, Objects and Inheritance :- Classes and Objects, data hiding, encapsulation, dynamic binding, message passing and polymorphism in C++ with suitable examples, advantages of OOP, Member Functions in a class. Private and Public Qualifiers and Data Security. Constructors and Destructors. Idea of Strings and Pointers.

UNIT-II

Introduction to Numerical methods: Computer algorithms, interpolations –cubic spline fitting, Derivation and related problems for Numerical differentiation – forward difference formula, Lagrange interpolation, Numerical integration by Simpson and Weddle's rules, random generators, Numerical solution of differential equations by Euler, predictor-corrector and Runge-Kutta methods, Feynman Newton Method.
Derivation and related problems for methods for solving simultaneous linear equations- gauss elimination method and Gauss-Seidel method.

Recommended Books:

1. Numerical Mathematical Analysis, J.B. Scarborough (Oxford Book Co.) 5th edition, (1964).
2. A first course in Computational Physics: P.L. DeVries (Wiley) (1994).
3. Schaum's outline of Programming with C, Byron Gottfried, McGraw-Hill
4. Computer Programming in C, V. Rajaraman (Prentice Hall of India Pvt. Ltd.) (2006)
5. Computational Physics: R.C. Verma, P.K. Ahluwalia and K.C. Sharma (New Age)(2005)
6. Computer Applications in Physics: S. Chandra (Narosa)(2008).

PHYSICS PRACTICALS

(PHY-DSC-17-Maj-604)

Max. Marks: 50 (40+10 internal assessment) **Total Credit: 2**

Teaching hours: 60 Hrs. / 4 Hrs. per week

General Guidelines for Physics Practical Examinations:

1. The distribution of marks is as follows:

(i) One full experiment requiring the student to discuss the theory and algorithm of the problem assigned along with designing and executing a C++ program based on it. 20

(ii) Viva-Voce 10

(iii) Record (Practical file) 10

(iv) Internal Assessment 10

Note for Examiners:

2. The marks scored under each head must be clearly written on the answer sheet.
3. There will be one session of 3 hours duration. Any one experiment is to be allotted by the external examiner.
4. Number of candidates in a group for practical examination should not exceed 12.
5. In a single group, no experiment to be allotted to more than three examinees in the group.

List of Practicals:

1. Study the charging and discharging of a capacitor in RC circuit with a DC source using Euler method. Graphically demonstrate the variation of charge with time for two values of time step size. Modify the program to include AC source instead of D.C. Source.
2. Study the growth and decay of current in RL circuit containing (a) DC source and (b) AC source using Runge Kutta method. Draw graphs between current and time in each case. Perform power analysis in the circuit for two values of time step size for the case.
3. Study graphically the motion of a falling spherical body under various effects of the medium (viscous drag, buoyancy and air drag) using Euler method.
4. Study graphically the path of a projectile with and without air drag, using FN method. Find the horizontal range and maximum height in either case.
5. Design a C++ program to study the motion of an artificial satellite.
6. Study of motion of a one-dimensional harmonic-oscillator without and with damping effect (use Euler method). Draw graphs showing the relations (a) velocity vs time (b) acceleration vs time (c) position vs time.
7. Obtain the energy eigen values of a quantum oscillator using Runge-Kutta method.
8. Monte-Carlo technique to simulate phenomenon of Nuclear radioactivity. Modify your program to a case when daughter nucleus is also unstable.
9. Study the motion of two coupled harmonic oscillators. Compare the numerical results with analytic results.

Books Suggested:

1. Schaum's outline of Programming with C, Byron Gottfried, McGraw-Hill.
2. Computational Physics: R.C. Verma, P.K. Ahluwalia and K.C. Sharma (New Age) (2005).
3. Computer Applications in Physics: S. Chandra (Narosa) (2008).

B.Sc. Semester 6

(PHY-DSC-18-Maj-605)

(Classical Mechanics)

Maximum marks: 100
Theory 100 marks [80 marks+20 marks (internal assessment 20%)]
Teaching hours: 60 Hrs. / 4 Hrs. per week
Total Credits: 4 (Theory)

Learning Outcomes

- Formulate and analyze mechanical systems using Lagrangian mechanics, incorporating constraints, generalized coordinates, dissipation functions, and variational principles to derive equations of motion.
- Apply Hamilton's principle and conservation laws to physical systems, understanding symmetry properties of space and time and extending variational formulations to non-holonomic systems.
- Analyze rigid body dynamics, including Euler angles, inertia tensor, principal axes, torque-free motion, and the dynamics of symmetric tops using Euler's equations.
- Solve advanced dynamical problems using Hamiltonian mechanics, including small oscillations, canonical transformations, Poisson brackets, Hamilton–Jacobi theory, and action–angle variables.

Note:

1. The theory paper will be of 100 marks including 20 marks for internal assessment and three-Hour duration.
2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of 10 short answer type questions covering the whole syllabus and the students have to attempt any 8 parts of the compulsory question. Each question will have a weightage of 16 marks. The candidates will attempt five questions in all, selecting two questions each from Unit I and II and the compulsory question from unit III.
3. The question paper is expected to contain problems with a weightage of 25-30%.
4. Use of non-programmable calculators will be allowed in the examination centre.
5. The Logarithmic table will be provided in the examination centre, if required.

Unit-I

Lagrangian Formulation: Mechanics of a system of particles: constraints of motion. Generalized coordinates, D'Alembert's Principle and Lagrange's velocity – dependent force and the dissipation function. Application of Lagrangian formulation.

Hamilton Principle : Calculus of variations. Hamilton principle. Lagrange's equation from Hamilton's principle. Extension to non-holonomic systems, advantages of variational principle formulation.

Rigid Body Motion : Independent co-ordinates of rigid body, orthogonal transformation. Eulerian angles and Euler's theorems. Coriolis force, angular momentum and kinetic energy of a rigid body, the inertia tensor, principal axis transformation. Euler equations of motion. Torque free motion of rigid body, motion of a symmetrical top.

Unit-II

Small Oscillations: Eigenvalue equation. Free vibrations. Normal Coordinates. Vibrations of a triatomic molecule.

Hamilton's Equations: Legendre Transformations. Hamilton's equations of motion. Cyclic-coordinates. Hamilton's equations from variational principle, principle of least action.

Canonical Transformation and Hamilton- Jacobi Theory: Canonical transformation and its example, Poission brackets. Equations of motion. Possion's bracket relations, infinitesimal canonical transformation. Conservation Theorems. Hamilton – Jacobi equations for principal and characteristic functions. Harmonic oscillator problem, Action angle variables for system with one degree of freedom.

Recommended Books:

1. Classical Mechanics: H. Goldstein (Narosa, New Delhi) (1992).
2. Classical Mechanics of Particles and Rigid Bodies: K.C. Gupta (Wiley Eastern, New Delhi) (2006).
3. Analytical Mechanics: L.N. Hand and J.D. Finch (Cambridge University Press, Cambridge) (1998).
4. Classical Mechanics: V.D. Barger and M.G. Olsson, (McGraw-Hill, New York) (1973).
5. Classical Mechanics: N.C. Rana and P.J. Joag (Tata McGraw Hill, New Delhi) (2004).

B.Sc. Semester 6
(PHY-DSC-19-Maj-606)
(Atomic and Molecular Physics)

Maximum marks: 100
(Theory 75 marks & Practical 25 marks)
Theory 75 marks [60 marks+15 marks (internal assessment 20%)]
Practical 25 marks [20 marks+05 marks (internal assessment 20%)]
Teaching hours: Theory-45 Hrs. / 3 Hrs. per week
Total Credits: 3(Theory)+1(Practical)

Learning Outcomes

- Analyze atomic spectra using vector models, spin-orbit interaction, and LS/JJ coupling schemes for multi-electron systems.
- Calculate energy splittings and interpret spectral lines resulting from Zeeman and Paschen-Back effects in magnetic fields.
- Evaluate rotational and vibrational spectra of diatomic molecules using rigid rotator and harmonic oscillator quantum models.
- Apply the Franck-Condon principle and Born-Oppenheimer approximation to explain intensity distributions in electronic and Raman spectra.
- Explain the operating principles and analytical applications of UV-Vis, FTIR, NMR, and ESR spectroscopic instrumentation techniques.

Note:

1. The theory paper will be of 75 marks including 15 marks for internal assessment and three-Hour duration.
2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of 8 short answer type questions covering the whole syllabus and the students have to attempt any 6 parts of the compulsory question. Each question will have a weightage of 12 marks. The candidates will attempt five questions in all, selecting two questions each from Unit I and II and the compulsory question from unit III.
3. The question paper is expected to contain problems with a weightage of 25-30%.
4. Use of non-programmable calculators will be allowed in the examination centre.
5. The Logarithmic table will be provided in the examination centre, if required.

UNIT-I

Vector model for one and two valance electron atoms; Spin-orbit interaction and fine structure of hydrogen, Lamb shift (qualitative), Spectroscopic notations for L-S and J-J couplings; Spectra of alkali and alkaline earth metals; Interaction energy in L-S and J-J coupling for two electron systems; Selection and Intensity rules for doublets and triplets. Exchange symmetry of atomic wavefunctions and Pauli's principle.

Natural breadth of spectral line, Line broadening mechanisms, The Zeeman Effect for two electron systems; Intensity rules for the Zeeman effect; The calculations of Zeeman patterns; Paschen-Back effect; LS coupling and Paschen-Back effect; Lande's factor in LS coupling.

UNIT II

Molecular spectra, Rotational spectra of diatomic molecules as a rigid and non-rigid rotator, Intensity of rotational lines, Effect of isotopic substitution, Vibrating diatomic molecule as a simple harmonic and an anharmonic oscillator, Diatomic vibrating rotator, The vibration-rotation spectrum of carbon monoxide, Rotational Raman spectra for diatomic molecules, Vibrational Raman spectra, Electronic structure of diatomic molecule, Electronic spectra of diatomic molecules, Frank-Condon principle. Born-Oppenheimer approximation, Rotational and Vibrational structures in electronic transitions of diatomic molecules, selection rules.

A brief account of Atomic Absorption and emission Spectrometers, UV-Vis Spectrometer-Outline of technique and instrumentation, Fourier transform spectroscopy and FTIR Spectrometer, Raman Spectrometer, Electron spin resonance, Nuclear magnetic resonance.

Recommended Books:

1. Introduction to Atomic Spectra: H.E. White-Auckland Mc Graw Hill, 1934.
2. Atomic and Molecular Spectra: Rajkumar (Kedarnath Ramnath Prakashan, Meerut).
3. Undergraduate Instrumental Analysis: James W. Robinson (Marcel Dekker, New York) 44(1970).
4. Fundamentals of Molecular spectroscopy: Banwell and McCash (Tata McGraw Hill) (1994).
5. Molecular Structure and Molecular Spectroscopy G. Aruldhas (PHI Learning) (2009)
6. Lasers and Non-linear Optics: B.B. Laud. (Wiley Eastern) (1991).
7. Spectra of diatomic molecules vol. I: Herzberg-New York, 1944.

PHYSICS PRACTICALS

(PHY-DSC-18-Maj-606)

Max. Marks: 25 (20+5 internal assessment) **Total Credit: 1**

Teaching hours: 30 Hrs. / 2 Hrs. per week

General Guidelines for Physics Practical Examinations:

1. The distribution of marks is as follows:

(i)	One full experiment requiring the student to take some data, analyze it and draw conclusions. (Candidates are expected to state their results with limits of error).	09
(ii)	One exercise based on experiment	04
(iii)	Viva-Voce	04
(iv)	Record (Practical file)	03
(v)	Internal Assessment	05

Note for Examiners:

2. The marks scored under each head must be clearly written on the answer sheet.
3. There will be one session of 3 hours duration. The paper will have two sections. In Section A, one experiment is to be allotted by the external examiner.
4. Section B will consist of exercises that will be set by the external examiner on the spot. The length of the exercises should be such that any of these could be completed in one hour.
5. The examiner should take care that the experiment allotted to an examinee from section A and the exercise allotted from section B are not directly related to each other.
6. The number of candidates in a group for practical examination should not exceed 12.
7. In a single group, no experiment is to be allotted to more than three examinees in the group.

Note: Each student will perform at least five experiments in the whole semester.

List of Practicals:

1. To determine charge to mass ratio (e/m) of an electron.
2. To determine the lonization Potential of mercury using Thyratron valve.
3. To determin he value of Rydberg's constant using hydrogen emission spectrum.
4. To determine the g-factor of free electron using ESR.
5. To determine the first excitation potential of gas by Frank-Hertz experiment.
6. To observe spectral band formation in Franck-Hertz Experiment and record the character-intic curve.
7. Determination of dissociation limit of iodine molecule by constant deviation spectrograph.

8. To study the absorption spectrum of iodine vapours and calculate electronic energy gap and vibrational energies.
9. Determination of velocity of light using modulated Laser method.
10. To determine the value of Planck's constant using different colour LED's.

Texts and Reference Books:

1. B.Sc. Practical Physics by C.L. Arora, S.Chand & Co. (2014).
2. A Laboratory Manual of Physics for Undergraduate Classes by D.P. Khandelwal.
3. Atomic and Molecular Spectra: Rajkumar (Kedarnath Ramnath Prakashan, Meerut).

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STATISTICS

NEP 2020 COLLEGES SCIENCE TEMPLATE: STUDENTS OPTING B.Sc., B.Sc. HONOURS AND B.Sc. HONOURS WITH RESEARCH
FOR SINGLE MAJOR AND SINGLE MINOR(Table 1*) and STUDENTS OPTING FOR MULTI-DISCIPLINARY/ INTER-DISCIPLINARY(Table 2*)
(FOR THE EXAMINATIONS OF 2026-27 ACCORDING TO NEP-2020)

Table 1						Table 2					
Sem	**Level of Course**	**Course Code**	**Course Name**	**Credits**	**Total Credits**	**Sem**	**Level of Course**	**Course Code**	**Course Name**	**Credits**	**Total Credits**
V	L-300	(STAT-DSC-9 Maj-501)	Nonparametric Statistical Inference	3+1=4	24	V	L-300	(STAT-DSC-6 Maj/Min-501)	Sampling Distributions	3+1=4(x3)	24
		(STAT-DSC-10 Maj-502)	Matrices	4				(STAT-DSC-7 Maj-502)	Applied Statistics-I	3+1=4 (X2)	
		(STAT-DSC-11 Maj-503)	Applied Statistics-II	3+1=4							
		(STAT-DSC-12 Maj-504)	Basic Statistics Using R Programming	3+1=4							
		(STAT-DSC-13 Maj-505)	Biostatistics	3+1=4							
		(STAT-INT/COM-506)	Internship/Community work	4				(STAT-INT/COM-506)	Internship /Community work	4	
VI	L-300	(STAT-DSC-14 Maj/Min-601)	Optimization Techniques	3+1=4(X2)	24	VI	L-300	(STAT-DSC-14 Maj/Min-601)	Optimization Techniques	3+1=4 (x3)	24
		(STAT-DSC-15 Maj-602)	Linear Models	3+1=4				(STAT-DSC-15 Maj-602)	Applied Statistics-II	3+1=4 (x3)	
		(STAT-DSC-16 Maj-603)	Demography	3+1=4							
		(STAT-DSC-17 Maj-604)	Advanced Statistics Using R Programming	3+1=4							
		(STAT-DSC-18 Maj-605)	Numerical Analysis	3+1=4							
The students successfully securing 144 credits will be awarded B.Sc.. Degree											

Syllabus under Single Major and Single Minor Scheme 2025-26

B.Sc. (Hons) STATISTICS

Semester-V

Course Code: (STAT-DSC-9 Maj-501)

Course Name: Non-parametric Statistical Inference

Credit: 4
L-3, P-1

Notes:

1. **Students are to be engaged to 5 hours of teaching per week (3 lectures for theory and 2 lectures for practical).**
2. **Practical examination will be held in the respective semester.**

Paper Title with code	Lecture(L)/ Credit	Practical(P)/ Credit	Marks Distribution	
Non-Parametric Statistical Inference (STAT-DSC-9 Maj-501)	Lecture :3 Credit: 3	Practical Lecture:2 Credit:1	Theory Max. Marks: 75 Theory: 60 Internal Assessment: 15 Time: 3 Hours	Practical Max. Marks: 25 Viva voce: 5 Record of the practical: 5 Written practical Exam: 15 Time: 3 Hours

NON-PARAMETRIC STATISTICAL INFERENCE

Learning Objectives: The learning objectives include:

1. Understand the Foundations of Nonparametric Methods.
2. Apply Nonparametric Tests to One-Sample, Two-Sample, and Paired-Sample Problems.
3. Analyze Location and Scale Differences Using Linear Rank Tests and ANOVA Alternatives.
4. Evaluate Measures of Association in Bivariate and Multivariate Contexts.

Learning Outcomes: After completing this course, students should have developed a clear understanding of:

1. Distinguish between Parametric and Nonparametric test and appreciate different measurementscales
2. Understands quantile and empirical distribution function and its utility
3. Use nonparametric test for both one/two samples problem, Test for randomness, Kolmogorov-Smirnov one sample and two sample tests, sign test, Wilcoxon signed rank test, run test. Mediantest, Mann Whitney U test, Kruskal-Wallis one-way analysis of variance by ranks.
4. Test association of bivariate samples using Kendall tau and Spearman's rank correlation.

Note: There will be nine questions in all. Question No.1 will be compulsory covering whole of the syllabus and comprising 5 to 8 short answer type questions. Rest of the eight questions will be set from the four units uniformly i.e. two from each unit. The candidate will be required to attempt five questions in all selecting one question from each unit and the compulsory one. All the questions will carry equal marks.

UNIT I

Nonparametric Tests:Introduction and Concept, Non-parametric tests-their advantages and disadvantages, comparison withparametric tests. Measurement scale-nominal, ordinal, interval and ratio. The quantile function, theempirical distribution function, Test for randomness based on total number of runs.

UNIT II

One-Sample, two-sample problem, and Paired-Sample Procedures: the sign test, treatment of ties in ranktests, Wilcoxon signed-rank test, Wald-Wolfowitz runs test, Kolmogorov-Smirnov one and two-sampletest, median test, and the Mann-Whitney U test.

UNIT III

Linear Rank Tests for the Location and Scale Problem:Definition of linear rank statistics, Wilcoxon rank-sum test; Tests of the Equality of k IndependentSamples: The Kruskal-Wallis one-way ANOVA test; Measures of Association for Bivariate Samples:definition of measures of association in a bivariate population, Kendall's Tau coefficient, Spearman'scoefficient of rank correlation.

UNIT IV

Measures of association in multiple classifications: Friedman's two-way ANOVA by ranks in a k x ntable, the Coefficient of Concordance of k sets of rankings of n objects, the Coefficient ofConcordance of k sets of incomplete rankings

Recommended Books:

1. Gibbons, J. D., and Chakraborti, S. (2020): Nonparametric statistical inference. CRC press.
2. Siegel, S. (1956). Nonparametric statistics for the behavioral sciences.McGraw-Hill.
3. Kloke, J., and McKean, J. W. (2014) : Nonparametric statistical methods using R. CRC Press.
4. Hollander, M., Wolfe, D. A., and Chicken, E. (2013): Nonparametric statistical methods (Vol.751). John Wiley & Sons.

(STAT-DSC-9 Maj-501): **NON-PARAMETRIC STATISTICAL INFERENCE (PRACTICAL)**

Notes:

(i) The Practical Question Paper will contain five questions. A student will be required to attempt any three questions, each of 5 marks, in three hours duration.

(ii) The practical of paper (STAT-DSC-9 Maj-501) will be based on the following topics:

1. Problem based on obtaining Quantile and Empirical Distribution
2. Problem based on Test for randomness.
3. Problem based on Sign test
4. Problem based on Wilcoxon Signed rank test
5. Problem based on Wald-Wolfowitz runs test,
6. Problem based on Kolmogorov-Smirnov one and two sample test,
7. Problem based on Median test and the Mann-Whitney U test.
8. Problem based on Wilcoxon rank-sum test
9. Problem based on The Kruskal-Wallis one-way ANOVA test.
10. Problem based on Kendall's Tau coefficient.
11. Problem based on Spearman's coefficient of rank correlation
12. Problem based on Friedman's two way ANONA by ranks in k×n table.
13. Problem based on Coefficient of Concordance of k sets of ranking on n objects.

B.Sc. (Hons) STATISTICS
Semester-V

Course Code: (STAT-DSC-10 Maj-502)
Course Name: Matrices

Credit: 4
L-4

Note: Students are to be engaged to 4 hours of teaching per week.

Paper Title with code	Lecture(L)/Credit	Marks Distribution
Matrices (STAT-DSC-10 Maj-502)	Lecture : 4 Credit : 4	Theory Max. Marks: 100 Theory: 80 Internal Assessment : 20 Time: 3 Hours

MATRICES

Course Objectives:

1. To develop a foundational understanding of the theory of equations.
2. To introduce the fundamental concepts of linear algebra.
3. To review and expand knowledge of matrix algebra and determinants.
4. To explore advanced topics in linear algebra**, including matrix rank, generalized inverses, characteristic roots and vectors, the Cayley-Hamilton theorem, and diagonalization of quadratic forms through orthogonal transformations.

Course Learning Outcomes:

After completing this course, students should have developed a clear understanding of:

1. Apply the Fundamental Theorem of Algebra and Root-Coefficient Relationships.
2. Demonstrate Proficiency in Vector Space Concepts.
3. Utilize Matrix Algebra and Determinants in Solving Linear Systems.
4. Determine Matrix Rank and Analyze Eigenvalues, Eigenvectors, and Quadratic Forms.

Note: There will be nine questions in all. Question No.1 will be compulsory covering whole of the syllabus and comprising 5 to 8 short answer type questions. Rest of the eight questions will be set from the four units uniformly i.e. two from each unit. The candidate will be required to attempt five questions in all selecting one question from each unit and the compulsory one. All the questions will carry equal marks.

UNIT I

Algebra of matrices - A review, theorems related to triangular, symmetric and skew symmetric matrices, idempotent matrices, Hermitian and skew Hermitian matrices, orthogonal matrices, singular and non-singular matrices and their properties. Trace of a matrix, unitary, involutory and nilpotent matrices. Adjoint and inverse of a matrix and related properties.

UNIT II

Determinants of Matrices: Definition, properties and applications of determinants for 3rd and higher orders, evaluation of determinants of order 3 and more using transformations. Symmetric and Skew symmetric determinants, Circulant determinants, Jacobi's Theorem, product of determinants. Use of determinants in solution to the system of linear equations, row reduction and echelon forms, the matrix equations AX=B, solution sets of linear equations, linear independence, Applications of linear equations, inverse of a matrix.

UNIT III

Theory of equations, statement of the fundamental theorem of algebra and its consequences.Relation between roots and coefficients or any polynomial equations.Solutions of cubic and biquadratic equations when some conditions on roots of equations are given.Evaluation of the symmetric polynomials and roots of cubic and biquadratic equations. Vector spaces, Subspaces, sum of subspaces, Span of a set, Linear dependence and independence, dimension and basis, dimension theorem.

UNIT IV

Rank of a matrix, row-rank, column-rank, standard theorems on ranks, rank of the sum and the product of two matrices. Generalized inverse (concept with illustrations).Partitioning of matrices and simple properties. Characteristic roots and Characteristic vector, Properties of characteristic roots, Cayley Hamilton theorem, Quadratic forms, Linear orthogonal transformation and their digitalization.

Recommended Books:

1. Cheney W. and Kincaid, D. (2012). Linear Algebra, Jones & Bartlett Learning.
2. Artin M. (1994). Algebra.Prentice Hall of India.
3. Hadley, G. (2002). Linear Algebra.Narosa Publishing House (Reprint).
4. Krishnamurthy, V., Mainra, V.P. and Arora, J.L. (2015). An Introduction to Linear Algebra, East West Press Pvt. Ltd., New Delhi.
5. Lay, D. C. (2000). Linear Algebra and its Applications, Addison Wesley.

B.Sc. (Hons) STATISTICS
Semester- V
Course Code: (STAT-DSC-11 Maj-503)
Course Name: Applied Statistics-II

Credit: 4
L-3, P-1

Notes:

1. **Students are to be engaged to 5 hours of teaching per week (3 lectures for theory and 2 lectures for practical).**
2. **Practical examination will be held in the respective semester.**

Paper Title with code	Lecture(L)/Credit	Practical(P)/ Credit	Marks Distribution	
Applied Statistics-II (STAT-DSC-11 Maj-503)	Lecture :3 Credit: 3	Practical Lecture:2 Credit:1	Theory Max. Marks: 75 Theory: 60 Internal Assessment : 15 Time: 3 Hours	Practical Max. Marks: 25 Viva voce: 5 Record of the practical: 5 Written practical Exam: 15 Time: 3 Hours

APPLIED STATISTICS- II

Objectives: The learning objectives include:

1. To familiarize students with the Indian statistical system and the role of official statistical agencies at central and state levels.
2. To understand the collection, classification, and interpretation of official statistics in sectors like population, agriculture, industry, trade, labor, transport, and finance.
3. To develop knowledge of Statistical Quality Control (SQC) methods, including control charts for variables and attributes, and their industrial applications.
4. To provide insights into acceptance sampling techniques, their plans, operating characteristics, and associated risks in quality inspection.

Learning Outcomes: After completing this course, students will be able to:

1. Explain the structure and functions of the Indian statistical system and interpret official statistics related to population, agriculture, industry, trade, labor, transport, and finance.
2. Apply Statistical Quality Control (SQC) techniques to analyze process variations and construct control charts for variables and attributes.
3. Interpret control charts (R, σ, p, d, c charts) for monitoring and improving product and process quality.

4. Design and evaluate acceptance sampling plans, understanding concepts such as AQL, LTPD, producer's risk, and consumer's risk.

Note: There will be nine questions in all. Question No.1 will be compulsory covering whole of the syllabus and comprising 5 to 8 short answer type questions. Rest of the eight questions will be set from the four units uniformly i.e. two from each unit. The candidate will be required to attempt five questions in all selecting one question from each unit and the compulsory one. All the questions will carry equal marks.

UNIT-I

Indian official statistics: Introduction, Indian statistical system, Statistical offices at the centre, Statistical offices in the states, Population statistics, Agricultural statistics, Industrial statistics, Trade statistics, Price statistics, Statistics of labor and employment, Statistics of transport and communication, Financial and banking statistics.

UNIT-II

Statistical Quality Control : Meaning and uses of Statistical Quality Control (SQC), causes of variations in quality, product and process control, control charts, 3 - σ control limits, control chart for variables- $\overline{X}$ and R chart, Interpretation of $\overline{X}$& R charts.

UNIT-III

Control chart for standard deviation (σ chart), control charts for attributes: control charts for number of defectives (d- chart), control charts for fraction defectives (p-chart) and control charts for number of defects per UNIT (c- chart), applications of c-chart, natural tolerance and specification limits.

UNIT-IV

Acceptance sampling :Acceptance sampling inspection plans, Acceptance quality level (AQL), lot tolerance proportion defective (LTPD), process average fraction defective, producer's and consumer's r isks, rectifying inspection plans, single and double sampling plans, their OC functions, concepts of AQL, LTPD, AOQL, average amount of inspection and ASN function.

Books recommended

S. No.	Title of Book	Name of author	Publisher
1.	Statistical Quality Control	Grant E.L.	McGraw Hill
2.	Statistical Methods in Quality Control	Cowden D.J.	Asia Pub. Society
3.	Fundamentals of Applied Statistics	Gupta S.C. & Kapoor V.K.	Sultan Chand & Sons
4.	Fundamentals of Statistics, Vol. II	Goon A.M., Gupta M.K. & Dasgupta B.	World Press Calcutta

(STAT-DSC-11 Maj-503): **APPLIED STATISTICS-II (PRACTICAL)**

Notes:

1. The Practical Question Paper will contain five questions. A student will be required to attempt any three questions, each of 5 marks, in three hours duration.
2. The practical of paper (STAT-DSC-11 Maj-503) will be based on the following topics:

    1. Problem based on Range and mean charts.
    2. Problem based on standard deviation chart.
    3. Problem based on d-chart.
    4. Problem based on p-chart.
    5. Problem based on c-chart.
    6. Problem based on single sampling plans.
    7. Problem based on double sampling plans.

B.Sc. (Hons) STATISTICS
Semester-V

Course Code: (STAT-DSC-12 Maj-504)
Course Name: Basic Statistics using R Programming

Credit: 4
L-3, P-1

Notes:

1. **Students are to be engaged to 5 hours of teaching per week (3 lectures for theory and 2 lectures for practical).**
2. **Practical examination will be held in the respective semester.**

Paper Title with code	Lecture(L)/ Credit	Practical(P)/ Credit	Marks Distribution	
Basic Statistics using R Programming (STAT-DSC-12 Maj-504)	Lecture :3 Credit: 3	Practical Lecture:2 Credit:1	Theory Max. Marks: 75 Theory: 60 Internal Assessment : 15 Time: 3 Hours	Practical Max. Marks: 25 Viva voce: 5 Record of the practical: 5 Written practical Exam: 15 Time: 3 Hours

BASIC STATISTICS USING R PROGRAMMING

Course Objectives:

The main objectives of this course are:

1. To Impart training in R programming, create different types of R objects and perform operations
2. To detail the construction of plots, various discrete and continuous probability distributions
3. To impart skills in analyzing univariate and bivariate data.

Course Outcomes:

On the successful completion of the course, student will be able:

1. To Impart training in R programming.
2. To create different types of R objects and perform operations, construction of plots.
3. To work on various discrete and continuous probability distributions.

Note : There will be nine questions in all. Question No. 1 will be compulsory covering whole of the syllabus and comprising 5 to 8 short answer type questions. Rest of the eight questions will be set from the four units uniformly i .e. two from each unit. The candidate will be required to attempt five questions in all selecting one question from each unit and the compulsory one. All the questions will carry equal marks.

UNIT-I

Introduction to R: Features, installation, variable, input of data, output, in-built functions, packages. Data types in R: vectors, matrices, array, lists, factors, data frame. Programming in R: decision making structures (if structure, switch statement), loops (For, While, Repeat), Import and export of data.

UNIT-II

Data Visualization and Descriptive Statistics: generating tables - Bar plots - Pie chart - Box plot – Histogram, frequency polygon, cumulative frequency curves or Ogives, stem and leaf chart, Scatter plot – line plots (single, multiple) – partitioning graphics window – adding title, labels and legends to plots - obtaining measures of central tendency, measures of dispersion, Skewness and kurtosis.

UNIT-III

Probability Distributions: Obtaining density, cumulative density and quantile values for discrete and continuous distributions: Binomial, Poisson, Exponential, and Normal, generating samples from discrete and continuous distributions - Plotting density and cumulative density curves - Q-Q plot.

UNIT-IV

Analyzing bivariate and multivariate data: Correlation analysis - Pearson, Spearman, multiple and partial correlations.
Regression Analysis: Fitting of linear, quadratic, exponential functions curves and multiple regressions with the interpretation of results.
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Recommended Books:

1. Motwani B. : Data Analytics with R, Wiley Publication.
2. Lander J.P.: R for Everyone, Pearsons Publication.
3. Pierre-Andre Cornillon et al. (2012): R for Statistics, CRC Press.
4. Jared P Lander (2014): R for Everyone: Advanced Analytics and Graphics, Pearson Education Inc.
5. Peter Dalgaard (2008): Introductory Statistics with R, 2nd edition, Springer.

(STAT-DSC-12 Maj-504): **BASIC STATISTICS USING R PROGRAMMING (PRACTICAL)**

Notes:

(i) The Practical Question Paper will contain five questions. A student will be required to attempt any three questions, each of 5 marks, in three hours duration.

(ii) The practical of paper (STAT-DSC-12 Maj-504) will be based on the following topics:

1. Graphical representation of data with bar-plot, pie-chart, stem & leaf chart and boxplot.
2. Histogram with equal and unequal class intervals, frequency polygon and QQ Plots.
3. Less than and more than Ogives.
4. Fitting of curve linear, quadratic, exponential functions,
5. Problem based on Scatter plots, Karl Pearson's correlation.
6. Problem based on Spearman Rank Correlation
7. Problem based on Multiple and Partial Correlation coefficients.
8. Problem based on Linear and multiple regression
9. Problem based on discrete distributions viz. Binomial, Poisson.
10. Problem based on continuous distribution viz. Exponential, and Normal.

B.Sc. (Hons) STATISTICS
Semester-V

Course Code: (STAT-DSC-13 Maj-505)
Course Name: Biostatistics

Credit: 4
L-3, P-1

Notes:

1. **Students are to be engaged to 5 hours of teaching per week (3 lectures for theory and 2 lectures for practical).**
2. **Practical examination will be held in the respective semester.**

Paper Title with code	Lecture(L)/ Credit	Practical(P)/ Credit	Marks Distribution	
Biostatistics (STAT-DSC-13 Maj-505)	Lecture :3 Credit: 3	Practical Lecture:2 Credit:1	Theory Max. Marks: 75 Theory: 60 Internal Assessment : 15 Time: 3 Hours	Practical Max. Marks: 25 Viva voce: 5 Record of the practical: 5 Written practical Exam: 15 Time: 3 Hours

BIOSTATISTICS

Course Objectives: Biostatistics is one area of Applied Statistics that is concerned with the application of statistical methods to medical, biological, epidemiological and health related aspects of all forms of life.
Course Learning Outcomes: After successful completion of this course, student will be able to:

1. Develop understanding of time-to-event data in Biomedical Sciences.
2. Understanding concepts of conditional, inverse probabilities as applied to survival data and survival patterns in presence and in absence of censoring.
3. Estimation of failure and hazard forms based on patient data records.
4. Formulate and interpret stochastic models for specific-disease data sets.
5. Comprehend basis and construction of clinical trials for different stages.
6. Analyses concept of Biometric genetics.

Note: There will be nine questions in all. Question No.1 will be compulsory covering whole of the syllabus and comprising 5 to 8 short answer type questions. Rest of the eight questions will be set from the four units uniformly i.e. two from each unit. The candidate will be required to attempt five questions in all selecting one question from each unit and the compulsory one. All the questions will carry equal marks.

UNIT I

Analysis of Medical, Epidemiologic and Clinical Data: Studying association between a disease and a characteristic: (a) Types of studies in Epidemiology and Clinical Research (i) Prospective study (ii) Retrospective study (iii) Cross-sectional data, (b) Dichotomous Response and Dichotomous Risk Factor: 2x2 Tables (c) Expressing relationship between a risk factor and a disease (d) Inference for relative risk and odds ratio for 2x2 table, Sensitivity, specificity and predictivities. Clinical Trials: Its Planning and its Four Phases.

UNIT II

Special Survival Features: Censoring and its types. Study time and Patient time. Survival Analysis: Survival Distributions and their Properties viz. Exponential, Weibull, Gamma, Rayleigh and Lognormal. Estimation of Survivor and Hazard Functions: Life Table, KaplanMeier and Nelson-Aalen Estimates. Estimating Median and Survival Times.Estimation of Mean survival time and variance for Type I and Type II Censored data with examples.

UNIT III

Cox-Proportional Hazards Model: Its Linear Component, Fitting, Hypothesis Tests. Estimating Hazard and Survivor Function. Kaplan Meier Estimate, Hazard and Cumulative Incidence Functions, Modelling. Cause Specific Hazard and Incidence, Model Checking. Sample Size Requirements for a Survival Study.

UNIT IV

Multiple Factor Hypothesis for Process of Heredity.Medelian Population: Gene Frequency and Genotype Frequency. Hardy Weinberg Law: Multiple Alleles, Two or More Pairs of Genes, Linkage of Genes, Sex Linked Genes. Influence of Gene Frequencies on Population Mean.Breeding Value of Genotypes.Dominance Deviation.

Recommended Books:

1. KleinbaumD.G.and Klein M.(2012): Survival Analysis, Springer, 3rd Edition.
2. Danial W.W. and Cross C.L. (2013):Biostatistics: A Foundation for Analysis in the Health Sciences. Wiley.
3. Cox, D.R. and Oakes, D. (1984). Analysis of Survival Data, Chapman and Hall.
4. Dabholkar A.R. (1999) Elements of Bio Metrical Genetics. Concept Publishing Co., New Delhi.
5. Ewens, W. J. and Grant, G.R. (2001). Statistical methods in Bio informatics: An Introduction, Springer.
6. Miller, R.G. (1981). Survival Analysis, John Wiley & Sons.

(STAT-DSC-13 Maj-505): BIOSTATISTICS (PRACTICAL)

Notes:

(i) The Practical Question Paper will contain five questions. A student will be required to attempt any three questions, each of 5 marks, in three hours duration.

(ii) The practical of paper (STAT-DSC-13 Maj-505) will be based on the following topics:

1. Interpreting clinical Trial data.
2. Sample size estimation in clinical Trials.
3. Plotting Survival and Hazard Curves for different parameter combinations in respect of some life time distributions.
4. Computing Kaplan-Meier estimates based on recorded surviving times with and without censoring.
5. Fitting of Cox-Proportional Hazard Model.
6. Hypothesis Formulation and their Testing for Cox-Proportional Hazard Model.
7. Estimation of Mean Survival Time and its variance for complete and survival data.
8. Random union among gametes.
9. Gene effect on population mea

B.Sc. (Hons) STATISTICS
Semester- VI
Course Code: (STAT-DSC-14 Maj/Min-601)
Course Name: Optimization Techniques

Credit: 4
L-3, P-1

Notes:

1. **Students are to be engaged to 5 hours of teaching per week (3 lectures for theory and 2 lectures for practical).**
2. **Practical examination will be held in the respective semester.**

Paper Title with code	Lecture(L)/ Credit	Practical(P)/ Credit	Marks Distribution	
Optimization Techniques (STAT-DSC-14 Maj/Min-601)	Lecture :3 Credit: 3	Practical Lecture:2 Credit:1	Theory Max. Marks: 75 Theory: 60 Internal Assessment : 15 Time: 3 Hours	Practical Max. Marks: 25 Viva voce: 5 Record of the practical: 5 Written practical Exam: 15 Time: 3 Hours

OPTIMIZATION TECHNIQUES

Objectives: The learning objectives of this course are as follows:

1. To develop the ability to formulate and solve Linear Programming Problems (LPP) using graphical and simplex-based methods, including duality concepts.
2. To equip students with techniques for solving transportation, assignment, and game theory problems for optimal decision-making in resource allocation and competitive situations.
3. To enable students to apply network analysis methods (CPM & PERT) for effective project planning, scheduling, and control.

Learning Outcomes: After successful completion of this course, students should be able to:

1. Formulate and solve Linear Programming Problems (LPP)** using graphical, simplex, Big-M, and Two-Phase methods, and apply the concept of duality.
2. Analyze and solve transportation and assignment problems** using standard methods such as NWC, LCM, VAM, MODI, and Hungarian method.

3. Apply game theory techniques (pure and mixed strategies, dominance rule, and graphical methods) to find optimal strategies in two-person zero-sum games.
4. Construct and analyze project networks using CPM and PERT, determining critical path, slack, float, and project scheduling parameters (EST, EFT, LST, LFT).

Note : There will be nine questions in all. Question No. 1 will be compulsory covering whole of the syllabus and comprising 5 to 8 short answer type questions. Rest of the eight questions will be set from the four units uniformly i .e. two from each unit. The candidate will be required to attempt f ive questions in all selecting one question from each unit and the compulsory one. All the questions will carry equal marks.

UNIT-I

Introduction to Linear Programming Problem (LPP): –Definition, general LPP, Formulation, Graphical solution of LPP – Simplex method – Big M-method and Two Phase method - Concepts of Duality – Conversion of Primal to Dual – Problems

UNIT-II

Transportation Problem- Initial Basic Solution- North West Corner Rule, Least Cost Method and Vogel's Approximation Method – Optimal Solution by Modified Distribution Method (MODI) - Assignment problem - Simple Problems

UNIT-III

Game Theory – Pure and Mixed strategies, saddle point - Dominance rule - Optimal Solution of two person zero sum games – Graphical solution of (2 x n) and (m x 2) games

UNIT-IV

Network analysis by CPM / PERT: Basic concepts: Construction of network - concepts of slack and float in network analysis - Determination of the floats and critical path, EST, EFT, LST and LFT.

Recommended Books:

1. Kanti Swarup, P.K. Gupta and Manmohan (2010): Operation Research, Sultan Chand and Sons.
2. S.D. Sharma (2003): Operations Research, Kedarnath Ramnath and Co.
3. Taha H.A. (2008): Operational Research: An Introduction,8/e, Pearson.
4. Hillier F.S. and Libermann G.J. (2004): Introduction to Operations Research, 7thEdition, McGraw Hill

(STAT-DSC-14 Maj/Min-601):**OPTIMIZATION TECHNIQUES (PRACTICAL)**

Notes:

(i) The Practical Question Paper will contain five questions. A student will be required to attempt any three questions, each of 5 marks, in three hours duration.

(ii) The practical of paper (STAT-DSC-14 Maj/MIN-601) will be based on the following topics:

1. Problem based on formulation of a LPP.
2. Problem based on Graphical method.
3. Problem based on simplex method of LPP.
4. Problem based on Big-M method.
5. Problem based on two-phase method.
6. Problem based on Duality.
7. Problem based on Transportation problem.
8. Problem based on Assignment problem.
9. Problem based on game theory.
10. Problem based on CPM/PERT networks.

B.SC. (Hons) STATISTICS
Semester- VI

Course Code: (STAT-DSC-15 Maj-602)
Course Name:Linear Models

Credit: 4
L-3, P-1

Notes:

1. **Students are to be engaged to 5 hours of teaching per week (3 lectures for theory and 2 lectures for practical).**
2. **Practical examination will be held in the respective semester.**

Paper Title with code	Lecture(L)/Credit	Practical(P)/ Credit	Marks Distribution	
Linear Models (STAT-DSC-15 Maj-602)	Lecture :3 Credit: 3	Practical Lecture:2 Credit:1	Theory Max. Marks: 75 Theory: 60 Internal Assessment : 15 Time: 3 Hours	Practical Max. Marks: 25 Viva voce: 5 Record of the practical: 5 Written practical Exam: 15 Time: 3 Hours

LINEAR MODELS

Course Objectives:
The learning objectives include developing a clear understanding of the fundamental concepts of linear models and a range of associated skills allowing the students to work effectively with them.
Learning Outcomes: After completing this course, students should have developed a clear understanding of:

1. Theory and estimation of Linear Models.
2. Gauss-Markov Theorem and its use.
3. Distribution of quadratic forms.
4. Simple and Multiple linear regression models and their applications.
5. Fitting of these models to real or synthetic data, derivation of confidence and prediction intervals, and a sound scientific interpretation of the results.
6. Techniques of Analysis of Variance and Covariance under fixed effects model.
7. Assessment of the quality of the fit using classical diagnostics, awareness of potential problems (outliers, etc.) and application of remedies to deal with them. Contents:

Note : There will be nine questions in all. Question No. 1 will be compulsory covering whole of the syllabus and comprising 5 to 8 short answer type questions. Rest of the eight questions will be set from the four units uniformly i .e. two from each unit. The candidate will be required to attempt f ive questions in all selecting one question from each unit and the compulsory one. All the questions will carry equal marks.

UNIT I

Gauss-Markov set up: Theory of linear estimation, Estimability of linear parametric functions, Method of least squares, Gauss-Markov theorem, Estimation of error variance. Distribution of quadratic forms.

UNIT II

Regression analysis: Simple Regression analysis, Estimation and hypothesis testing in case of simple and multiple regression analysis, Confidence intervals and Prediction intervals, Concept of model matrix and its use in estimation. Effect of orthogonal columns in the X matrix, Partial F-test and Sequential F-test, Bias in regression estimates.

UNIT III

Analysis of Variance and Covariance: Definition of fixed, random and mixed effect models, analysis of variance and covariance in one-way classified data for fixed effect models, analysis of variance in two-way classified data with equal number of observations per cell for fixed effect models.

UNIT IV

Model checking: Prediction from a fitted model, Residuals and Outliers, Lack of fit and pure error, Violation of usual assumptions concerning normality, Homoscedasticity and collinearity, Diagnostics using quantile-quantile plots. Model Building: Techniques for Variable selection. Polynomial Regression models: Orthogonal Polynomials.

Recommended Books:

1. Gujrati, D.N. and Porter, D.C. (2010): Essentials of Econometrics, 4th Ed, McGraw Hill International Edition.
2. Draper, N. R. and Smith, H. (1998): Applied Regression Analysis, 3rd Ed., John Wiley and Sons.
3. Montgomery, D. C., Peck, E. A. and Vining, G. G. (2004): Introduction to Linear Regression Analysis, 3rd Ed., John Wiley and Sons.
4. Rencher, A. C. and Schaalje, G. B. (2008): Linear Models in Statistics, 2nd Ed., John Wiley and Sons.
5. Weisberg, S. (2005): Applied Linear Regression, 3rd Ed., John Wiley and Sons.

(STAT-DSC-15 Maj-602): **LINEAR MODELS (PRACTICAL)**

Notes:

(i) The Practical Question Paper will contain five questions. A student will be required to attempt any three questions, each of 5 marks, in three hours duration.

(ii) The practical of paper (STAT-DSC-15 Maj-602) will be based on the following topics:

1. Estimability when X is a full rank matrix.
2. Estimability when X is not a full rank matrix.
3. Distribution of Quadratic forms.
4. Simple Linear Regression.
5. Multiple Regression.
6. Tests for Linear Hypothesis.
7. Bias in regression estimates and Lack of fit.
8. Stepwise regression procedure.
9. Analysis of Variance of a one-way classified data.
10. Analysis of Variance of a two-way classified data with one observation per cell.
11. Analysis of Variance of a two-way classified data with m (> 1) observations per cell.
12. Analysis of Covariance of a one-way classified data.
13. Residual Analysis and Orthogonal Polynomials.

B.Sc. (Hons) STATISTICS
Semester- VI

Course Code: (STAT-DSC-16 Maj-603)
Course Name: Demography

Credit: 4
L-3, P-1

Notes:

1. **Students are to be engaged to 5 hours of teaching per week (3 lectures for theory and 2 lectures for practical).**
2. **Practical examination will be held in the respective semester.**

Paper Title with code	Lecture(L)/Credit	Practical(P)/ Credit	Marks Distribution	
Demography (STAT-DSC-16 Maj-603)	Lecture :3 Credit: 3	Practical Lecture:2 Credit:1	Theory Max. Marks: 75 Theory: 60 Internal Assessment : 15 Time: 3 Hours	Practical Max. Marks: 25 Viva voce: 5 Record of the practical: 5 Written practical Exam: 15 Time: 3 Hours

DEMOGRAPHY

Objectives: The learning objectives of this course are as follows:

1. To collect valid Demographic data using different methods.
2. To learn basic measures of Mortality, Fertility, Population Growth and life tables.
3. To learn about measures of marriages and migration.

Learning Outcomes: After successful completion of this course, students should be able to:

1. Understand vital statistics and its sources.
2. Understand the basic measures of Mortality.
3. Understand the concept of Life Tables and their construction.
4. Understand the basic measures of Fertility and population growth.
5. Apply measures of marriages and migration.

Note : There will be nine questions in all. Question No. 1 will be compulsory covering whole of the syllabus and comprising 5 to 8 short answer type questions. Rest of the eight questions will be set from the four units uniformly i .e. two from each unit. The candidate will be required to attempt f ive questions in all selecting one question from each unit and the compulsory one. All the questions will carry equal marks.

UNIT I

Introduction to Vital Statistics: Definition sources of collecting data on vital statistics, errors in the census, and registration data. Measurement of population, rate, and the ratio of vital events. Measurements of Mortality: Crude Death Rate (CDR), Specific Death Rate (SDR), Standardized Death Rates (STDR) and Infant Mortality Rate (IMR).

UNIT II

Life(Mortality) Tables: Assumption, description, construction of Life Tables, and Uses of Life, Expectation of life, Central Mortality Rates, and Force of Mortality. Abridged life table: Definition, methods of construction of abridged life table: Reed-Merrell Method and Greville's Method.

UNIT III

Measurements of Fertility: Crude Birth Rate (CBR), General Fertility Rate (GFR), Specific Fertility Rate (SFR), and TotalFertility Rate (TFR). Measurement of Population Growth: Crude rates of natural increase, Pearl's Vital Index, Gross Reproduction Rate (GRR), and Net Reproduction Rate (NRR).

UNIT IV

Introduction, Sources and Quality of Nuptiality Data, General, Specific, Total and Standardized Marriage rates, Mean Age at Marriage, Measures of Migration: Concept of mobility and migration, sources of data, types of migration, Internal & international migrations; measures of internal migration.

Recommended Books:

1. Gun, A.M., Gupta, M.K. and Dasgupta, B. (2008). Fundamentals of Statistics, Vol. II,9thEd., World Press.
2. Biswas, S. (1988). Stochastic Processes in Demography & Application, Wiley Eastern Ltd.
3. Mukhopadhyay, P. (1999). Applied Statistics, Books and Allied (P) Ltd.
4. Keyfitz, N. and Beekman,J.A. (1985). Demography through Problems. S-Verlag, NewYork.

STAT-DSC-16 Maj-603**: DEMOGRAPHY (PRACTICAL)**

Notes:

(i) The Practical Question Paper will contain five questions. A student will be required to attempt any three questions, each of 5 marks, in three hours duration.

(ii) The practical of paper (STAT-DSC-16 Maj-603) will be based on the following topics:

1. To calculate CDR and Age Specific death rate for a given set of data.
2. To find a standardized death rate by (i) Direct method and (ii) Indirect method.
3. To construct a complete life table.
4. To fill in the missing entries in a life table.
5. To calculate CBR, GFR, SFR, TFR for a given set of data.
6. To calculate Crude rate of Natural Increase and Pearle's Vital Index for a given set of data.
7. Calculate GRR and NRR for a given set of data and compare them.

B.Sc. (Hons) STATISTICS
Semester-VI

Course Code: (STAT- DSC-17 Maj-604)
Course Name: Advanced Statistics using R Programming

Credit: 4
L-3, P-1

Notes:

1. **Students are to be engaged to 5 hours of teaching per week (3 lectures for theory and 2 lectures for practical).**
2. **Practical examination will be held in the respective semester.**

Paper Title with code	Lecture(L)/ Credit	Practical(P)/ Credit	Marks Distribution	
			Theory	Practical
Advanced Statistics using R Programming (STAT-DSC-17 Maj-604)	Lecture :3 Credit: 3	Practical Lecture:2 Credit:1	Max. Marks: 75 Theory: 60 Internal Assessment : 15 Time: 3 Hours	Max. Marks: 25 Viva voce: 5 Record of the practical: 5 Written practical Exam: 15 Time: 3 Hours

ADVANCED STATISTICS USING R PROGRAMMING

Course Objectives: The main objective of this course is to allow the students to learn the advanced techniques of modelling real data from diverse discipline using R Programming. Also, they can utilize Nonparametric/distribution free tests and its role for different type of data.Students learn and apply different types of parametric and non-parametric test in real life using R.

Learning Outcomes: After completing this course, students should have developed a clear understanding of:

1. Distinguish between Parametric and Nonparametric test and appreciate different measurement scales.
2. Use nonparametric test for one/two and more than two samples problem using R.
3. Understand and apply different types of regression model using R Programming.

Note : There will be nine questions in all. Question No. 1 will be compulsory covering whole of the syllabus and comprising 5 to 8 short answer type questions. Rest of the eight questions will be set from the four units uniformly i .e. two from each unit. The candidate will be required to attempt f ive questions in all selecting one question from each unit and the compulsory one. All the questions will carry equal marks.

UNIT-I

Sampling distributions: Chi Square, t and F distribution, Hypothesis testing of mean and variance: one-sample and two-sample t, F and chi-square test, Levene test. Testing of proportion for one-sample and two sample problem.

UNIT-II

Linear Models: Fixed, random and mixed effects models, ANOVA: one-way and two-ways, ANOCOVA. Regression Models: Simple and Multiple Linear Regression, Forward, Backward and stepwise regression, Residual analysis. Diagnostics and tests for violations of model assumptions: Multicollinearity, Autocorrelation and Homoscedasticity.

UNIT-III

Nonparametric Tests: One-Sample, two-sample problem, and Paired-Sample Procedures: the sign test, treatment of ties in rank tests, Wilcoxon signed-rank test, Wald-Wolfowitz runs test, Kolmogorov-Smirnov one and two-sample test, median test, and the Mann-Whitney U test.

UNIT-IV

Nonparametric Tests (Continued): Tests for equality of k independent samples: Kruskal-Wallis one way ANOVA test, Measures of Association for bivariate samples: Kendall's Tau coefficient. Measures of association in multiple classifications: Friedman's two-way ANOVA by ranks in a k x n table, the Coefficient of Concordance of k sets of rankings of n objects.

Recommended Books:

1. Lander J.P.: R for Everyone, Pearsons Publication.
2. Pierre-Andre Cornillon et al. (2012): R for Statistics, CRC Press.
3. Gibbons, J. D., and Chakraborti, S. (2020): Nonparametric statistical inference. CRC press.
4. Kloke, J., and McKean, J. W. (2014) : Nonparametric statistical methods using R. CRC Press.
5. Hollander, M., Wolfe, D. A., and Chicken, E. (2013): Nonparametric statistical methods (Vol.751). John Wiley & Sons.
6. Agresti, A. (2015). Foundations of Linear and Generalized Linear Models, John Wiley, New Jersey.
7. Chatterjee, S. and Hadi, A.S. (2012). Regression Analysis by Example, 5th ed., John Wiley, New Jersey.
8. Fox, J. and Weisberg, S. (2011). An R Companion to Applied Regression, 2nd ed., Sage.
9. Montgomery, D.C. (2001). Designs and Analysis of Experiments, John Wiley & Sons, New York.
10. Weisberg, S. (2014). Applied Linear Regression, 4th ed., John Wiley,

(STAT-DSC-17 Maj-604): **ADVANCED STATISTICS USING R PROGRAMMING (PRACTICAL)**

Notes:

(i) The Practical Question Paper will contain five questions. A student will be required to attempt any three questions, each of 5 marks, in three hours duration.

(ii) The practical of paper (STAT-DSC-17 Maj-604) will be based on the following topics:

1. Problem based on one/two sample t-test using R.
2. Problem based on F-test using R.
3. Problem based on chi-square test.
4. Problem based on one and two sample proportions.
5. Problem based on one way and two way ANONA.
6. Problem based on ANOCOVA.
7. Fitting of Linear Regression, Forward, Backward and stepwise regression.
8. Problem based on Muticollineaity.
9. Problem based on one sample nonparametric tests.
10. Problem based on two sample nonparametric tests.
11. Problem based on The Kruskal-Wallis one-way ANOVA test.
12. Problem based on Kendall's Tau coefficient and Spearman's coefficient of rank correlation
13. Problem based on Friedman's two way ANONA by ranks in k×n table.
14. Problem based on Coefficient of Concordance of k sets of ranking on n objects.

B.SC. (Hons) STATISTICS
Semester- VI
Course Code: (STAT-DSC-18 Maj-605)
Course Name: Numerical Analysis

Credit: 4
L-3, P-1

Notes:

1. **Students are to be engaged to 5 hours of teaching per week (3 lectures for theory and 2 lectures for practical).**
2. **Practical examination will be held in the respective semester.**

Paper Title with code	Lecture(L)/Credit	Practical(P)/ Credit	Marks Distribution	
Numerical Analysis (STAT-DSC-18 Maj-605)	Lecture :3 Credit: 3	Practical Lecture:2 Credit:1	Theory Max. Marks: 75 Theory: 60 Internal Assessment : 15 Time: 3 Hours	Practical Max. Marks: 25 Viva voce: 5 Record of the practical: 5 Written practical Exam: 15 Time: 3 Hours

NUMERICAL ANALYSIS

Course Objectives: To comprehend various computational techniques to find approximate value for possible root(s) of non-algebraic equations, to find the approximate solutions of system of linear equations and ordinary differential equations.

Course Learning Outcomes: The course will enable the students to learn the following:

1. Some numerical methods to find the zeroes of nonlinear functions of a single variable and solution of a system of linear equations, up to a certain given level of precision.
2. Interpolation techniques to compute the values for a tabulated function at points not in the table.
3. Applications of numerical differentiation and integration to convert differential equations into difference equations for numerical solutions.

Note : There will be nine questions in all. Question No. 1 will be compulsory covering whole of the syllabus and comprising 5 to 8 short answer type questions. Rest of the eight questions will be set from the four units uniformly i .e. two from each unit. The candidate will be required to attempt f ive questions in all selecting one question from each unit and the compulsory one. All the questions will carry equal marks.

UNIT-I

Methods for Solving Algebraic and Transcendental Equations: Algorithms, Convergence, Bisection method, False position method, Fixed point iteration method, Newton's method, Secant method

UNIT-II

Techniques to Solve Linear Systems: Partial and scaled partial pivoting, LU decomposition and its applications, Iterative methods: Gauss-Jacobi, Gauss-Seidel and SOR methods.

UNIT-III

Finite differences, shift operators and their relationship,finding the missing terms. Interpolation with equal intervals:Newton'forward and backward formulae. Interpolation with unequal intervals:Newton's divided differences, Lagrange.

UNIT-IV

Numerical Differentiation and Integration: First order and higher order approximation for first derivative, Approximation for second derivative. Numerical integration by closed Newton-Cotes formula: trapezoidal rule, Simpson's rule and its error analysis. Euler's method to solve ODE's.

Recommended Books:

1. S.R.K., Iyengar, & Jain, R. K. (2009). Numerical Methods. New Age International Publisher, India.
2. Bradie, Brian. (2006). A Friendly Introduction to Numerical Analysis. Pearson Education, India. Dorling Kindersley (India) Pvt. Ltd. Third impression.
3. Gerald, C. F., & Wheatley, P. O. (2008). Applied Numerical Analysis (7th ed.). Pearson Education. India.
4. Jain, M. K., Iyengar, S. R. K., & Jain, R. K. (2012). Numerical Methods for Scientific and Engineering Computation. (6th ed.). New Age International Publisher, India.

(STAT-DSC-18 Maj-605)**: NUMERICAL ANALYSIS (PRACTICAL)**

Notes:

(i) The Practical Question Paper will contain five questions. A student will be required to attempt any three questions, each of 5 marks, in three hours duration.

(ii) The practical of paper (STAT-DSC-17 Maj-605) will be based on the following topics:

1. Bisection method
2. Newton Raphson method
3. Secant method
4. Regula Falsi method
5. LU decomposition method
6. Gauss-Jacobi method
7. SOR method
8. Gauss-Seidel method
9. Lagrange interpolation
10. Newton interpolation
11. Trapezoidal rule
12. Simpson's rule
13. Euler’s method

NEP 2020 COLLEGES ARTS B.SC., B.SC. HONOURS AND B.SC. HONOURS WITH RESEARCH (STATISTICS):MULTI-DISCIPLINARY/ INTER-DISCIPLINARY (TABLE 2)

B.SC. (Hons) STATISTICS
Semester-V

Course Code: (STAT-DSC-6 Maj/Min-501)
Course Name: Sampling Distribution

Credit: 4
L-3, P-1

Notes:

1. **Students are to be engaged to 5 hours of teaching per week (3 lectures for theory and 2 lectures for practical).**
2. **Practical examination will be held in the respective semester.**

Paper Title with code	Lecture(L)/ Credit	Practical(P)/ Credit	Marks Distribution	
Sampling Distributions (STAT-DSC-6 Maj/Min-501)	Lecture :3 Credit: 3	Practical Lecture:2 Credit:1	Theory Max. Marks: 75 Theory: 60 Internal Assessment : 15 Time: 3 Hours	Practical Max. Marks: 25 Viva voce: 5 Record of the practical: 5 Written practical Exam: 15 Time: 3 Hours

SAMPLING DISTRIBUTIONS

Objectives: The learning objectives of this course are as follows:

1. To introduce the modes of convergence and their relation to limit laws, with a focus on the central limit theorem.
2. To introduce the concept of sampling distributions and their applications in statistical inference.
3. To describe the statistical ideas behind the procedure of hypothesis testing.
4. To explain the assumptions and conditions under which to apply different tests of hypothesis about population parameters and draw appropriate conclusions from the analysis.

Learning Outcomes: After successful completion of this course, students should be able to:

1. Understand the basics of convergence theory and its importance in limit laws.
2. Apply the concept of the central limit theorem and the relevance of the theorem in inferential statistics.
3. Analyze data by using suitable hypothesis testing procedures in real-life applications related to large and small samples.
4. Apply the knowledge of the idea of sampling distributions and appreciate their importance
5. in the field of statistics.
6. Integrate the knowledge of various sampling distributions like chi-square, t, and F distributions in hypothesis testing problems.

Note : There will be nine questions in all. Question No. 1 will be compulsory covering whole of the syllabus and comprising 5 to 8 short answer type questions. Rest of the eight questions will be set from the four units uniformly i .e. two from each unit. The candidate will be required to attempt f ive questions in all selecting one question from each unit and the compulsory one. All the questions will carry equal marks.

UNIT I

Generating functions and Central Limit Theorem: Moment generating function, its limitations, properties, cumulants and their properties, characteristic function and its properties,Chebyshev's inequality, Weak Law of Large Numbers (WLLN),along with examples and applications.

Basic idea and relevance of Central Limit Theorem (CLT), De-Moivre Laplace theorem,Lindeberg Levy theorem, Liapunov Theorem (only statement), and applications of CLT.

UNIT II

Sampling Distributions and Test of Hypotheses: Concepts of parameter, statistic, sampling distribution of a statistic, standard error. Samplingdistribution of sample mean, standard errors of the sample mean, sample variance, and sampleproportion.

Null and alternative hypotheses, level of significance, Type I and Type II errors, theirprobabilities and critical region, determination of sample size, confidence intervals, and pvalue.

Tests of significance and confidence intervals for - single proportion, difference of twoproportions, single mean, difference of two means, and difference of two standard deviations.

UNIT III

Exact Sampling Distribution: Chi-Square distribution: Definition and derivation of the probability distribution of Chi-squaredistribution with n degrees of freedom, nature of the curve for different degrees of freedom,mean, mode, variance, moment generating function, cumulantgenerating function, additiveproperty, and limiting form of the Chi-square distribution, Applications of Chi-Squaredistribution.

UNIT IV

Exact Sampling Distributions (continued): Student's t-statistic and Fishers t-statistic: definition and derivation of their samplingdistributions, nature and characteristics of graph of t distribution, moments, limiting form andapplications of the t distribution.

F-statistic: Definition and derivation of the sampling distribution, the graph of F distribution,moments, and applications of the F distribution. Relationship between t, F, and Chi-squaredistributions.

Books recommended

- Gupta, S. C. and Kapoor, V. K. (2020). Fundamentals of Mathematical Statistics, Twelfth Edition, S. Chand and Sons. Delhi.
- Goon, A.M., Gupta, M.K. and Dasgupta, B. (2016). An Outline of Statistical Theory,Volume I, The World Press, Kolkata.
- Mukhopadhyay, P. (2016). Mathematical Statistics, Books and Allied, India.
- Hogg, R.V., Tanis, E.A. and Rao, J.M. (2009). Probability and Statistical Inference,Seventh Edition, Pearson Education, New Delhi.
- Miller, I. and Miller, M. (2006). John E. Freund's Mathematical Statistics with Applications, Eight Edition, Pearson Education, Asia.
- Johnson, R.A. and Bhattacharya, G.K. (2001). Statistics-Principles and Methods, Fourth Edition, John Wiley and Sons.

SUGGESTED READINGS

- Bhat, B.R. (2016). Modern Probability Theory- An Introductory Textbook, FourthEdition, New Age International Publishers.
- Rohatgi, V. K and Saleh M. E. (2015). An Introduction to Probability and Statistics,Third Edition, John Wiley and Sons, Inc., New Jersey.
- Mood, A.M. Graybill, F.A. and Boes, D.C. (2007). Introduction to the Theory ofStatistics, Third Edition, (Reprint), Tata McGraw-Hill Pub. Co. Ltd.

STAT-DSC-6 Maj/Min-501): **SAMPLING DISTRIBUTIONS (PRACTICAL)**

Notes:

(i) The Practical Question Paper will contain five questions. A student will be required to attempt any three questions, each of 5 marks, in three hours duration.

(ii) The practical of paper (STAT-DSC-6 Maj/Min-501) will be based on the following topics:

1. Problem based on moment generating function and characteristic function.
2. Problem based on Chebyshev's inequality and Weak Law of Large Numbers.
3. To apply large sample test of significance for single mean and to obtained confidence interval.
4. To apply large sample test of significance for difference between two means and standard deviations.
5. To apply large sample test of significance for single proportion and to obtained confidence interval.
6. To apply large sample test of significance for difference between two
7. Proportions and to obtained confidence interval.
8. To apply Chi-square test for testing the single variance.
9. To apply Chi-square test for goodness of fit.
10. To apply Chi-square test for independence of attributes.
11. Testing based on a 2 x 2 contingency table without and with Yates' corrections.
12. To apply *t*-test for testing single mean and difference between means and to obtain their confidence intervals.
13. To apply paired *t*-test for difference between two means.
14. To apply test of significance of sample correlation coefficient.
15. To apply F-test for testing difference of two variances.

Semester- V

Course Code: (STAT-DSC-7 Maj-602)

Course Name: Applied Statistics-I

Credit: 4
L-3, P-1

Notes:

1. **Students are to be engaged to 5 hours of teaching per week (3 lectures for theory and 2 lectures for practical).**
2. **Practical examination will be held in the respective semester.**

Paper Title with code	Lecture(L)/Credit	Practical(P)/ Credit	Marks Distribution	
Applied Statistics-I (STAT-DSC-7 Maj-602)	Lecture :3 Credit: 3	Practical Lecture:2 Credit:1	Theory Max. Marks: 75 Theory: 60 Internal Assessment : 15 Time: 3 Hours	Practical Max. Marks: 25 Viva voce: 5 Record of the practical: 5 Written practical Exam: 15 Time: 3 Hours

APPLIED STATISTICS-I

Objectives: The Learning Objectives of this course are as follows:

1. Introduce the concept of time series, its components, and their estimation.
2. Introduce the application of time series.
3. Introduce the concept, formulation, and application of index numbers.

Learning Outcomes: After completion of this course, the students will be able to:

1. Understand the concepts of time series and index numbers.
2. Formulate, solve, and analyses the use of time series and index numbers for real-world problems.
3. Apply index number and time series concepts for making future planning and policy related to business.

Note : There will be nine questions in all. Question No. 1 will be compulsory covering whole of the syllabus and comprising 5 to 8 short answer type questions. Rest of the eight questions will be set from the four units uniformly i .e. two from each unit. The candidate will be required to attempt f ive questions in all selecting one question from each unit and the compulsory one. All the questions will carry equal marks.

UNIT-I

Index Numbers : Definition, problems involved in the construction of index numbers, types of index numbers, methods of construction of price index numbers – unweighted and weighted aggregates method, simple average of price relatives, weighted average of price relatives, quantity and value indexes, errors in index numbers, Uses and limitations of index numbers.

UNIT-II

Index Numbers (Continued): Tests of consistency of index numbers-time reversal test, factor reversal test and circular test, chain base index numbers, uses and limitations of chin base index numbers, base shifting, splicing and deflating of index numbers, cost of living or consumer price index numbers, its constructionand uses.

UNIT-III

Time Series Analysis : Definition, components of time series-trend, seasonal variations, cyclic variations, irregular component, illustrations, additive and multiplicative models, determination of trend: graphic method, semi-averages method, method of curve fitting by principle of least squares- fitting of straight line, second degree parabola and exponential curves, growth curves and their fittings, moving average method and uses of time series.

UNIT-IV

Time Series Analysis (Continued): Analysis of seasonal fluctuations, construction of seasonal indices using method of simple averages, ratio to trend method and ratio to moving average method, link relative method, Deseasonalisation of data, Measurement of cyclic fluctuations, measurement of irregular or random variations, Serial correlation and correlogram and forcasting in time series.

Books recommended

S. No.	Title of Book	Name of author	Publisher
1.	Applied General Statistics	Croxton F.E., Cowden D.J. &Kelin S.	Prentice Hall
2.	Fundamentals of Applied Statistics	Gupta S.C. & Kapoor V.K.	Sultan Chand & Sons

(STAT-DSC-7Maj-502): **APPLIED STATISTICS-I (PRACTICAL)**

Notes:

(i) The Practical Question Paper will contain five questions. A student will be required to attempt any three questions, each of 5 marks, in three hours duration.

(ii) The practical of paper (STAT-DSC-11 Maj-602) will be based on the following topics:

1. Construction of price and quantity index numbers by simple aggregate method.
2. Construction of price and quantity index numbers by Laspeyres, Paasche's, Drobish-Bowley, Marshall-Edgeworth, Walsch and Fisher's Formula.
3. Construction of price and quantity index numbers by simple and weighted average of price relatives.
4. Construction of consumer price index number by family budget method and Aggregate expenditure method
5. Time Reversal Test and Factor Reversal Test Measurement of trend by using the least square method with the help of semi-average method.
6. Problem based on chain base index numbers.
7. Problem based on cost of living index numbers.
8. Problem based on base shifting, splicing and deflation of index numbers.
9. Fit the trend with the help of a straight line by using semi-average method.
10. Fit the trend with the help of a straight line by using least square method.
11. Fit the trend with the help second degree parabola curve by using least square method.
12. Fit the trend with the help of an exponential curve by using least square method.
13. Measure the trend by using moving average method.
14. Measurement of seasonal variations using simple Average method.
15. Measurement of seasonal variations using simple ratio to trend method.
16. Measurement of seasonal variations using ratio to moving average method.
17. Measurement of seasonal variations using link relative method.

B.SC. (Hons) STATISTICS
Semester- VI

Course Code: (STAT-DSC-14 Maj/Min-601)
Course Name: Optimization Techniques

Credit: 4
L-3, P-1

Notes:

1. **Students are to be engaged to 5 hours of teaching per week (3 lectures for theory and 2 lectures for practical).**
2. **Practical examination will be held in the respective semester.**

Paper Title with code	Lecture(L)/ Credit	Practical(P)/ Credit	Marks Distribution	
Optimization Techniques (STAT-DSC-14 Maj/Min-601)	Lecture :3 Credit: 3	Practical Lecture:2 Credit:1	Theory Max. Marks: 75 Theory: 60 Internal Assessment : 15 Time: 3 Hours	Practical Max. Marks: 25 Viva voce: 5 Record of the practical: 5 Written practical Exam: 15 Time: 3 Hours

OPTIMIZATION TECHNIQUES

Objectives: The learning objectives of this course are as follows:

1. To develop the ability to formulate and solve Linear Programming Problems (LPP) using graphical and simplex-based methods, including duality concepts.
2. To equip students with techniques for solving transportation, assignment, and game theory problems for optimal decision-making in resource allocation and competitive situations.
3. To enable students to apply network analysis methods (CPM & PERT) for effective project planning, scheduling, and control.

Learning Outcomes: After successful completion of this course, students should be able to:

1. Formulate and solve Linear Programming Problems (LPP)** using graphical, simplex, Big-M, and Two-Phase methods, and apply the concept of duality.
2. Analyze and solve transportation and assignment problems** using standard methods such as NWC, LCM, VAM, MODI, and Hungarian method.
3. Apply game theory techniques (pure and mixed strategies, dominance rule, and graphical methods) to find optimal strategies in two-person zero-sum games.
4. Construct and analyze project networks using CPM and PERT, determining critical path, slack, float, and project scheduling parameters (EST, EFT, LST, LFT).

Note : There will be nine questions in all. Question No. 1 will be compulsory covering whole of the syllabus and comprising 5 to 8 short answer type questions. Rest of the eight questions will be set from the four units uniformly i .e. two from each unit. The candidate will be required to attempt f ive questions in all selecting one question from each unit and the compulsory one. All the questions will carry equal marks.

UNIT-I

Introduction to Linear Programming Problem (LPP): –Definition, general LPP, Formulation, Graphical solution of LPP – Simplex method – Big M-method and Two Phase method - Concepts of Duality – Conversion of Primal to Dual – Problems

UNIT-II

Transportation Problem- Initial Basic Solution- North West Corner Rule, Least Cost Method and Vogel's Approximation Method – Optimal Solution by Modified Distribution Method (MODI) - Assignment problem - Simple Problems

UNIT-III

Game Theory – Pure and Mixed strategies, saddle point - Dominance rule - Optimal Solution of two person zero sum games – Graphical solution of (2 x n) and (m x 2) games

UNIT-IV

Network analysis by CPM / PERT: Basic concepts: Construction of network - concepts of slack and float in network analysis - Determination of the floats and critical path, EST, EFT, LST and LFT.

Recommended Books:

1. Kanti Swarup, P.K. Gupta and Manmohan (2010): Operation Research, Sultan Chand and Sons.
2. S.D. Sharma (2003): Operations Research, Kedarnath Ramnath and Co.
3. Taha H.A. (2008): Operational Research: An Introduction,8/e, Pearson.
4. Hillier F.S. and Libermann G.J. (2004): Introduction to Operations Research, 7thEdition, McGraw Hill

(STAT-DSC-14 Maj/Min-601)**: OPTIMIZATION TECHNIQUES (PRACTICAL)**

Notes:

(i) The Practical Question Paper will contain five questions. A student will be required to attempt any three questions, each of 5 marks, in three hours duration.

(ii) The practical of paper (STAT-DSC-14 Maj/Min-601) will be based on the following topics:

1. Problem based on formulation of a LPP.
2. Problem based on Graphical method.
3. Problem based on simplex method of LPP.
4. Problem based on Big-M method.
5. Problem based on two-phase method.
6. Problem based on Duality.
7. Problem based on Transportation problem.
8. Problem based on Assignment problem.
9. Problem based on game theory.
10. Problem based on CPM/PERT networks.

B.Sc. (Hons.) STATISTICS
Semester- VI

Course Code: (STAT-DSC-15 Maj/Min-602)
Course Name: Applied Statistics-II

Credit: 4
L-3, P-1

Notes:

1. **Students are to be engaged to 5 hours of teaching per week (3 lectures for theory and 2 lectures for practical).**
2. **Practical examination will be held in the respective semester.**

Paper Title with code	Lecture(L)/Credit	Practical(P)/ Credit	Marks Distribution	
Applied Statistics-II (STAT-DSC-15 Maj/Min-602)	Lecture :3 Credit: 3	Practical Lecture:2 Credit:1	Theory Max. Marks: 75 Theory: 60 Internal Assessment : 15 Time: 3 Hours	Practical Max. Marks: 25 Viva voce: 5 Record of the practical: 5 Written practical Exam: 15 Time: 3 Hours

APPLIED STATISTICS-II

Objectives: The learning objectives include:

1. To familiarize students with the Indian statistical system and the role of official statistical agencies at central and state levels.
2. To understand the collection, classification, and interpretation of official statistics in sectors like population, agriculture, industry, trade, labor, transport, and finance.
3. To develop knowledge of Statistical Quality Control (SQC) methods, including control charts for variables and attributes, and their industrial applications.
4. To provide insights into acceptance sampling techniques, their plans, operating characteristics, and associated risks in quality inspection.

Learning Outcomes: After completing this course, students will be able to:

1. Explain the structure and functions of the Indian statistical system and interpret official statistics related to population, agriculture, industry, trade, labor, transport, and finance.
2. Apply Statistical Quality Control (SQC) techniques to analyze process variations and construct control charts for variables and attributes.
3. Interpret control charts (R, σ, p, d, c charts) for monitoring and improving product and process quality.
4. Design and evaluate acceptance sampling plans, understanding concepts such as AQL, LTPD, producer's risk, and consumer's risk.

Note: There will be nine questions in all. Question No.1 will be compulsory covering whole of the syllabus and comprising 5 to 8 short answer type questions. Rest of the eight questions will be set from the four units uniformly i.e. two from each unit. The candidate will be required to attempt five questions in all selecting one question from each unit and the compulsory one. All the questions will carry equal marks.

UNIT-I

Indian official statistics: Introduction, Indian statistical system, Statistical offices at the centre, Statistical offices in the states, Population statistics, Agricultural statistics, Industrial statistics, Trade statistics, Price statistics, Statistics of labor and employment, Statistics of transport and communication, Financial and banking statistics.

UNIT-II

Statistical Quality Control : Meaning and uses of Statistical Quality Control (SQC), causes of variations in quality, product and process control, control charts, 3 - σ control limits, control chart for variables- $\overline{X}$ and R chart, Interpretation of $\overline{X}$& R charts.

UNIT-III

Control chart for standard deviation (σ chart), control charts for attributes: control charts for number of defectives (d- chart), control charts for fraction defectives (p- chart) and control charts for number of defects per UNIT (c- chart), applications of c- chart, natural tolerance and specification limits.

UNIT-IV

Acceptance sampling :Acceptance sampling inspection plans, Acceptance quality level (AQL), lot tolerance proportion defective (LTPD), process average fraction defective, producer's and consumer's r isks, rectifying inspection plans, single and double sampling plans, their OC functions, concepts of AQL, LTPD, AOQL, average amount of inspection and ASN function.

Books recommended

S. No.	Title of Book	Name of author	Publisher
1.	Statistical Quality Control	Grant E.L.	McGraw Hill
2.	Statistical Methods in Quality Control	Cowden D.J.	Asia Pub. Society
3.	Fundamentals of Applied Statistics	Gupta S.C. & Kapoor V.K.	Sultan Chand & Sons
4.	Fundamentals of Statistics, Vol. II	Goon A.M., Gupta M.K. & Dasgupta B.	World Press Calcutta

(STAT-DSC-15 Maj/Min-602): **APPLIED STATISTICS-II (PRACTICAL)**

Notes:

(i) The Practical Question Paper will contain five questions. A student will be required to attempt any three questions, each of 5 marks, in three hours duration.

(ii) The practical of paper (STAT-DSC-15 Maj/Min-602) will be based on the following topics:

1. Problem based on Range and mean charts.
2. Problem based on standard deviation chart.
3. Problem based on d-chart.
4. Problem based on p-chart.
5. Problem based on c-chart.
6. Problem based on single sampling plans.
7. Problem based on double sampling plans.

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ZOOLOGY

STUDENTS OPTING FOR SINGLE MAJOR AND SINGLE MINOR (Table1*) AND
STUDENTS OPTING FOR MULTI-DISCIPLINARY/ INTER-DISCIPLINARY (Table 2*)
B.Sc., B.Sc. HONOURS AND B.Sc.HONOURS WITH RESEARH IN ZOOLOGY

Table1*						Table2*					
Sem	**Level**	**CourseCode**	**CourseName**	**Credits Th(L) + Pr(P)**	**Total Credits**	**Sem**	**Level**	**Course Code**	**Course Name**	**Credits Th(L) + Pr(P)**	**Total Credits**
V	L-300	(ZOO-DSC-9 Maj-501)	Cell Biology	3+1=4	24	V	L-300	(ZOO-DSC-9 Maj/Min-501)	Cell Biology	3+1=4(x3)	24
		(ZOO-DSC-10 Maj-502)	Economic Zoology	3+1=4				(ZOO-DSC-10 Maj502)	Economic Zoology	3+1=4(x2)	
		(ZOO-DSC-11 Maj-503)	Systematics and Taxonomy	3+1=4							
		(ZOO-DSC-12 Maj-504)	Basic Techniques in Cell Biology	3+1=4							
		(ZOO-DSC-13 Maj-505)	Insect Vectors and Diseases	3+1=4							
		(ZOO-INT/COM-506)	Internship/ Community Work	4				(ZOO- INT/COM-506)	Internship/ Community Work	4	
VI	L-300	(ZOO-DSC-14 Maj/Min-601)	Principles of Genetics	3+1=4 (x2)	24	VI	L-300	(ZOO-DSC-14 Maj/Min-601)	Principles of Genetics	3+1=4 (x3)	24
		(ZOO-DSC-15Maj-602)	Molecular Biology	3+1=4				(ZOO-DSC-15 Maj/Min602)	Molecular Biology	3+1=4 (x3)	
		(ZOO-DSC-16 Maj-603)	Molecular Tools and Techniques	3+1=4							
		(ZOO-DSC-17 Maj-604)	Evolutionary Biology	3+1=4							
		(ZOO-DSC-18 Maj-605)	Biodiversity and Conservation	3+1=4							
72 (Maj)+32(Min)+9(MDC)+8(AEC)+6(VAC)+9(SEC)+4(INT/COM)+4(PBI/HCP)					**144**	**36x2(Maj)+32(Min)+9(MDC)+8(AEC)+6(VAC)+9(SEC)+4(INT/COM)+4(PBI/HCP)**					**144**
The student successfully securing 144 credits will be awarded B.Sc. degree											

Notes:

1. In the first semester student is offered three different subjects A, B & C.
2. At the end of second semester each student has the option to choose any one of the two streams namely Single Major or Multi-Disciplinary. The student choosing Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student who chooses multi-disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3-year degree) shall be eligible for 4 YEAR DEGREE B.Sc. HONOURS with research, on the condition that the Institution is a PU approved Research Centre
5. The student opting for Single Major must secure 50% credits of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for Minor in a particular subject, the student must attain at least 12 credits in that subject(s).
6. The student exiting the programme after one year securing 48 credits will be awarded UG Certificate provided he/she secures 4 credits from summer internship/community work. Further, the student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secure 4 credits from summer internship. The student successfully securing 144 credits will be awarded BA degree.
7. Language paper chosen as MIL-1 and MIL-2 should be in the same subject.
8. Once Major and Minor papers are chosen in 3rd semester, the same will continue as Major and Minor subject in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write Dissertation of 08 credits in one of the two Major subjects.

Semester- V (L-300)

Course Code: ZOO-DSC-9 Maj-501
Course Title: Cell Biology

Total Hours= 75 (Theory= 45 + Practical= 30) **Credits= 4 (L=3 + P=1)**
Total Marks= 100 (Theory=75 + Practical=25)
Theory Marks= 75 {67 (Theory examination) + 08 (Internal assessment)}
Practical Marks= 25 {22 (Practical examination) + 03 (Internal assessment)}
Duration for final examination: 3 hours

Theory

Learning Outcomes:
The students will be able to:

- *Acquire comprehensive knowledge of cell structure and organization.*
- *Understand the principles and applications of tissue fixation in histological studies.*
- *Develop a clear understanding of cell organelles and their ultrastructure.*
- *Explain the roles of various cell organelles in cellular metabolism and physiological processes.*

UNIT- 1

Microscopy: Principles and types of microscopes –simple microscope, compound microscope, light microscope, and phase contrast microscope.

Fixation and Staining: Principles of tissue fixation;fixatives(Carnoy's and Bouin's) and their significance;Staining procedures (single and double staining)

Overview of cells: General structure and organization of prokaryotic and eukaryotic cells. Structure of viruses and viroids.

UNIT- 2

Plasma membrane: Structure and organization of the plasma membrane with reference to the Fluid Mosaic Model.

Transport across membranes – passive transport and active transport.

Endoplasmic reticulum: Structure and functions of rough endoplasmic reticulum (RER) and smooth endoplasmic reticulum (SER).

Golgi complex: Structure and functions of the Golgi apparatus.

UNIT- 3

Ribosomes: Types, structure, and functions of ribosomes.

Lysosomes: Structure, functions, and polymorphism of lysosomes.

Mitochondria: Structure and functions of mitochondria; semi-autonomous nature of mitochondria.

UNIT- 4

Nucleus: Structure and functions of the nucleus, including nuclear envelope, nuclear pore complex, and nucleolus.

Cytoskeleton: Structure and functions of microtubules and microfilaments

Cell Division: Cell Cycle, Mitosis,Meiosis and their significance.

Instructions For Paper Setter and candidates

Nine questions are to be set. Question No.1 is compulsory, consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1.5 marks each. Two questions of 13 marks each are to be set from each Unit. One question is to be attempted from each Unit. In all, five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts. To qualify for a Course, the candidate must obtain a minimum of 40% marks.

ZOO-DSC-9 Maj-501: Cell Biology

Practical

1. Study and identification of parts of a Simple/dissection Microscope and Compound microscope.
2. Laboratory preparation of Carnoy's fixative and Bouin's fixative
3. Identification and study of the following cell organelles through photographs, charts, and models: Mitochondria, Golgi complex, Endoplasmic reticulum, Ribosomes and Nucleus
4. Study of different stages of mitosis and meiosis through permanent slides/photographs/charts/models.
5. Preparation of temporary stained squash of onion root tip to study various stages of mitosis.

SUGGESTED READINGS

1. Dhami, P.S., & Dhami, J.K. (2015). ***Textbook of Zoology***. Pardeep Publications.
2. Sabharwal, A. (2021). ***Modern's Zoology***. Modern Publishers.
3. Powar, C.B. (1999). ***Cell Biology***. Himalaya Publishing House, Mumbai.
4. Singh, S., & Tomar, C.B. (2008). ***Cell Biology***. Rastogi Publications.
5. Rastogi, V.B. (2023). ***Cell Biology***. MedTech Science Press.
6. Cooper, G.M., & Hausman, R.E. (2007). ***The Cell: A Molecular Approach*** (4th ed.). Sinauer Associates Inc.
7. Karp, G. (2001). ***Cell and Molecular Biology: Concepts and Experiments***. John Wiley & Sons.
8. De Robertis, E.D.P., & De Robertis, E.M.F. (1995). ***Cell and Molecular Biology*** (8th ed.). W.B. Saunders Co., Philadelphia.

Course Code: ZOO-DSC-10 Maj-502
Course Title: Economic Zoology

Total Hours= 75 (Theory= 45 + Practical= 30) **Credits= 4 (L=3 + P=1)**
Total Marks= 100 (Theory=75 + Practical=25)
Theory Marks= 75 {67 (Theory examination) + 08 (Internal assessment)}
Practical Marks= 25 {22 (Practical examination) + 03 (Internal assessment)}
Duration for final examination: 3 hours

Theory

Learning Outcomes:
The students will be able to:

- *Understand the scope and significance of aquaculture and allied farming practices.*
- *Describe basic techniques of sericulture, lac culture, butterfly, and livestock farming.*
- *Identify important insect pests and diseases transmitted by them.*
- *Explain pest control methods and principles of Integrated Pest Management (IPM).*

UNIT- 1

Pisciculture: Composite fish culture – principle, culture of the Indian major carps (***Catla catla***, ***Labeo rohita***, ***Cirrhinus mrigala***) and exotic carps (grass carp, silver carp, common carp), Integrated fish culture – with paddy, poultry, cattle and piggery.
Aquaculture: Extensive and intensive fish culture, semi-intensive fish culture, cage culture, pen culture, capture and culture fisheries, freshwater prawn farming – principle and practices.
Pearl culture: Definition and importance; species used; basic principle of pearl formation and implantation.

UNIT- 2
Sericulture: Definition and importance; species of silkworm (mulberry and non-mulberry: Eri, Muga, Tassar); basic requirements of sericulture; overview of silk reeling.
Lac culture: Definition and importance; major lac insect species; host plants; basic rearing practices; economic importance of lac.
Butterfly farming: Definition, importance, and basic requirements

UNIT- 3
Dairy (Milch animal) farming: Definition and importance; major indigenous and exotic breeds of cow and buffalo (examples only); basic housing and nutritional management
Goat farming: Importance; major breeds (examples only); basic rearing and feeding practices, pasteurized goat milk.
Sheep farming: Importance; major breeds (examples only); basic rearing and feeding practices; types of sheep wool (fine, medium, long, and coarse).

UNIT- 4

Medically and veterinary important arthropods: Study of life cycle and economic importance of housefly, mosquito (***Aedes***, ***Anopheles***, ***Culex***), sand fly, tsetse fly, warble fly, tick *(Hyalomma)*, and mite *(Sarcoptes)*

Insect Control Methods: Physical and cultural methods; chemical control including organochlorines (DDT, BHC), organophosphates (malathion), botanical insecticides (nicotine, pyrethrum), and fumigants (aluminium phosphide); biological control using predators, parasites, and microbial agents (bacteria, viruses)

Integrated Pest Management (IPM): Concept, components, and importance of IPM.

Instructions For Paper Setter and candidates

Nine questions are to be set. Question No.1 is compulsory, consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1.5 marks each. Two questions of 13 marks each are to be set from each Unit. One question is to be attempted from each Unit. In all, five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts. To qualify for a Course, the candidate must obtain a minimum of 40% marks.

ZOO-DSC-10 Maj-502: Economic Zoology

Practical

1. Study and identification of aquatic organisms: Fish – Indian major carps (***Catla catla***, ***Labeo rohita***, ***Cirrhinus mrigala***), prawn, and oyster through specimens/models.
2. Study of life cycles and economic importance of silkworm, lac insect, butterfly, housefly, mosquito (***Aedes***, ***Anopheles***, ***Culex***), sand fly, tsetse fly, warble fly, tick *(Hyalomma)*, and mite *(Sarcoptes)*through specimens/photographs/charts/models.
3. Identification of different types of natural silk (mulberry, Eri, Muga, Tassar) through samples/photographs.
4. Identification of different types of wool (fine, medium, long, and coarse) through samples/photographs.
5. Identification of major breeds of cow, buffalo, goat, and sheep through charts/models/photographs.
6. Measurement of milk density using a lactometer.
7. Demonstration of maintenance of a fish aquarium.
8. Field visit to a fish farm or dairy farm.
9. Project report

SUGGESTED READINGS

1. Agarwal, S.C., & Johal, M.S. (1997). ***Fishery Development***. Narendra Publishing House, Delhi.
2. Atwal, A.S., & Dhaliwal, G.S. (2009). ***Agricultural Pests of South Asia and Their Management***. Kalyani Publishers.
3. Banerjee, C.C. (2018). ***A Text Book of Animal Husbandry***. Oxford & IBH Publishing Co.

4. Dent, D. (1995). ***Integrated Pest Management***. Chapman & Hall, London.
5. Gupta, R., & Amrutkar, R. (2019). ***Comprehensive Book on Goat Farming***. Jaya Publishing House.
6. Jhingran, V.G. (1991). ***Fish and Fisheries of India***. Hindustan Publishing Corporation of India, Delhi.
7. Kumar, R., & Nigam, P. (2008). ***Economic and Applied Entomology***. Emkay Publishing, New Delhi.
8. Sandhu, A.K. (2019). ***A Textbook of Indian Inland Fisheries and Aquaculture***. Mahindra Publishing House, Chandigarh.
9. Srivastava, K.P., & Dhaliwal, G.S. (2010). ***A Textbook of Applied Entomology*** (Vol. I & II). Kalyani Publishers, Ludhiana.
10. Shukla, G.S., & Upadhyay, V.B. (2011). ***Economic Zoology***. Rastogi Publications, Meerut.
11. Nigam, H.C. (2009). ***Economic Zoology***. Vishal Publishing Co., Jalandhar.
12. Srivastava, C.B.L. (2014). ***A Textbook of Applied Entomology***. Kalyani Publishers, Ludhiana.
13. Pillai, T.V.R. (2013). ***Aquaculture: Principles and Practices***. Blackwell Publishing.
14. Rath, R.K. (2011). ***Freshwater Aquaculture***. Scientific Publishers, Jodhpur.
15. Ghosh, A. (2012). ***Sericulture and Silk Industry***. Daya Publishing House, New Delhi.
16. Nair, K.K. (2007). ***Lac Culture in India***. Indian Council of Agricultural Research (ICAR), New Delhi.
17. Banerjee, G.C. (2016). ***A Textbook of Animal Husbandry***. Oxford & IBH Publishing Co.

Course Code: ZOO-DSC-11 Maj -503
Course Title: Systematics and Taxonomy

Total Hours= 75 (Theory= 45 + Practical= 30) **Credits= 4 (L=3 + P=1)**
Total Marks= 100 (Theory=75 + Practical=25)
Theory Marks= 75 {67 (Theory examination) + 08 (Internal assessment)}
Practical Marks= 25 {22 (Practical examination) + 03 (Internal assessment)}
Duration for final examination: 3 hours

Theory

Learning Outcomes:
The students will be able to:

- *Understand the concepts of biosystematics and taxonomy*
- *Understand the importance of taxonomy in conservation of Biodiversity*
- *Learn thetechniques of collection, preservation, curation, and identification of animal specimens*

UNIT- 1

Definition, basic concepts of biosystematics taxonomy and classification.

Goals and importance of taxonomy

History and theories of classification

Trends in biosystematics: Chemotaxonomy, cytotaxonomy and molecular taxonomy

UNIT- 2

Different mechanisms of speciation, biological mechanism of genetic incompatibility

Species concepts:Typological, Nominalistic and Biological species concepts, Difficulties in applying biological species concept

Polytypic species, subspecies and other infra-specific categories (Variety, Race, cline, intrapopulation, variants)

Analysis of Intrapopulation variations: non-genetic and genetic

UNIT- 3

Taxonomic Characters:Morphological,physiological, molecular, behavioral, ecological, geographical characters, Weighting and relative value of characters

Taxonomic procedures: Taxonomic collections, preservation, curation, process of identification,Different types of taxonomic keys (their merits and demerits).

UNIT- 4

International code of Zoological Nomenclature (ICZN): Rules of nomenclature (Uni-, Bi-, Tri- nominalism, use of author name, tautonyms, synonyms, homonyms), typification and type specimen (Holotype, allotype, lectotype, Isotype, neotype, paratype), Designationof types (holotype, lectotype and neotype)

Taxonomic publications: kinds of publications, major features of taxonomic publication, form and style of taxonomic articles and review of manuscript

Instructions For Paper Setter and candidates

Nine questions are to be set. Question No.1 is compulsory, consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1.5 marks each. Two questions of 13 marks each are to be set from each Unit. One question is to be attempted from each Unit. In all, five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts. To qualify for a Course, the candidate must obtain a minimum of 40% marks.

ZOO-DSC-11 Maj -503: Systematics and Taxonomy

Practical

1. Identification of different fishes using taxonomic keys
2. Identification of any insect species using taxonomic keys
3. Demonstration of collection, preservation and storage of invertebrate species (by videos, models and charts).
4. Demonstration of collection, preservation and storage of vertebrate species (by videos, models and charts).
5. Visit to zoology museum and report preparation.

SUGGESTED READINGS

1. E. Mayr, Peter D. Ashlock.*Principles of Systematic Zoology*.
2. V.C. Kapoor. *Theory and Practice of Animal taxonomy*.
3. M. Kato. *The Biology of Biodiversity*, Springer.
4. J.C. Avise. *Molecular Markers, Natural History and Evolution*, Chapman & Hall, New York.
5. E.O. Wilson. *Biodiversity*, Academic Press, Washington.
6. G.G. Simpson. *Principle of animal taxonomy*, Oxford IBH Publishing Company.
7. E. Mayer. *Elements of Taxonomy*.
8. E.O. Wilson. *The Diversity of Life (The College Edition)*, W.W. Northem& Co.
9. B.K. Tikadar. *Threatened Animals of India*, ZSI Publication, Calcutta.
10. R.E. Blackwelder.*Taxonomy*, John Wiley & Sons, New York (1967).

Course Code: ZOO-DSC-12 Maj -504
Course Title: Basic Techniques in Cell Biology

Total Hours= 75 (Theory= 45 + Practical= 30) **Credits= 4 (L=3 + P=1)**
Total Marks= 100 (Theory=75 + Practical=25)
Theory Marks= 75 {67 (Theory examination) + 08 (Internal assessment)}
Practical Marks= 25 {22 (Practical examination) + 03 (Internal assessment)}
Duration for final examination: 3 hours

Theory

Learning Outcomes:
The students will be able to:

- *Understand the principles and applications of basic cell biology techniques*
- *Identify common laboratory instruments*
- *Learn the use, care and maintenance of common Laboratory Instruments*
- *Develop practical skills required for advanced biological studies and research*

UNIT- 1

Tissue fixation, Staining and Microtomy:
Fixation: Types of fixatives (Carnoy's, Bouin's, Formaldehyde, Zenker)
Staining: Different types of stains (Leishman& Giemsa) and Staining techniques (H & E staining)
Microtomy: Tissue preparation and procedure
Study of good laboratory practices (GLP).

UNIT- 2

Introduction to Microscopy:
Magnification, resolution and contrast
Principle and applications of Light microscopy, Phase-contrast microscopy, Fluorescence microscopy
Advanced Microscopy: Electron microscopy (SEM and TEM) and Confocal Microscopy.

UNIT- 3

Cell fractionation and Centrifugation
Tissue homogenization: Cell lysis by mechanical enzymatic and detergent methods
Centrifugation: Principle and applications of differential and density gradient centrifugation
Cell fractionation, Cryopreservation.

UNIT- 4

Introduction to cell culture:
Basic design of cell culture laboratory and Biosafety levels
Various methods of Sterilization (Autoclave and Hot Air oven)
Types of culture: Primary culture and established cell lines
Cell viability test by Dye exclusion method, MTT assay.

Instructions For Paper Setter and candidates

Nine questions are to be set. Question No.1 is compulsory, consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1.5 marks each. Two questions of 13 marks each are to be set from each Unit. One question is to be attempted from each Unit. In all, five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts. To qualify for a Course, the candidate must obtain a minimum of 40% marks.

ZOO-DSC-12 Maj -504: Basic Techniques in Cell Biology

Practical

1. Laboratory safety practices.
2. Study of parts of compound microscope and it's handling.
3. Preparation of blood smear and Giemsa staining.
4. Microtomy: Tissue fixation, embedding, section cutting.
5. Double tissue staining by H & E.
6. Demonstration of centrifugation.
7. Study of instruments related to cell culturing: Autoclave, Laminar Hood, CO_2Incubator and Inverted Microscope.
8. Visit to cell culture laboratory and Electron Microscope facility.

SUGGESTED READINGS

1. Bruce Alberts, Karen Hopkin, Alexander Johnson, David Morgan, Martin Raff, Keith Roberts & Peter Walter. *Essential Cell Biology*. 6th Edition. W.W. Norton & Company, 2023.
2. Geoffrey M. Cooper & Robert E. Hausman. *The Cell: A Molecular Approach*. 9th Edition. Oxford University Press, 2022.
3. Gerald Karp. *Cell and Molecular Biology: Concepts and Experiments*. 9th Edition. Wiley, 2023.
4. Lodish, H., Berk, A., Kaiser, C. A., et al. *Molecular Cell Biology*. 10th Edition. W.H. Freeman, 2024.
5. De Robertis, E.D.P. & De Robertis, E.M.F. *Cell and Molecular Biology*. 9th Edition. Lippincott Williams & Wilkins, 2021.
6. C.B. Powar. *Cell Biology*. Himalaya Publishing House, 2025.
7. P.S. Verma & V.K. Agarwal. *Cell Biology, Genetics, Molecular Biology, Evolution and Ecology*. S. Chand Publishing, Latest Edition.
8. S. C. Rastogi. *Cell Biology*. New Age International Publishers, Latest Edition.
9. S.K. Sawhney & Randhir Singh. *Practical Manual of Cell Biology*. Pointer Publishers, 2000.
10. J. Jayaraman. *Laboratory Manual in Biochemistry*. New Age International Publishers, 2020.
11. R. Ian Freshney. *Culture of Animal Cells: A Manual of Basic Technique and Specialized Applications*. 8th Edition. Wiley-Blackwell, 2021.

Course Code: ZOO-DSC-13 Maj -505
Course Title: Insect Vectors and Diseases

Total Hours= 75 (Theory= 45 + Practical= 30) **Credits= 4 (L=3 + P=1)**
Total Marks= 100 (Theory=75 + Practical=25)
Theory Marks= 75 {67 (Theory examination) + 08 (Internal assessment)}
Practical Marks= 25 {22 (Practical examination) + 03 (Internal assessment)}
Duration for final examination: 3 hours

Theory

Learning Outcomes:
The students will be able to:

- *Identify different insects and classify them based on their morphology and behaviour.*
- *Describe host–parasite relationships and the role of host reservoirs in parasite transmission.*
- *Explain the different modes of parasite transmission by insect vectors.*
- *Recognize the importance of insect vectors in the spread and control of parasitic diseases.*

UNIT- 1

Introduction to Insects:General features of insects; Classification of insects up to Order; General features of insect vector orders: Hemiptera, Anoplura, Diptera, Siphonaptera; **Morphological features of insect vectors:** Head, types of antennae, mouthparts, legs, wings; Morphological characteristics and identification of larval and pupal stages of major insect vectors (***Mosquitoes and House flies***)

UNIT- 2

Concept of Vectors: Definition and types of insect vectors (mechanical and biological); Insect reservoirs; Host-vector relationship; Vectorial capacity; Host Specificity; Modes of disease transmission (Vertical and horizontal transmission; mechanical and biological transmission; non-persistent, semi-persistent, and persistent transmission).

UNIT- 3

Insectsas disease Vectors:

Mosquitoes: Life cycle and mosquito-borne diseases—Malaria, Dengue, Chikungunya, Filariasis, and Viral encephalitis (causative agent and symptoms).
Sand flies: Life cycle and sand fly-borne diseases—Visceral and Cutaneous leishmaniasis (causative agent and symptoms).
Fleas: Life cycle, and flea-borne diseases—Plague and Murine Typhus (causative agent and symptoms).
Human lice: Life cycle and louse-borne diseases—Epidemic Typhus and Relapsing fever (causative agent and symptoms).
Deer flies: Life cycle and deer fly-borne diseases—Tularemia and Loiasis (causative agent and symptoms).
House flies:Life cycle and house fly-borne diseases—Cholera and Typhoid (causative agent and symptoms).

UNIT- 4

Prevention and Control: General vector control measures, community hygiene and sanitation, Integrated Vector Management (IVM), Offensive and defensive measures for the prevention and control of diseases transmitted by mosquitoes, sand flies, fleas, human lice, deer flies, and house flies.

Instructions For Paper Setter and candidates

Nine questions are to be set. Question No.1 is compulsory, consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1.5 marks each. Two questions of 13 marks each are to be set from each Unit. One question is to be attempted from each Unit. In all, five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts. To qualify for a Course, the candidate must obtain a minimum of 40% marks.

ZOO-DSC-13 Maj -505: Insect Vectors and Diseases

Practical

1. Study of different types of mouthparts, antennae, and legs of insects using permanent slides/specimens.
2. Study of larval and pupal stages of major insect vectors using slides/specimens.
3. Identification of major insect vectors using permanent slides/specimens/photographs:
    - Mosquitoes (*Aedes, Culex,* and *Anopheles*)
    - Sand fly (*Phlebotomus*)
    - Flea (*Xenopsylla cheopis*)
    - Human lice (*Pediculus humanus capitis, Pediculus humanus corporis,* and *Pthirus pubis*)
    - Deer fly (*Chrysops*)
    - House fly (*Musca domestica*)
4. Study of different diseases transmitted by above insect vectors as mentioned in theory syllabus using photographs/videos.
5. Study of national public health programmes for the prevention and control of vector-borne diseases.
6. Preparation of a project report on any one disease transmitted by insect vector.
7. Field visit/Laboratory visit to a public health laboratory, insect museum, vector surveillance unit, or mosquito breeding site.

SUGGESTED READINGS

1. Burgess, N.R.H. & Cowan, G.O. *A Colour Atlas of Medical Entomology*. Springer, 1993.
2. Mullen, G.R. & Durden, L.A. *Medical and Veterinary Entomology*. 3rd Edition. Academic Press, 2019.
3. Service, M.W. *A Guide to Medical Entomology*. Macmillan Press, 1980.
4. Service, M.W. *Medical Entomology for Students*. 5th Edition. Cambridge University Press, 2012.
5. Eldridge, B.F. & Edman, J.D. *Medical Entomology: A Textbook on Public Health and Veterinary Problems Caused by Arthropods*. Springer, 2004.
6. Marquardt, W.C. (Ed.). *Biology of Disease Vectors*. 2nd Edition. Elsevier Academic Press, 2005.
7. Ahmad, S. Rehan. *Insect, Vectors & Diseases*. Purushottam Publishers, 2020.
8. Bhushan, Brijender. *Insect Vectors and Diseases*. Trueman Publications, 2020.
9. Awasthi, V.B. *Introduction to General and Applied Entomology*. Scientific Publishers, Latest Edition.
10. David, B. Vasantharaj & Kumaraswami, T. *Elements of Economic Entomology*. Popular Book Depot, Latest Edition.
11. Verma, P.S. & Agarwal, V.K. *A Textbook of Entomology*. S. Chand Publishing, Latest Edition.
12. Chapman, R.F. *The Insects: Structure and Function*. 5th Edition. Cambridge University Press, 2013.

Semester- VI (L-300)

Course Code: ZOO-DSC-14 Maj/Min -601
Course Title: Principles of Genetics

Total Hours= 75 (Theory= 45 + Practical= 30) **Credits= 4 (L=3 + P=1)**
Total Marks= 100 (Theory=75 + Practical=25)
Theory Marks= 75 {67 (Theory examination) + 08 (Internal assessment)}
Practical Marks= 25 {22 (Practical examination) + 03 (Internal assessment)}
Duration for final examination: 3 hours

Theory

Learning Outcomes:
The students will be able to:

- *Understand foundational concepts of Genetics*
- *Differentiate between Mendelian and Non-Mendelian patterns of inheritance*
- *Explain chromosomal behavior in inheritance*
- *Apply genetic principles to real-world contexts*

UNIT- 1

Basic terminology in Genetics: gene, allele, locus, genotype, phenotype, homozygous, heterozygous, pure lines, test cross, back cross

Mendel's experiments and laws of inheritance

Monohybrid and dihybrid crosses

Non-allelic gene interactions: Modified Mendelian ratios (9:7, 9:3:4, 12:3:1, 9:6:1, 15:1)

UNIT- 2

Non-Mendelian inheritance: Incomplete dominance (Snapdragon flowers), Codominance (Human blood group type AB), Multiple alleles (Eye color in *Drosophila*), Sex-linked inheritance (Haemophilia), Polygenic inheritance (Skin color in humans), Pleiotropy (Phenylketonuria), Lethal alleles (Coat color in Mice), Cytoplasmic inheritance (Kappa particles in *Paramecium*)

UNIT- 3

Chromosomal theory of inheritance

Linkage: Characteristics, Types (Complete & Incomplete) and Significance

Crossing over: Types (Single & Double), Mechanism (Synapsis, Synaptonemal complex, Tetrad formation, Crossing over & Terminalization) and Significance

UNIT- 4

Mutations: Spontaneous and induced mutations, Physical mutagens (Ionizing & Non-ionizing radiations), Chemical mutagens (Base analogs & Alkylating agents)

Human genetic disorders: Causes and symptoms of Cystic fibrosis**,** Sickle cell anemia,Down's syndrome,Turner's syndrome, Klinefelter syndrome

Population genetics: Gene frequencies, Hardy-Weinberg Principle.

Instructions For Paper Setter and candidates

Nine questions are to be set. Question No.1 is compulsory, consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1.5 marks each. Two questions of 13 marks each are to be set from each Unit. One question is to be attempted from each Unit. In all, five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts. To qualify for a Course, the candidate must obtain a minimum of 40% marks.

ZOO-DSC-14 Maj/Min-601: Principles of Genetics

Practical

1. Demonstration of Monohybrid cross and Dihybrid cross, and verification of Mendel's law of segregation and law of independent assortment using colored beads.
2. Identification of different stages of meiosis in onion buds/Study of stages of meiosis through permanent slides.
3. Preparation of sex-chromatin/Barr body from buccal smear.
4. Analysis of human karyotype photographs and identification of chromosomal disorders.
5. Analysis of fingerprint and palm print patterns through dermatoglyphics.
6. Demonstration of Hardy-Weinberg principle using colored beads.
7. Project report

SUGGESTED READINGS

1. Gardner, Eldon John, Michael J. Simmons, and D. Peter Snustad. *Principles of Genetics*. 8th ed., Wiley, 1991.
2. Pierce, Benjamin A. *Genetics: A Conceptual Approach*. 7th ed., W. H. Freeman & Co., 2020. Snustad, D. Peter, and Michael J. Simmons. *Principles of Genetics*. 7th ed., Wiley, 2016.
3. Klug, William S., Michael Cummings, Charlotte A. Spencer, Michael A. Palladino, and Darrell Killian. *Essentials of Genetics*. 10th ed., Pearson Education, 2020.
4. Dhami, P.S. and Dhami, J.K. *A Textbook of Zoology (Developmental Biology, Genetics and Cell Biology)* Pradeep Publications, 2022.
5. Srivastava, H.N. *Cell Biology and Genetics.* Pradeep Publications, 2022.
6. Gupta, P.K. *Genetics: A Textbook for University Students*. 5th ed., Rastogi Publications, 2019.
7. Rastogi, Veer Bala, and Nirvika Rastogi. *Fundamentals of Genetics*. 1st ed., MedTech Science Press, 2022.
8. Rathoure, Ashok Kumar, and Meena Srivastava. *Cell Biology and Genetics*. Daya Publishing House, 2020.
9. Randhawa, S. S. *Text Book of Genetics*. S. Vikas & Company (PV Books), 2013.
10. Singh, B.D. *Fundamentals of Genetics*. 6th ed., MedTech Science Press, 2025.

Course Code: ZOO-DSC-15 Maj-602
Course Title: Molecular Biology

Total Hours= 75 (Theory= 45 + Practical= 30) **Credits= 4 (L=3 + P=1)**
Total Marks= 100 (Theory=75 + Practical=25)
Theory Marks= 75 {67 (Theory examination) + 08 (Internal assessment)}
Practical Marks= 25 {22 (Practical examination) + 03 (Internal assessment)}
Duration for final examination: 3 hours

Theory

Learning Outcomes:
The students will be able to:

- *Understand the fundamental concepts and scope of Molecular Biology.*
- *Explain the structure and functionof DNA and RNA.*
- *Describe molecular mechanisms such as replication, transcription, translation, and gene regulation.*
- *Understand how gene expression and mutations affect cells and organisms.*

UNIT- 1

Nucleic acids as genetic material:Griffith's experiment; Avery, MacLeod and McCarty's experiment; Hershey–Chase experiment

Structure and function of nucleic acids: nucleosides, nucleotides, purines, and pyrimidines; biological importance of nucleotides

DNA structure: Watson and Crick model; A, B, and Z forms; supercoiled and relaxed DNA; Denaturation and renaturation; melting temperature (Tm); hyperchromic effect.

RNA: Structure and types of RNA

Central Dogma of molecular biology

UNIT- 2

Genome organization: structure of a gene, coding and regulatory sequences, introns, exons; Nucleosome concept and packaging of DNA into higher-order structures.

DNA Replication: Meselson–Stahl experiment (proof of semiconservative DNA replication); Mechanism of DNA replication in prokaryotes and eukaryotes.

UNIT- 3

Transcription in prokaryotes and eukaryotes: promoter, initiation, elongation, and termination.

Components of protein synthesis machinery: mRNA, tRNA, ribosomes.

Translation: aminoacyl-tRNA synthetases, charging of tRNA, ribosome assembly.

Mechanism of protein synthesis in prokaryotes and eukaryotes: initiation, elongation, and termination.

UNIT- 4

Regulation of gene expression in prokaryotes: lac operon (negative and positive regulation).
Regulation of gene expression in eukaryotes.
Mutations: Molecular basis of mutations; types of mutations (missense, nonsense, silent, point, and frameshift).

Instructions For Paper Setter and candidates

Nine questions are to be set. Question No.1 is compulsory, consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1.5 marks each. Two questions of 13 marks each are to be set from each Unit. One question is to be attempted from each Unit. In all, five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts. To qualify for a Course, the candidate must obtain a minimum of 40% marks.

ZOO-DSC-15 Maj-602: Molecular Biology

Practical

1. Study of DNA and RNA structures through models.
2. Demonstration of paper chromatography for the separationof amino acids.
3. **Introduction to Molecular Biology Apparatus (Demonstration/Video):**
   - Micropipettes
   - Microcentrifuge
   - Agarose Gel Electrophoresis**Apparatus**
   - Polyacrylamide Gel Electrophoresis**Apparatus**
   - **Spectrophotometer**
   - **PCR (Polymerase Chain Reaction) Apparatus**

4. Demonstration of Agarose Gel Electrophoresis to separate DNA fragments (video/model if lab setup unavailable).

5. Demonstration ofSDS-PAGE (Polyacrylamide Gel Electrophoresis) to separate proteins (video/model if lab setup unavailable).

SUGGESTED READINGS

1. Watson, J.D., Baker, T.A., Bell, S.P., Gann, A., Levine, M., & Losick, R. (2014). ***Molecular Biology of the Gene***, 6th Edition. Cold Spring Harbor Laboratory Press, Pearson.
2. Alberts, B., Johnson, A.D., Lewis, J., Raff, M., Roberts, K., & Walter, P. (2014). ***Molecular Biology of the Cell***, 6th Edition. Garland Science.
3. Krebs, J., Goldstein, E., & Kilpatrick, S. (2013). ***Lewin's Essential Genes***, 3rd Edition. Jones and Bartlett Learning.
4. Rastogi, S.C. (2012). ***Cell and Molecular Biology***. New Age International.
5. Karp, G. (2013). ***Cell and Molecular Biology: Concepts and Experiments***, 7th Edition. Wiley.

6. Lewin, B. (2008). ***Gene IX***. Peterson Publications / Panima.
7. Malacinski, G.M., & Freifelder, D. (1998). ***Essentials of Molecular Biology***, 3rd Edition. Jones & Bartlett Publishers.
8. Verma, P.S. (2001). ***Cell Biology, Genetics, Molecular Biology, Evolution and Ecology***.
9. Chandar, N., & Viselli, S. (2010). ***Cell and Molecular Biology: Lippincott's Illustrated Reviews***. Wolters Kluwer Health / Lippincott Williams & Wilkins.
10. Vidyavathi, N. (2013). ***Molecular Biology***.
11. Kumar, P. (2024). ***Biophysics and Molecular Biology: Tools and Techniques***, 5th Edition.
12. Primrose, S.B. (2001). ***Molecular Biotechnology***. Panima.

Course Code: ZOO-DSC-16 Maj-603
Course Title: Molecular Tools and Techniques

Total Hours= 75 (Theory= 45 + Practical= 30) **Credits= 4 (L=3 + P=1)**
Total Marks= 100 (Theory=75 + Practical=25)
Theory Marks= 75 {67 (Theory examination) + 08 (Internal assessment)}
Practical Marks= 25 {22 (Practical examination) + 03 (Internal assessment)}
Duration for final examination: 3 hours

Theory

Learning Outcomes:
The students will be able to:

- *Understand the basic molecular tools used in biological laboratories*
- *Explain the principles of common molecular techniques*
- *Describe the applications of molecular tools and techniques in biological research*
- *Interpret the basic results of molecular biology experiments*

UNIT- 1

Introduction to cloning vectors: Brief overview of types of vectors (plasmids, phagemids, cosmids); Basic characteristics of a vector: Origin of replication, selectable marker, Multiple cloning sites; Commonly used plasmid vectors (pBR322, pUC18-19)
Gene cloning: Principle, procedure (Chemical transformation by $CaCl_2$ method, Electroporation, Antibiotic selection) and applications.

UNIT- 2

Isolation of DNA (phenol-chloroform-isoamyl alcohol extraction method), RNA (guanidinium thiocyanate-phenol-chloroform extraction method) and proteins (Buffer extraction method): Principles and methodology
Quantification of DNA and RNA concentration
Principle, procedure and applications of Agarose Gel Electrophoresis, Polyacrylamide Gel Electrophoresis (PAGE), Polymerase Chain Reaction (PCR) (Conventional and RT-PCR)

UNIT- 3

Basic recombinant DNA techniques: cutting and joining DNA molecules, restriction modification systems, various enzymes used in recombinant DNA technology
Recombinant DNA Technology: Principle, procedure and applicationsin medicine and agriculture
Blotting techniques (Southern, Northern and Western blotting): Principle, procedure and applications

UNIT- 4

Principle, procedure and applications of Restriction Fragment Length Polymorphism, DNA fingerprinting and DNA sequencing (Sanger Sequencing)

Importance of bioinformatics in molecular biology

Introduction to biological databases: GenBank and NCBI; Differences between primary and secondary databases.

Instructions For Paper Setter and candidates

Nine questions are to be set. Question No.1 is compulsory, consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1.5 marks each. Two questions of 13 marks each are to be set from each Unit. One question is to be attempted from each Unit. In all, five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts. To qualify for a Course, the candidate must obtain a minimum of 40% marks.

ZOO-DSC-16 Maj-603: Molecular Tools and Techniques
Practical

1. Demonstration of molecular biology laboratory instruments: Micropipette, Centrifuge, pH meter, Water bath, Vortex mixer Spectrophotometer, Autoclave, Laminar airflow cabinet
2. Handling and calibration of micropipettes.
3. Laboratory Demonstration (video demonstration where laboratory facilities are not available) of the following:
   - Isolation of DNA, RNA and Proteins
   - Measurement of DNA concentration using a spectrophotometer
   - Agarose gel electrophoresis
   - Polyacrylamide Gel Electrophoresis (PAGE)
   - Polymerase Chain Reaction (PCR)
4. Study of the following through charts/models/videos:
   - DNA fingerprinting
   - DNA sequencing
   - Blotting techniques (Southern, Northern and Western)

SUGGESTED READINGS

1. Brown, T.A. *Gene Cloning and DNA Analysis: An Introduction*. 8th ed., Wiley-Blackwell, 2021.
2. Green, Michael R., and Joseph Sambrook. *Molecular Cloning: A Laboratory Manual*. 4th ed., Cold Spring Harbor Laboratory Press, 2019.
3. Primrose, S.B., and Richard Twyman. *Principles of Gene Manipulation and Genomics*. 8th ed., Wiley-Blackwell, 2013.
4. Wilson, Keith, and John Walker. *Principles and Techniques of Biochemistry and Molecular Biology*. 8th ed., Cambridge University Press, 2018.
5. Glick, Bernard R., Jack J. Pasternak, and Cheryl L. Patten. *Molecular Biotechnology: Principles and Applications of Recombinant DNA*. 5th ed., ASM Press, 2017.
6. Gupta, P.K. *Genetics: A Textbook for University Students*. 5th ed., Rastogi Publications, 2019.
7. Rastogi, S.C. *Cell and Molecular Biology*. 3rd ed., New Age International Publishers, 2019.
8. Verma, P.S., and V.K. Agarwal. *Cell Biology, Genetics, Molecular Biology, Evolution and Ecology*. S. Chand Publishing, Latest edition.
9. Kumar, P. *Biophysics and Molecular Biology: Tools and Techniques*. 5th ed., Pathfinder Publication, 2024.

Course Code: ZOO-DSC-17 Maj-604
Course Title: Evolutionary Biology

Total Hours= 75 (Theory= 45 + Practical= 30) **Credits= 4 (L=3 + P=1)**
Total Marks= 100 (Theory=75 + Practical=25)
Theory Marks= 75 {67 (Theory examination) + 08 (Internal assessment)}
Practical Marks= 25 {22 (Practical examination) + 03 (Internal assessment)}
Duration for final examination: 3 hours

Theory

Learning Outcomes:
The students will be able to:

- *Mechanism and process of origin and evolution of life on earth*
- *History of evolutionary thinking and evidences of evolution*
- *Mass extinctions, geological time scale and evolution of man*
- *Hardy Weinberg equilibrium and its applications in population genetics*

UNIT- 1

Origin of Life: Chemogeny, RNA world, Biogeny, Origin of photosynthesis, Origin of eukaryotic cell.
Theories of Evolution: Lamarckism, Darwinism, Neo-Darwinism
Causes and effects of mass extinction, K-T Extinction

UNIT- 2

Evolution and Speciation: Microevolution and Macroevolution; Species concept; Isolating mechanisms; Modes of speciation (Allopatric, Sympatric); Adaptive radiation (Galápagos finches); Sources of genetic variation and their role in evolution.

UNIT- 3

Evidence of Evolution: Types of Fossils (unaltered and altered); Geological time scale; Evolution of horse; Human evolution (Dryopithecus to Homo sapiens); molecular evidence (Universal genetic code, Molecular clock, Neutral theory)

UNIT- 4

Population Genetics:Hardy–Weinberg Law (Assumptions, limitations, applications); Genetic drift (Founder and Bottleneck effects); Role of Mutation and Migration in changing allele frequencies; Natural selection; Types of selection (Kin, Sexual, Density-dependent).

Instructions For Paper Setter and candidates

Nine questions are to be set. Question No.1 is compulsory, consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1.5 marks each. Two questions of 13 marks each are to be set from each Unit. One question is to be attempted from each Unit. In all, five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts. To qualify for a Course, the candidate must obtain a minimum of 40% marks.

ZOO-DSC-17 Maj-604: Evolutionary Biology

Practical

1. Study of fossils from models/ pictures
2. Study of homology and analogy using suitable specimens
3. Study and verification of Hardy-Weinberg Law by chi square analysis
4. Demonstration of role of natural selection and genetic drift in changing allele frequencies using simulation studies
5. Graphical representation and interpretation of height/weight data of a sample of 100 humans in relation to their age and sex.

SUGGESTED READINGS

1. Hall, B.K. and Hallgrimson, B (2007). *Evolution,* IV Edition. Jones and Barlett Publishers. ISBN No.- 9780763700669.
2. Snustad, D.P. & Simmons, M. J. (2011). *Principles of Genetics*, VII edition. John Wiley & Sons Inc. ISBN No.- 9781118129210.
3. Pierce, B.A. (2017). *Genetics: A Conceptual Approach*, VI edition. W.H. Freeman and Company. ISBN No.- 978-1-319-15391-5.
4. Rastogi, V.B. (2014). *Organic Evolution*, MedTec, Daryaganji, New Delhi. ISBN No.- 9789381714515.
5. Futuyma, D.J., Kirkpatrick, M. (2018). *Evolution*, IV edition, Oxford University Press, ISBN No.- 9781605357409.
6. Ridley, M. (2004). *Evolution,* III Edition. Blackwell publishing. ISBN No.- 9781405103459.
7. Pevsner, J. (2009). *Bioinformatics and Functional Genomics*, II Edition Wiley-Blackwell. ISBN No.- 9780470085851.

Course Code: ZOO-DSC-18 Maj-605
Course Title: Biodiversity and Conservation

Total Hours= 75 (Theory= 45 + Practical= 30) **Credits= 4 (L=3 + P=1)**
Total Marks= 100 (Theory=75 + Practical=25)
Theory Marks= 75 {67 (Theory examination) + 08 (Internal assessment)}
Practical Marks= 25 {22 (Practical examination) + 03 (Internal assessment)}
Duration for final examination: 3 hours

Theory

Learning Outcomes:
The students will be able to:

- *Understand the Importance of Biodiversity Conservation.*
- *Learn about different causes of biodiversity loss and ways to conserve them by different methods.*
- *Know about the role of Government and NGOs in Biodiversity Conservation.*
- *Various modern technologies used for conservation of Biodiversity.*

UNIT- 1

Biodiversity: Meaning, values and ethics of biodiversity, importance of biodiversity
Types of Biodiversity: Genetic, species and ecological biodiversity, Alpha, Beta and Gama biodiversity, Shannon Diversity Index, Simpson's dominance index.
Causes and Depletion of Biodiversity: Natural and Man-made.
Hotspots of biodiversity, Role of systemics in Biodiversity, Extinction of species.

UNIT- 2

Value of Biodiversity: Monetizing the value of Biodiversity, Intrinsic value, Ethical and Aesthetic values, intellectual value., Ecocentrism.
Threats to Biodiversity: Habitat Destruction, fragmentation, degradation and biodiversity loss.
Causes, patterns and consequences of biodiversity loss of major land and Aquatic systems; Invasive species, human-wildlife conflicts.

UNIT- 3

Natural Resources and Conservation: Strategies of conservation, natural reserves, classification of natural resources.
Conservation of Biodiversity: In-Situ and Ex-Situ conservation, Role of people for biodiversity conservation, Wildlife Conservation.
Efforts for Wildlife Conservation: Wildlife Protection Act (1972), Selection criteria for protection of habitats and hotspots. Project Tiger, Project Sangai, Crocodile breeding project, Gir Lion Project, Swamp dear. IUCN Categories and Red Data Book, CITES and Kyoto Agreement (1997).

UNIT- 4

Modern technology in Conservation Biology (Camera trapping, Acoustic monitoring, Remote sensing with LIDAR).
Role of NGOs in conservation, National and International movements, reproductive parameters in conservation (Breeding habitats, mating systems, inbreeding depressions, genetic bottlenecks, genetic constraints, captive breeding), Sensor technology, Thermal drones, Ecotourism.

Instructions For Paper Setter and candidates

Nine questions are to be set. Question No.1 is compulsory, consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1.5 marks each. Two questions of 13 marks each are to be set from each Unit. One question is to be attempted from each Unit. In all, five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts. To qualify for a Course, the candidate must obtain a minimum of 40% marks.

ZOO-DSC-18 Maj-605: Biodiversity and Conservation

Practical

1. Quantitative analysis of species richness, evenness and dominance.
2. Techniques for photographing animals.
3. Identification of big cats: Lion, Tiger, Cheetah, leopard, and Jaguar.
4. Demonstration of basic equipment needed in wildlife studies- Use,care and maintenance (Compass, binoculars, various types of cameras and lenses).
5. Familiarization of study of animal evidences in field: Identification of animals through pugmarks, hoof marks and scats.
6. Field trip to biodiversity hot spots/Zoo/National Park.
7. Project report: Identification of mammalian fauna, avian fauna through direct and indirect evidences seen on a field trip to a National Park/Zoo/Hotspot.

SUGGESTED READINGS

1. Banerjee, K. *Biodiversity Conservation in Managed and Protected Areas*. Agrobios, India, 2002.
2. Singh, J.S., Singh, S.P. & Gupta, S.R. *Ecology, Environmental Science and Conservation*. S. Chand Publishing, Latest Edition.
3. Verma, P.S. & Agarwal, V.K. *Environmental Biology: Principles of Ecology*. S. Chand Publishing, Latest Edition.
4. Arora, M.P. *Ecology*. Himalaya Publishing House, Latest Edition.
5. Sharma, P.D. *Ecology and Environment*. Rastogi Publications, Latest Edition.
6. Bharucha, E. *Textbook of Environmental Studies for Undergraduate Courses*. Universities Press, Latest Edition.
7. Trivedi, P.C. *Biodiversity: Conservation and Utilization*. Pointer Publishers, Latest Edition.
8. Sinclair, A.R.E., Fryxell, J.M. & Caughley, G. *Wildlife Ecology, Conservation, and Management*. 2nd Edition. Wiley-Blackwell, 2006.
9. Groom, M.J., Meffe, G.K. & Carroll, C.R. *Principles of Conservation Biology*. 3rd Edition. Sinauer Associates, 2006.
10. Primack, R.B. *Essentials of Conservation Biology*. 6th Edition. Sinauer Associates, 2014.
11. Sodhi, N.S. & Ehrlich, P.R. (Eds.). *Conservation Biology for All*. Oxford University Press, 2010.
12. Magurran, A.E. *Measuring Biological Diversity*. Blackwell Publishing, 2004.

Morris, D. *Animal Watching: A Field Guide*. Crown Publishers, 1990

Semester- V (L-300)

Course Code: ZOO-DSC-9 Maj/Min-501
Course Title: Cell Biology

Total Hours= 75 (Theory= 45 + Practical= 30) **Credits= 4 (L=3 + P=1)**
Total Marks= 100 (Theory=75 + Practical=25)
Theory Marks= 75 {67 (Theory examination) + 08 (Internal assessment)}
Practical Marks= 25 {22 (Practical examination) + 03 (Internal assessment)}
Duration for final examination: 3 hours

Theory

Learning Outcomes:
The students will be able to:

- *Acquire comprehensive knowledge of cell structure and organization.*
- *Understand the principles and applications of tissue fixation in histological studies.*
- *Develop a clear understanding of cell organelles and their ultrastructure.*
- *Explain the roles of various cell organelles in cellular metabolism and physiological processes.*

UNIT- 1

Microscopy: Principles and types of microscopes – simple microscope, compound microscope, light microscope, and phase contrast microscope.

Fixation and Staining: Principles of tissue fixation; fixatives(Carnoy's and Bouin's) and their significance;Staining procedures (single and double staining)

Overview of cells: General structure and organization of prokaryotic and eukaryotic cells. Structure of viruses and viroids.

UNIT- 2

Plasma membrane: Structure and organization of the plasma membrane with reference to the Fluid Mosaic Model.

Transport across membranes – passive transport and active transport.

Endoplasmic reticulum: Structure and functions of rough endoplasmic reticulum (RER) and smooth endoplasmic reticulum (SER).

Golgi complex: Structure and functions of the Golgi apparatus.

UNIT- 3

Ribosomes: Types, structure, and functions of ribosomes.

Lysosomes: Structure, functions, and polymorphism of lysosomes.

Mitochondria: Structure and functions of mitochondria; semi-autonomous nature of mitochondria.

UNIT- 4

Nucleus: Structure and functions of the nucleus, including nuclear envelope, nuclear pore complex, and nucleolus.

Cytoskeleton: Structure and functions of microtubules and microfilaments

Cell Division: Cell Cycle, Mitosis, Meiosis and their significance.

Instructions For Paper Setter and candidates

Nine questions are to be set. Question No.1 is compulsory, consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1.5 marks each. Two questions of 13 marks each are to be set from each Unit. One question is to be attempted from each Unit. In all, five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts. To qualify for a Course, the candidate must obtain a minimum of 40% marks.

ZOO-DSC-9 Maj/Min-501: Cell Biology

Practical

1. Study and identification of parts of a Simple/dissection Microscope and Compound microscope.
2. Laboratory preparation of Carnoy's fixative and Bouin's fixative
3. Identification and study of the following cell organelles through photographs, charts, and models: Mitochondria, Golgi complex, Endoplasmic reticulum, Ribosomes and Nucleus
4. Study of different stages of mitosis and meiosis through permanent slides/photographs/charts/models.
5. Preparation of temporary stained squash of onion root tip to study various stages of mitosis.

SUGGESTED READINGS

1. Dhami, P.S., & Dhami, J.K. (2015). ***Textbook of Zoology***. Pardeep Publications.
2. Sabharwal, A. (2021). ***Modern's Zoology***. Modern Publishers.
3. Powar, C.B. (1999). ***Cell Biology***. Himalaya Publishing House, Mumbai.
4. Singh, S., & Tomar, C.B. (2008). ***Cell Biology***. Rastogi Publications.
5. Rastogi, V.B. (2023). ***Cell Biology***. MedTech Science Press.
6. Cooper, G.M., & Hausman, R.E. (2007). ***The Cell: A Molecular Approach*** (4th ed.). Sinauer Associates Inc.
7. Karp, G. (2001). ***Cell and Molecular Biology: Concepts and Experiments***. John Wiley & Sons.
8. De Robertis, E.D.P., & De Robertis, E.M.F. (1995). ***Cell and Molecular Biology*** (8th ed.). W.B. Saunders Co., Philadelphia.

Course Code: ZOO-DSC-10 Maj-502
Course Title: Economic Zoology

Total Hours= 75 (Theory= 45 + Practical= 30) **Credits= 4 (L=3 + P=1)**
Total Marks= 100 (Theory=75 + Practical=25)
Theory Marks= 75 {67 (Theory examination) + 08 (Internal assessment)}
Practical Marks= 25 {22 (Practical examination) + 03 (Internal assessment)}
Duration for final examination: 3 hours

Theory

Learning Outcomes:
The students will be able to:

- *Understand the scope and significance of aquaculture and allied farming practices.*
- *Describe basic techniques of sericulture, lac culture, butterfly, and livestock farming.*
- *Identify important insect pests and diseases transmitted by them.*
- *Explain pest control methods and principles of Integrated Pest Management (IPM).*

UNIT- 1

Pisciculture: Composite fish culture – principle, culture of the Indian major carps (***Catla catla***, ***Labeo rohita***, ***Cirrhinus mrigala***) and exotic carps (grass carp, silver carp, common carp), Integrated fish culture – with paddy, poultry, cattle and piggery.

Aquaculture: Extensive and intensive fish culture, semi-intensive fish culture, cage culture, pen culture, capture and culture fisheries, freshwater prawn farming – principle and practices.

Pearl culture: Definition and importance; species used; basic principle of pearl formation and implantation.

UNIT- 2

Sericulture: Definition and importance; species of silkworm (mulberry and non-mulberry: Eri, Muga, Tassar); basic requirements of sericulture; overview of silk reeling.

Lac culture: Definition and importance; major lac insect species; host plants; basic rearing practices; economic importance of lac.

Butterfly farming: Definition, importance, and basic requirements

UNIT- 3

Dairy (Milch animal) farming: Definition and importance; major indigenous and exotic breeds of cow and buffalo (examples only); basic housing and nutritional management

Goat farming: Importance; major breeds (examples only); basic rearing and feeding practices, pasteurized goat milk.

Sheep farming: Importance; major breeds (examples only); basic rearing and feeding practices; types of sheep wool (fine, medium, long, and coarse).

UNIT- 4

Medically and veterinary important arthropods: Study of life cycle and economic importance of housefly, mosquito (***Aedes***, ***Anopheles***, ***Culex***), sand fly, tsetse fly, warble fly, tick *(Hyalomma)*, and mite *(Sarcoptes)*

Insect Control Methods: Physical and cultural methods; chemical control including organochlorines (DDT, BHC), organophosphates (malathion), botanical insecticides (nicotine, pyrethrum), and fumigants (aluminium phosphide); biological control using predators, parasites, and microbial agents (bacteria, viruses)

Integrated Pest Management (IPM): Concept, components, and importance of IPM.

Instructions For Paper Setter and candidates

Nine questions are to be set. Question No.1 is compulsory, consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1.5 marks each. Two questions of 13 marks each are to be set from each Unit. One question is to be attempted from each Unit. In all, five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts. To qualify for a Course, the candidate must obtain a minimum of 40% marks.

ZOO-DSC-10 Maj-502: Economic Zoology

Practical

1. Study and identification of aquatic organisms: Fish – Indian major carps (***Catla catla***, ***Labeo rohita***, ***Cirrhinus mrigala***), prawn, and oyster through specimens/models.
2. Study of life cycles and economic importance of silkworm, lac insect, butterfly, housefly, mosquito (***Aedes***, ***Anopheles***, ***Culex***), sand fly, tsetse fly, warble fly, tick *(Hyalomma)*, and mite *(Sarcoptes)*through specimens/photographs/charts/models.
3. Identification of different types of natural silk (mulberry, Eri, Muga, Tassar) through samples/photographs.
4. Identification of different types of wool (fine, medium, long, and coarse) through samples/photographs.
5. Identification of major breeds of cow, buffalo, goat, and sheep through charts/models/photographs.
6. Measurement of milk density using a lactometer.
7. Demonstration of maintenance of a fish aquarium.
8. Field visit to a fish farm or dairy farm.
9. Project report

SUGGESTED READINGS

1. Agarwal, S.C., & Johal, M.S. (1997). ***Fishery Development***. Narendra Publishing House, Delhi.
2. Atwal, A.S., & Dhaliwal, G.S. (2009). ***Agricultural Pests of South Asia and Their Management***. Kalyani Publishers.
3. Banerjee, C.C. (2018). ***A Text Book of Animal Husbandry***. Oxford & IBH Publishing Co.
4. Dent, D. (1995). ***Integrated Pest Management***. Chapman & Hall, London.
5. Gupta, R., & Amrutkar, R. (2019). ***Comprehensive Book on Goat Farming***. Jaya Publishing House.
6. Jhingran, V.G. (1991). ***Fish and Fisheries of India***. Hindustan Publishing Corporation of India, Delhi.
7. Kumar, R., & Nigam, P. (2008). ***Economic and Applied Entomology***. Emkay Publishing, New Delhi.
8. Sandhu, A.K. (2019). ***A Textbook of Indian Inland Fisheries and Aquaculture***. Mahindra Publishing House, Chandigarh.
9. Srivastava, K.P., & Dhaliwal, G.S. (2010). ***A Textbook of Applied Entomology*** (Vol. I & II). Kalyani Publishers, Ludhiana.
10. Shukla, G.S., & Upadhyay, V.B. (2011). ***Economic Zoology***. Rastogi Publications, Meerut.
11. Nigam, H.C. (2009). ***Economic Zoology***. Vishal Publishing Co., Jalandhar.
12. Srivastava, C.B.L. (2014). ***A Textbook of Applied Entomology***. Kalyani Publishers, Ludhiana.
13. Pillai, T.V.R. (2013). ***Aquaculture: Principles and Practices***. Blackwell Publishing.
14. Rath, R.K. (2011). ***Freshwater Aquaculture***. Scientific Publishers, Jodhpur.
15. Ghosh, A. (2012). ***Sericulture and Silk Industry***. Daya Publishing House, New Delhi.
16. Nair, K.K. (2007). ***Lac Culture in India***. Indian Council of Agricultural Research (ICAR), New Delhi.
17. Banerjee, G.C. (2016). ***A Textbook of Animal Husbandry***. Oxford & IBH Publishing Co.

Semester- VI (L-300)

Course Code: ZOO-DSC-14 Maj/Min -601
Course Title: Principles of Genetics

Total Hours= 75 (Theory= 45 + Practical= 30) **Credits= 4 (L=3 + P=1)**
Total Marks= 100 (Theory=75 + Practical=25)
Theory Marks= 75 {67 (Theory examination) + 08 (Internal assessment)}
Practical Marks= 25 {22 (Practical examination) + 03 (Internal assessment)}
Duration for final examination: 3 hours

Theory

Learning Outcomes:
The students will be able to:

- *Understand foundational concepts of Genetics*
- *Differentiate between Mendelian and Non-Mendelian patterns of inheritance*
- *Explain chromosomal behavior in inheritance*
- *Apply genetic principles to real-world contexts*

UNIT- 1

Basic terminology in Genetics: gene, allele, locus, genotype, phenotype, homozygous, heterozygous, pure lines, test cross, back cross

Mendel's experiments and laws of inheritance

Monohybrid and dihybrid crosses

Non-allelic gene interactions: Modified Mendelian ratios (9:7, 9:3:4, 12:3:1, 9:6:1, 15:1)

UNIT- 2

Non-Mendelian inheritance: Incomplete dominance (Snapdragon flowers), Codominance (Human blood group type AB), Multiple alleles (Eye color in *Drosophila*), Sex-linked inheritance (Haemophilia), Polygenic inheritance (Skin color in humans), Pleiotropy (Phenylketonuria), Lethal alleles (Coat color in Mice), Cytoplasmic inheritance (Kappa particles in *Paramecium*)

UNIT- 3

Chromosomal theory of inheritance
Linkage: Characteristics, Types (Complete & Incomplete) and Significance
Crossing over: Types (Single & Double), Mechanism (Synapsis, Synaptonemal complex, Tetrad formation, Crossing over & Terminalization) and Significance

UNIT- 4

Mutations: Spontaneous and induced mutations, Physical mutagens (Ionizing & Non-ionizing radiations), Chemical mutagens (Base analogs & Alkylating agents)
Human genetic disorders: Causes and symptoms of Cystic fibrosis**,** Sickle cell anemia,Down's syndrome,Turner's syndrome, Klinefelter syndrome
Population genetics: Gene frequencies, Hardy-Weinberg Principle.

Instructions For Paper Setter and candidates

Nine questions are to be set. Question No.1 is compulsory, consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1.5 marks each. Two questions of 13 marks each are to be set from each Unit. One question is to be attempted from each Unit. In all, five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts. To qualify for a Course, the candidate must obtain a minimum of 40% marks.

ZOO-DSC-14 Maj/Min-601: Principles of Genetics

Practical

1. Demonstration of Monohybrid cross and Dihybrid cross, and verification of Mendel's law of segregation and law of independent assortment using colored beads.
2. Identification of different stages of meiosis in onion buds/Study of stages of meiosis through permanent slides.
3. Preparation of sex-chromatin/Barr body from buccal smear.
4. Analysis of human karyotype photographs and identification of chromosomal disorders.
5. Analysis of fingerprint and palm print patterns through dermatoglyphics.
6. Demonstration of Hardy-Weinberg principle using colored beads.
7. Project report

SUGGESTED READINGS

1. Gardner, Eldon John, Michael J. Simmons, and D. Peter Snustad. *Principles of Genetics*. 8th ed., Wiley, 1991.
2. Pierce, Benjamin A. *Genetics: A Conceptual Approach*. 7th ed., W. H. Freeman & Co., 2020. Snustad, D. Peter, and Michael J. Simmons. *Principles of Genetics*. 7th ed., Wiley, 2016.
3. Klug, William S., Michael Cummings, Charlotte A. Spencer, Michael A. Palladino, and Darrell Killian. *Essentials of Genetics*. 10th ed., Pearson Education, 2020.
4. Dhami, P.S. and Dhami, J.K. *A Textbook of Zoology (Developmental Biology, Genetics and Cell Biology)* Pradeep Publications, 2022.
5. Srivastava, H.N. *Cell Biology and Genetics.* Pradeep Publications, 2022.
6. Gupta, P.K. *Genetics: A Textbook for University Students*. 5th ed., Rastogi Publications, 2019.
7. Rastogi, Veer Bala, and Nirvika Rastogi. *Fundamentals of Genetics*. 1st ed., MedTech Science Press, 2022.
8. Rathoure, Ashok Kumar, and Meena Srivastava. *Cell Biology and Genetics*. Daya Publishing House, 2020.
9. Randhawa, S. S. *Text Book of Genetics*. S. Vikas & Company (PV Books), 2013.
10. Singh, B.D. *Fundamentals of Genetics*. 6th ed., MedTech Science Press, 2025.

Course Code: ZOO-DSC-15 Maj/Min-602
Course Title: Molecular Biology

Total Hours= 75 (Theory= 45 + Practical= 30) **Credits= 4 (L=3 + P=1)**
Total Marks= 100 (Theory=75 + Practical=25)
Theory Marks= 75 {67 (Theory examination) + 08 (Internal assessment)}
Practical Marks= 25 {22 (Practical examination) + 03 (Internal assessment)}
Duration for final examination: 3 hours

Theory

Learning Outcomes:
The students will be able to:

- *Understand the fundamental concepts and scope of Molecular Biology.*
- *Explain the structure and functionof DNA and RNA.*
- *Describe molecular mechanisms such as replication, transcription, translation, and gene regulation.*
- *Understand how gene expression and mutations affect cells and organisms.*

UNIT- 1

Nucleic acids as genetic material:Griffith's experiment; Avery, MacLeod and McCarty's experiment; Hershey–Chase experiment

Structure and function of nucleic acids: nucleosides, nucleotides, purines, and pyrimidines; biological importance of nucleotides

DNA structure: Watson and Crick model; A, B, and Z forms; supercoiled and relaxed DNA; Denaturation and renaturation; melting temperature (Tm); hyperchromic effect.

RNA: Structure and types of RNA

Central Dogma of molecular biology

UNIT- 2

Genome organization: structure of a gene, coding and regulatory sequences, introns, exons; Nucleosome concept and packaging of DNA into higher-order structures.

DNA Replication: Meselson–Stahl experiment (proof of semiconservative DNA replication); Mechanism of DNA replication in prokaryotes and eukaryotes.

UNIT- 3

Transcription in prokaryotes and eukaryotes: promoter, initiation, elongation, and termination.

Components of protein synthesis machinery: mRNA, tRNA, ribosomes.

Translation: aminoacyl-tRNA synthetases, charging of tRNA, ribosome assembly.

Mechanism of protein synthesis in prokaryotes and eukaryotes: initiation, elongation, and termination.

UNIT- 4

Regulation of gene expression in prokaryotes: lac operon (negative and positive regulation).
Regulation of gene expression in eukaryotes.
Mutations: Molecular basis of mutations; types of mutations (missense, nonsense, silent, point, and frameshift).

Instructions For Paper Setter and candidates

Nine questions are to be set. Question No.1 is compulsory, consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1.5 marks each. Two questions of 13 marks each are to be set from each Unit. One question is to be attempted from each Unit. In all, five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts. To qualify for a Course, the candidate must obtain a minimum of 40% marks.

ZOO-DSC-15 Maj/Min-602: Molecular Biology

Practical

1. Study of DNA and RNA structures through models.
2. Demonstration of paper chromatography for the separation of amino acids.
3. **Introduction to Molecular Biology Apparatus (Demonstration/Video):**
   - Micropipettes
   - Microcentrifuge
   - Agarose Gel Electrophoresis **Apparatus**
   - Polyacrylamide Gel Electrophoresis **Apparatus**
   - **Spectrophotometer**
   - **PCR (Polymerase Chain Reaction) Apparatus**

4. Demonstration of Agarose Gel Electrophoresis to separate DNA fragments (video/model if lab setup unavailable).

5. Demonstration ofSDS-PAGE (Polyacrylamide Gel Electrophoresis) to separate proteins (video/model if lab setup unavailable).

SUGGESTED READINGS

1. Watson, J.D., Baker, T.A., Bell, S.P., Gann, A., Levine, M., & Losick, R. (2014). ***Molecular Biology of the Gene***, 6th Edition. Cold Spring Harbor Laboratory Press, Pearson.
2. Alberts, B., Johnson, A.D., Lewis, J., Raff, M., Roberts, K., & Walter, P. (2014). ***Molecular Biology of the Cell***, 6th Edition. Garland Science.
3. Krebs, J., Goldstein, E., & Kilpatrick, S. (2013). ***Lewin's Essential Genes***, 3rd Edition. Jones and Bartlett Learning.
4. Rastogi, S.C. (2012). ***Cell and Molecular Biology***. New Age International.

5. Karp, G. (2013). ***Cell and Molecular Biology: Concepts and Experiments***, 7th Edition. Wiley.
6. Lewin, B. (2008). ***Gene IX***. Peterson Publications / Panima.
7. Malacinski, G.M., & Freifelder, D. (1998). ***Essentials of Molecular Biology***, 3rd Edition. Jones & Bartlett Publishers.
8. Verma, P.S. (2001). ***Cell Biology, Genetics, Molecular Biology, Evolution and Ecology***.
9. Chandar, N., & Viselli, S. (2010). ***Cell and Molecular Biology: Lippincott's Illustrated Reviews***. Wolters Kluwer Health / Lippincott Williams & Wilkins.
10. Vidyavathi, N. (2013). ***Molecular Biology***.
11. Kumar, P. (2024). ***Biophysics and Molecular Biology: Tools and Techniques***, 5th Edition.

Primrose, S.B. (2001). ***Molecular Biotechnology***.

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